

# CyberLab Reference Guide

Field quick-reference for the kit  
capability → tool → command → every option explained

Generated from the repository. Markdown is the source of truth.  
[github.com/project-cyberlab/cyberlab-reference-guide](https://github.com/project-cyberlab/cyberlab-reference-guide)

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#### Alphabetical Tool Index

# CyberLab Reference Guide

A field quick-reference for the kit. You open it mid-mission or mid-exercise when you need a **capability** and do not remember which tool provides it — it routes you to the tool, the exact command, and **every option, explained**.

## Start here

| You are...                                                  | Go to                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------------|
| Facing a problem — " <i>I need to carve deleted files</i> " | <a href="#">Capability Index</a> — 46 capabilities                |
| Asking what the kit even has                                | <a href="#">Kit Tool List</a> — 1,006 tools, VM → category → tool |
| Wanting a finished page                                     | <a href="#">fls</a> · <a href="#">vol</a> · <a href="#">yara</a>  |
| Carrying it into the field                                  | <a href="#">PDF</a> — 372 pages                                   |
| Wanting to know how pages are built and why                 | <a href="#">The Format</a>                                        |

## Where it stands

|                             |                                                      |
|-----------------------------|------------------------------------------------------|
| Kit inventory               | <b>1,006</b> tools, 5 platforms, 54 categories       |
| Captured from a real binary | <b>945</b> commands                                  |
| Capabilities                | <b>46</b> , across 10 investigation phases           |
| Tool pages                  | <b>135</b> , every one with a complete options table |
| Flags with usage guidance   | <b>153</b> across 20 reviewed tools                  |
| Mined invocations           | <b>360</b> real commands from cyberlab               |
| Citations checked           | <b>255</b> URLs — 245 live                           |
| Linter                      | <b>0 errors</b> , 269 warnings (tracked debt)        |

## What makes this different

Every DFIR quick-reference in the field either organises by tool (so you must already know the tool) or skips flag documentation. The most complete one in public circulation states outright that it "*assumes operator familiarity*."

This guide inverts that:

- **Capability-first.** You search by what you need to *do*, not what the binary is called.
- **Options completely listed out** — every flag, with *what it does* and *when you'd use it*.
- **Tools attached to commands**, each tagged with the VM that carries it, so you know whether you can run it on the box in front of you.

## Scope is binding

Only tools in `catalog/kit-tools.json` may be documented. It is generated from the kit VMs' own upstream manifests:

| Platform         | Tools | Source of truth                                                                  |
|------------------|-------|----------------------------------------------------------------------------------|
| Kali Linux       | 403   | <code>kali-meta</code> <code>debian/control</code> ( <code>kali-tools-*</code> ) |
| REMnux           | 268   | <code>REMnux/docs</code> <code>discover-the-tools</code>                         |
| SIFT Workstation | 162   | <code>teamdfir/sift-saltstack</code>                                             |
| FLARE-VM         | 137   | <code>mandiant/flare-vm</code> <code>config.xml</code>                           |
| Security Onion   | 36    | kit tracker CSV                                                                  |

## Every flag is real

Options are **captured from the running binary** ( `--help` / `man`, inside the kit container), stored raw under `capture/`, and enforced by `scripts/lint.py`: a documented flag absent from a capture is a **build error**.

This answers a measured failure in the predecessor project, where fabricated CLI flags reached ~44 of 61 modules. Rationale in `docs/FORMAT.md`.

Citations are liveness-checked too — see `capture/SOURCES.md`.

## Rebuild everything

```
python scripts/build_kit_list.py      # kit inventory from upstream manifests
python scripts/candidates.py          # candidate command names
# scripts/probe_container.sh runs inside the kit container on rick
python scripts/merge_coverage.py      # merge probe results -> coverage report
python scripts/mine_cyberlab.py       # candidate invocations (read-only)
python scripts/generate_pages.py      # tool pages (never overwrites hand-written)
python scripts/build_index.py         # capability index
python scripts/lint.py                # gate: 0 errors required
python scripts/validate_sources.py    # citation liveness
python scripts/build_pdf.py           # print render
```

## Relationship to cyberlab

Separate repository, deliberately. The `cyberlab training lab` is **on hold** and receives no changes from this work; it is read-only input for mining.

## Layout

|            |                                                                |
|------------|----------------------------------------------------------------|
| catalog/   | kit tool list – the binding scope (generated)                  |
| reference/ | the guide: INDEX.md + one page per tool, grouped by capability |
| capture/   | raw --help/man output, committed as evidence                   |
| docs/      | FORMAT.md (how pages are built), PLAN.md (roadmap)             |
| scripts/   | build, probe, mine, generate, lint, validate, render           |



# Capability Index

---

**Start here when you have a problem.** Find what you need to *do*; it points you at the tool and its page.

Phases follow [NIST SP 800-86](#) (collect → examine → analyse → report), ordered the way an investigation actually runs.

Legend: **bold** = captured from a real binary, page can be written · plain = in the kit but not captured yet (needs a booted VM or has no CLI).

## Phases

---

- [Acquire & preserve](#)
- [Examine the filesystem](#)
- [Build the timeline](#)
- [Windows artifacts](#)
- [Memory forensics](#)
- [Network analysis](#)
- [Malware triage — static](#)
- [Malware triage — documents](#)
- [Reverse engineering](#)
- [Decode & deobfuscate](#)
- [Report & support](#)

## Acquire & preserve

---

### Image a disk, volume or device

[dc3dd](#), [dcfldd](#), [dd](#), [ewfacquire](#), [affconvert](#), [guymager](#)

### Verify evidence integrity with hashes

[rahash2](#), [ssdeep](#), [sha256sum](#), [md5sum](#), [sigtool](#), [hashdeep](#), [md5deep](#), [sha256deep](#)

### Inspect or mount a forensic image container

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [affcat](#), [img\\_stat](#), [ntfs-3g](#), [vshadowinfo](#), [bdeinfo](#)

### Capture live network traffic

[dumpcap](#), [tshark](#), [tcpdump](#)

## Examine the filesystem

---

### See the partition and volume layout

[mmls](#), [fsstat](#), [img\\_stat](#), [testdisk](#)

List files and directories, including deleted ones

[fls](#), [ffind](#), [ils](#), [tsk\\_recover](#)

Recover deleted or lost files

[tsk\\_recover](#), [icat](#), [photorec](#), [testdisk](#), [blkls](#), [ext4magic](#), [extundelete](#), [ext3grep](#)

Carve files out of unstructured data

[foremost](#), [scalpel](#), [binwalk](#), [bulk\\_extractor](#), [tcpxtract](#), [magicrescue](#)

Inspect metadata for one file or inode

[istat](#), [ils](#), [file](#), [stat](#), [exiftool](#), [trid](#), [magika](#)

Search raw data for a pattern

[rafind2](#), [strings](#), [grep](#), [xxd](#), [bulk\\_extractor](#), [lightgrep](#)

## Build the timeline

---

Build a super-timeline from many artifact sources

[log2timeline.py](#), [psort.py](#), [pinfo.py](#), [psteal.py](#)

Build a filesystem MAC-time timeline

[fls](#), [mactime](#), [tsk\\_gettimes](#)

## Windows artifacts

---

Parse registry hives

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#), [regipy-diff](#), [RECcmd](#), [hivexget](#), [hivexml](#)

Parse Windows event logs

[evtxexport](#), [evtxinfo](#), [EvtxECmd](#), [chainsaw](#), [hayabusa](#), [evtx\\_dump](#)

Parse ESE / SRUM / Amcache databases

[esedbexport](#), [esedbinfo](#), [SrumsECmd](#), [AmcacheParser](#)

Parse execution and persistence artifacts

[PECmd](#), [AppCompatCacheParser](#), [MFTECmd](#), [AmcacheParser](#), [analyzeMFT](#)

## Parse mail stores

[pffexport](#), [pffinfo](#), [readpst](#)

## Memory forensics

---

### Analyse a memory image

[vol](#), [volatility3](#), [volshell](#), [vol.py](#), [rekall](#)

### Recover encryption keys from memory

[aeskeyfind](#), [rsakeyfind](#), [bulk\\_extractor](#)

## Network analysis

---

### Read and filter packet captures

[tshark](#), [capinfos](#), [ngrep](#), [tcpflow](#)

### Split, merge or repair capture files

[editcap](#), [mergcap](#), [reordercap](#)

### Extract files and payloads from traffic

[tcpxtract](#), [tcpflow](#), [foremost](#)

### Detect intrusions in traffic

[suricata](#), [suricatasc](#), [zeek](#), [zeek-cut](#), [rita](#)

### Probe or scan hosts and services

[nmap](#), [nping](#), [arp-scan](#), [ncat](#), [netdiscover](#)

### Simulate network services for detonation

[inetsim](#), [fakedns](#), [fakenet](#)

## Malware triage — static

---

### Identify what a file actually is

[file](#), [die](#), [diec](#), [trid](#), [magika](#), [exiftool](#)

## Scan with signatures for known-bad

[yara](#), [yarak](#), [clamscan](#), [freshclam](#), [sigtool](#), [clamscan](#)

## Identify capabilities in a binary

[capa](#), [floss](#)

## Extract strings, including obfuscated ones

[floss](#), [strings](#), [base64dump.py](#), [numbers-to-string.py](#)

## Inspect PE / ELF structure

[rabin2](#), [readelf](#), [readelf.py](#), [objdump](#), [pescan](#), [pecheck.py](#), [peframe](#)

## Detect and reverse packing

[upx](#), [die](#), [diec](#), [binwalk](#), [7za](#), [unzip](#)

## Compare or cluster samples

[ssdeep](#), [radiff2](#), [bytehist](#)

# Malware triage — documents

---

## Analyse Office documents and macros

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#), [oleobj](#), [mraptor](#), [msodde](#), [pcodedmp](#), [pyxswf](#), [xlmdEOFfuscator](#), [runxldr2.py](#), [vipermonkey](#)

## Analyse RTF documents and embedded objects

[rtfobj](#), [oleobj](#), [rtfdump.py](#)

## Analyse PDFs

[pdfid](#), [pdfid.py](#), [pdf-parser](#), [pdf-parser.py](#), [peepdf](#), [qpdf](#)

## Analyse other container formats

[7za](#), [unzip](#), [msOFFcrypto-tool](#), [onenoteanalyzer](#)

# Reverse engineering

---

## Disassemble and explore a binary

[r2](#), [rabin2](#), [rasm2](#), [objdump](#), [vivbin](#), [vdbbin](#), [rizin](#), [ghidra](#), [cutter](#)

## Diff two binaries

[radiff2](#), [bindiff](#)

## Emulate or instrument execution

[frida](#), [frida-trace](#), [frida-ps](#), [frida-discover](#), [frida-kill](#), [frida-ls-devices](#), [speakeasy](#), [qiling](#), [scdbg](#), [unicorn](#)

## Analyse shellcode

[rasm2](#), [xortool](#), [scdbg](#), [shellcode2exe](#)

## Decode & deobfuscate

---

### Decode, decrypt or transform encoded data

[cyberchef](#), [base64dump.py](#), [rax2](#), [xxd](#), [openssl](#), [numbers-to-string.py](#)

### Break simple obfuscation

[xortool](#), [floss](#), [xlmdeobfuscator](#), [de4dot](#)

### Crack passwords and hashes

[hashcat](#), [john](#), [hydra](#), [unshadow](#), [zip2john](#), [rar2john](#), [fcrackzip](#), [hashid](#)

### Find hidden data

[binwalk](#), [ssdeep](#), [steghide](#), [stegosuite](#)

## Report & support

---

### Fetch and verify external references

[curl](#), [wget](#)

### Inspect files by hand

[xxd](#), [ezhexviewer](#), [less](#), [hexdump](#)

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## Coverage of this index

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- **46 capabilities** across 11 phases
- **147 captured tools** are reachable through a capability
- **5 named kit tools map to no capability** (below)
- **2 capabilities have no captured tool** behind them yet

## Capabilities with nothing captured yet

These are real needs the kit may cover with GUI or Windows-only tools that a Linux container cannot capture.

- Windows artifacts / Parse mail stores
- Network analysis / Detect intrusions in traffic

## Named kit tools not yet mapped to a capability

These are named in the kit manifests, captured from a real binary, and reach no capability in this index. Each is a taxonomy gap to close or an explicit out-of-scope decision. Listed rather than silently dropped, because a missing capability is otherwise invisible.

`aircrack-ng`, `cabextract`, `ccrypt`, `cryptsetup`, `dislocker`

## Container-provided commands (not named in the kit manifests)

Discovered by walking the container's `PATH`. They ship in a kit image but no kit manifest names them, so the great majority is OS plumbing rather than investigative tooling. Kept for completeness only.

► 807 container-provided commands

# The Format

How every page in this guide is built, and why. Derived from studying the reference works that already do this well, then fixing what they each get wrong for our use case: **a field guide you open mid-mission when you need a capability and do not remember the tool.**

## 1. What the research said

| Source                                                         | What it does well                                                                                              | What it does not give us                                                               |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <a href="#">tldr-pages style guide</a>                         | Curated examples ( $\leq 8$ ), imperative mood, <code>{{placeholder}}</code> syntax, machine-linted            | Deliberately <b>incomplete</b> — omits most options by design. No capability index.    |
| <a href="#">POSIX Utility Conventions ch.12 + man-pages(7)</a> | Unambiguous synopsis grammar: <code>[]</code> optional, <code>\ </code> exclusive, <code>...</code> repeatable | Reference-shaped, not task-shaped. You must already know the tool's name.              |
| <a href="#">SANS DFIR cheat sheets</a>                         | Organised by <b>investigative task</b> — the right primary axis                                                | Print-first, not greppable, not exhaustive on flags                                    |
| <a href="#">Jai Minton DFIR cheatsheet</a>                     | 200+ scannable anchors; artifact paths and ATT&CK IDs as headings                                              | <i>"Most entries lack detailed flag documentation... assumes operator familiarity"</i> |
| <a href="#">navi / cheat.sh</a>                                | Machine-parseable, taggable, directly executable entries                                                       | No explanation layer at all                                                            |
| <a href="#">Progressive disclosure (Nielsen, 1995)</a>         | Layer information: essentials first, depth on demand                                                           | —                                                                                      |

**The gap we fill:** every existing DFIR quick-reference assumes you already know the tool and its flags. None of them explains the options. That is exactly the layer this guide exists to provide, and it is why "completely listed out" is a hard requirement rather than a stretch goal.

## 2. Two entry points, one dataset

You arrive at this guide in one of two states:

1. **"I have a problem."** — *I need to recover deleted files from this image.* → **Capability Index:** capability → tool → command → options. This is the primary path and the reason the guide exists.
2. **"What do I have?"** — *Is there anything in the kit for Go binaries?* → **Kit Index:** VM → category → tool. This is `catalog/KIT-TOOLS.md`, generated from upstream manifests.

Both views are projections of the same JSON, so they can never disagree.

## 3. Scope is binding

A tool may appear in this guide **only** if it is in `catalog/kit-tools.json`, which is generated from the kit's own VM manifests. If it is not in the kit, it does not get a page — no matter how good the tool is.

Every command entry is tagged with the **VM(s) that carry it**, so in the field you instantly know whether you can actually run it on the box in front of you.

---

## 4. The page format

---

Every tool page has these sections, in this order. Sections are layered by progressive disclosure: a reader in a hurry stops after §3; a reader who needs precision continues to §4.

### Header

```
# <tool>
**Kit:** REMnux · SIFT    **Category:** Filesystem analysis    **Version:** 4.11.1
**Docs:** <upstream URL>    **Verified:** 2026-07-29 from `cyberlab-aio:v1`
```

The version and the verification date are not decoration — they are the claim that the options below were read off the real binary on a known date.

### 1. Purpose — one line, imperative

List files and directory names in a disk image, including deleted entries.

### 2. Synopsis — POSIX notation, verbatim from the tool

```
fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/]
    [-o imgoffset] [-z ZONE] [-s seconds] image [images] [inode]
```

### 3. Common invocations — ≤8, task-titled, copy-paste ready

Each example is titled by **the task it accomplishes**, not by the flag it uses, because the reader is searching by intent. Placeholders use tldr syntax so they are visually obvious and machine-checkable:

```
# List deleted files in a partition, recursively, with full paths
fls -rpd -o {{2048}} {{path/to/image.dd}}
```

### 4. Options — COMPLETE, one row per flag

Every flag the binary accepts. Not the useful ones — **all** of them.



| Flag            | Argument | What it does                 | When you would use it                         |
|-----------------|----------|------------------------------|-----------------------------------------------|
| <code>-d</code> | —        | Display deleted entries only | Narrowing a recovery hunt to what was removed |

The fourth column is the one that makes this a guide rather than a man page dump. It answers *why would I reach for this*, which is what you actually need under time pressure.

## 5. Gotchas

Real operational traps, each one earned. Examples already known on this project: `timeout-guard` `vol`, `binwalk`, `clamscan` and `radare2` in any harness — a synthetic memory image once pinned `vol` at 99% CPU indefinitely.

## 6. See also

Sibling tools for the same capability, so a dead end reroutes instead of stalling.

## 5. Verification: why the options can be trusted

The predecessor project shipped **fabricated CLI flags in roughly 44 of its 61 modules** — invented options for `yara`, `clamscan`, `PE-bear`, `scdbg` and others — because content was generated from a language model rather than read off the tools. That failure is the reason this guide exists in its current form.

So the rule here is mechanical, not aspirational:

1. Run the tool's `--help` / `-h` / man page **inside the kit container or VM**.
2. Store the raw output under `capture/<tool>.help.txt`, committed.
3. Generate/check the options table **against that capture**.
4. `scripts/lint.py` fails the build if a documented flag is absent from the capture, or if a captured flag is missing from the page.

A flag that cannot be produced from a real binary cannot ship. This is the same discipline `tldr-lint` applies to formatting, applied instead to truth.

**Worked proof.** The captured `fls` usage line advertises `[-adDFlhpruvV]`, but the body of the same output documents `-P`, `-B`, `-S` and `-k` as well. Anyone transcribing the synopsis — or working from memory — silently loses four real options. Only a capture catches that.

## 6. Sources and citation validation

Two different claims need two different kinds of evidence, and they must not be confused:

- **What a flag is** → the binary. Captured, committed, linter-enforced (§5). No online source overrides a real `--help`.
- **What a tool is for, and how practitioners actually use it** → authoritative upstream documentation and reputable DFIR publishers.

Acceptable sources for the second kind, in preference order: the project's own docs/repo (sleuthkit.org, volatilityfoundation.org, docs.remnux.org, ericzimmerman.github.io, Microsoft Learn), then recognised practitioner sources (SANS, thedfirreport.com, Red Canary, Mandiant/Unit42), then MITRE ATT&CK for technique mapping.

**Every cited URL is liveness-checked before it ships.** The predecessor project shipped hallucinated citations that looked plausible and 404'd; it already has a hardened, allowlisted, deadline-guarded fetcher (`cyberlab_retrieval.py`, with `verify_url`) built precisely for this, and this guide reuses that discipline. A dead or unreachable citation fails the build rather than degrading quietly.

Version skew is a real hazard: upstream docs describe the *current* release, while the kit runs pinned versions (TSK 4.11.1 in the container vs 4.15.0 in the tracker, for example). Where docs and capture disagree, **the capture wins** and the discrepancy is worth a line in Gotchas.

## 7. Output

---

- **Markdown** is the source of truth: greppable, diffable, reviewable, renders on GitHub.
- **PDF/print render** is generated from the markdown for field use. It is a build artifact, never hand-edited, so the two can never drift.

# Implementation Plan

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Goal: a field quick-reference covering the kit's tools, capability-indexed, with every command's options completely listed and explained, verified against real binaries.

Status legend:  done ·  next ·  pending

---

## Phase 0 — Foundations

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-  **Scope fixed.** `catalog/kit-tools.json` + `KIT-TOOLS.md` : 1,006 tools, 5 platforms, 54 categories, each from an upstream machine-readable manifest.
-  **Format designed** from six reference works — `docs/FORMAT.md`.
-  **Verification proven.** `cyberlab-aio:v1` on rick has `fls`, `clamscan`, `binwalk`, `yara`, `vol`, `tshark`, `exiftool`, `log2timeline.py`.
-  **Worked example** — `reference/examine-the-filesystem/fls.md`, 22 options from a real capture.

## Phase 1 — Capability taxonomy

---

The primary index. Draft ~40–60 capabilities phrased as **problems**, not tool names, e.g. "*Recover deleted files from an image*", "*Find what persisted on a host*", "*Decode an obfuscated payload*", "*Prove lateral movement*".

- Seed from the categories already carried by upstream: REMnux's 11 capability categories, Kali's 12 `kali-tools-*` groups, Security Onion's 9, and SANS investigative-task ordering.
- Map every one of the 1,006 kit tools to  $\geq 1$  capability. Tools that map to none are reported, not silently dropped — an unmapped tool is either a taxonomy gap or genuinely out of scope, and both need a decision.
- **Deliverable:** `reference/INDEX.md` — capability  $\rightarrow$  tools  $\rightarrow$  page links.

**Open question for review:** should the capability taxonomy follow an existing public framework (NIST/DFIR phases, MITRE ATT&CK tactics) for familiarity, or a kit-native one tuned to what we actually carry? Recommendation: kit-native primary, with ATT&CK IDs as secondary tags where they genuinely apply — forcing forensic tools into attacker tactics distorts both.

## Phase 1b — Mine cyberlab as a source

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The cyberlab training lab already contains a large body of commands and prior exhaustive research. It is **read-only input** here — cyberlab is on hold and receives no changes.

- `scripts/mine_cyberlab.py` : extract every fenced command block from the 61 modules, plus `docs/CURRICULUM_ROADMAP.md`, `docs/TEST_REPORT.md` and `docs/CONTENT_QUALITY_AUDIT.md`, and the `catalog/` research.
- Produce `capture/cyberlab-candidates.json` : tool  $\rightarrow$  observed invocations, with the module each came from.
- These become **candidate invocations** for the "Common invocations" section — real workflows someone already thought through, which is exactly the part that is expensive to invent.

**Mine the cleaned branch, not `main`.** The content-quality audit found fabricated CLI flags in ~44 of 61 modules, concentrated in the generated "enrichment tail". Branch `audit/content-quality-p0-p1` (PR #1) already removed that tail — ~11,000 lines — so it is the higher-signal source. Modules 57 (acquisition) and 58 (eventlog) were rated the cleanest and are worth reading first.

**Candidates are never trusted.** Every mined invocation still has to pass the Phase 3 linter against a real capture. Mining saves the *thinking*, not the *verification* — a mined command with an invented flag must fail exactly as loudly as one written from scratch.

## Phase 2 — Capture pipeline

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`scripts/capture_help.py` : for each in-scope tool, run `--help / -h / man` in the right container, store raw output under `capture/<tool>.help.txt`.

- Containers: `cyberlab-aio:v1` and `dfir-aio:v4` on rick (Docker active, 830 GB free).
- **Always timeout-guard** — `vol` once pinned 99% CPU indefinitely on a synthetic image. Hard per-tool timeout, failure recorded not retried blindly.
- Emit a coverage report: which kit tools are covered by a container, which need a booted VM (Kali, FLARE-VM Windows guests), which are GUI-only and have no CLI surface to document.

**Reality check:** the 1,006-tool list includes GUI-only tools (Autopsy, Wireshark GUI, PE-bear) and appliance services (Elasticsearch, Kibana) with no meaningful command surface. Expect the genuinely CLI-documentable set to be substantially smaller. Phase 2's report turns that from a guess into a number before any page is written.

## Phase 3 — Page generation + linter

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- Generate a page skeleton per tool from its capture: header, synopsis, complete options table. Mechanical, deterministic — no model invention.
- Human/model adds only the *judgement* columns: purpose, "when you'd use it", invocations, gotchas.
- `scripts/lint.py` gates the build:
- every documented flag exists in the capture;
- every captured flag appears in the page (**completeness**, the hard requirement);
- every tool is in `kit-tools.json`;
- format rules hold (imperative mood, ≤8 examples, placeholder syntax).

## Phase 4 — Print render

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Markdown → PDF for field carry. Build artifact only, never hand-edited. Per-capability booklets rather than one 1,000-page brick.

## Phase 5 — Prioritised content ► (20 of 135 tools reviewed)

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Do not attempt all tools at once. Order by field value:

1. Core forensics: TSK suite, Volatility, Plaso, RegRipper, bulk\_extractor

2. Network: Zeek, Suricata, tshark, tcpdump
  3. Malware triage: YARA, ClamAV, capa, FLOSS, Detect-It-Easy
  4. Live response / hunting: Velociraptor, osquery, Hayabusa, Chainsaw
  5. Offensive-side as needed for exercises
- 

## Risks

| Risk                                                                                                | Mitigation                                                           |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Fabricated flags (killed the predecessor project)                                                   | Capture-or-it-does-not-ship, enforced by linter                      |
| Scale — 1,006 tools is a lot of pages                                                               | Phase 2 coverage report right-sizes it; Phase 5 prioritises          |
| Kit VMs powered down ( <code>kali</code> , <code>range-linux-web</code> , <code>redinfra01</code> ) | Containers cover most Linux tooling; boot VMs only for the remainder |
| Version drift between capture and deployed kit                                                      | Version + capture date in every page header                          |
| GUI-only tools with no CLI                                                                          | Identify in Phase 2; document workflow, not flags                    |

# Kit Tool List

The binding scope for the quick-reference guide. A tool absent from this list is **not in the kit** and must not be documented in the guide.

Every entry is derived from an upstream machine-readable manifest; none is written from memory. Sources are listed at the end.

## Contents

- [REMnux](#) — 268 tools, 31 categories
- [Kali Linux](#) — 403 tools, 12 categories
- [FLARE-VM](#) — 137 tools, 1 categories
- [SIFT Workstation](#) — 162 tools, 1 categories
- [Security Onion](#) — 36 tools, 9 categories

**Total: 1006 tools across 5 platforms.**

## REMnux

### Analyze Documents / Email Messages

| Tool          | Command(s)               | Purpose                                                                        |
|---------------|--------------------------|--------------------------------------------------------------------------------|
| emldump.py    | —                        | Parse and analyze EML files.                                                   |
| mail-parser   | —                        | Parse raw SMTP and .MSG email messages and generate a parsed object from them. |
| msg-extractor | <code>extract_msg</code> | Extract emails and attachments from MSG files.                                 |
| msgconvert    | —                        | Convert MSG files to MBOX files.                                               |

### Analyze Documents / General

| Tool          | Command(s)             | Purpose                                                                                |
|---------------|------------------------|----------------------------------------------------------------------------------------|
| base64dump.py | —                      | Locate and decode strings encoded in Base64 and other common encodings.                |
| Tesseract OCR | <code>tesseract</code> | Examine images to identify and extract text using optical character recognition (OCR). |

### Analyze Documents / Microsoft Office

| Tool       | Command(s)                                                                                                       | Purpose                                                      |
|------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| EvilClippy | —                                                                                                                | Modify aspects of Microsoft Office documents.                |
| Hachoir    | <code>hachoir-grep</code> , <code>hachoir-metadata</code> , <code>hachoir-strip</code> , <code>hachoir-wx</code> | View, edit, and carve contents of various binary file types. |
| libolecf   | <code>olecfexport</code> , <code>olecfinfo</code> , <code>olecfmount</code> , <code>etc</code>                   | Microsoft Office OLE2 compound documents.                    |

| Tool                 | Command(s)                                                                                                                                                                                                                                                                                             | Purpose                                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| msoffcrypto-crack.py | —                                                                                                                                                                                                                                                                                                      | Recover the password of an encrypted Microsoft Office document.                                                 |
| msoffcrypto-tool     | —                                                                                                                                                                                                                                                                                                      | Decrypt a Microsoft Office file with password, intermediate key, or private key which generated its escrow key. |
| msoffice-crypt       | —                                                                                                                                                                                                                                                                                                      | Encrypt and decrypt OOXML Microsoft Office documents.                                                           |
| oledump.py           | —                                                                                                                                                                                                                                                                                                      | Analyze OLE2 Structured Storage files.                                                                          |
| olefile              | —                                                                                                                                                                                                                                                                                                      | Python package to parse, read and write MS OLE2 files.                                                          |
| oletools             | <code>mraptor</code> , <code>msodde</code> , <code>olebrowse</code> , <code>oledir</code> , <code>oleid</code> , <code>olemap</code> , <code>olemeta</code> , <code>oleobj</code> , <code>oletimes</code> , <code>olevba</code> , <code>pyxswf</code> , <code>rtfobj</code> , <code>ezhexviewer</code> | Microsoft Office OLE2 compound documents.                                                                       |
| onedump.py           | —                                                                                                                                                                                                                                                                                                      | Extract and analyze embedded files from OneNote documents.                                                      |
| pcode2code           | —                                                                                                                                                                                                                                                                                                      | Decompile VBA macro p-code from Microsoft Office documents.                                                     |
| pcodedmp             | —                                                                                                                                                                                                                                                                                                      | Disassemble VBA p-code.                                                                                         |
| rtfdump.py           | —                                                                                                                                                                                                                                                                                                      | Analyze a suspicious RTF file.                                                                                  |
| SSView               | <code>ssview</code>                                                                                                                                                                                                                                                                                    | Analyze OLE2 Structured Storage files.                                                                          |
| XLMMacroDeobfuscator | <code>xlmdeobfuscator</code> , <code>runxldr2.py</code>                                                                                                                                                                                                                                                | Deobfuscate XLM macros (also known as Excel 4.0 macros) from Microsoft Office files.                            |
| xmldump.py           | —                                                                                                                                                                                                                                                                                                      | Extract contents of XML files, in particular OOXML-formatted Microsoft Office documents.                        |
| zipdump.py           | —                                                                                                                                                                                                                                                                                                      | Analyze zip-compressed files.                                                                                   |

## Analyze Documents / Pdf

| Tool          | Command(s)                                                                                                 | Purpose                                                            |
|---------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Origamindee   | <code>pdfcop</code> , <code>pdfdecompress</code> , <code>pdfdecrypt</code> , <code>pdfextract</code> , etc | Parse, modify, generate PDF files.                                 |
| pdf-parser.py | —                                                                                                          | Examine elements of the PDF file.                                  |
| pdfid.py      | —                                                                                                          | Identify suspicious elements of the PDF file.                      |
| pdfresurrect  | —                                                                                                          | Extract previous versions of content from PDF files.               |
| pdftk-java    | <code>pdftk</code>                                                                                         | Edit, create, and examine PDF files.                               |
| pdftool.py    | —                                                                                                          | Analyze PDF files to identify incremental updates to the document. |
| peepdf-3      | —                                                                                                          | Examine elements of the PDF file.                                  |

| Tool | Command(s) | Purpose                                           |
|------|------------|---------------------------------------------------|
| qpdf | —          | Manipulate (merge, convert, transform) PDF files. |

## Dynamically Reverse-Engineer Code / Elf Files

| Tool                 | Command(s) | Purpose                                                              |
|----------------------|------------|----------------------------------------------------------------------|
| edb                  | —          | An AArch32/x86/x86-64 debugger, well suited for debugging ELF files. |
| GNU Project Debugger | —          | Multi-language debugger.                                             |
| ltrace               | —          | Trace library calls and signals.                                     |
| strace               | —          | Trace process' system calls and signals.                             |

## Dynamically Reverse-Engineer Code / General

| Tool                                  | Command(s)                                                                                                                                                           | Purpose                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Frida                                 | <code>frida</code> , <code>frida-ps</code> , <code>frida-trace</code> , <code>frida-discover</code> , <code>frida-ls-devices</code> , <code>frida-kill</code>        | Trace the execution of a process to analyze its behavior.                                                                |
| r2pipe                                | —                                                                                                                                                                    | Examine binary files, including disassembling and debugging.                                                             |
| radare2                               | <code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code> | Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A |
| Wine                                  | <code>wine</code>                                                                                                                                                    | Run Windows applications.                                                                                                |
| x64dbg Automate MCP (OpenCode skills) | —                                                                                                                                                                    | Drive x64dbg on a remote Windows VM from OpenCode on REMnux, with eight AI commands for tracing, unpacking, state snapsh |

## Dynamically Reverse-Engineer Code / Scripts

| Tool                    | Command(s)                                    | Purpose                                                                                                     |
|-------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| box-js                  | <code>box-js</code> , <code>box-export</code> | Analyze suspicious JavaScript scripts.                                                                      |
| JavaScript Deobfuscator | —                                             | Deobfuscate JavaScript by removing common obfuscation techniques such as string arrays and proxy functions. |
| js_unshroud             | —                                             | Monitor and deobfuscate JavaScript behavior in a headless browser to analyze malicious web pages.           |
| JStillery               | <code>jstillery</code>                        | Deobfuscate JavaScript scripts using AST and Partial Evaluation techniques.                                 |
| objects.js              | —                                             | Emulate common browser and PDF viewer objects, methods, and properties when deobfuscating JavaScript.       |
| PowerShell Core         | <code>pwsh</code>                             | Run PowerShell scripts and commands.                                                                        |
| Rhino Debugger          | <code>rhino-debugger</code>                   | GUI JavaScript debugger.                                                                                    |
| SpiderMonkey            | <code>js</code>                               | Execute and deobfuscate JavaScript using Mozilla's standalone JavaScript engine.                            |



| Tool                   | Command(s)                                   | Purpose                                                                                               |
|------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------|
| SpiderMonkey (Patched) | <code>js-ascii</code> , <code>js-file</code> | Execute and deobfuscate JavaScript using a patched version of Mozilla's standalone JavaScript engine. |
| STPyV8                 | —                                            | Python3 and JavaScript interop engine, fork of the original PyV8 project.                             |
| Webcrack               | —                                            | Deobfuscate, unminify, and unpack bundled JavaScript, including scripts protected with obfuscator.io. |

## Dynamically Reverse-Engineer Code / Shellcode

| Tool              | Command(s)             | Purpose                                                                                         |
|-------------------|------------------------|-------------------------------------------------------------------------------------------------|
| libemu            | —                      | A library for x86 code emulation and shellcode detection.                                       |
| Qiling            | —                      | Emulate code execution of PE files, shellcode, etc. for a variety of OS and hardware platforms. |
| runsc             | —                      | Run shellcode to trace and analyze its execution.                                               |
| scdbg             | —                      | Analyze shellcode by emulating its execution.                                                   |
| shcode2exe        | —                      | Convert 32 and 64-bit shellcode to a Windows executable file.                                   |
| shellcode2exe.bat | —                      | Convert 32 and 64-bit shellcode to a Windows executable file.                                   |
| Speakeasy         | —                      | Emulate code execution, including shellcode, Windows drivers, and Windows PE files.             |
| XORSearch         | <code>xorsearch</code> | Locate and decode strings obfuscated using common techniques.                                   |

## Examine Static Properties / .Net

| Tool       | Command(s) | Purpose                                                 |
|------------|------------|---------------------------------------------------------|
| dnfile     | —          | Analyze static properties of .NET files.                |
| dotnetfile | —          | Analyze static properties of .NET files.                |
| monodis    | —          | Disassemble and extract resources from .NET assemblies. |

## Examine Static Properties / Deobfuscation

| Tool                                                    | Command(s)                                                                                    | Purpose                                                                            |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1768.py                                                 | —                                                                                             | Analyze Cobalt Strike beacons.                                                     |
| Balbuzard                                               | <code>balbuzard</code> , <code>bbcrack</code> , <code>bbharvest</code> , <code>bbtrans</code> | Extract and deobfuscate patterns from suspicious files.                            |
| base64dump.py                                           | —                                                                                             | Locate and decode strings encoded in Base64 and other common encodings.            |
| brxor.py                                                | —                                                                                             | Bruteforce XOR'ed strings to find those that are English words.                    |
| Chepy                                                   | <code>chepy</code>                                                                            | Decode and otherwise analyze data using this command-line tool and Python library. |
| Cobalt Strike Configuration Extractor (CSCE) and Parser | <code>csce</code> , <code>list-cs-settings</code>                                             | Analyze Cobalt Strike beacons.                                                     |
| cs-analyze-processdump.py                               | —                                                                                             | Analyze Cobalt Strike beacon process dumps to detect sleep mask encoding.          |

| Tool                   | Command(s)             | Purpose                                                                            |
|------------------------|------------------------|------------------------------------------------------------------------------------|
| cs-decrypt-metadata.py | —                      | Decrypt Cobalt Strike metadata.                                                    |
| cs-extract-key.py      | —                      | Extract AES and HMAC keys from Cobalt Strike beacon process memory.                |
| cut-bytes.py           | —                      | Cut out a part of a data stream.                                                   |
| CyberChef              | <code>cyberchef</code> | Decode and otherwise analyze data using this browser app.                          |
| DC3-MWCP               | <code>mwcp</code>      | Parsing configuration information from malware.                                    |
| ex_pe_xor.py           | —                      | Search an XOR'ed file for indications of executable binaries.                      |
| FLOSS                  | <code>floss</code>     | Extract and deobfuscate strings from PE executables.                               |
| format-bytes.py        | —                      | Decompose structured binary data with format strings.                              |
| hex-to-bin.py          | —                      | Convert hexadecimal text dumps to binary data.                                     |
| Malchive               | —                      | Perform static analysis of various aspects of malicious code.                      |
| NoMoreXOR.py           | —                      | Help guess a file's 256-byte XOR by using frequency analysis.                      |
| numbers-to-string.py   | —                      | Translate number sequences into ASCII characters.                                  |
| re-search.py           | —                      | Search files using regular expressions from a built-in library or custom patterns. |
| sets.py                | —                      | Perform set operations on lines or bytes in text files.                            |
| strdeob.pl             | —                      | Locate and decode stack strings in executable files.                               |
| translate.py           | —                      | Translate bytes according to a Python expression.                                  |
| unicode                | —                      | Display Unicode character properties.                                              |
| unXOR                  | —                      | Deobfuscate XOR'ed files.                                                          |
| xor-kpa.py             | —                      | Implement a XOR known plaintext attack.                                            |
| xorBruteForcer.py      | —                      | Bruteforce an XOR-encoded file.                                                    |
| XORSearch              | <code>xorsearch</code> | Locate and decode strings obfuscated using common techniques.                      |
| xorsearch.py           | —                      | Search for XOR, ROL, ROT, and SHIFT encoded strings with YARA and regex support.   |
| XORStrings             | —                      | Search for XOR encoded strings in a file.                                          |
| xortool                | —                      | Analyze XOR-encoded data.                                                          |

## Examine Static Properties / Elf Files

| Tool       | Command(s)              | Purpose                                                                             |
|------------|-------------------------|-------------------------------------------------------------------------------------|
| pyelftools | <code>readelf.py</code> | Python library for parsing and analyzing ELF files and DWARF debugging information. |

## Examine Static Properties / General

| Tool                 | Command(s)                                                                                                          | Purpose                                                                                |
|----------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 7-Zip                | <code>7za</code>                                                                                                    | Compress and decompress files using a variety of algorithms.                           |
| binwalk              | —                                                                                                                   | Extract and analyze firmware images.                                                   |
| bulk_extractor       | —                                                                                                                   | Extract interesting strings from binary files.                                         |
| ClamAV               | <code>clamscan</code> , <code>freshclam</code>                                                                      | Scan files for malware signatures.                                                     |
| Detect-It-Easy       | —                                                                                                                   | Determine types of files and examine file properties.                                  |
| disitool             | <code>disitool.py</code>                                                                                            | Manipulate embedded digital signatures.                                                |
| DroidLysis           | <code>droidlysis</code>                                                                                             | Perform static analysis of Android applications.                                       |
| ExifTool             | <code>exiftool</code>                                                                                               | Tool to read from, write to, and edit EXIF metadata of various file types.             |
| file                 | —                                                                                                                   | Identify file type using "magic" numbers.                                              |
| file-magic.py        | —                                                                                                                   | Identify file types using the Python magic module.                                     |
| Hachoir              | <code>hachoir-grep</code> , <code>hachoir-metadata</code> ,<br><code>hachoir-strip</code> , <code>hachoir-wx</code> | View, edit, and carve contents of various binary file types.                           |
| Hash ID              | <code>hash-id.py</code>                                                                                             | Identify different types of hashes.                                                    |
| LIEF                 | —                                                                                                                   | Parse and analyze PE, ELF, MachO, DEX, OAT, VDEX, ART, and DWARF executable formats.   |
| Magika               | —                                                                                                                   | Identify file type using signatures.                                                   |
| Malcat Lite          | —                                                                                                                   | Analyze binary files using a hex editor, disassembler, and file dissector.             |
| msitools             | —                                                                                                                   | Create, inspect and extract Windows Installer (.msi) files.                            |
| Name-That-Hash       | <code>nth</code>                                                                                                    | Identify different types of hashes.                                                    |
| numbers-to-string.py | —                                                                                                                   | Convert decimal numbers to strings.                                                    |
| re-search.py         | —                                                                                                                   | Search the file for built-in regular expressions of common suspicious artifacts.       |
| signsrch             | —                                                                                                                   | Find patterns of common encryption, compression, or encoding algorithms.               |
| Sleuth Kit           | —                                                                                                                   | Analyze disk images and recover files from them.                                       |
| ssdeep               | —                                                                                                                   | Compute Context Triggered Piecewise Hashes (CTPH), also known as fuzzy hashes.         |
| strings.py           | —                                                                                                                   | Extract ASCII and Unicode strings from binary files with length sorting and filtering. |
| thefuzz              | —                                                                                                                   | Fuzzy String Matching in Python.                                                       |
| TrID                 | <code>trid</code> , <code>tridupdate</code>                                                                         | Identify file type using signatures.                                                   |
| wxHexEditor          | —                                                                                                                   | Hex editor.                                                                            |

| Tool             | Command(s) | Purpose                                                                                                        |
|------------------|------------|----------------------------------------------------------------------------------------------------------------|
| Yara Rules       | —          | Scan a file with YARA rules to identify capabilities and behaviors (packer detection, anti-debug, networking). |
| YARA-Forge Rules | —          | Scan files with curated YARA rules from 45+ sources for malware family identification.                         |

## Examine Static Properties / Go

| Tool    | Command(s)           | Purpose                                                                                               |
|---------|----------------------|-------------------------------------------------------------------------------------------------------|
| GoReSym | <code>GoReSym</code> | Extract metadata and symbols from Go binaries, including stripped ones.                               |
| Redress | <code>redress</code> | Analyze stripped Go binaries to recover symbols, types, source structure, and integrate with Radare2. |

## Examine Static Properties / Pe Files

| Tool                  | Command(s)                                                                                                                                                                                                                                   | Purpose                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| bearparser            | <code>bearcommander</code>                                                                                                                                                                                                                   | Parse PE file contents.                                                     |
| debloat               | —                                                                                                                                                                                                                                            | Remove junk contents from bloated Windows executables.                      |
| disitool.py           | —                                                                                                                                                                                                                                            | Extract, delete, copy, and inject digital signatures in PE files.           |
| dllcharacteristics.py | —                                                                                                                                                                                                                                            | Read and set DLL characteristics of a PE file.                              |
| Manalyze              | —                                                                                                                                                                                                                                            | Perform static analysis of suspicious PE files.                             |
| PE Tree               | <code>pe-tree</code>                                                                                                                                                                                                                         | Examine contents and structure of PE files.                                 |
| pecheck.py            | —                                                                                                                                                                                                                                            | Analyze static properties of PE files.                                      |
| pedump                | —                                                                                                                                                                                                                                            | Statically analyze PE files and extract their components (e.g., resources). |
| pefile                | —                                                                                                                                                                                                                                            | Python library for analyzing static properties of PE files.                 |
| PEframe               | <code>peframe</code>                                                                                                                                                                                                                         | Statically analyze PE and Microsoft Office files.                           |
| pev                   | <code>pestr</code> , <code>readpe</code> , <code>pedis</code> , <code>pehash</code> , <code>pescan</code> , <code>peldd</code> , <code>peres</code>                                                                                          | Analyze PE files and extract strings from them.                             |
| PortEx                | <code>portex</code>                                                                                                                                                                                                                          | Statically analyze PE files.                                                |
| readpe (formerly pev) | <code>readpe</code> , <code>pestr</code> , <code>pedis</code> , <code>pehash</code> , <code>pescan</code> , <code>pesec</code> , <code>peldd</code> , <code>pepack</code> , <code>peres</code> , <code>ofs2rva</code> , <code>rva2ofs</code> | Analyze PE files and extract strings from them.                             |

## Explore Network Interactions / Connecting

| Tool            | Command(s)           | Purpose                                                                                                             |
|-----------------|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Anomy           | <code>anomy</code>   | A wrapper around wget, ssh, sftp, ftp, and telnet to route these connections through Tor to anonymize your traffic. |
| cURL            | <code>curl</code>    | Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool. |
| EPIC IRC Client | <code>epic5</code>   | Examine IRC activities with this IRC client.                                                                        |
| GNU Wget        | <code>wget</code>    | Interact with servers via HTTP, HTTPS, FTP, and FTPS using this command-line tool.                                  |
| netcat          | <code>nc</code>      | Read and write data across network connections.                                                                     |
| thug            | —                    | Examine suspicious website using this low-interaction honeyclient.                                                  |
| tor             | —                    | Obfuscate your origins by routing traffic through a network of anonymizing nodes.                                   |
| Unfurl          | —                    | Deconstruct and decode data from a URL.                                                                             |
| zbarimg         | <code>zbarimg</code> | Decode QR codes and barcodes from image files.                                                                      |

## Explore Network Interactions / Monitoring

| Tool                         | Command(s)                                                            | Purpose                                                                                                                  |
|------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Burp Suite Community Edition | <code>burpsuite</code>                                                | Investigate website interactions using this web proxy.                                                                   |
| cs-parse-traffic.py          | —                                                                     | Decrypt and parse Cobalt Strike beacon network traffic.                                                                  |
| mitmproxy                    | <code>mitmproxy</code> , <code>mitmdump</code> , <code>mitmweb</code> | Investigate website interactions using this web proxy.                                                                   |
| monitor-network              | —                                                                     | Monitor traffic on the first active network interface using tshark, printing a live summary to the screen or saving it t |
| Network Miner Free Edition   | <code>networkminer</code>                                             | Examine network traffic and carve PCAP capture files.                                                                    |
| ngrep                        | —                                                                     | Look for patterns in network traffic.                                                                                    |
| PolarProxy                   | <code>polarproxy</code>                                               | Intercept and decrypt TLS traffic.                                                                                       |
| tcpdump                      | —                                                                     | Capture and analyze network traffic with this command-line sniffer.                                                      |
| tcpflow                      | —                                                                     | Analyze the flow of network traffic.                                                                                     |
| tcpick                       | —                                                                     | Capture and analyze network traffic with this command-line sniffer.                                                      |
| tcpxtract                    | —                                                                     | Extract files from network traffic.                                                                                      |
| tshark                       | —                                                                     | Capture and analyze network traffic with this console-based sniffer.                                                     |
| wireshark                    | —                                                                     | Capture and analyze network traffic with this sniffer.                                                                   |

## Explore Network Interactions / Services

| Tool           | Command(s) | Purpose                                                                                            |
|----------------|------------|----------------------------------------------------------------------------------------------------|
| accept-all-ips | —          | Accept connections to all IPv4 and IPv6 addresses and redirect it to the corresponding local port. |
| dnsresolver.py | —          | DNS resolver tool for dynamic analysis with wildcard and tracking support.                         |

| Tool       | Command(s)           | Purpose                                                               |
|------------|----------------------|-----------------------------------------------------------------------|
| fakedns    | —                    | Respond to DNS queries with the specified IP address.                 |
| fakemail   | —                    | Intercept and examine SMTP email activity with this fake SMTP server. |
| FakeNet-NG | —                    | Emulate common network services and interact with malware.            |
| INetSim    | <code>inetsim</code> | Emulate common network services and interact with malware.            |
| inspired 3 | —                    | Examine IRC activity with this IRC server.                            |
| netcat     | <code>nc</code>      | Read and write data across network connections.                       |
| Nginx      | —                    | Web server.                                                           |

## Gather And Analyze Data / General

| Tool              | Command(s)                        | Purpose                                                                                           |
|-------------------|-----------------------------------|---------------------------------------------------------------------------------------------------|
| DeXRAY            | <code>dexray</code>               | Extract and decode data from antivirus quarantine files.                                          |
| dissect           | —                                 | Perform a variety of forensics and incident response tasks using this DFIR framework and toolset. |
| dnslib            | —                                 | Python library to encode/decode DNS wire-format packets.                                          |
| ioc_parser        | —                                 | Extract IOCs from security report PDFs.                                                           |
| ipwhois           | —                                 | Retrieve and parse whois data for IP addresses.                                                   |
| malwoverview      | <code>malwoverview</code>         | Query public repositories of malware data (e.g., VirusTotal, HybridAnalysis).                     |
| nsrlookup         | —                                 | Look up MD5 file hashes in the NIST National Software Reference Library (NSRL).                   |
| pdnstool          | —                                 | Query passive DNS databases for DNS data.                                                         |
| Scalpel           | —                                 | Carve contents out of binary files, such as partitions.                                           |
| time-decode       | —                                 | Decode and encode date and timestamps.                                                            |
| virustotal-search | <code>virustotal-search.py</code> | Search VirusTotal for file hashes.                                                                |
| virustotal-submit | <code>virustotal-submit.py</code> | Submit files to VirusTotal.                                                                       |
| Yara              | <code>yara</code>                 | Identify and classify malware samples using Yara rules.                                           |
| YARA-X            | —                                 | Scan files using YARA rules, the next generation of YARA written in Rust.                         |

## General Utilities / General

| Tool       | Command(s)           | Purpose                                                                                                             |
|------------|----------------------|---------------------------------------------------------------------------------------------------------------------|
| 7-Zip      | <code>7za</code>     | Compress and decompress files using a variety of algorithms.                                                        |
| cabextract | —                    | Extract Microsoft cabinet (cab) files.                                                                              |
| cURL       | <code>curl</code>    | Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool. |
| Docker     | —                    | Run and manage containers.                                                                                          |
| Firefox    | <code>firefox</code> | Web browser.                                                                                                        |

| Tool             | Command(s)                                                 | Purpose                                                                                                                  |
|------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| GNOME Calculator | <code>galculator</code>                                    | Calculator.                                                                                                              |
| IBus             | <code>ibus-setup</code>                                    | Adjust input methods for the GUI.                                                                                        |
| Info-ZIP         | <code>zip</code> , <code>unzip</code>                      | Compress and decompress files using the zip algorithm.                                                                   |
| myip             | —                                                          | Determine the IP address of the default network interface.                                                               |
| myjson-filter.py | —                                                          | Filter data formatted using the JSON format used by Didier Stevens' tools.                                               |
| nasm             | —                                                          | An x86-64 assembler.                                                                                                     |
| Nautilus         | —                                                          | Graphical file manager.                                                                                                  |
| OpenSSH          | <code>sftp</code> , <code>ssh</code> ,<br><code>etc</code> | Initiate and receive SSH and SFTP connections.                                                                           |
| PowerShell Core  | <code>pwsh</code>                                          | Run PowerShell scripts and commands.                                                                                     |
| RAR              | <code>rar</code>                                           | Compress and decompress files using a variety of algorithms.                                                             |
| REMnux Installer | —                                                          | Install and update the REMnux distro.                                                                                    |
| restrict-egress  | —                                                          | Restrict outbound network access to an allowlist of domains and CIDRs using an nftables default-deny egress policy. It i |
| sortcanon.py     | —                                                          | Sort text files using canonicalization functions built into this tool.                                                   |
| SQLite           | <code>sqlite3</code>                                       | Manage and interact with SQL database files.                                                                             |
| sshpass          | <code>sshpass</code>                                       | Supply a password to SSH non-interactively for automated logins.                                                         |
| texteditor.py    | —                                                          | Edit text files from the command line using search-and-replace commands.                                                 |
| unrar-free       | <code>unrar</code>                                         | Decompress files using a variety of algorithms.                                                                          |
| Wine             | <code>wine</code>                                          | Run Windows applications.                                                                                                |

## Investigate System Interactions / General

| Tool                   | Command(s)           | Purpose                                                                            |
|------------------------|----------------------|------------------------------------------------------------------------------------|
| ProcDOT                | <code>procdot</code> | Visualize and examine the output of Process Monitor.                               |
| ProcmonMCP             | —                    | MCP server that lets AI assistants analyze Process Monitor (Procmon) XML captures. |
| sandfly-processdecloak | —                    | Find hidden processes on the local Linux system.                                   |
| Unhide                 | —                    | Find hidden processes or connections on the local Linux system.                    |

## Perform Memory Forensics / General

| Tool                 | Command(s)              | Purpose                                              |
|----------------------|-------------------------|------------------------------------------------------|
| AESKeyFinder         | <code>aeskeyfind</code> | Find 128-bit and 256-bit AES keys in a memory image. |
| bulk_extractor       | —                       | Extract interesting strings from binary files.       |
| RSAKeyFinder         | <code>rsakeyfind</code> | Find BER-encoded RSA private keys in a memory image. |
| Volatility Framework | —                       | Memory forensics tool and framework.                 |

## Statically Analyze Code / .Net

| Tool   | Command(s) | Purpose                               |
|--------|------------|---------------------------------------|
| de4dot | —          | Deobfuscate and unpack .NET programs. |
| ILSpy  | —          | Examine and decompile .NET programs.  |

## Statically Analyze Code / Android

| Tool                  | Command(s)                                        | Purpose                                                                                     |
|-----------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------|
| androguard            | —                                                 | Examine Android files.                                                                      |
| AndroidProjectCreator | —                                                 | Convert an Android APK application file into an Android Studio project for easier analysis. |
| APKiD                 | <code>apkid</code>                                | Identify compilers, packers, and obfuscators used to protect Android APK and DEX files.     |
| apktool               | —                                                 | Reverse-engineer Android APK files.                                                         |
| baksmali              | —                                                 | Disassembler for the dex format used by Dalvik, Android's Java VM implementation.           |
| dex2jar               | <code>dex-tools</code> , <code>d2j-dex2jar</code> | Examine Dalvik Executable (dex) files.                                                      |
| DroidLysis            | <code>droidlysis</code>                           | Perform static analysis of Android applications.                                            |
| JADX                  | <code>jadx</code> , <code>jadx-gui</code>         | Generate Java source code from Dalvik Executable (dex) and Android APK files.               |

## Statically Analyze Code / General

| Tool           | Command(s)                                                                                                                                                           | Purpose                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Cutter         | —                                                                                                                                                                    | Reverse engineering platform powered by Rizin.                                                                           |
| Detect-It-Easy | —                                                                                                                                                                    | Determine types of files and examine file properties.                                                                    |
| Ghidra         | —                                                                                                                                                                    | Software reverse engineering tool suite.                                                                                 |
| objdump        | —                                                                                                                                                                    | Disassemble binary files.                                                                                                |
| Qiling         | —                                                                                                                                                                    | Emulate code execution of PE files, shellcode, etc. for a variety of OS and hardware platforms.                          |
| radare2        | <code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code> | Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A |
| Vivisect       | <code>vivbin</code> , <code>vdbbin</code>                                                                                                                            | Statically examine and emulate binary files.                                                                             |

## Statically Analyze Code / Java

| Tool            | Command(s)                 | Purpose                                    |
|-----------------|----------------------------|--------------------------------------------|
| cfr             | —                          | Java decompiler.                           |
| Java IDX Parser | <code>idx_parser.py</code> | Analyze Java IDX files.                    |
| Javassist       | —                          | Java bytecode engineering toolkit/library. |



| Tool                   | Command(s)           | Purpose                   |
|------------------------|----------------------|---------------------------|
| JD-GUI Java Decompiler | <code>jd-gui</code>  | Java decompiler with GUI. |
| Procyon                | <code>procyon</code> | Java decompiler.          |

## Statically Analyze Code / Pe-Files

| Tool                                 | Command(s)              | Purpose                                                                             |
|--------------------------------------|-------------------------|-------------------------------------------------------------------------------------|
| binee (Binary Emulation Environment) | —                       | Analyze I/O operations of a suspicious PE file by emulating its execution.          |
| capa                                 | —                       | Detect suspicious capabilities in PE files.                                         |
| Malchive                             | —                       | Perform static analysis of various aspects of malicious code.                       |
| mbcscan                              | <code>mbcscan.py</code> | Scan a PE file to list the associated Malware Behavior Catalog (MBC) details.       |
| Speakeasy                            | —                       | Emulate code execution, including shellcode, Windows drivers, and Windows PE files. |

## Statically Analyze Code / Python

| Tool                  | Command(s)                               | Purpose                                                                                            |
|-----------------------|------------------------------------------|----------------------------------------------------------------------------------------------------|
| Decompyle++           | <code>pycdas</code> , <code>pycdc</code> | Python bytecode disassembler and decompiler.                                                       |
| PyInstaller Extractor | <code>pyinstxtractor.py</code>           | Extract contents of a PyInstaller-generated PE files.                                              |
| pyinstxtractor-ng     | <code>pyinstxtractor-ng</code>           | Extract contents of PyInstaller-generated executables without requiring a matching Python version. |
| uncompyle6            | <code>uncompyle6</code>                  | Python cross-version bytecode decompiler for Python 1.0 through 3.8.                               |

## Statically Analyze Code / Scripts

| Tool                      | Command(s)                 | Purpose                                                                                                         |
|---------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| Autolt-Ripper             | <code>autoit-ripper</code> | Extract Autolt scripts embedded in PE binaries.                                                                 |
| decode-vbe.py             | —                          | Decode encoded VBS scripts (VBE).                                                                               |
| GootLoaderAutoJsDecode.py | —                          | Statically deobfuscate GootLoader (GOOTLOADER) malicious JScript to recover the payload and extract C2 domains. |
| JS Beautifier             | <code>js-beautify</code>   | Reformat JavaScript scripts for easier analysis.                                                                |

## Statically Analyze Code / Unpacking

| Tool     | Command(s)                                     | Purpose                                                                         |
|----------|------------------------------------------------|---------------------------------------------------------------------------------|
| binwalk  | —                                              | Extract and analyze firmware images.                                            |
| Bytehist | <code>bytehist</code>                          | Generate byte-usage-histograms for all types of files with a focus on PE files. |
| ClamAV   | <code>clamscan</code> , <code>freshclam</code> | Scan files for malware signatures.                                              |
| TrID     | <code>trid</code> , <code>tridupdate</code>    | Identify file type using signatures.                                            |

| Tool | Command(s)       | Purpose                   |
|------|------------------|---------------------------|
| UPX  | <code>upx</code> | Pack and unpack PE files. |

## Use Artificial Intelligence / General

| Tool                                  | Command(s)                                                                                                                                                           | Purpose                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| GhidraAssistMCP                       | —                                                                                                                                                                    | MCP server for AI-assisted reverse engineering in Ghidra.                                                                |
| OpenCode                              | —                                                                                                                                                                    | Open-source AI coding agent for the terminal.                                                                            |
| ProcmonMCP                            | —                                                                                                                                                                    | MCP server that lets AI assistants analyze Process Monitor (Procmon) XML captures.                                       |
| radare2                               | <code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code> | Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A |
| REMnux MCP Server                     | <code>remnux-mcp-server</code>                                                                                                                                       | MCP server for using the REMnux malware analysis toolkit via AI assistants.                                              |
| x64dbg Automate MCP (OpenCode skills) | —                                                                                                                                                                    | Drive x64dbg on a remote Windows VM from OpenCode on REMnux, with eight AI commands for tracing, unpacking, state snapsh |

## View Or Edit Files / General

| Tool               | Command(s)                                                                                    | Purpose                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| dos2unix           | <code>dos2unix</code> , <code>mac2unix</code> , <code>unix2dos</code> , <code>unix2mac</code> | Convert text files with Windows or macOS line breaks to Unix line breaks and vice versa.         |
| Evince             | <code>evince</code>                                                                           | View documents in a variety of formats, including PDF.                                           |
| feh                | —                                                                                             | View images.                                                                                     |
| ImageMagick        | <code>display</code> , <code>convert</code> , <code>mogrify</code> , <code>etc</code>         | View and manipulate image and related files.                                                     |
| Mermaid Viewer     | —                                                                                             | View Mermaid diagrams, such as AI-generated code-analysis workflow diagrams, in a local browser. |
| SciTE              | —                                                                                             | Edit text files.                                                                                 |
| VBinDiff           | <code>vbindiff</code>                                                                         | Compare binary files.                                                                            |
| Visual Studio Code | <code>code</code>                                                                             | Powerful source code editor.                                                                     |
| wxHexEditor        | —                                                                                             | Hex editor.                                                                                      |

## Kali Linux

### Crypto Stego

| Tool       | Command(s)              | Purpose |
|------------|-------------------------|---------|
| aesfix     | <code>aesfix</code>     |         |
| aeskeyfind | <code>aeskeyfind</code> |         |

| Tool       | Command(s) | Purpose |
|------------|------------|---------|
| ccrypt     | ccrypt     |         |
| steghide   | steghide   |         |
| stegosuite | stegosuite |         |
| stegsnow   | stegsnow   |         |

## Database

| Tool           | Command(s)     | Purpose |
|----------------|----------------|---------|
| jsql-injection | jsql-injection |         |
| mdbtools       | mdbtools       |         |
| osscanner      | osscanner      |         |
| sidguesser     | sidguesser     |         |
| sqldict        | sqldict        |         |
| sqlitebrowser  | sqlitebrowser  |         |
| sqlmap         | sqlmap         |         |
| sqlninja       | sqlninja       |         |
| sqlsus         | sqlsus         |         |
| tnscmd10g      | tnscmd10g      |         |

## Exploitation

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| armitage             | armitage             |         |
| beef-xss             | beef-xss             |         |
| exploitdb            | exploitdb            |         |
| metasploit-framework | metasploit-framework |         |
| msfpc                | msfpc                |         |
| set                  | set                  |         |
| shellnoob            | shellnoob            |         |
| sqlmap               | sqlmap               |         |
| termineter           | termineter           |         |

## Forensics

| Tool         | Command(s)   | Purpose |
|--------------|--------------|---------|
| 7zip         | 7zip         |         |
| afflib-tools | afflib-tools |         |
| apktool      | apktool      |         |

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| autopsy            | autopsy            |         |
| binwalk            | binwalk            |         |
| binwalk3           | binwalk3           |         |
| bulk-extractor     | bulk-extractor     |         |
| bytecode-viewer    | bytecode-viewer    |         |
| cabextract         | cabextract         |         |
| chkrootkit         | chkrootkit         |         |
| creddump7          | creddump7          |         |
| dc3dd              | dc3dd              |         |
| dcfldd             | dcfldd             |         |
| ddrescue           | ddrescue           |         |
| dumpzilla          | dumpzilla          |         |
| edb-debugger       | edb-debugger       |         |
| ewf-tools          | ewf-tools          |         |
| exifprobe          | exifprobe          |         |
| exiv2              | exiv2              |         |
| ext3grep           | ext3grep           |         |
| ext4magic          | ext4magic          |         |
| extundelete        | extundelete        |         |
| fcrackzip          | fcrackzip          |         |
| firmware-mod-kit   | firmware-mod-kit   |         |
| foremost           | foremost           |         |
| forensic-artifacts | forensic-artifacts |         |
| forensics-colorize | forensics-colorize |         |
| galleta            | galleta            |         |
| gdb                | gdb                |         |
| gpart              | gpart              |         |
| gparted            | gparted            |         |
| grokevt            | grokevt            |         |
| guymager           | guymager           |         |
| hashdeep           | hashdeep           |         |
| inetsim            | inetsim            |         |
| jadx               | jadx               |         |
| jasvasnoop         | jasvasnoop         |         |

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| libhivex-bin       | libhivex-bin       |         |
| libsmali-java      | libsmali-java      |         |
| lvm2               | lvm2               |         |
| lynis              | lynis              |         |
| mac-robber         | mac-robber         |         |
| magicrescue        | magicrescue        |         |
| md5deep            | md5deep            |         |
| mdbtools           | mdbtools           |         |
| memdump            | memdump            |         |
| metacam            | metacam            |         |
| missidentify       | missidentify       |         |
| myrescue           | myrescue           |         |
| nasm               | nasm               |         |
| nasty              | nasty              |         |
| ollydbg            | ollydbg            |         |
| parted             | parted             |         |
| pasco              | pasco              |         |
| pdf-parser         | pdf-parser         |         |
| pdfid              | pdfid              |         |
| plaso              | plaso              |         |
| polenum            | polenum            |         |
| pst-utils          | pst-utils          |         |
| python3-capstone   | python3-capstone   |         |
| python3-dfdatetime | python3-dfdatetime |         |
| python3-dfvfs      | python3-dfvfs      |         |
| python3-dfwinreg   | python3-dfwinreg   |         |
| python3-distorm3   | python3-distorm3   |         |
| radare2            | radare2            |         |
| readpe             | readpe             |         |
| recoverdm          | recoverdm          |         |
| recoverjpeg        | recoverjpeg        |         |
| recstudio          | recstudio          |         |
| reglookup          | reglookup          |         |
| regripper          | regripper          |         |

| Tool          | Command(s)    | Purpose |
|---------------|---------------|---------|
| rephrase      | rephrase      |         |
| rifiuti       | rifiuti       |         |
| rifiuti2      | rifiuti2      |         |
| rizin-cutter  | rizin-cutter  |         |
| rkhunter      | rkhunter      |         |
| rsakeyfind    | rsakeyfind    |         |
| rz-ghidra     | rz-ghidra     |         |
| safecopy      | safecopy      |         |
| samdump2      | samdump2      |         |
| scalpel       | scalpel       |         |
| scrounge-ntfs | scrounge-ntfs |         |
| sleuthkit     | sleuthkit     |         |
| sqlitebrowser | sqlitebrowser |         |
| ssdeep        | ssdeep        |         |
| tcpdump       | tcpdump       |         |
| tcpflow       | tcpflow       |         |
| tcpick        | tcpick        |         |
| tcpreplay     | tcpreplay     |         |
| truecrack     | truecrack     |         |
| undbx         | undbx         |         |
| unhide        | unhide        |         |
| unrar         | unrar         |         |
| upx-ucl       | upx-ucl       |         |
| vinetto       | vinetto       |         |
| wce           | wce           |         |
| winregfs      | winregfs      |         |
| wireshark     | wireshark     |         |
| xmount        | xmount        |         |
| xplico        | xplico        |         |
| yara          | yara          |         |

## Fuzzing

| Tool  | Command(s) | Purpose |
|-------|------------|---------|
| afl++ | afl++      |         |
| sfuzz | sfuzz      |         |

| Tool  | Command(s) | Purpose |
|-------|------------|---------|
| spike | spike      |         |
| wfuzz | wfuzz      |         |

## Information Gathering

| Tool        | Command(s)  | Purpose |
|-------------|-------------|---------|
| Otrace      | 0trace      |         |
| arping      | arping      |         |
| braa        | braa        |         |
| dmitry      | dmitry      |         |
| dnsenum     | dnsenum     |         |
| dnsmap      | dnsmap      |         |
| dnsrecon    | dnsrecon    |         |
| dnstracer   | dnstracer   |         |
| dnswalk     | dnswalk     |         |
| enum4linux  | enum4linux  |         |
| fierce      | fierce      |         |
| firewalk    | firewalk    |         |
| fping       | fping       |         |
| fragrouter  | fragrouter  |         |
| ftester     | ftester     |         |
| hping3      | hping3      |         |
| ike-scan    | ike-scan    |         |
| intrace     | intrace     |         |
| irpas       | irpas       |         |
| lbd         | lbd         |         |
| legion      | legion      |         |
| maltego     | maltego     |         |
| masscan     | masscan     |         |
| metagoofil  | metagoofil  |         |
| nbtscan     | nbtscan     |         |
| ncat        | ncat        |         |
| netdiscover | netdiscover |         |
| netmask     | netmask     |         |
| nmap        | nmap        |         |
| onesixtyone | onesixtyone |         |

| Tool           | Command(s)     | Purpose |
|----------------|----------------|---------|
| p0f            | p0f            |         |
| qsslcaudit     | qsslcaudit     |         |
| recon-ng       | recon-ng       |         |
| smbmap         | smbmap         |         |
| smtp-user-enum | smtp-user-enum |         |
| snmpcheck      | snmpcheck      |         |
| ssldump        | ssldump        |         |
| sslh           | sslh           |         |
| sslscan        | sslscan        |         |
| sslyze         | sslyze         |         |
| swaks          | swaks          |         |
| thc-ipv6       | thc-ipv6       |         |
| theharvester   | theharvester   |         |
| tlssled        | tlssled        |         |
| twofi          | twofi          |         |
| unicornscan    | unicornscan    |         |
| urlcrazy       | urlcrazy       |         |
| wafw00f        | wafw00f        |         |
| zenmap         | zenmap         |         |

## Passwords

| Tool                | Command(s)          | Purpose |
|---------------------|---------------------|---------|
| cewl                | cewl                |         |
| chntpw              | chntpw              |         |
| cisco-auditing-tool | cisco-auditing-tool |         |
| cmospwd             | cmospwd             |         |
| crackle             | crackle             |         |
| creddump7           | creddump7           |         |
| crunch              | crunch              |         |
| fcrackzip           | fcrackzip           |         |
| freerdp3-x11        | freerdp3-x11        |         |
| gpp-decrypt         | gpp-decrypt         |         |
| hash-identifier     | hash-identifier     |         |
| hashcat             | hashcat             |         |
| hashcat-utils       | hashcat-utils       |         |



| Tool             | Command(s)       | Purpose |
|------------------|------------------|---------|
| hashid           | hashid           |         |
| hydra            | hydra            |         |
| hydra-gtk        | hydra-gtk        |         |
| john             | john             |         |
| johnny           | johnny           |         |
| maskprocessor    | maskprocessor    |         |
| medusa           | medusa           |         |
| mimikatz         | mimikatz         |         |
| ncrack           | ncrack           |         |
| onesixtyone      | onesixtyone      |         |
| ophcrack         | ophcrack         |         |
| ophcrack-cli     | ophcrack-cli     |         |
| pack             | pack             |         |
| pack2            | pack2            |         |
| passing-the-hash | passing-the-hash |         |
| patator          | patator          |         |
| pdfcrack         | pdfcrack         |         |
| pipal            | pipal            |         |
| polenum          | polenum          |         |
| rainbowcrack     | rainbowcrack     |         |
| rarcrack         | rarcrack         |         |
| rcracki-mt       | rcracki-mt       |         |
| rsmangler        | rsmangler        |         |
| samdump2         | samdump2         |         |
| seclists         | seclists         |         |
| sipcrack         | sipcrack         |         |
| sipvicious       | sipvicious       |         |
| smbmap           | smbmap           |         |
| sqldict          | sqldict          |         |
| statsprocessor   | statsprocessor   |         |
| sucrack          | sucrack          |         |
| thc-pptp-bruter  | thc-pptp-bruter  |         |
| truecrack        | truecrack        |         |
| twofi            | twofi            |         |

| Tool      | Command(s) | Purpose |
|-----------|------------|---------|
| wordlists | wordlists  |         |

## Post Exploitation

| Tool         | Command(s)   | Purpose |
|--------------|--------------|---------|
| cymothoa     | cymothoa     |         |
| dbd          | dbd          |         |
| dns2tcp      | dns2tcp      |         |
| exe2hexbat   | exe2hexbat   |         |
| iodine       | iodine       |         |
| laudanum     | laudanum     |         |
| mimikatz     | mimikatz     |         |
| miredo       | miredo       |         |
| nishang      | nishang      |         |
| powersploit  | powersploit  |         |
| proxychains4 | proxychains4 |         |
| proxytunnel  | proxytunnel  |         |
| ptunnel      | ptunnel      |         |
| pwnat        | pwnat        |         |
| sbd          | sbd          |         |
| shellter     | shellter     |         |
| sslh         | sslh         |         |
| stunnel4     | stunnel4     |         |
| udptunnel    | udptunnel    |         |
| veil         | veil         |         |
| webacoo      | webacoo      |         |
| weeveily     | weeveily     |         |

## Reverse Engineering

| Tool            | Command(s)      | Purpose |
|-----------------|-----------------|---------|
| apktool         | apktool         |         |
| bytecode-viewer | bytecode-viewer |         |
| clang           | clang           |         |
| dex2jar         | dex2jar         |         |
| edb-debugger    | edb-debugger    |         |
| jadx            | jadx            |         |

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| javasnoop            | javasnoop            |         |
| jd-gui               | jd-gui               |         |
| metasploit-framework | metasploit-framework |         |
| ollydbg              | ollydbg              |         |
| radare2              | radare2              |         |
| recstudio            | recstudio            |         |
| rizin                | rizin                |         |
| rizin-cutter         | rizin-cutter         |         |
| rz-ghidra            | rz-ghidra            |         |

## Sniffing Spoofing

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| above              | above              |         |
| bettercap          | bettercap          |         |
| darkstat           | darkstat           |         |
| dnscraf            | dnscraf            |         |
| driftnet           | driftnet           |         |
| dsniff             | dsniff             |         |
| ettercap-graphical | ettercap-graphical |         |
| ferret-sidejack    | ferret-sidejack    |         |
| fiked              | fiked              |         |
| hamster-sidejack   | hamster-sidejack   |         |
| hexinject          | hexinject          |         |
| isr-evilgrade      | isr-evilgrade      |         |
| macchanger         | macchanger         |         |
| mitmproxy          | mitmproxy          |         |
| netsniff-ng        | netsniff-ng        |         |
| rebind             | rebind             |         |
| responder          | responder          |         |
| sniffjoke          | sniffjoke          |         |
| sslsniff           | sslsniff           |         |
| sslsplit           | sslsplit           |         |
| tcpflow            | tcpflow            |         |
| tcpreplay          | tcpreplay          |         |
| wifi-honey         | wifi-honey         |         |

| Tool      | Command(s) | Purpose |
|-----------|------------|---------|
| wireshark | wireshark  |         |

## Vulnerability

| Tool                   | Command(s)             | Purpose |
|------------------------|------------------------|---------|
| afl++                  | afl++                  |         |
| bed                    | bed                    |         |
| cisco-auditing-tool    | cisco-auditing-tool    |         |
| cisco-global-exploiter | cisco-global-exploiter |         |
| cisco-ocs              | cisco-ocs              |         |
| cisco-torch            | cisco-torch            |         |
| copy-router-config     | copy-router-config     |         |
| dhcpig                 | dhcpig                 |         |
| enumiax                | enumiax                |         |
| gvm                    | gvm                    |         |
| iaxflood               | iaxflood               |         |
| inviteflood            | inviteflood            |         |
| legion                 | legion                 |         |
| lynis                  | lynis                  |         |
| nikto                  | nikto                  |         |
| nmap                   | nmap                   |         |
| ohrwurm                | ohrwurm                |         |
| peass                  | peass                  |         |
| protos-sip             | protos-sip             |         |
| rtpbreak               | rtpbreak               |         |
| rtpflood               | rtpflood               |         |
| rtpinsertsound         | rtpinsertsound         |         |
| rtpmixsound            | rtpmixsound            |         |
| sctpscan               | sctpscan               |         |
| sfuzz                  | sfuzz                  |         |
| siege                  | siege                  |         |
| siparmyknife           | siparmyknife           |         |
| sipp                   | sipp                   |         |
| sipsak                 | sipsak                 |         |
| sipvicious             | sipvicious             |         |
| slowhttpptest          | slowhttpptest          |         |

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| spike              | spike              |         |
| t50                | t50                |         |
| thc-ssl-dos        | thc-ssl-dos        |         |
| unix-privesc-check | unix-privesc-check |         |
| voiphopper         | voiphopper         |         |
| yersinia           | yersinia           |         |

## Web

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| apache-users         | apache-users         |         |
| apache2              | apache2              |         |
| beef-xss             | beef-xss             |         |
| burpsuite            | burpsuite            |         |
| cadaver              | cadaver              |         |
| commix               | commix               |         |
| cutycapt             | cutycapt             |         |
| davtest              | davtest              |         |
| default-mysql-server | default-mysql-server |         |
| dirb                 | dirb                 |         |
| dirbuster            | dirbuster            |         |
| dotdotpwn            | dotdotpwn            |         |
| eyewitness           | eyewitness           |         |
| ferret-sidejack      | ferret-sidejack      |         |
| ftester              | ftester              |         |
| hakrawler            | hakrawler            |         |
| hamster-sidejack     | hamster-sidejack     |         |
| heartleech           | heartleech           |         |
| httpprint            | httpprint            |         |
| httrack              | httrack              |         |
| hydra                | hydra                |         |
| hydra-gtk            | hydra-gtk            |         |
| jboss-autopwn        | jboss-autopwn        |         |
| joomscan             | joomscan             |         |
| jsql-injection       | jsql-injection       |         |
| laudanum             | laudanum             |         |

| Tool            | Command(s)      | Purpose |
|-----------------|-----------------|---------|
| lbd             | lbd             |         |
| maltego         | maltego         |         |
| medusa          | medusa          |         |
| mitmproxy       | mitmproxy       |         |
| ncrack          | ncrack          |         |
| nikto           | nikto           |         |
| nishang         | nishang         |         |
| nmap            | nmap            |         |
| osscanner       | osscanner       |         |
| owasp-mantra-ff | owasp-mantra-ff |         |
| padbuster       | padbuster       |         |
| paros           | paros           |         |
| patator         | patator         |         |
| php             | php             |         |
| php-mysql       | php-mysql       |         |
| proxychains4    | proxychains4    |         |
| proxytunnel     | proxytunnel     |         |
| qsslcaudit      | qsslcaudit      |         |
| redsocks        | redsocks        |         |
| sidguesser      | sidguesser      |         |
| siege           | siege           |         |
| skipfish        | skipfish        |         |
| slowhttptest    | slowhttptest    |         |
| sqldict         | sqldict         |         |
| sqlitebrowser   | sqlitebrowser   |         |
| sqlmap          | sqlmap          |         |
| sqlninja        | sqlninja        |         |
| sqlsus          | sqlsus          |         |
| ssldump         | ssldump         |         |
| sslh            | sslh            |         |
| sslscan         | sslscan         |         |
| sslsniff        | sslsniff        |         |
| sslsplit        | sslsplit        |         |
| sslyze          | sslyze          |         |

| Tool        | Command(s)  | Purpose |
|-------------|-------------|---------|
| stunnel4    | stunnel4    |         |
| thc-ssl-dos | thc-ssl-dos |         |
| tlssled     | tlssled     |         |
| tnscmd10g   | tnscmd10g   |         |
| uniscan     | uniscan     |         |
| wafw00f     | wafw00f     |         |
| wapiti      | wapiti      |         |
| watobo      | watobo      |         |
| webacoo     | webacoo     |         |
| webscarab   | webscarab   |         |
| webshells   | webshells   |         |
| weevely     | weevely     |         |
| wfuzz       | wfuzz       |         |
| whatweb     | whatweb     |         |
| wireshark   | wireshark   |         |
| wpscan      | wpscan      |         |
| xsser       | xsser       |         |
| zaproxy     | zaproxy     |         |

## FLARE-VM

### All Packages (choco)

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| 010editor          | 010editor          |         |
| 7zip               | 7zip               |         |
| advanced-installer | advanced-installer |         |
| angr               | angr               |         |
| apimonitor         | apimonitor         |         |
| apktool            | apktool            |         |
| asar               | asar               |         |
| autoit-ripper      | autoit-ripper      |         |
| binaryninja        | binaryninja        |         |
| bindiff            | bindiff            |         |
| blobrunner         | blobrunner         |         |

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| blobrunner64         | blobrunner64         |         |
| bytecodeviewer       | bytecodeviewer       |         |
| capa                 | capa                 |         |
| capa-explorer-web    | capa-explorer-web    |         |
| chrome.extensions    | chrome.extensions    |         |
| cmdr                 | cmdr                 |         |
| codetrack            | codetrack            |         |
| cryptotester         | cryptotester         |         |
| cutter               | cutter               |         |
| cyberchef            | cyberchef            |         |
| cygwin               | cygwin               |         |
| de4dot-cex           | de4dot-cex           |         |
| dependencywalker     | dependencywalker     |         |
| dex2jar              | dex2jar              |         |
| didier-stevens-beta  | didier-stevens-beta  |         |
| didier-stevens-suite | didier-stevens-suite |         |
| die                  | die                  |         |
| dll-to-exe           | dll-to-exe           |         |
| dnlib                | dnlib                |         |
| dnspyex              | dnspyex              |         |
| dotdumper            | dotdumper            |         |
| dotnet3.5            | dotnet3.5            |         |
| exeinfope            | exeinfope            |         |
| explorersuite        | explorersuite        |         |
| extreme_dumper       | extreme_dumper       |         |
| ezviewer             | ezviewer             |         |
| fakenet-ng           | fakenet-ng           |         |
| fiddler              | fiddler              |         |
| file                 | file                 |         |
| floss                | floss                |         |
| garbageman           | garbageman           |         |
| ghidra               | ghidra               |         |
| goresym              | goresym              |         |
| gostringungarbler    | gostringungarbler    |         |



| Tool                     | Command(s)               | Purpose |
|--------------------------|--------------------------|---------|
| hashmyfiles              | hashmyfiles              |         |
| hollowshunter            | hollowshunter            |         |
| hxd                      | hxd                      |         |
| ida.plugin.capa          | ida.plugin.capa          |         |
| ida.plugin.comida        | ida.plugin.comida        |         |
| ida.plugin.delphihelper  | ida.plugin.delphihelper  |         |
| ida.plugin.dereferencing | ida.plugin.dereferencing |         |
| ida.plugin.diaphora      | ida.plugin.diaphora      |         |
| ida.plugin.flare         | ida.plugin.flare         |         |
| ida.plugin.flare-emu     | ida.plugin.flare-emu     |         |
| ida.plugin.hashdb        | ida.plugin.hashdb        |         |
| ida.plugin.hrtng         | ida.plugin.hrtng         |         |
| ida.plugin.ifl           | ida.plugin.ifl           |         |
| ida.plugin.xray          | ida.plugin.xray          |         |
| ida.plugin.xrefer        | ida.plugin.xrefer        |         |
| idafree                  | idafree                  |         |
| idr                      | idr                      |         |
| ifpstools                | ifpstools                |         |
| ilspy                    | ilspy                    |         |
| innoextract              | innoextract              |         |
| innounp                  | innounp                  |         |
| internet_detector        | internet_detector        |         |
| ipython                  | ipython                  |         |
| isd                      | isd                      |         |
| js-beautify              | js-beautify              |         |
| js-deobfuscator          | js-deobfuscator          |         |
| keystone                 | keystone                 |         |
| libraries.python3        | libraries.python3        |         |
| magika                   | magika                   |         |
| malware-jail             | malware-jail             |         |
| map                      | map                      |         |
| microsoft-office         | microsoft-office         |         |
| nasm                     | nasm                     |         |
| net-reactor-slayer       | net-reactor-slayer       |         |

| Tool                       | Command(s)                 | Purpose |
|----------------------------|----------------------------|---------|
| nmap                       | nmap                       |         |
| notepadplusplus            | notepadplusplus            |         |
| notepadpp.plugin.compare   | notepadpp.plugin.compare   |         |
| notepadpp.plugin.jstool    | notepadpp.plugin.jstool    |         |
| notepadpp.plugin.xmltools  | notepadpp.plugin.xmltools  |         |
| obfuscator-io-deobfuscator | obfuscator-io-deobfuscator |         |
| offvis                     | offvis                     |         |
| onenoteanalyzer            | onenoteanalyzer            |         |
| pdbresym                   | pdbresym                   |         |
| pdfstreamdumper            | pdfstreamdumper            |         |
| pe_unmapper                | pe_unmapper                |         |
| pebear                     | pebear                     |         |
| peid                       | peid                       |         |
| pesieve                    | pesieve                    |         |
| pestudio                   | pestudio                   |         |
| pkg-unpacker               | pkg-unpacker               |         |
| pma-labs                   | pma-labs                   |         |
| procdot                    | procdot                    |         |
| processdump                | processdump                |         |
| psnotify                   | psnotify                   |         |
| pycdas                     | pycdas                     |         |
| pycdc                      | pycdc                      |         |
| pylingual                  | pylingual                  |         |
| rat-king-parser            | rat-king-parser            |         |
| recaf                      | recaf                      |         |
| reg_export                 | reg_export                 |         |
| regcool                    | regcool                    |         |
| regshot                    | regshot                    |         |
| resourcehacker             | resourcehacker             |         |
| rundotnetdll               | rundotnetdll               |         |
| sctdbg                     | sctdbg                     |         |
| sclauncher                 | sclauncher                 |         |
| sclauncher64               | sclauncher64               |         |
| sfextract                  | sfextract                  |         |

| Tool                     | Command(s)               | Purpose |
|--------------------------|--------------------------|---------|
| shellcode_launcher       | shellcode_launcher       |         |
| sysinternals             | sysinternals             |         |
| systeminformer           | systeminformer           |         |
| ttd                      | ttd                      |         |
| uncompyle6               | uncompyle6               |         |
| uniextract2              | uniextract2              |         |
| unpyc3                   | unpyc3                   |         |
| upx                      | upx                      |         |
| vb-decompiler-lite       | vb-decompiler-lite       |         |
| vbdec                    | vbdec                    |         |
| vcbuildtools             | vcbuildtools             |         |
| vcredis-all              | vcredis-all              |         |
| vscode                   | vscode                   |         |
| vscode.extension.jupyter | vscode.extension.jupyter |         |
| vscode.extension.python  | vscode.extension.python  |         |
| windbg                   | windbg                   |         |
| windows-terminal         | windows-terminal         |         |
| wireshark                | wireshark                |         |
| x64dbg                   | x64dbg                   |         |
| x64dbg.plugin.dbgchild   | x64dbg.plugin.dbgchild   |         |
| x64dbg.plugin.ollydumpex | x64dbg.plugin.ollydumpex |         |
| x64dbg.plugin.scyllahide | x64dbg.plugin.scyllahide |         |
| x64dbg.plugin.x64dbgpy   | x64dbg.plugin.x64dbgpy   |         |
| yara                     | yara                     |         |

## SIFT Workstation

### Forensic Packages

| Tool                       | Command(s)                 | Purpose |
|----------------------------|----------------------------|---------|
| absent                     | absent                     |         |
| aeskeyfind                 | aeskeyfind                 |         |
| afflib-tools               | afflib-tools               |         |
| aircrack-ng                | aircrack-ng                |         |
| android-sdk-platform-tools | android-sdk-platform-tools |         |

| Tool               | Command(s)         | Purpose |
|--------------------|--------------------|---------|
| arp-scan           | arp-scan           |         |
| autopsy            | autopsy            |         |
| avfs               | avfs               |         |
| aws-cli            | aws-cli            |         |
| bless              | bless              |         |
| blt                | blt                |         |
| bulk-extractor     | bulk-extractor     |         |
| cabextract         | cabextract         |         |
| ccrypt             | ccrypt             |         |
| chromium-browser   | chromium-browser   |         |
| clamav             | clamav             |         |
| claude-code        | claude-code        |         |
| cmospwd            | cmospwd            |         |
| cryptcat           | cryptcat           |         |
| cryptsetup         | cryptsetup         |         |
| dc3dd              | dc3dd              |         |
| dcfldd             | dcfldd             |         |
| default-jre        | default-jre        |         |
| disktype           | disktype           |         |
| dislocker          | dislocker          |         |
| docker             | docker             |         |
| dos2unix           | dos2unix           |         |
| dotnet             | dotnet             |         |
| driftnet           | driftnet           |         |
| dsniff             | dsniff             |         |
| e2fsprogs          | e2fsprogs          |         |
| ent                | ent                |         |
| epic5              | epic5              |         |
| etherape           | etherape           |         |
| ettercap-graphical | ettercap-graphical |         |
| ewf-tools          | ewf-tools          |         |
| exfat-extras       | exfat-extras       |         |
| exfat-fuse         | exfat-fuse         |         |
| exif               | exif               |         |

| Tool        | Command(s)  | Purpose |
|-------------|-------------|---------|
| extundelete | extundelete |         |
| fdupes      | fdupes      |         |
| feh         | feh         |         |
| file        | file        |         |
| flex        | flex        |         |
| foremost    | foremost    |         |
| gawk        | gawk        |         |
| gdb         | gdb         |         |
| gddrescue   | gddrescue   |         |
| ghex        | ghex        |         |
| graphviz    | graphviz    |         |
| grepcidr    | grepcidr    |         |
| gthumb      | gthumb      |         |
| gzrt        | gzrt        |         |
| hashdeep    | hashdeep    |         |
| hexedit     | hexedit     |         |
| hydra       | hydra       |         |
| hydra-gtk   | hydra-gtk   |         |
| init        | init        |         |
| ipython3    | ipython3    |         |
| jq          | jq          |         |
| kdiff3      | kdiff3      |         |
| kpartx      | kpartx      |         |
| lft         | lft         |         |
| lvm2        | lvm2        |         |
| magnus      | magnus      |         |
| mdadm       | mdadm       |         |
| mtd-utils   | mtd-utils   |         |
| nbd-client  | nbd-client  |         |
| nbtscan     | nbtscan     |         |
| netcat      | netcat      |         |
| netpbm      | netpbm      |         |
| netsed      | netsed      |         |
| netwox      | netwox      |         |

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| nfdump               | nfdump               |         |
| ngrep                | ngrep                |         |
| nikto                | nikto                |         |
| ntfs-3g              | ntfs-3g              |         |
| okular               | okular               |         |
| onboard              | onboard              |         |
| open-iscsi           | open-iscsi           |         |
| openjdk              | openjdk              |         |
| ophcrack             | ophcrack             |         |
| ophcrack-cli         | ophcrack-cli         |         |
| orca                 | orca                 |         |
| outguess             | outguess             |         |
| p0f                  | p0f                  |         |
| p7zip-full           | p7zip-full           |         |
| patch                | patch                |         |
| pdftk-java           | pdftk-java           |         |
| perl                 | perl                 |         |
| pev                  | pev                  |         |
| pff-tools            | pff-tools            |         |
| phonon               | phonon               |         |
| pkg-config           | pkg-config           |         |
| plaso-tools          | plaso-tools          |         |
| powershell           | powershell           |         |
| pst-utils            | pst-utils            |         |
| pv                   | pv                   |         |
| python-flowgrep      | python-flowgrep      |         |
| python3-debian       | python3-debian       |         |
| python3-dfvfs        | python3-dfvfs        |         |
| python3-fuse         | python3-fuse         |         |
| python3-keyrings-alt | python3-keyrings-alt |         |
| python3-m2crypto     | python3-m2crypto     |         |
| python3-magic        | python3-magic        |         |
| python3-pefile       | python3-pefile       |         |
| python3-plaso        | python3-plaso        |         |

| Tool                    | Command(s)              | Purpose |
|-------------------------|-------------------------|---------|
| python3-pypff           | python3-pypff           |         |
| python3-pyqt5           | python3-pyqt5           |         |
| python3-pytsk3          | python3-pytsk3          |         |
| python3-redis           | python3-redis           |         |
| python3-setuptools-rust | python3-setuptools-rust |         |
| python3-tk              | python3-tk              |         |
| python3-tsk             | python3-tsk             |         |
| python3-virtualenv      | python3-virtualenv      |         |
| python3-wxgtk4          | python3-wxgtk4          |         |
| python3-xlswriter       | python3-xlswriter       |         |
| python3-yara            | python3-yara            |         |
| qemu                    | qemu                    |         |
| qemu-utils              | qemu-utils              |         |
| radare2                 | radare2                 |         |
| rar                     | rar                     |         |
| rsakeyfind              | rsakeyfind              |         |
| safecopy                | safecopy                |         |
| samdump2                | samdump2                |         |
| scalpel                 | scalpel                 |         |
| silversearcher-ag       | silversearcher-ag       |         |
| sleuthkit               | sleuthkit               |         |
| socat                   | socat                   |         |
| squashfs-tools          | squashfs-tools          |         |
| ssdeep                  | ssdeep                  |         |
| ssldump                 | ssldump                 |         |
| sslsniff                | sslsniff                |         |
| stunnel4                | stunnel4                |         |
| swig                    | swig                    |         |
| tcl                     | tcl                     |         |
| tcpflow                 | tcpflow                 |         |
| tcpick                  | tcpick                  |         |
| tcpreplay               | tcpreplay               |         |
| tcpslice                | tcpslice                |         |
| tcpstat                 | tcpstat                 |         |

| Tool                 | Command(s)           | Purpose |
|----------------------|----------------------|---------|
| tcptrace             | tcptrace             |         |
| tcptrack             | tcptrack             |         |
| tcpxtract            | tcpxtract            |         |
| testdisk             | testdisk             |         |
| tofrodo              | tofrodo              |         |
| transmission         | transmission         |         |
| ugrep                | ugrep                |         |
| unity-control-center | unity-control-center |         |
| unrar                | unrar                |         |
| upx-ucl              | upx-ucl              |         |
| vbindiff             | vbindiff             |         |
| virtuoso-minimal     | virtuoso-minimal     |         |
| vmfs-tools           | vmfs-tools           |         |
| winbind              | winbind              |         |
| wine                 | wine                 |         |
| wireshark            | wireshark            |         |
| xdot                 | xdot                 |         |
| xfsprogs             | xfsprogs             |         |
| xmount               | xmount               |         |
| zenity               | zenity               |         |
| zlib1g-dev           | zlib1g-dev           |         |

## Security Onion

### DB/Search

| Tool          | Command(s) | Purpose                                             |
|---------------|------------|-----------------------------------------------------|
| Elasticsearch | —          | NoSQL index for logs/metadata/alerts                |
| ILM           | —          | Index Lifecycle Management (aging/rolling/deletion) |
| Kibana        | —          | Search analytics & dashboards                       |

### Detection

| Tool             | Command(s) | Purpose                                  |
|------------------|------------|------------------------------------------|
| ATT&CK Navigator | —          | Visual defensive-coverage matrix         |
| ElastAlert 2     | —          | Continuous alert-indicator scanner       |
| Playbook         | —          | Custom detections mapped to MITRE ATT&CK |



| Tool              | Command(s) | Purpose                                           |
|-------------------|------------|---------------------------------------------------|
| Sigma / Sigma-CLI | —          | Translate YAML detection rules to Elastic queries |
| so-idstools       | —          | Manage/update Suricata ET rules                   |

## Diagnostics

| Tool     | Command(s) | Purpose                                |
|----------|------------|----------------------------------------|
| Grafana  | —          | Infra/node-health dashboards           |
| InfluxDB | —          | Time-series store for hardware metrics |
| Telegraf | —          | Hardware health/CPU/mem/disk agent     |

## File Analysis

| Tool        | Command(s) | Purpose                                       |
|-------------|------------|-----------------------------------------------|
| DomainStats | —          | Domain-age checks to flag malicious sites     |
| FreqServer  | —          | Entropy analysis to detect DGAs               |
| Strelka     | —          | Real-time file extraction & YARA/file carving |

## Host/EDR

| Tool          | Command(s) | Purpose                                           |
|---------------|------------|---------------------------------------------------|
| Elastic Agent | —          | Host log/metric/audit collector                   |
| Elastic Fleet | —          | Central agent config & deployment                 |
| OpenCanary    | —          | Network honeypot / decoy services                 |
| Osquery       | —          | Live endpoint queries via SQL                     |
| Wazuh         | —          | HIDS - File Integrity Monitoring & rootkit checks |

## IR/AI

| Tool                         | Command(s) | Purpose                                     |
|------------------------------|------------|---------------------------------------------|
| CyberChef                    | —          | Visual data decode/parse/analysis           |
| Onion AI Assistant           | —          | AI triage assistance for analysts           |
| Security Onion Console (SOC) | —          | Web UI for threat hunting & grid management |
| TheHive                      | —          | Collaborative IR & case management          |

## Identity/Infra

| Tool             | Command(s) | Purpose                                |
|------------------|------------|----------------------------------------|
| ManagerHype      | —          | Hypervisor integration for VM mapping  |
| Nginx (so-nginx) | —          | Reverse proxy for internal web traffic |
| Ory Kratos & Dex | —          | IAM - SSO/auth/sessions                |

| Tool             | Command(s) | Purpose                                        |
|------------------|------------|------------------------------------------------|
| so-apt-cacher-ng | —          | Local package caching proxy                    |
| so-firewall      | —          | iptables/nftables wrapper for inter-node comms |

## Log Routing

| Tool                | Command(s) | Purpose                                   |
|---------------------|------------|-------------------------------------------|
| Filebeat            | —          | Lightweight log shipper                   |
| Logstash            | —          | Data pipeline - ECS normalize & enrich    |
| Redis               | —          | Log ingestion queue/broker                |
| Rsyslog / Syslog-ng | —          | Ingest traditional syslog from appliances |

## Network Visibility

| Tool         | Command(s) | Purpose                                              |
|--------------|------------|------------------------------------------------------|
| Sensoroni    | —          | Backend API agent retrieving/extracting PCAPs        |
| Stenographer | —          | High-speed raw full packet capture daemon            |
| Suricata     | —          | Signature-based IDS/IPS & packet capture             |
| Zeek         | —          | Network traffic analyzer & protocol metadata logging |

## Sources

- `remnux_docs` — <https://raw.githubusercontent.com/REMnux/docs/master/discover-the-tools>
- `kali_control` — <https://gitlab.com/kalilinux/packages/kali-meta/-/raw/kali/master/debian/control>
- `flare_config` — <https://raw.githubusercontent.com/mandiant/flare-vm/main/config.xml>

# Tool Reference

---

One page per tool. Every option below was read off the real binary and is checked by the linter.

# affcat

**Kit:** Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** affcat version 3.7.20 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

*TODO: one-line imperative purpose.*

## Synopsis

```
affcat [options] infile [... more infiles]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 7 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                       | When you would use it |
|------|----------|--------------------------------------------------------------------|-----------------------|
| -q   | —        | --- quiet; don't print to STDERR if a page is skipped              |                       |
| -n   | —        | --- noisy; tell when pages are skipped.                            |                       |
| -l   | —        | --- List all of the segment names                                  |                       |
| -L   | —        | --- List segment names, lengths, and args                          |                       |
| -d   | —        | --- debug. Print the page numbers to stderr as data goes to stdout |                       |
| -b   | —        | --- Output BADFALG for bad blocks (default is NULLs)               |                       |
| -v   | —        | --- Just print the version number and exit.                        |                       |

## Gotchas

*TODO: operational traps.*

## See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [img\\_stat](#), [ntfs-3g](#), [vshadowinfo](#)

# affconvert

---

**Kit:** Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** affconvert version 3.7.20

**Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Please, see more info in manpage.

## Synopsis

---

```
affconvert [options] file1 [... files]
```

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

[dc3dd](#) , [dcfldd](#) , [dd](#) , [ewfacquire](#)

# affinfo

---

**Kit:** Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** affinfo version 3.7.20 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

(may be repeated)

## Synopsis

---

```
affinfo [options] infile
-a = print ALL segments (normally data segments are suppressed)
-b = print how many bad blocks in each segment (implies -a)
-i = identify the files, don't do info on them.
-w = wide output; print more than 1 line if necessary.
-s segment = Just print information about 'segment'.
```

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affcat](#), [img\\_stat](#), [ntfs-3g](#), [vshadowinfo](#)

# bdeinfo

**Kit:** SIFT Workstation (libyal) **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: bdeinfo

## Synopsis

```
bdeinfo [ -k keys ] [ -o offset ] [ -p password ]
[ -r password ] [ -s filename ] [ -hvV ] source
```

## Common invocations

```
# from cyberlab 06-windows-artifact-libs
bdeinfo -V
# from cyberlab 06-windows-artifact-libs
bdeinfo exercise/bitlocker.raw
```

## Options

All 8 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                                    | When you would use it |
|------|----------|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| -h   | —        | shows this help                                                                                                 |                       |
| -k   | —        | the full volume encryption key and tweak key formatted in base16 and separated by a : character e.g. FVEK:TWEAK |                       |
| -o   | —        | specify the volume offset in bytes                                                                              |                       |
| -p   | —        | specify the password/passphrase                                                                                 |                       |
| -r   | —        | specify the recovery password                                                                                   |                       |
| -s   | —        | specify the file containing the startup key. typically this file has the extension .BEK                         |                       |

| Flag            | Argument | What it does             | When you would use it |
|-----------------|----------|--------------------------|-----------------------|
| <code>-v</code> | —        | verbose output to stderr |                       |
| <code>-V</code> | —        | print version            |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`ewfinfo`, `ewfmount`, `ewfverify`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `ntfs-3g`



# dc3dd

**Kit:** Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** dc3dd (dc3dd) 7.2.646  
**Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

dc3dd [OPTION 1] [OPTION 2] ... [OPTION N]

## Synopsis

-----

## Common invocations

```
# from cyberlab 57-forensic-acquisition
sudo dc3dd if=/dev/sdX of=/evidence/case01.dd hash=sha256 log=/evidence/case01.log
```

## Options

All 3 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                        | When you would use it |
|-----------|----------|-------------------------------------|-----------------------|
| --help    | —        | display this help and exit          |                       |
| --version | —        | output version information and exit |                       |
| --flags   | —        | display compile-time flags and exit |                       |

## Gotchas

*TODO: operational traps.*

## See also

[dcfldd](#), [dd](#), [ewfacquire](#), [affconvert](#)

# dcfldd

**Kit:** Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** dcfldd (dcfldd) 1.9 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Enhanced version of dd for forensics and security.

## Synopsis

```
dcfldd [OPTION]...
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 2 options parsed from the captured help text. The final column is filled in by review.

| Flag                   | Argument | What it does                        | When you would use it |
|------------------------|----------|-------------------------------------|-----------------------|
| <code>--help</code>    | —        | display this help and exit          |                       |
| <code>--version</code> | —        | output version information and exit |                       |

## Gotchas

*TODO: operational traps.*

## See also

`dc3dd`, `dd`, `ewfacquire`, `affconvert`

# dd

**Kit:** Base OS — present on every Linux image **Capability:** Image a disk, volume or device **Version:** dd (coreutils) 9.1  
**Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

or: dd OPTION

## Synopsis

```
dd [OPERAND]...  
or: dd OPTION
```

## Common invocations

```
# from cyberlab 01-disk-forensics  
dd if=/dev/zero of=exercise/sample.dd bs=1M count=10  
# from cyberlab 51-linux-triage-workflow  
dd if=/dev/zero of=trriage_sample.raw bs=1M count=8
```

## Options

All 2 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                        | When you would use it |
|-----------|----------|-------------------------------------|-----------------------|
| --help    | —        | display this help and exit          |                       |
| --version | —        | output version information and exit |                       |

## Gotchas

TODO: operational traps.

## See also

[dc3dd](#), [dcfldd](#), [ewfacquire](#), [affconvert](#)

# dumpcap

**Kit:** REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Capture live network traffic **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw Docs](#): <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Capture and analyze network traffic with this sniffer.

## Synopsis

```
dumpcap [options] ...
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 48 options parsed from the captured help text. The final column is filled in by review.

| Flag                               | Argument       | What it does                                                                                                           | When you would use it |
|------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-i</code>                    | interface      | name or idx of interface (def: first non-loopback), or for remote capturing, use one of these formats: rpcap:/// TCP@: |                       |
| <code>--interface</code>           | interface      | name or idx of interface (def: first non-loopback), or for remote capturing, use one of these formats: rpcap:/// TCP@: |                       |
| <code>--ifname</code>              | name           | name to use in the capture file for a pipe from which we're capturing                                                  |                       |
| <code>--ifdescr</code>             | description    | description to use in the capture file for a pipe from which we're capturing                                           |                       |
| <code>-f</code>                    | capture filter | packet filter in libpcap filter syntax                                                                                 |                       |
| <code>-s</code>                    | snaplen        | packet snapshot length (def: appropriate maximum)                                                                      |                       |
| <code>--snapshot-length</code>     | snaplen        | packet snapshot length (def: appropriate maximum)                                                                      |                       |
| <code>-p</code>                    | —              | don't capture in promiscuous mode                                                                                      |                       |
| <code>--no-promiscuous-mode</code> | —              | don't capture in promiscuous mode                                                                                      |                       |

| Flag                                 | Argument        | What it does                                                                                                                                                                                             | When you would use it |
|--------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-I</code>                      | —               | capture in monitor mode, if available                                                                                                                                                                    |                       |
| <code>--monitor-mode</code>          | —               | capture in monitor mode, if available                                                                                                                                                                    |                       |
| <code>-B</code>                      | buffer size     | size of kernel buffer in MiB (def: 2MiB)                                                                                                                                                                 |                       |
| <code>--buffer-size</code>           | buffer size     | size of kernel buffer in MiB (def: 2MiB)                                                                                                                                                                 |                       |
| <code>-y</code>                      | link type       | link layer type (def: first appropriate)                                                                                                                                                                 |                       |
| <code>--linktype</code>              | link type       | link layer type (def: first appropriate)                                                                                                                                                                 |                       |
| <code>-D</code>                      | —               | print list of interfaces and exit                                                                                                                                                                        |                       |
| <code>--list-interfaces</code>       | —               | print list of interfaces and exit                                                                                                                                                                        |                       |
| <code>-L</code>                      | —               | print list of link-layer types of iface and exit                                                                                                                                                         |                       |
| <code>--list-data-link-types</code>  | —               | print list of link-layer types of iface and exit                                                                                                                                                         |                       |
| <code>--list-time-stamp-types</code> | —               | print list of timestamp types for iface and exit                                                                                                                                                         |                       |
| <code>-d</code>                      | —               | print generated BPF code for capture filter                                                                                                                                                              |                       |
| <code>-S</code>                      | —               | print statistics for each interface once per second                                                                                                                                                      |                       |
| <code>-M</code>                      | —               | for -D, -L, and -S, produce machine-readable output                                                                                                                                                      |                       |
| <code>-c</code>                      | packet count    | stop after n packets (def: infinite)                                                                                                                                                                     |                       |
| <code>-a</code>                      | autostop cond.  | duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM kB files:NUM - stop after NUM files packets:NUM - stop after NUM packets                                                   |                       |
| <code>--autostop</code>              | autostop cond.  | duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM kB files:NUM - stop after NUM files packets:NUM - stop after NUM packets                                                   |                       |
| <code>-w</code>                      | filename        | name of file to save (def: tempfile)                                                                                                                                                                     |                       |
| <code>-g</code>                      | —               | enable group read access on the output file(s)                                                                                                                                                           |                       |
| <code>-b</code>                      | ringbuffer opt. | duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM kB files:NUM - ringbuffer: replace after NUM files packets:NUM - ringbuffer: replace after NUM packets in |                       |
| <code>--ring-buffer</code>           | ringbuffer opt. | duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM kB files:NUM - ringbuffer: replace after NUM files packets:NUM - ringbuffer: replace after NUM packets in |                       |
| <code>-n</code>                      | —               | use pcapng format instead of pcap (default)                                                                                                                                                              |                       |
| <code>-P</code>                      | —               | use libpcap format instead of pcapng                                                                                                                                                                     |                       |
| <code>--capture-comment</code>       | comment         | add a capture comment to the output file (only for pcapng)                                                                                                                                               |                       |
| <code>--temp-dir</code>              | directory       | write temporary files to this directory (default: /tmp)                                                                                                                                                  |                       |

| Flag                       | Argument     | What it does                                                      | When you would use it |
|----------------------------|--------------|-------------------------------------------------------------------|-----------------------|
| <code>--log-level</code>   | level        | sets the active log level ("critical", "warning", etc.)           |                       |
| <code>--log-fatal</code>   | level        | sets level to abort the program ("critical" or "warning")         |                       |
| <code>--log-domains</code> | [!]list      | comma separated list of the active log domains                    |                       |
| <code>--log-debug</code>   | [!]list      | comma separated list of domains with "debug" level                |                       |
| <code>--log-noisy</code>   | [!]list      | comma separated list of domains with "noisy" level                |                       |
| <code>--log-file</code>    | path         | file to output messages to (in addition to stderr)                |                       |
| <code>-N</code>            | packet_limit | maximum number of packets buffered within dumpcap                 |                       |
| <code>-C</code>            | byte_limit   | maximum number of bytes used for buffering packets within dumpcap |                       |
| <code>-t</code>            | —            | use a separate thread per interface                               |                       |
| <code>-q</code>            | —            | don't report packet capture counts                                |                       |
| <code>-v</code>            | —            | print version information and exit                                |                       |
| <code>--version</code>     | —            | print version information and exit                                |                       |
| <code>-h</code>            | —            | display this help and exit                                        |                       |
| <code>--help</code>        | —            | display this help and exit                                        |                       |

## Gotchas

*TODO: operational traps.*

## See also

[tshark](#)

# ewfacquire

**Kit:** Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Acquire a disk, volume or device into an EWF/E01 evidence container, with the case metadata and hashes stored inside the image itself.

## Synopsis

```
ewfacquire [ -A codepage ] [ -b number_of_sectors ]
[ -B number_of_bytes ] [ -c compression_values ]
[ -C case_number ] [ -d digest_type ] [ -D description ]
[ -e examiner_name ] [ -E evidence_number ] [ -f format ]
[ -g number_of_sectors ] [ -l log_filename ]
[ -m media_type ] [ -M media_flags ] [ -N notes ]
```

## Common invocations

```
# from cyberlab 57-forensic-acquisition
sudo ewfacquire -t /evidence/case01 -f encase6 -c deflate -S 2G /dev/sdX
```

## Options

All 32 options parsed from the captured help text; 30 reviewed with usage guidance.

| Flag            | Argument | What it does                                                                                                                                                                                             | When you would use it                                                                                              |
|-----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>-A</code> | —        | codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125 | Header codepage, for non-ASCII metadata.                                                                           |
| <code>-b</code> | —        | specify the number of sectors to read at once (per chunk), options: 16, 32, 64 (default), 128, 256, 512, 1024, 2048, 4096, 8192, 16384 or 32768                                                          | Sectors per chunk. Larger chunks read faster but lose more data to each unreadable sector.                         |
| <code>-B</code> | —        | specify the number of bytes to acquire (default is all bytes)                                                                                                                                            | Acquire a fixed number of bytes rather than the whole device — with <code>-o</code> , the way to image one region. |

| Flag            | Argument | What it does                                                                                                                                                                                                | When you would use it                                                                                                                                                             |
|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-c</code> | —        | specify the compression values as: level or method:level<br>compression method options: deflate (default), bzip2 (bzip2 is only supported by EWF2 formats) compression level options: none (default), empty | Trade size against time. <code>best</code> on a slow USB source is often faster overall than <code>none</code> , because the bottleneck is the read, not the CPU.                 |
| <code>-C</code> | —        | specify the case number (default is case_number).                                                                                                                                                           | Case number, stored in the image header.                                                                                                                                          |
| <code>-d</code> | —        | calculate additional digest (hash) types besides md5, options: sha1, sha256                                                                                                                                 | Add sha1 or sha256 alongside the default md5. Worth doing once, at acquisition: md5 alone is increasingly challenged, and rehashing later means reading the whole image again.    |
| <code>-D</code> | —        | specify the description (default is description).                                                                                                                                                           | Description, stored in the image header.                                                                                                                                          |
| <code>-e</code> | —        | specify the examiner name (default is examiner_name).                                                                                                                                                       | Examiner name, stored in the image header.                                                                                                                                        |
| <code>-E</code> | —        | specify the evidence number (default is evidence_number).                                                                                                                                                   | Evidence number, stored in the image header.                                                                                                                                      |
| <code>-f</code> | —        | specify the EWF file format to write to, options: ewf, smart, ftk, encase2, encase3, encase4, encase5, encase6 (default), encase7, encase7-v2, linen5, linen6, linen7, ewfx                                 | Choose the EWF variant. <code>encase6</code> is the safe default for interoperability; pick the format the tool that will read it expects, not the newest one available.          |
| <code>-g</code> | —        | specify the number of sectors to be used as error granularity                                                                                                                                               | Error granularity — how much data around a bad sector is discarded. Lower values preserve more of a failing disk at the cost of speed.                                            |
| <code>-h</code> | —        | shows this help                                                                                                                                                                                             |                                                                                                                                                                                   |
| <code>-l</code> | —        | logs acquiry errors and the digest (hash) to the log_filename                                                                                                                                               | Write the acquisition log, including errors and hashes, to a file. This is the record you cite later; do not skip it.                                                             |
| <code>-m</code> | —        | specify the media type, options: fixed (default), removable, optical, memory                                                                                                                                | Record the media type (fixed, removable, optical, memory) in the header.                                                                                                          |
| <code>-M</code> | —        | specify the media flags, options: logical, physical (default)                                                                                                                                               | Record whether this is a physical device or a logical volume.                                                                                                                     |
| <code>-N</code> | —        | specify the notes (default is notes).                                                                                                                                                                       | Free-text notes, stored in the image header. These five metadata flags are the reason to choose E01 over a raw <code>dd</code> image: the container describes its own provenance. |
| <code>-o</code> | —        | specify the offset to start to acquire (default is 0)                                                                                                                                                       | Start at an offset rather than sector 0.                                                                                                                                          |
| <code>-p</code> | —        | specify the process buffer size (default is the chunk size)                                                                                                                                                 | Process buffer size — a throughput tuning knob.                                                                                                                                   |
| <code>-P</code> | —        | specify the number of bytes per sector (default is 512) (use this to override the automatic bytes per sector detection)                                                                                     | Override bytes-per-sector. Needed on 4Kn drives where the 512-byte assumption is wrong.                                                                                           |
| <code>-q</code> | —        | quiet shows minimal status information                                                                                                                                                                      | Minimal status output, for logs.                                                                                                                                                  |
| <code>-r</code> | —        | specify the number of retries when a read error occurs (default is 2)                                                                                                                                       | Read retries before giving up on a sector. Raise it for a dying drive; lower it when retries are heating a drive that may not survive the acquisition.                            |
| <code>-R</code> | —        | resume acquiry at a safe point                                                                                                                                                                              | Resume an interrupted acquisition at a safe point, instead of restarting a multi-hour read.                                                                                       |



| Flag            | Argument | What it does                                                                                                                                                      | When you would use it                                                                                                                                                     |
|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-s</code> | —        | swap byte pairs of the media data (from AB to BA) (use this for big to little endian conversion and vice versa)                                                   | Swap byte pairs for a big-endian source. Rare, and wrong unless you know the source endianness differs.                                                                   |
| <code>-S</code> | —        | specify the segment file size in bytes (default is 1.4 GiB) (minimum is 1.0 MiB, maximum is 7.9 EiB for encase6 and encase7 format and 1.9 GiB for other formats) | Segment size. The 1.4 GiB default suits FAT32 and optical media; raise it on a modern filesystem to avoid hundreds of fragments.                                          |
| <code>-t</code> | —        | specify the target file (without extension) to write to                                                                                                           | Set the output name, without extension — <code>ewfacquire</code> appends <code>.E01</code> , <code>.E02</code> ... itself. The one flag you always pass.                  |
| <code>-T</code> | —        | specify the file containing the table of contents (TOC) of an optical disc. The TOC file must be in the CUE format.                                               | Supply a CUE file when imaging optical media, so the track layout is preserved.                                                                                           |
| <code>-u</code> | —        | unattended mode (disables user interaction)                                                                                                                       | Unattended — suppresses the interactive prompts. Required for any scripted or headless acquisition.                                                                       |
| <code>-v</code> | —        | verbose output to stderr                                                                                                                                          | Verbose diagnostics to stderr — worth capturing alongside <code>-l</code> when a source is throwing read errors.                                                          |
| <code>-V</code> | —        | print version                                                                                                                                                     |                                                                                                                                                                           |
| <code>-w</code> | —        | zero sectors on read error (mimic EnCase like behavior)                                                                                                           | Zero unreadable sectors instead of aborting, the way EnCase does. Keeps offsets aligned so the filesystem still parses.                                                   |
| <code>-x</code> | —        | use the chunk data instead of the buffered read and write functions.                                                                                              | Bypass the buffered read/write path.                                                                                                                                      |
| <code>-2</code> | —        | specify the secondary target file (without extension) to write to                                                                                                 | Write a second copy in the same pass. Two independent copies for the cost of one read, which matters when the source is failing and may not survive a second acquisition. |

## Gotchas

- The case metadata flags default to literal placeholder strings — `case_number`, `examiner_name`, `evidence_number`. Omit them and the image ships with those placeholders recorded as fact, which is worse than an empty field because it looks filled in.
- `-t` takes the name **without** an extension. Passing `image.E01` produces `image.E01.E01`.
- Acquisition is not verification. `ewfacquire` records hashes; proving the image still matches them later is `ewfverify`.

## See also

`dc3dd`, `dcfldd`, `dd`, `affconvert`

# ewfexport

**Kit:** Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Convert an EWF/E01 image to raw, or to another EWF format, including extracting a subset of it.

## Synopsis

```
ewfexport [ -A codepage ] [ -b number_of_sectors ]
[ -B number_of_bytes ] [ -c compression_values ]
[ -d digest_type ] [ -f format ] [ -l log_filename ]
[ -o offset ] [ -p process_buffer_size ]
[ -S segment_file_size ] [ -t target ] [ -hqsuvVwx ] ewf_files
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 19 options parsed from the captured help text; 17 reviewed with usage guidance.

| Flag            | Argument | What it does                                                                                                                                                                                                | When you would use it                                                      |
|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <code>-A</code> | —        | codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125    | Header codepage.                                                           |
| <code>-b</code> | —        | specify the number of sectors to read at once (per chunk), options: 16, 32, 64 (default), 128, 256, 512, 1024, 2048, 4096, 8192, 16384 or 32768 (not used for raw and files formats)                        | Sectors per chunk.                                                         |
| <code>-B</code> | —        | specify the number of bytes to export (default is all bytes)                                                                                                                                                | Number of bytes to export — with <code>-o</code> , extracts one partition. |
| <code>-c</code> | —        | specify the compression values as: level or method:level<br>compression method options: deflate (default), bzip2 (bzip2 is only supported by EWF2 formats) compression level options: none (default), empty | Compression for an EWF target.                                             |

| Flag            | Argument | What it does                                                                                                                                                                                             | When you would use it                                                                                                                           |
|-----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-d</code> | —        | calculate additional digest (hash) types besides md5, options: sha1, sha256 (not used for raw and files format)                                                                                          | Calculate an additional digest over the exported data.                                                                                          |
| <code>-f</code> | —        | specify the output format to write to, options: raw (default), files (restricted to logical volume files), ewf, smart, encase1, encase2, encase3, encase4, encase5, encase6, encase7, encase7-v2, linen5 | Output format — <code>raw</code> for tools that cannot read EWF, or another EWF variant for compatibility.                                      |
| <code>-h</code> | —        | shows this help                                                                                                                                                                                          |                                                                                                                                                 |
| <code>-l</code> | —        | logs export errors and the digest (hash) to the log_filename                                                                                                                                             | Log the export.                                                                                                                                 |
| <code>-o</code> | —        | specify the offset to start the export (default is 0)                                                                                                                                                    | Start offset, to export a region rather than the whole image.                                                                                   |
| <code>-p</code> | —        | specify the process buffer size (default is the chunk size)                                                                                                                                              | Process buffer size.                                                                                                                            |
| <code>-q</code> | —        | quiet shows minimal status information                                                                                                                                                                   | Minimal output.                                                                                                                                 |
| <code>-s</code> | —        | swap byte pairs of the media data (from AB to BA) (use this for big to little endian conversion and vice versa)                                                                                          | Swap byte pairs.                                                                                                                                |
| <code>-S</code> | —        | specify the segment file size in bytes (default is 1.4 GiB) (minimum is 1.0 MiB, maximum is 7.9 EiB for raw, encase6 and encase7 format and 1.9 GiB for other formats) (not used for files format)       | Segment size for the output.                                                                                                                    |
| <code>-t</code> | —        | specify the target file to export to, use - for stdout (default is export) stdout is only supported for the raw format                                                                                   | Target name. <code>-</code> writes to stdout, which lets the image be piped straight into another tool without staging a full raw copy on disk. |
| <code>-u</code> | —        | unattended mode (disables user interaction)                                                                                                                                                              | Unattended, for scripted conversion.                                                                                                            |
| <code>-v</code> | —        | verbose output to stderr                                                                                                                                                                                 | Verbose diagnostics to stderr.                                                                                                                  |
| <code>-V</code> | —        | print version                                                                                                                                                                                            |                                                                                                                                                 |
| <code>-w</code> | —        | zero sectors on checksum error (mimic EnCase like behavior)                                                                                                                                              | Zero sectors that cannot be read.                                                                                                               |
| <code>-x</code> | —        | use the chunk data instead of the buffered read and write functions.                                                                                                                                     | Bypass the buffered read/write path.                                                                                                            |

## Gotchas

- Exporting to raw discards the metadata and hashes that justified using E01. Keep the original: the raw copy is a working artefact, not the evidence.
- A raw export needs the full uncompressed size in free space. E01 compression routinely hides a 2 TB image inside 700 GB.

## See also

---

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [affinfo](#), [affcat](#), [img\\_stat](#), [ntfs-3g](#), [vshadowinfo](#)

**Kit:** Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Show the metadata, hashes and acquisition details recorded inside an EWF/E01 image.

Synopsis

```
ewfinfo [ -A codepage ] [ -d date_format ] [ -f format ]
[ -himvVx ] ewf_files
```

Common invocations

```
# from cyberlab 57-forensic-acquisition
ewfinfo /evidence/case01.E01
```

Options

All 9 options parsed from the captured help text; 7 reviewed with usage guidance.

| Flag | Argument | What it does                                                                                                                                                                                             | When you would use it                                                                                                                            |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| -A   | —        | codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125 | Header codepage, for images with non-ASCII metadata.                                                                                             |
| -d   | —        | specify the date format, options: ctime (default), dm (day/month), md (month/day), iso8601                                                                                                               | Date format. iso8601 is the unambiguous choice for a report.                                                                                     |
| -e   | —        | only show EWF read error information                                                                                                                                                                     | Read errors recorded at acquisition. Check this before trusting a clean-looking image: unreadable sectors are noted here, not in the filesystem. |
| -f   | —        | specify the output format, options: text (default), dfxml                                                                                                                                                | Emit DFXML instead of text, when the output feeds a tool rather than a person.                                                                   |
| -h   | —        | shows this help                                                                                                                                                                                          |                                                                                                                                                  |

| Flag            | Argument | What it does                      | When you would use it                                      |
|-----------------|----------|-----------------------------------|------------------------------------------------------------|
| <code>-i</code> | —        | only show EWF acquiry information | Acquisition details only — who imaged it, when, with what. |
| <code>-m</code> | —        | only show EWF media information   | Media details only — geometry, sector size, media type.    |
| <code>-v</code> | —        | verbose output to stderr          | Verbose diagnostics to stderr.                             |
| <code>-V</code> | —        | print version                     |                                                            |

## Gotchas

- This reads the header only. It reports the hash that was recorded at acquisition; it does not recompute it, so it cannot tell you the image is still intact. Use `ewfverify` for that.

## See also

`ewfmount`, `ewfverify`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `ntfs-3g`, `vshadowinfo`

# ewfmount

**Kit:** Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: ewfmount

## Synopsis

```
ewfmount [ -f format ] [ -X extended_options ] [ -hvV ] image mount_point
```

## Common invocations

```
# from cyberlab 57-forensic-acquisition
sudo ewfmount /evidence/case01.E01 /mnt/ewf
```

## Options

All 5 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                 | When you would use it |
|------|----------|----------------------------------------------------------------------------------------------|-----------------------|
| -f   | —        | specify the input format, options: raw (default), files (restricted to logical volume files) |                       |
| -h   | —        | shows this help                                                                              |                       |
| -v   | —        | verbose output to stderr, while ewfmount will remain running in the foreground               |                       |
| -V   | —        | print version                                                                                |                       |
| -X   | —        | extended options to pass to sub system                                                       |                       |

## Gotchas

TODO: operational traps.

## See also

[ewfinfo](#) , [ewfverify](#) , [ewfexport](#) , [affinfo](#) , [affcat](#) , [img\\_stat](#) , [ntfs-3g](#) , [vshadowinfo](#)

# ewfverify

**Kit:** Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Recompute an EWF/E01 image's hashes and check them against the values stored at acquisition.

## Synopsis

```
ewfverify [ -A codepage ] [ -d digest_type ] [ -f format ]
[ -l log_filename ] [ -p process_buffer_size ]
[ -hqVWwx ] ewf_files
```

## Common invocations

```
# from cyberlab 57-forensic-acquisition
ewfverify /evidence/case01.E01
```

## Options

All 11 options parsed from the captured help text; 9 reviewed with usage guidance.

| Flag | Argument | What it does                                                                                                                                                                                             | When you would use it                                                                                             |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| -A   | —        | codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125 | Header codepage.                                                                                                  |
| -d   | —        | calculate additional digest (hash) types besides md5, options: sha1, sha256                                                                                                                              | Also verify an additional digest such as sha256, when one was recorded at acquisition.                            |
| -f   | —        | specify the input format, options: raw (default), files (restricted to logical volume files)                                                                                                             | Output format.                                                                                                    |
| -h   | —        | shows this help                                                                                                                                                                                          |                                                                                                                   |
| -l   | —        | logs verification errors and the digest (hash) to the log_filename                                                                                                                                       | Write the verification result to a log file — the artefact worth keeping with the case, not just terminal output. |



| Flag            | Argument | What it does                                                         | When you would use it                |
|-----------------|----------|----------------------------------------------------------------------|--------------------------------------|
| <code>-p</code> | —        | specify the process buffer size (default is the chunk size)          | Process buffer size.                 |
| <code>-q</code> | —        | quiet shows minimal status information                               | Minimal output, for scripted checks. |
| <code>-v</code> | —        | verbose output to stderr                                             | Verbose diagnostics to stderr.       |
| <code>-V</code> | —        | print version                                                        |                                      |
| <code>-w</code> | —        | zero sectors on checksum error (mimic EnCase like behavior)          | Wipe sectors that could not be read. |
| <code>-x</code> | —        | use the chunk data instead of the buffered read and write functions. | Bypass the buffered read/write path. |

## Gotchas

- This reads every byte, so it takes as long as the acquisition did. Budget for that rather than discovering it mid-deadline.
- A pass proves the image matches what was recorded **at acquisition**. It says nothing about whether the acquisition captured the device correctly — a disk failing mid-image produces a verifiable image of incomplete data.

## See also

`ewfinfo`, `ewfmount`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `ntfs-3g`, `vshadowinfo`

# img\_stat

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container; See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Analyze disk images and recover files from them.

## Synopsis

```
img_stat [-tvV] [-i imgtype] [-b dev_sector_size] image
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 5 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument        | What it does                                                             | When you would use it |
|------|-----------------|--------------------------------------------------------------------------|-----------------------|
| -t   | —               | display type only                                                        |                       |
| -i   | imgtype         | The format of the image file (use '-i list' for list of supported types) |                       |
| -b   | dev_sector_size | The size (in bytes) of the device sectors                                |                       |
| -v   | —               | verbose output to stderr                                                 |                       |
| -V   | —               | Print version                                                            |                       |

## Gotchas

*TODO: operational traps.*

## See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [affcat](#), [ntfs-3g](#), [vshadowinfo](#)

# md5sum

**Kit:** Base OS — present on every Linux image **Capability:** Verify evidence integrity with hashes **Version:** md5sum (GNU coreutils) 9.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Print or check MD5 (128-bit) checksums.

## Synopsis

```
md5sum [OPTION]... [FILE]...
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 17 options parsed from the captured help text. The final column is filled in by review.

| Flag             | Argument | What it does                                                               | When you would use it |
|------------------|----------|----------------------------------------------------------------------------|-----------------------|
| -b               | —        | read in binary mode                                                        |                       |
| --binary         | —        | read in binary mode                                                        |                       |
| -c               | —        | read checksums from the FILEs and check them                               |                       |
| --check          | —        | read checksums from the FILEs and check them                               |                       |
| --tag            | —        | create a BSD-style checksum                                                |                       |
| -t               | —        | read in text mode (default)                                                |                       |
| --text           | —        | read in text mode (default)                                                |                       |
| -z               | —        | end each output line with NUL, not newline, and disable file name escaping |                       |
| --zero           | —        | end each output line with NUL, not newline, and disable file name escaping |                       |
| --ignore-missing | —        | don't fail or report status for missing files                              |                       |
| --quiet          | —        | don't print OK for each successfully verified file                         |                       |

| Flag                   | Argument | What it does                                          | When you would use it |
|------------------------|----------|-------------------------------------------------------|-----------------------|
| <code>--status</code>  | —        | don't output anything, status code shows success      |                       |
| <code>--strict</code>  | —        | exit non-zero for improperly formatted checksum lines |                       |
| <code>-w</code>        | —        | warn about improperly formatted checksum lines        |                       |
| <code>--warn</code>    | —        | warn about improperly formatted checksum lines        |                       |
| <code>--help</code>    | —        | display this help and exit                            |                       |
| <code>--version</code> | —        | output version information and exit                   |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`rahash2`, `ssdeep`, `sha256sum`, `sigtool`

# ntfs-3g

---

**Kit:** SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** ntfs-3g 2022.10.3

**Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

ntfs-3g 2022.10.3 integrated FUSE 28 - Third Generation NTFS Driver

## Synopsis

---

```
ntfs-3g [-o option[,...]] <device|image_file> <mount_point>
```

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`ewfinfo`, `ewfmount`, `ewfverify`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `vshadowinfo`

# rahash2

**Kit:** REMnux **Capability:** Verify evidence integrity with hashes **Version:** rahash2 6.1.9 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

## Synopsis

```
rahash2 [-BehjkLqRrvX] [-b S] [-a A] [-c H] [-E A] [-s S] [-f O] [-t O] [file] ...
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 23 options parsed from the captured help text. The final column is filled in by review.

| Flag            | Argument | What it does                                             | When you would use it |
|-----------------|----------|----------------------------------------------------------|-----------------------|
| <code>-a</code> | algo     | comma separated list of algorithms (default is 'sha256') |                       |
| <code>-b</code> | bsize    | specify the size of the block (instead of full file)     |                       |
| <code>-B</code> | —        | show per-block hash                                      |                       |
| <code>-c</code> | hash     | compare with this hash                                   |                       |
| <code>-e</code> | —        | swap endian (use little endian)                          |                       |
| <code>-E</code> | algo     | encrypt. Use -S to set key and -I to set IV              |                       |
| <code>-D</code> | algo     | decrypt. Use -S to set key and -I to set IV              |                       |

| Flag            | Argument | What it does                                                                                                                    | When you would use it |
|-----------------|----------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-f</code> | from     | start hashing at given address                                                                                                  |                       |
| <code>-i</code> | num      | repeat hash N iterations (f.ex: 3DES)                                                                                           |                       |
| <code>-I</code> | iv       | use give initialization vector (IV) (hexa or s:string)                                                                          |                       |
| <code>-j</code> | —        | output in json                                                                                                                  |                       |
| <code>-J</code> | —        | new simplified json output (same as -jj)                                                                                        |                       |
| <code>-S</code> | seed     | use given seed (hexa or s:string) use ^ to prefix (key for -E) (- will slurp the key from stdin, the @ prefix points to a file) |                       |
| <code>-k</code> | —        | show hash using the openssh's randomkey algorithm                                                                               |                       |
| <code>-q</code> | —        | run in quiet mode (-qq to show only the hash)                                                                                   |                       |
| <code>-L</code> | —        | list muta plugins (combines with -q, used by -a, -E and -D)                                                                     |                       |
| <code>-r</code> | —        | output radare commands                                                                                                          |                       |
| <code>-R</code> | —        | output radare2 sdb commands (k file.=...)                                                                                       |                       |
| <code>-s</code> | string   | hash this string instead of files                                                                                               |                       |
| <code>-t</code> | to       | stop hashing at given address                                                                                                   |                       |
| <code>-x</code> | hexstr   | hash this hexpair string instead of files                                                                                       |                       |
| <code>-X</code> | —        | output in hexpairs instead of binary/plain                                                                                      |                       |
| <code>-v</code> | —        | show version information                                                                                                        |                       |

## Gotchas

TODO: operational traps.

## See also

`ssdeep`, `sha256sum`, `md5sum`, `sigtool`

# sha256sum

**Kit:** Base OS — present on every Linux image **Capability:** Verify evidence integrity with hashes **Version:** sha256sum (GNU coreutils) 9.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Print or check SHA256 (256-bit) checksums.

## Synopsis

```
sha256sum [OPTION]... [FILE]...
```

## Common invocations

```
# from cyberlab 01-disk-forensics
sha256sum /tmp/recovered.txt
# from cyberlab 04-registry-analysis
sha256sum exercise/SYSTEM_sample.hive
# from cyberlab 05-file-carving
sha256sum /tmp/ak_tcp/*.jpg
# from cyberlab 10-malicious-documents
sha256sum exercise/sample.doc
# from cyberlab 21-yara-authoring
sha256sum exercise/eicar_sample.txt
# from cyberlab 26-metasploit-workflow
sha256sum exercise/scan.xml
# from cyberlab 32-remnux-static-triage
sha256sum exercise/sample.exe
# from cyberlab 33-binwalk-firmware
sha256sum firmware.bin
```

## Options

All 17 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does        | When you would use it |
|------|----------|---------------------|-----------------------|
| -b   | —        | read in binary mode |                       |



| Flag                          | Argument | What it does                                                               | When you would use it |
|-------------------------------|----------|----------------------------------------------------------------------------|-----------------------|
| <code>--binary</code>         | —        | read in binary mode                                                        |                       |
| <code>-c</code>               | —        | read checksums from the FILEs and check them                               |                       |
| <code>--check</code>          | —        | read checksums from the FILEs and check them                               |                       |
| <code>--tag</code>            | —        | create a BSD-style checksum                                                |                       |
| <code>-t</code>               | —        | read in text mode (default)                                                |                       |
| <code>--text</code>           | —        | read in text mode (default)                                                |                       |
| <code>-z</code>               | —        | end each output line with NUL, not newline, and disable file name escaping |                       |
| <code>--zero</code>           | —        | end each output line with NUL, not newline, and disable file name escaping |                       |
| <code>--ignore-missing</code> | —        | don't fail or report status for missing files                              |                       |
| <code>--quiet</code>          | —        | don't print OK for each successfully verified file                         |                       |
| <code>--status</code>         | —        | don't output anything, status code shows success                           |                       |
| <code>--strict</code>         | —        | exit non-zero for improperly formatted checksum lines                      |                       |
| <code>-w</code>               | —        | warn about improperly formatted checksum lines                             |                       |
| <code>--warn</code>           | —        | warn about improperly formatted checksum lines                             |                       |
| <code>--help</code>           | —        | display this help and exit                                                 |                       |
| <code>--version</code>        | —        | output version information and exit                                        |                       |

## Gotchas

*TODO: operational traps.*

## See also

`rahash2`, `ssdeep`, `md5sum`, `sigtool`

# sigtool

**Kit:** REMnux · SIFT Workstation **Capability:** Verify evidence integrity with hashes; Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw Docs:](#) <https://www.clamav.net>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Scan files for malware signatures.

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 35 options parsed from the captured help text. The final column is filled in by review.

| Flag              | Argument | What it does                                                         | When you would use it |
|-------------------|----------|----------------------------------------------------------------------|-----------------------|
| --help            | —        | Show this help                                                       |                       |
| -h                | —        | Show this help                                                       |                       |
| --version         | —        | Print version number and exit                                        |                       |
| -V                | —        | Print version number and exit                                        |                       |
| --quiet           | —        | Be quiet, output only error messages                                 |                       |
| --debug           | —        | Enable debug messages                                                |                       |
| --stdout          | —        | Write to stdout instead of stderr. Does not affect 'debug' messages. |                       |
| --hex-dump        | —        | Convert data from stdin to a hex string and print it on stdout       |                       |
| --md5             | FILES    | Generate MD5 checksum from stdin or MD5 sigs for FILES               |                       |
| --sha1            | FILES    | Generate SHA1 checksum from stdin or SHA1 sigs for FILES             |                       |
| --sha256          | FILES    | Generate SHA256 checksum from stdin or SHA256 sigs for FILES         |                       |
| --mdb             | FILES    | Generate .mdb (section hash) sigs                                    |                       |
| --imp             | FILES    | Generate .imp (import table hash) sigs                               |                       |
| --html-normalise  | FILE     | Create normalised parts of HTML file                                 |                       |
| --ascii-normalise | FILE     | Create normalised text file from ascii source                        |                       |

| Flag                          | Argument        | What it does                                                                                                                                                                                            | When you would use it |
|-------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--utf16-decode</code>   | FILE            | Decode UTF16 encoded files                                                                                                                                                                              |                       |
| <code>--info</code>           | FILE            | -i FILE Print database information                                                                                                                                                                      |                       |
| <code>--max-bad-sigs</code>   | NUMBER          | Maximum number of mismatched signatures When building a CVD. Default: 3000                                                                                                                              |                       |
| <code>--flevel</code>         | FLEVEL          | Specify a custom flevel. Default: 213                                                                                                                                                                   |                       |
| <code>--cvd-version</code>    | NUMBER          | Specify the version number to use for the build. Default is to use the value+1 from the current CVD in --datadir. If no datafile is found the default behaviour is to prompt for a version number, this |                       |
| <code>--no-cdiff</code>       | —               | Don't generate .cdiff file                                                                                                                                                                              |                       |
| <code>--unsigned</code>       | —               | Create unsigned database file (.cud)                                                                                                                                                                    |                       |
| <code>--hybrid</code>         | —               | Create a hybrid (standard and bytecode) database file                                                                                                                                                   |                       |
| <code>--print-certs</code>    | FILE            | Print Authenticode details from a PE                                                                                                                                                                    |                       |
| <code>--server</code>         | ADDR            | ClamAV Signing Service address                                                                                                                                                                          |                       |
| <code>--datadir</code>        | DIR             | Use DIR as default database directory                                                                                                                                                                   |                       |
| <code>--unpack</code>         | FILE            | -u FILE Unpack a CVD/CLD file                                                                                                                                                                           |                       |
| <code>--unpack-current</code> | SHORTNAME       | Unpack local CVD/CLD into cwd                                                                                                                                                                           |                       |
| <code>--list-sigs</code>      | FILE (optional) | -l[FILE] List signature names                                                                                                                                                                           |                       |
| <code>--find-sigs</code>      | REGEX           | -fREGEX Find signatures matching REGEX                                                                                                                                                                  |                       |
| <code>--decode-sigs</code>    | —               | Decode signatures from stdin                                                                                                                                                                            |                       |
| <code>--vba</code>            | FILE            | Extract VBA/Word6 macro code                                                                                                                                                                            |                       |
| <code>--vba-hex</code>        | FILE            | Extract Word6 macro code with hex values                                                                                                                                                                |                       |
| <code>--run-cdiff</code>      | FILE            | -r FILE Execute update script FILE in cwd                                                                                                                                                               |                       |
| <code>--tempdir</code>        | DIRECTORY       | Create temporary files in DIRECTORY                                                                                                                                                                     |                       |

## Gotchas

*TODO: operational traps.*

## See also

[rahash2](#), [ssdeep](#), [sha256sum](#), [md5sum](#), [yara](#), [yarak](#), [clamscan](#), [freshclam](#)

# ssdeep

---

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Verify evidence integrity with hashes; Compare or cluster samples; Find hidden data **Version:** 2.14.1 **Captured:** cyberlab-aio via -h on 2026-08-01 [raw Docs:](https://ssdeep-project.github.io/ssdeep/index.html) <https://ssdeep-project.github.io/ssdeep/index.html>

← [Capability index](#) · [Kit tool list](#)

## Purpose

---

Compute Context Triggered Piecewise Hashes (CTPH), also known as fuzzy hashes.

## Synopsis

---

```
ssdeep [-m file] [-k file] [-dpgvrsblcxa] [-t val] [-h|-V] [FILES]
```

## Common invocations

---

```
# from cyberlab 08-malware-static-triage
ssdeep -V
# from cyberlab 08-malware-static-triage
ssdeep exercise/sample.bin
# from cyberlab 32-remnux-static-triage
ssdeep sample.exe > baseline.txt
# from cyberlab 32-remnux-static-triage
ssdeep -m baseline.txt sample_mod.exe
# from cyberlab 32-remnux-static-triage
ssdeep exercise/sample.exe > exercise/baseline.txt
# from cyberlab 32-remnux-static-triage
ssdeep -m exercise/baseline.txt exercise/sample_mod.exe
```

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

[rahash2](#), [sha256sum](#), [md5sum](#), [sigtool](#), [radiff2](#), [binwalk](#)

# tshark

**Kit:** REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Capture live network traffic; Read and filter packet captures **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Read, filter and dissect network captures from the command line.

## Synopsis

```
tshark [options] ...
```

## Common invocations

```
# from cyberlab 07-network-pcap
tshark --version | head -n 1
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -q -z io,phs | head -n 30
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'http.request' -T fields -e ip.dst -e http.host -e http.request.uri
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'dns.flags.response == 0' -T fields -e dns.qry.name | sort -u
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'http.request' -T fields -e http.host -e http.request.uri
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -c 20
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -Y "dns.flags.response == 0" -T fields -e dns.qry.name
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -Y "http.request" -T fields -e http.host -e http.request.uri
```

## Options

All 75 options parsed from the captured help text; 12 reviewed with usage guidance.

| Flag                                 | Argument        | What it does                                                                                                                                                                                                      | When you would use it                                                 |
|--------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <code>-i</code>                      | interface       | name or idx of interface (def: first non-loopback)                                                                                                                                                                | Capture live from an interface instead of reading a file.             |
| <code>--interface</code>             | interface       | name or idx of interface (def: first non-loopback)                                                                                                                                                                |                                                                       |
| <code>-f</code>                      | capture filter  | packet filter in libpcap filter syntax                                                                                                                                                                            | Apply a <i>capture</i> filter (BPF syntax) before packets are stored. |
| <code>-s</code>                      | snaplen         | packet snapshot length (def: appropriate maximum)                                                                                                                                                                 |                                                                       |
| <code>--snapshot-length</code>       | snaplen         | packet snapshot length (def: appropriate maximum)                                                                                                                                                                 |                                                                       |
| <code>-p</code>                      | —               | don't capture in promiscuous mode                                                                                                                                                                                 |                                                                       |
| <code>--no-promiscuous-mode</code>   | —               | don't capture in promiscuous mode                                                                                                                                                                                 |                                                                       |
| <code>-I</code>                      | —               | capture in monitor mode, if available                                                                                                                                                                             |                                                                       |
| <code>--monitor-mode</code>          | —               | capture in monitor mode, if available                                                                                                                                                                             |                                                                       |
| <code>-B</code>                      | buffer size     | size of kernel buffer (def: 2MB)                                                                                                                                                                                  |                                                                       |
| <code>--buffer-size</code>           | buffer size     | size of kernel buffer (def: 2MB)                                                                                                                                                                                  |                                                                       |
| <code>-y</code>                      | link type       | link layer type (def: first appropriate)                                                                                                                                                                          |                                                                       |
| <code>--linktype</code>              | link type       | link layer type (def: first appropriate)                                                                                                                                                                          |                                                                       |
| <code>-D</code>                      | —               | print list of interfaces and exit                                                                                                                                                                                 |                                                                       |
| <code>--list-interfaces</code>       | —               | print list of interfaces and exit                                                                                                                                                                                 |                                                                       |
| <code>-L</code>                      | —               | print list of link-layer types of iface and exit                                                                                                                                                                  |                                                                       |
| <code>--list-data-link-types</code>  | —               | print list of link-layer types of iface and exit                                                                                                                                                                  |                                                                       |
| <code>--list-time-stamp-types</code> | —               | print list of timestamp types for iface and exit                                                                                                                                                                  |                                                                       |
| <code>-c</code>                      | packet count    | stop after n packets (def: infinite)                                                                                                                                                                              | Stop after N packets — a fast way to sample a huge capture.           |
| <code>-a</code>                      | autostop cond.  | duration:NUM - stop after NUM seconds<br>filesize:NUM - stop this file after NUM KB<br>files:NUM - stop after NUM files<br>packets:NUM - stop after NUM packets                                                   |                                                                       |
| <code>--autostop</code>              | autostop cond.  | duration:NUM - stop after NUM seconds<br>filesize:NUM - stop this file after NUM KB<br>files:NUM - stop after NUM files<br>packets:NUM - stop after NUM packets                                                   |                                                                       |
| <code>-b</code>                      | ringbuffer opt. | duration:NUM - switch to next file after NUM secs<br>filesize:NUM - switch to next file after NUM KB<br>files:NUM - ringbuffer: replace after NUM files<br>packets:NUM - switch to next file after NUM packets in |                                                                       |

| Flag                             | Argument           | What it does                                                                                                                                                                                                      | When you would use it                                                                                                  |
|----------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <code>--ring-buffer</code>       | ringbuffer opt.    | duration:NUM - switch to next file after NUM secs<br>filesize:NUM - switch to next file after NUM KB<br>files:NUM - ringbuffer: replace after NUM files<br>packets:NUM - switch to next file after NUM packets in |                                                                                                                        |
| <code>-r</code>                  | infile             | set the filename to read from (or '-' for stdin)                                                                                                                                                                  | Read from a capture file — the normal forensic mode.                                                                   |
| <code>--read-file</code>         | infile             | set the filename to read from (or '-' for stdin)                                                                                                                                                                  |                                                                                                                        |
| <code>-2</code>                  | —                  | perform a two-pass analysis                                                                                                                                                                                       | Two-pass analysis, so fields needing later context resolve.                                                            |
| <code>-M</code>                  | packet count       | perform session auto reset                                                                                                                                                                                        |                                                                                                                        |
| <code>-R</code>                  | read filter        | packet Read filter in Wireshark display filter syntax (requires -2)                                                                                                                                               |                                                                                                                        |
| <code>--read-filter</code>       | read filter        | packet Read filter in Wireshark display filter syntax (requires -2)                                                                                                                                               |                                                                                                                        |
| <code>-Y</code>                  | display filter     | packet display filter in Wireshark display filter syntax                                                                                                                                                          | Apply a <i>display</i> filter (Wireshark syntax) after dissection.                                                     |
| <code>--display-filter</code>    | display filter     | packet display filter in Wireshark display filter syntax                                                                                                                                                          |                                                                                                                        |
| <code>-n</code>                  | —                  | disable all name resolutions (def: "mNd" enabled, or as set in preferences)                                                                                                                                       | Disable name resolution. Also stops DNS lookups leaking from an evidence host — use it by default on an investigation. |
| <code>-N</code>                  | name resolve flags | enable specific name resolution(s): "mnNtdv"                                                                                                                                                                      |                                                                                                                        |
| <code>-H</code>                  | hosts file         | read a list of entries from a hosts file, which will then be written to a capture file. (Implies -W n)                                                                                                            |                                                                                                                        |
| <code>--enable-protocol</code>   | proto_name         | enable dissection of proto_name                                                                                                                                                                                   |                                                                                                                        |
| <code>--disable-protocol</code>  | proto_name         | disable dissection of proto_name                                                                                                                                                                                  |                                                                                                                        |
| <code>--enable-heuristic</code>  | short_name         | enable dissection of heuristic protocol                                                                                                                                                                           |                                                                                                                        |
| <code>--disable-heuristic</code> | short_name         | disable dissection of heuristic protocol                                                                                                                                                                          |                                                                                                                        |
| <code>-w</code>                  | outfile -          | write packets to a pcapng-format file named "outfile" (or '-' for stdout)                                                                                                                                         | Write the (filtered) packets to a new capture file.                                                                    |
| <code>--capture-comment</code>   | comment            | add a capture file comment, if supported                                                                                                                                                                          |                                                                                                                        |
| <code>-C</code>                  | config profile     | start with specified configuration profile                                                                                                                                                                        |                                                                                                                        |
| <code>-F</code>                  | output file type   | set the output file type, default is pcapng an empty "-F" option will list the file types                                                                                                                         |                                                                                                                        |



| Flag                                   | Argument       | What it does                                                                                                                                                                                            | When you would use it                                                |
|----------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <code>-V</code>                        | —              | add output of packet tree (Packet Details)                                                                                                                                                              |                                                                      |
| <code>-O</code>                        | protocols      | Only show packet details of these protocols, comma separated                                                                                                                                            |                                                                      |
| <code>-P</code>                        | —              | print packet summary even when writing to a file                                                                                                                                                        |                                                                      |
| <code>--print</code>                   | —              | print packet summary even when writing to a file                                                                                                                                                        |                                                                      |
| <code>-S</code>                        | separator      | the line separator to print between packets                                                                                                                                                             |                                                                      |
| <code>-x</code>                        | —              | add output of hex and ASCII dump (Packet Bytes)                                                                                                                                                         | Hex and ASCII dump of packet contents.                               |
| <code>--hexdump</code>                 | hexoption      | add hexdump, set options for data source and ASCII dump all dump all data sources (-x default) frames dump only frame data source ascii include ASCII dump text (-x default) delimit delimit ASCII dump |                                                                      |
| <code>-j</code>                        | protocolfilter | protocols layers filter if -T ek pdml json selected (e.g. "ip ip.flags text", filter does not expand child nodes, unless child is specified also in the filter)                                         |                                                                      |
| <code>-J</code>                        | protocolfilter | top level protocol filter if -T ek pdml json selected (e.g. "http tcp", filter which expands all child nodes)                                                                                           |                                                                      |
| <code>-e</code>                        | field          | field to print if -Tfields selected (e.g. tcp.port, _ws.col.info) this option can be repeated to print multiple fields                                                                                  | With <code>-T fields</code> , name each field to print. Repeatable.  |
| <code>-l</code>                        | —              | flush standard output after each packet                                                                                                                                                                 |                                                                      |
| <code>-q</code>                        | —              | be more quiet on stdout (e.g. when using statistics)                                                                                                                                                    | Suppress per-packet output, for use with <code>-z</code> statistics. |
| <code>-Q</code>                        | —              | only log true errors to stderr (quieter than -q)                                                                                                                                                        |                                                                      |
| <code>-g</code>                        | —              | enable group read access on the output file(s)                                                                                                                                                          |                                                                      |
| <code>-W</code>                        | n              | Save extra information in the file, if supported. n = write network address resolution information                                                                                                      |                                                                      |
| <code>-U</code>                        | tap_name       | PDU export mode, see the man page for details                                                                                                                                                           |                                                                      |
| <code>-z</code>                        | statistics     | various statistics, see the man page for details                                                                                                                                                        | Run a statistics tap (conversations, endpoints, protocol tree).      |
| <code>--export-tls-session-keys</code> | keyfile        | export TLS Session Keys to a file named "keyfile"                                                                                                                                                       |                                                                      |
| <code>--color</code>                   | —              | color output text similarly to the Wireshark GUI, requires a terminal with 24-bit color support Also supplies color attributes to pdml and psml formats (Note that attributes are nonstandard)          |                                                                      |
| <code>--no-duplicate-keys</code>       | —              | If -T json is specified, merge duplicate keys in an object into a single key with as value a json array containing all values                                                                           |                                                                      |
| <code>--temp-dir</code>                | directory      | write temporary files to this directory (default: /tmp)                                                                                                                                                 |                                                                      |
| <code>--log-level</code>               | level          | sets the active log level ("critical", "warning", etc.)                                                                                                                                                 |                                                                      |

| Flag                       | Argument | What it does                                                                                          | When you would use it |
|----------------------------|----------|-------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--log-fatal</code>   | level    | sets level to abort the program ("critical" or "warning")                                             |                       |
| <code>--log-domains</code> | [!]list  | comma separated list of the active log domains                                                        |                       |
| <code>--log-debug</code>   | [!]list  | comma separated list of domains with "debug" level                                                    |                       |
| <code>--log-noisy</code>   | [!]list  | comma separated list of domains with "noisy" level                                                    |                       |
| <code>--log-file</code>    | path     | file to output messages to (in addition to stderr)                                                    |                       |
| <code>-h</code>            | —        | display this help and exit                                                                            |                       |
| <code>--help</code>        | —        | display this help and exit                                                                            |                       |
| <code>-v</code>            | —        | display version info and exit                                                                         |                       |
| <code>--version</code>     | —        | display version info and exit                                                                         |                       |
| <code>-K</code>            | keytab   | keytab file to use for kerberos decryption                                                            |                       |
| <code>-G</code>            | report   | dump one of several available reports and exit<br>default report="fields" use "-G help" for more help |                       |

## Gotchas

- `-f` and `-Y` are different languages. `-f` is BPF and applies at capture time; `-Y` is the Wireshark display filter and applies to a file. Passing display syntax to `-f` fails, sometimes silently.
- Name resolution is on by default and will emit DNS queries from the analysis host. Use `-n` when touching evidence.
- `-T fields` prints nothing useful without at least one `-e`.

## See also

`dumpcap`, `capinfos`, `ngrep`, `tcpflow`

# vshadowinfo

**Kit:** SIFT Workstation (libyal) **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: vshadowinfo

## Synopsis

```
vshadowinfo [ -o offset ] [ -ahvV ] source
```

## Common invocations

```
# from cyberlab 06-windows-artifact-libs
vshadowinfo -V
# from cyberlab 06-windows-artifact-libs
vshadowinfo exercise/volume.raw
```

## Options

All 5 options parsed from the captured help text. The final column is filled in by review.

| Flag            | Argument | What it does                       | When you would use it |
|-----------------|----------|------------------------------------|-----------------------|
| <code>-a</code> | —        | shows allocation information       |                       |
| <code>-h</code> | —        | shows this help                    |                       |
| <code>-o</code> | —        | specify the volume offset in bytes |                       |
| <code>-v</code> | —        | verbose output to stderr           |                       |
| <code>-V</code> | —        | print version                      |                       |

## Gotchas

*TODO: operational traps.*

## See also

ewfinfo , ewfmount , ewfverify , ewfexport , affinfo , affcat , img\_stat , ntfs-3g

# log2timeline.py

**Kit:** Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - log2timeline version 20260512 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Extract timestamped events from evidence into a Plaso storage file, the first half of a super-timeline.

## Synopsis

```
log2timeline.py [-h] [--troubles] [-V] [--artifact_definitions PATH]
[--custom_artifact_definitions PATH] [--data PATH]
[--archives TYPES]
[--artifact_filters ARTIFACT_FILTERS]
[--artifact_filters_file PATH]
[--extract_winreg_binary] [--preferred_year YEAR]
```

## Common invocations

```
# from cyberlab 03-timeline-analysis
log2timeline.py --version
# from cyberlab 03-timeline-analysis
log2timeline.py --status_view none --storage-file /tmp/case.plaso exercise/sample_fs/
# from cyberlab 49-intrusion-timeline-case
log2timeline.py --storage-file timeline.plaso "$IMAGE"
```

## Options

All 100 options parsed from the captured help text; 8 reviewed with usage guidance.

| Flag       | Argument | What it does                      | When you would use it |
|------------|----------|-----------------------------------|-----------------------|
| -h         | —        | Show this help message and exit.  |                       |
| --help     | —        | Show this help message and exit.  |                       |
| --troubles | —        | Show troubleshooting information. |                       |
| -V         | —        | Show the version information.     |                       |
| --version  | —        | Show the version information.     |                       |

| Flag                                       | Argument         | What it does                                                                                                                                                                                             | When you would use it |
|--------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--artifact_definitions</code>        | PATH             | Path to a directory or file containing artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or Windows |                       |
| <code>--artifact-definitions</code>        | PATH             | Path to a directory or file containing artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or Windows |                       |
| <code>--custom_artifact_definitions</code> | PATH             | Path to a directory or file containing custom artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or  |                       |
| <code>--custom-artifact-definitions</code> | PATH             | Path to a directory or file containing custom artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or  |                       |
| <code>--data</code>                        | PATH             | Path to a directory containing the data files.                                                                                                                                                           |                       |
| <code>--archives</code>                    | TYPES            | Define a list of archive and storage media image types for which to process embedded file entries, such as TAR (archive.tar) or ZIP (archive.zip). This is a comma separated list where each entry is th |                       |
| <code>--artifact_filters</code>            | ARTIFACT_FILTERS | Names of forensic artifact definitions, provided on the command command line (comma separated). Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions proj |                       |
| <code>--artifact-filters</code>            | ARTIFACT_FILTERS | Names of forensic artifact definitions, provided on the command command line (comma separated). Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions proj |                       |
| <code>--artifact_filters_file</code>       | PATH             | Names of forensic artifact definitions, provided in a file with one artifact name per line. Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions project. |                       |

| Flag                                   | Argument    | What it does                                                                                                                                                                                             | When you would use it                           |
|----------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <code>--artifact-filters_file</code>   | PATH        | Names of forensic artifact definitions, provided in a file with one artifact name per line. Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions project. |                                                 |
| <code>--extract_winreg_binary</code>   | —           | Extract binary Windows Registry values. WARNING: This can make processing significantly slower.                                                                                                          |                                                 |
| <code>--extract-winreg-binary</code>   | —           | Extract binary Windows Registry values. WARNING: This can make processing significantly slower.                                                                                                          |                                                 |
| <code>--preferred_year</code>          | YEAR        | When a format's timestamp does not include a year, e.g. syslog, use this as the initial year instead of attempting auto-detection.                                                                       |                                                 |
| <code>--preferred-year</code>          | YEAR        | When a format's timestamp does not include a year, e.g. syslog, use this as the initial year instead of attempting auto-detection.                                                                       |                                                 |
| <code>--skip_compressed_streams</code> | —           | Skip processing file content within compressed streams, such as syslog.gz and syslog.bz2.                                                                                                                |                                                 |
| <code>--skip-compressed-streams</code> | —           | Skip processing file content within compressed streams, such as syslog.gz and syslog.bz2.                                                                                                                |                                                 |
| <code>-f</code>                        | FILE_FILTER | List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a   | Use a file filter to limit what gets processed. |
| <code>--filter-file</code>             | FILE_FILTER | List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a   |                                                 |
| <code>--filter_file</code>             | FILE_FILTER | List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a   |                                                 |
| <code>--file-filter</code>             | FILE_FILTER | List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a   |                                                 |

| Flag                                  | Argument                 | What it does                                                                                                                                                                                             | When you would use it                                                     |
|---------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <code>--file_filter</code>            | FILE_FILTER              | List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a   |                                                                           |
| <code>--hasher_file_size_limit</code> | SIZE                     | Define the maximum file size in bytes that hashers should process. Any larger file will be skipped. A size of 0 represents no limit.                                                                     |                                                                           |
| <code>--hasher-file-size-limit</code> | SIZE                     | Define the maximum file size in bytes that hashers should process. Any larger file will be skipped. A size of 0 represents no limit.                                                                     |                                                                           |
| <code>--hashers</code>                | HASHER_LIST              | Define a list of hashers to use by the tool. This is a comma separated list where each entry is the name of a hasher, such as "md5,sha256". "all" indicates that all hashers should be enabled. "none" d | Compute hashes during extraction, saving a second pass.                   |
| <code>--parsers</code>                | PARSER_FILTER_EXPRESSION | Define which presets, parsers and/or plugins to use, or show possible values. The expression is a comma separated string where each element is a preset, parser or plugin name. Each element can be prep | Restrict to specific parsers. A targeted run is minutes instead of hours. |
| <code>--yara_rules</code>             | PATH                     | Path to a file containing Yara rules definitions.                                                                                                                                                        |                                                                           |
| <code>--yara-rules</code>             | PATH                     | Path to a file containing Yara rules definitions.                                                                                                                                                        |                                                                           |
| <code>--partitions</code>             | PARTITIONS               | Define partitions to be processed. A range of partitions can be defined as: "3..5". Multiple partitions can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be comb |                                                                           |
| <code>--partition</code>              | PARTITIONS               | Define partitions to be processed. A range of partitions can be defined as: "3..5". Multiple partitions can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be comb |                                                                           |
| <code>--volumes</code>                | VOLUMES                  | Define volumes to be processed. A range of volumes can be defined as: "3..5". Multiple volumes can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be combined as:  |                                                                           |
| <code>--volume</code>                 | VOLUMES                  | Define volumes to be processed. A range of volumes can be defined as: "3..5". Multiple volumes can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be combined as:  |                                                                           |



| Flag                                       | Argument     | What it does                                                                                                                                                                                             | When you would use it            |
|--------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <code>--codepage</code>                    | CODEPAGE     | The preferred codepage, which is used for decoding single-byte or multi-byte character extracted strings.                                                                                                |                                  |
| <code>--language</code>                    | LANGUAGE_TAG | The preferred language, which is used for extracting and formatting Windows EventLog message strings. Use " --language list" to see a list of supported language tags. The en-US (LCID 0x0409) language  |                                  |
| <code>--no_extract_winevt_resources</code> | —            | Do not extract Windows EventLog resources such as event message template strings. By default Windows EventLog resources will be extracted when a Windows EventLog parser is enabled.                     |                                  |
| <code>--no-extract-winevt-resources</code> | —            | Do not extract Windows EventLog resources such as event message template strings. By default Windows EventLog resources will be extracted when a Windows EventLog parser is enabled.                     |                                  |
| <code>-z</code>                            | TIME_ZONE    | preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT | Time zone of the source machine. |
| <code>--zone</code>                        | TIME_ZONE    | preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT |                                  |
| <code>--timezone</code>                    | TIME_ZONE    | preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT |                                  |
| <code>--no_vss</code>                      | —            | Do not scan for Volume Shadow Snapshots (VSS). This means that Volume Shadow Snapshots (VSS) are not processed. WARNING: this option is deprecated use --vss_stores=none instead.                        |                                  |
| <code>--no-vss</code>                      | —            | Do not scan for Volume Shadow Snapshots (VSS). This means that Volume Shadow Snapshots (VSS) are not processed. WARNING: this option is deprecated use --vss_stores=none instead.                        |                                  |

| Flag                                 | Argument   | What it does                                                                                                                                                                                             | When you would use it                                                          |
|--------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <code>--vss_only</code>              | —          | Do not process the current volume if Volume Shadow Snapshots (VSS) have been selected.                                                                                                                   |                                                                                |
| <code>--vss-only</code>              | —          | Do not process the current volume if Volume Shadow Snapshots (VSS) have been selected.                                                                                                                   |                                                                                |
| <code>--vss_stores</code>            | VSS_STORES | Define Volume Shadow Snapshots (VSS) (or stores) that need to be processed. A range of snapshots can be defined as: "3..5". Multiple snapshots can be defined as: "1,3,5" (a list of comma separated val | Also process Volume Shadow Copies — often where the pre-attack state survives. |
| <code>--vss-stores</code>            | VSS_STORES | Define Volume Shadow Snapshots (VSS) (or stores) that need to be processed. A range of snapshots can be defined as: "3..5". Multiple snapshots can be defined as: "1,3,5" (a list of comma separated val |                                                                                |
| <code>-d</code>                      | —          | Enable debug output.                                                                                                                                                                                     |                                                                                |
| <code>--debug</code>                 | —          | Enable debug output.                                                                                                                                                                                     |                                                                                |
| <code>-q</code>                      | —          | Disable informational output.                                                                                                                                                                            |                                                                                |
| <code>--quiet</code>                 | —          | Disable informational output.                                                                                                                                                                            |                                                                                |
| <code>-u</code>                      | —          | Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.                                                                                    |                                                                                |
| <code>--unattended</code>            | —          | Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.                                                                                    |                                                                                |
| <code>--info</code>                  | —          | Print out information about supported plugins and parsers.                                                                                                                                               |                                                                                |
| <code>--use_markdown</code>          | —          | Output lists in Markdown format use in combination with "--hashers list", "--parsers list" or "--timezone list"                                                                                          |                                                                                |
| <code>--use-markdown</code>          | —          | Output lists in Markdown format use in combination with "--hashers list", "--parsers list" or "--timezone list"                                                                                          |                                                                                |
| <code>--no_dependencies_check</code> | —          | Disable the dependencies check.                                                                                                                                                                          |                                                                                |
| <code>--no-dependencies-check</code> | —          | Disable the dependencies check.                                                                                                                                                                          |                                                                                |
| <code>--logfile</code>               | FILENAME   | Path of the file in which to store log messages, by default this file will be named: "log2timeline-YYYYMMDDThmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".         |                                                                                |

| Flag                                | Argument    | What it does                                                                                                                                                                                      | When you would use it                                      |
|-------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <code>--log_file</code>             | FILENAME    | Path of the file in which to store log messages, by default this file will be named: "log2timeline-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz". |                                                            |
| <code>--log-file</code>             | FILENAME    | Path of the file in which to store log messages, by default this file will be named: "log2timeline-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz". |                                                            |
| <code>--status_view</code>          | TYPE        | The processing status view mode: "file", "linear", "none" or "window".                                                                                                                            | Change progress display; <code>none</code> for clean logs. |
| <code>--status-view</code>          | TYPE        | The processing status view mode: "file", "linear", "none" or "window".                                                                                                                            |                                                            |
| <code>--status_view_file</code>     | PATH        | The name of the status view file.                                                                                                                                                                 |                                                            |
| <code>--status-view-file</code>     | PATH        | The name of the status view file.                                                                                                                                                                 |                                                            |
| <code>--status_view_interval</code> | SECONDS     | Number of seconds to update the status view.                                                                                                                                                      |                                                            |
| <code>--status-view-interval</code> | SECONDS     | Number of seconds to update the status view.                                                                                                                                                      |                                                            |
| <code>--buffer_size</code>          | BUFFER_SIZE | The buffer size for the output (defaults to 196MiB).                                                                                                                                              |                                                            |
| <code>--buffer-size</code>          | BUFFER_SIZE | The buffer size for the output (defaults to 196MiB).                                                                                                                                              |                                                            |
| <code>--bs</code>                   | BUFFER_SIZE | The buffer size for the output (defaults to 196MiB).                                                                                                                                              |                                                            |
| <code>--queue_size</code>           | QUEUE_SIZE  | The maximum number of queued items per worker (defaults to 125000)                                                                                                                                |                                                            |
| <code>--queue-size</code>           | QUEUE_SIZE  | The maximum number of queued items per worker (defaults to 125000)                                                                                                                                |                                                            |
| <code>--single_process</code>       | —           | Indicate that the tool should run in a single process.                                                                                                                                            |                                                            |
| <code>--single-process</code>       | —           | Indicate that the tool should run in a single process.                                                                                                                                            |                                                            |

| Flag                                | Argument  | What it does                                                                                                                                                                                             | When you would use it |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--process_memory_limit</code> | SIZE      | Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process |                       |
| <code>--process-memory-limit</code> | SIZE      | Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process |                       |
| <code>--temporary_directory</code>  | DIRECTORY | Path to the directory that should be used to store temporary files created during processing.                                                                                                            |                       |
| <code>--temporary-directory</code>  | DIRECTORY | Path to the directory that should be used to store temporary files created during processing.                                                                                                            |                       |
| <code>--vfs_back_end</code>         | TYPE      | The preferred dfVFS back-end: "auto", "fsext", "fsfat", "fshfs", "fsntfs", "tsk" or "vsgpt".                                                                                                             |                       |
| <code>--vfs-back-end</code>         | TYPE      | The preferred dfVFS back-end: "auto", "fsext", "fsfat", "fshfs", "fsntfs", "tsk" or "vsgpt".                                                                                                             |                       |
| <code>--worker_memory_limit</code>  | SIZE      | Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc |                       |
| <code>--worker-memory-limit</code>  | SIZE      | Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc |                       |
| <code>--worker_timeout</code>       | MINUTES   | Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the |                       |

| Flag                                 | Argument       | What it does                                                                                                                                                                                             | When you would use it                                 |
|--------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <code>--worker-timeout</code>        | MINUTES        | Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the |                                                       |
| <code>--workers</code>               | WORKERS        | Number of worker processes. The default is the number of available system CPUs minus one, for the main (foreman) process.                                                                                | Number of worker processes; tune to the host's cores. |
| <code>--sigsegv_handler</code>       | —              | Enables the SIGSEGV handler. WARNING this functionality is experimental and will a deadlock worker process if a real segfault is caught, but not signal SIGSEGV. This functionality is therefore primari |                                                       |
| <code>--sigsegv-handler</code>       | —              | Enables the SIGSEGV handler. WARNING this functionality is experimental and will a deadlock worker process if a real segfault is caught, but not signal SIGSEGV. This functionality is therefore primari |                                                       |
| <code>--profilers</code>             | PROFILERS_LIST | List of profilers to use by the tool. This is a comma separated list where each entry is the name of a profiler. Use "--profilers list" to list the available profilers.                                 |                                                       |
| <code>--profiling_directory</code>   | DIRECTORY      | Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.                                                  |                                                       |
| <code>--profiling-directory</code>   | DIRECTORY      | Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.                                                  |                                                       |
| <code>--profiling_sample_rate</code> | SAMPLE_RATE    | Profiling sample rate (defaults to a sample every 1000 files).                                                                                                                                           |                                                       |
| <code>--profiling-sample-rate</code> | SAMPLE_RATE    | Profiling sample rate (defaults to a sample every 1000 files).                                                                                                                                           |                                                       |
| <code>--storage_file</code>          | PATH           | The path of the storage file. If not specified, one will be made in the form -.plaso                                                                                                                     | Where the .plaso output goes.                         |
| <code>--storage-file</code>          | PATH           | The path of the storage file. If not specified, one will be made in the form -.plaso                                                                                                                     |                                                       |
| <code>--storage_format</code>        | FORMAT         | Format of the storage file, the default is: sqlite. Supported options: sqlite                                                                                                                            |                                                       |

| Flag                               | Argument | What it does                                                                      | When you would use it |
|------------------------------------|----------|-----------------------------------------------------------------------------------|-----------------------|
| <code>--storage-format</code>      | FORMAT   | Format of the storage file, the default is: sqlite. Supported options: sqlite     |                       |
| <code>--task_storage_format</code> | FORMAT   | Format for task storage, the default is: sqlite. Supported options: redis, sqlite |                       |
| <code>--task-storage-format</code> | FORMAT   | Format for task storage, the default is: sqlite. Supported options: redis, sqlite |                       |

## Gotchas

- This produces a .plaso database, **not** a timeline you can read. `psort.py` is the second half; running only this looks like nothing happened.
- A full run on a disk image can take hours and tens of GB. Scope with `--parsers` unless you genuinely need everything.

## See also

`psort.py` , `pinfo.py`

# mactime

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Build a filesystem MAC-time timeline **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Turn a TSK body file into a human-readable chronological timeline.

## Common invocations

```
# from cyberlab 01-disk-forensics
mactime -b /tmp/bodyfile.txt -d > /tmp/timeline.csv
# from cyberlab 03-timeline-analysis
mactime -V
# from cyberlab 03-timeline-analysis
mactime -b /tmp/body.txt -d > /tmp/mactime.csv
# from cyberlab 22-sleuthkit-mastery
mactime -b exercise/bodyfile.txt -d > exercise/timeline.csv
# from cyberlab 22-sleuthkit-mastery
mactime -b exercise/bodyfile.txt -d | tail
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt > timeline.csv
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt | awk -F',' ' $3 ~ /b/' | sort
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt | awk -F',' ' $3 ~ /b/' | sort | head -n 1
```

## Options

All 9 options parsed from the captured help text; 8 reviewed with usage guidance.

| Flag            | Argument | What it does                                         | When you would use it                                                       |
|-----------------|----------|------------------------------------------------------|-----------------------------------------------------------------------------|
| <code>-b</code> | —        | Specifies the body file location, else STDIN is used | Read the body file produced by <code>fls -m</code> — the normal input.      |
| <code>-d</code> | —        | Output in comma delimited format                     | Emit CSV rather than the default text, for spreadsheets or further tooling. |
| <code>-h</code> | —        | Display a header with session information            | Produce HTML output for a report.                                           |

| Flag            | Argument | What it does                                                                                               | When you would use it                                                            |
|-----------------|----------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <code>-y</code> | —        | Dates are displayed in ISO 8601 format                                                                     | Print dates ISO-style (year first), which sorts correctly.                       |
| <code>-m</code> | —        | Dates have month as number instead of word (does not work with <code>-y</code> )                           | Print month numerically instead of by name.                                      |
| <code>-z</code> | —        | Specify the timezone the data came from (in the local system format) (does not work with <code>-y</code> ) | Time zone of the evidence machine. Getting this wrong shifts the whole timeline. |
| <code>-g</code> | —        | Specifies the group file location, else GIDs are used                                                      | Map group IDs to names using a supplied group file.                              |
| <code>-p</code> | —        | Specifies the password file location, else UIDs are used                                                   | Map user IDs to names using a supplied passwd file.                              |
| <code>-V</code> | —        | Prints the version to STDOUT                                                                               |                                                                                  |

## Gotchas

- `mactime` reports in the time zone you give it, not the one embedded in the evidence. An unstated `-z` silently produces a plausible, wrong timeline — state it explicitly every run.
- A date range is passed as a trailing argument ( `mactime -b body.txt 2026-01-01..2026-02-01` ), not as a flag.

## See also

`fls`, `tsk_gettimes`



# pininfo.py

**Kit:** Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - pininfo version 20260512 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Shows information about a Plaso storage file, for example how it was collected, what information was extracted from a source, etc.

## Synopsis

```
pininfo.py [-h] [--troubles] [-V] [--logfile FILENAME]
[--process_memory_limit SIZE] [--compare STORAGE_FILE]
[--output_format FORMAT] [--hash TYPE] [--report TYPE]
[--sections SECTIONS_LIST] [-v] [-w OUTPUTFILE]
[PATH]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 20 options parsed from the captured help text. The final column is filled in by review.

| Flag       | Argument | What it does                                                                                                                                                                                  | When you would use it |
|------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -h         | —        | Show this help message and exit.                                                                                                                                                              |                       |
| --help     | —        | Show this help message and exit.                                                                                                                                                              |                       |
| --troubles | —        | Show troubleshooting information.                                                                                                                                                             |                       |
| -V         | —        | Show the version information.                                                                                                                                                                 |                       |
| --version  | —        | Show the version information.                                                                                                                                                                 |                       |
| --logfile  | FILENAME | Path of the file in which to store log messages, by default this file will be named: "pininfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz". |                       |

| Flag                                | Argument      | What it does                                                                                                                                                                                             | When you would use it |
|-------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--log_file</code>             | FILENAME      | Path of the file in which to store log messages, by default this file will be named: "pinfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".              |                       |
| <code>--log-file</code>             | FILENAME      | Path of the file in which to store log messages, by default this file will be named: "pinfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".              |                       |
| <code>--process_memory_limit</code> | SIZE          | Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process |                       |
| <code>--process-memory-limit</code> | SIZE          | Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process |                       |
| <code>--compare</code>              | STORAGE_FILE  | The path of the storage file to compare against.                                                                                                                                                         |                       |
| <code>--output_format</code>        | FORMAT        | Format of the output, the default is: text. Supported options: json, markdown, text.                                                                                                                     |                       |
| <code>--output-format</code>        | FORMAT        | Format of the output, the default is: text. Supported options: json, markdown, text.                                                                                                                     |                       |
| <code>--hash</code>                 | TYPE          | Type of hash to output in file_hashes report. Supported options: md5, sha1, sha256                                                                                                                       |                       |
| <code>--report</code>               | TYPE          | Report on specific information. Supported options: browser_search, chrome_extension, environment_variables, file_hashes, list, none, windows_services, winevt_providers                                  |                       |
| <code>--sections</code>             | SECTIONS_LIST | List of sections to output. This is a comma separated list where each entry is the name of a section. Use " --sections list" to list the available sections and " --sections all" to show all available  |                       |
| <code>-v</code>                     | —             | Print verbose output.                                                                                                                                                                                    |                       |
| <code>--verbose</code>              | —             | Print verbose output.                                                                                                                                                                                    |                       |
| <code>-w</code>                     | OUTPUTFILE    | Output filename.                                                                                                                                                                                         |                       |
| <code>--write</code>                | OUTPUTFILE    | Output filename.                                                                                                                                                                                         |                       |

## Gotchas

*TODO: operational traps.*

## See also

[log2timeline.py](#), [psort.py](#)

# psort.py

**Kit:** Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - psort version 20260512 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Filter, sort and output the events in a Plaso storage file.

## Synopsis

```
psort.py [-h] [--troubles] [-V] [--analysis PLUGIN_LIST]
[--process_memory_limit SIZE]
[--temporary_directory DIRECTORY] [--worker_memory_limit SIZE]
[--worker_timeout MINUTES] [--logfile FILENAME] [-d] [-q] [-u]
[--status_view TYPE] [--status_view_file PATH]
[--status_view_interval SECONDS] [--slice DATE_TIME]
```

## Common invocations

```
# from cyberlab 03-timeline-analysis
psort.py --version
# from cyberlab 03-timeline-analysis
psort.py -o l2tcsv -w /tmp/timeline.csv /tmp/case.plaso
# from cyberlab 49-intrusion-timeline-case
psort.py -o l2tcsv -w super_timeline.csv timeline.plaso "date > '2024-01-10 00:00:00' AND date < '2024-01-10 00:00:00'"
```

## Options

All 59 options parsed from the captured help text; 4 reviewed with usage guidance.

| Flag       | Argument | What it does                      | When you would use it |
|------------|----------|-----------------------------------|-----------------------|
| -h         | —        | Show this help message and exit.  |                       |
| --help     | —        | Show this help message and exit.  |                       |
| --troubles | —        | Show troubleshooting information. |                       |
| -V         | —        | Show the version information.     |                       |

| Flag                                | Argument    | What it does                                                                                                                                                                                             | When you would use it                                     |
|-------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <code>--version</code>              | —           | Show the version information.                                                                                                                                                                            |                                                           |
| <code>--analysis</code>             | PLUGIN_LIST | A comma separated list of analysis plugin names to be loaded or "--analysis list" to see a list of available plugins.                                                                                    | Run analysis plugins (tagging, sessionizing) over events. |
| <code>--process_memory_limit</code> | SIZE        | Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process |                                                           |
| <code>--process-memory-limit</code> | SIZE        | Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process |                                                           |
| <code>--temporary_directory</code>  | DIRECTORY   | Path to the directory that should be used to store temporary files created during processing.                                                                                                            |                                                           |
| <code>--temporary-directory</code>  | DIRECTORY   | Path to the directory that should be used to store temporary files created during processing.                                                                                                            |                                                           |
| <code>--worker_memory_limit</code>  | SIZE        | Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc |                                                           |
| <code>--worker-memory-limit</code>  | SIZE        | Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc |                                                           |
| <code>--worker_timeout</code>       | MINUTES     | Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the |                                                           |
| <code>--worker-timeout</code>       | MINUTES     | Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the |                                                           |
| <code>--logfile</code>              | FILENAME    | Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".               |                                                           |

| Flag                                | Argument   | What it does                                                                                                                                                                                             | When you would use it     |
|-------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <code>--log_file</code>             | FILENAME   | Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".               |                           |
| <code>--log-file</code>             | FILENAME   | Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".               |                           |
| <code>-d</code>                     | —          | Enable debug output.                                                                                                                                                                                     |                           |
| <code>--debug</code>                | —          | Enable debug output.                                                                                                                                                                                     |                           |
| <code>-q</code>                     | —          | Disable informational output.                                                                                                                                                                            | Quiet, for scripted runs. |
| <code>--quiet</code>                | —          | Disable informational output.                                                                                                                                                                            |                           |
| <code>-u</code>                     | —          | Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.                                                                                    |                           |
| <code>--unattended</code>           | —          | Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.                                                                                    |                           |
| <code>--status_view</code>          | TYPE       | The processing status view mode: "file", "linear", "none" or "window".                                                                                                                                   |                           |
| <code>--status-view</code>          | TYPE       | The processing status view mode: "file", "linear", "none" or "window".                                                                                                                                   |                           |
| <code>--status_view_file</code>     | PATH       | The name of the status view file.                                                                                                                                                                        |                           |
| <code>--status-view-file</code>     | PATH       | The name of the status view file.                                                                                                                                                                        |                           |
| <code>--status_view_interval</code> | SECONDS    | Number of seconds to update the status view.                                                                                                                                                             |                           |
| <code>--status-view-interval</code> | SECONDS    | Number of seconds to update the status view.                                                                                                                                                             |                           |
| <code>--slice</code>                | DATE_TIME  | Date and time to create a time slice around. This parameter, if defined, will display all events that happened X minutes before and after the defined date, where X is controlled by the --slice_size op |                           |
| <code>--slice_size</code>           | SLICE_SIZE | Defines the slice size. In the case of a regular time slice it defines the number of minutes the slice size should be. In the case of the --slicer it determines the number of events before and after a |                           |

| Flag                             | Argument          | What it does                                                                                                                                                                                                          | When you would use it |
|----------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--slice-size</code>        | SLICE_SIZE        | Defines the slice size. In the case of a regular time slice it defines the number of minutes the slice size should be. In the case of the <code>--slicer</code> it determines the number of events before and after a |                       |
| <code>--slicer</code>            | —                 | Create a time slice around every filter match. This parameter, if defined will save all X events before and after a filter match has been made. X is defined by the <code>--slice_size</code> parameter.              |                       |
| <code>--data</code>              | PATH              | Path to a directory containing the data files.                                                                                                                                                                        |                       |
| <code>-a</code>                  | —                 | By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.                                                                                      |                       |
| <code>--include_all</code>       | —                 | By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.                                                                                      |                       |
| <code>--include-all</code>       | —                 | By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.                                                                                      |                       |
| <code>--language</code>          | LANGUAGE_TAG      | The preferred language, which is used for extracting and formatting Windows EventLog message strings. Use " <code>--language list</code> " to see a list of supported language tags. The en-US (LCID 0x0409) language |                       |
| <code>--additional_fields</code> | ADDITIONAL_FIELDS | Defines additional fields to be included in the output besides the default fields. Multiple additional field names can be defined as a list of comma separated values. Output formats that support addit              |                       |
| <code>--additional-fields</code> | ADDITIONAL_FIELDS | Defines additional fields to be included in the output besides the default fields. Multiple additional field names can be defined as a list of comma separated values. Output formats that support addit              |                       |
| <code>--custom_fields</code>     | CUSTOM_FIELDS     | Defines custom fields to be included in the output besides the default fields. A custom field is defined as "name:value". Multiple custom field names can be defined as list of comma separated values.               |                       |
| <code>--custom-fields</code>     | CUSTOM_FIELDS     | Defines custom fields to be included in the output besides the default fields. A custom field is defined as "name:value". Multiple custom field names can be defined as list of comma separated values.               |                       |

| Flag                                        | Argument       | What it does                                                                                                                                                                                             | When you would use it                                                                                |
|---------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <code>--custom_formatter_definitions</code> | PATH           | Path to a file containing custom event formatter definitions, which is a .yaml file. Custom event formatter definitions can be used to customize event messages and override the built-in event formatte |                                                                                                      |
| <code>--custom-formatter-definitions</code> | PATH           | Path to a file containing custom event formatter definitions, which is a .yaml file. Custom event formatter definitions can be used to customize event messages and override the built-in event formatte |                                                                                                      |
| <code>--dynamic_time</code>                 | —              | Indicate that the output should use dynamic time. Output formats that support dynamic time are: dynamic                                                                                                  |                                                                                                      |
| <code>--dynamic-time</code>                 | —              | Indicate that the output should use dynamic time. Output formats that support dynamic time are: dynamic                                                                                                  |                                                                                                      |
| <code>--output_time_zone</code>             | TIME_ZONE      | time zone of date and time values written to the output, if supported by the output format. Use "list" to see a list of available time zones. Output formats that support an output time zone are: dynam |                                                                                                      |
| <code>--output-time-zone</code>             | TIME_ZONE      | time zone of date and time values written to the output, if supported by the output format. Use "list" to see a list of available time zones. Output formats that support an output time zone are: dynam |                                                                                                      |
| <code>-o</code>                             | FORMAT         | The output format. Use "-o list" to see a list of available output formats.                                                                                                                              | Output format — <code>l2tcsv</code> , <code>dynamic</code> , <code>json</code> , or a timeline tool. |
| <code>--output_format</code>                | FORMAT         | The output format. Use "-o list" to see a list of available output formats.                                                                                                                              |                                                                                                      |
| <code>--output-format</code>                | FORMAT         | The output format. Use "-o list" to see a list of available output formats.                                                                                                                              |                                                                                                      |
| <code>-w</code>                             | OUTPUT_FILE    | Output filename.                                                                                                                                                                                         | Write output to a file rather than stdout.                                                           |
| <code>--write</code>                        | OUTPUT_FILE    | Output filename.                                                                                                                                                                                         |                                                                                                      |
| <code>--fields</code>                       | FIELDS         | Defines which fields should be included in the output.                                                                                                                                                   |                                                                                                      |
| <code>--profilers</code>                    | PROFILERS_LIST | List of profilers to use by the tool. This is a comma separated list where each entry is the name of a profiler. Use "--profilers list" to list the available profilers.                                 |                                                                                                      |

| Flag                                 | Argument    | What it does                                                                                                                                            | When you would use it |
|--------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--profiling_directory</code>   | DIRECTORY   | Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory. |                       |
| <code>--profiling-directory</code>   | DIRECTORY   | Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory. |                       |
| <code>--profiling_sample_rate</code> | SAMPLE_RATE | Profiling sample rate (defaults to a sample every 1000 files).                                                                                          |                       |
| <code>--profiling-sample-rate</code> | SAMPLE_RATE | Profiling sample rate (defaults to a sample every 1000 files).                                                                                          |                       |

## Gotchas

- The date filter is a positional argument, not a flag. Filtering to the incident window is what makes a multi-million-event timeline usable.
- Output ordering follows the storage file, so always sort or filter explicitly rather than assuming chronology.

## See also

`log2timeline.py`, `pinfo.py`



# tsk\_gettimes

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Build a filesystem MAC-time timeline **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://www.sleuthkit.org/sleuthkit>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Analyze disk images and recover files from them.

## Synopsis

```
tsk_gettimes [-vVm] [-i imgtype] [-b dev_sector_size] [-z zone] [-s seconds] image [image]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 7 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument        | What it does                                                              | When you would use it |
|------|-----------------|---------------------------------------------------------------------------|-----------------------|
| -i   | imgtype         | The format of the image file (use '-i list' for supported types)          |                       |
| -b   | dev_sector_size | The size (in bytes) of the device sectors                                 |                       |
| -m   | —               | Calculate MD5 hash in output (slow)                                       |                       |
| -v   | —               | verbose output to stderr                                                  |                       |
| -V   | —               | Print version                                                             |                       |
| -z   | —               | Time zone of original machine (i.e. EST5EDT or GMT) (only useful with -l) |                       |
| -s   | seconds         | Time skew of original machine (in seconds) (only useful with -l & -m)     |                       |

## Gotchas

*TODO: operational traps.*

## See also

[fls](#), [mactime](#)

# cyberchef

---

**Kit:** REMnux · FLARE-VM · Security Onion **Capability:** Decode, decrypt or transform encoded data **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#) **Docs:** <https://github.com/gchq/CyberChef/>

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Decode and otherwise analyze data using this browser app.

## Common invocations

---

```
# from cyberlab 09-deobfuscation
cyberchef &
# from cyberlab 25-cyberchef-recipes
cyberchef --help | head -n 5
# from cyberlab 25-cyberchef-recipes
cyberchef --recipe "From Base64" --input encoded_payload.ps1.txt 2>/dev/null | cyberchef --recipe "XOR"
# from cyberlab 25-cyberchef-recipes
cyberchef --recipe "From Base64" --input encoded_payload.ps1.txt | cyberchef --recipe "XOR({'option': '"
```

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

[base64dump.py](#), [rax2](#), [xxd](#), [openssl](#), [numbers-to-string.py](#)

# hashcat

**Kit:** Kali Linux **Capability:** Crack passwords and hashes **Version:** v6.2.6 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

hashcat (v6.2.6) starting in help mode

## Synopsis

```
hashcat [options]... hash|hashfile|hccapxfile [dictionary|mask|directory]...
```

## Common invocations

```
# from cyberlab 11-offensive-kali
hashcat --version
# from cyberlab 11-offensive-kali
hashcat -b -m 0 2>&1 | head -n 8
# from cyberlab 11-offensive-kali
hashcat -m 0 -a 0 exercise/lab_hash.txt /usr/share/wordlists/rockyou.txt
# from cyberlab 11-offensive-kali
hashcat -m 0 --show exercise/lab_hash.txt
# from cyberlab 40-password-cracking
hashcat prints a version string like "v6.2.6".
# from cyberlab 40-password-cracking
hashcat -m 0 -a 0 exercise/hash.txt exercise/wordlist.txt --potfile-path exercise/hc.pot
# from cyberlab 40-password-cracking
hashcat -m 0 --show exercise/hash.txt --potfile-path exercise/hc.pot
# from cyberlab 40-password-cracking
hashcat -m 0 -a 0 -r /usr/share/hashcat/rules/best64.rule exercise/hash.txt exercise/wordlist.txt --po
```

## Options

All 143 options parsed from the captured help text. The final column is filled in by review.

| Flag                               | Argument | What it does                                                                          | When you would use it |
|------------------------------------|----------|---------------------------------------------------------------------------------------|-----------------------|
| <code>-m</code>                    | —        | Num   Hash-type, references below (otherwise autodetect)   -m 1000                    |                       |
| <code>--hash-type</code>           | —        | Num   Hash-type, references below (otherwise autodetect)   -m 1000                    |                       |
| <code>-a</code>                    | —        | Num   Attack-mode, see references below   -a 3                                        |                       |
| <code>--attack-mode</code>         | —        | Num   Attack-mode, see references below   -a 3                                        |                       |
| <code>-V</code>                    | —        | Print version                                                                         |                       |
| <code>--version</code>             | —        | Print version                                                                         |                       |
| <code>-h</code>                    | —        | Print help                                                                            |                       |
| <code>--help</code>                | —        | Print help                                                                            |                       |
| <code>--quiet</code>               | —        | Suppress output                                                                       |                       |
| <code>--hex-charset</code>         | —        | Assume charset is given in hex                                                        |                       |
| <code>--hex-salt</code>            | —        | Assume salt is given in hex                                                           |                       |
| <code>--hex-wordlist</code>        | —        | Assume words in wordlist are given in hex                                             |                       |
| <code>--force</code>               | —        | Ignore warnings                                                                       |                       |
| <code>--status</code>              | —        | Enable automatic update of the status screen                                          |                       |
| <code>--status-json</code>         | —        | Enable JSON format for status output                                                  |                       |
| <code>--status-timer</code>        | —        | Num   Sets seconds between status screen updates to X   --status-timer=1              |                       |
| <code>--stdin-timeout-abort</code> | —        | Num   Abort if there is no input from stdin for X seconds   --stdin-timeout-abort=300 |                       |
| <code>--machine-readable</code>    | —        | Display the status view in a machine-readable format                                  |                       |
| <code>--keep-guessing</code>       | —        | Keep guessing the hash after it has been cracked                                      |                       |
| <code>--self-test-disable</code>   | —        | Disable self-test functionality on startup                                            |                       |
| <code>--loopback</code>            | —        | Add new plains to induct directory                                                    |                       |
| <code>--markov-hcstat2</code>      | —        | File   Specify hcstat2 file to use   --markov-hcstat2=my.hcstat2                      |                       |
| <code>--markov-disable</code>      | —        | Disables markov-chains, emulates classic brute-force                                  |                       |
| <code>--markov-classic</code>      | —        | Enables classic markov-chains, no per-position                                        |                       |
| <code>--markov-inverse</code>      | —        | Enables inverse markov-chains, no per-position                                        |                       |
| <code>-t</code>                    | —        | Num   Threshold X when to stop accepting new markov-chains   -t 50                    |                       |
| <code>--markov-threshold</code>    | —        | Num   Threshold X when to stop accepting new markov-chains   -t 50                    |                       |
| <code>--runtime</code>             | —        | Num   Abort session after X seconds of runtime   --runtime=10                         |                       |
| <code>--session</code>             | —        | Str   Define specific session name   --session=mysession                              |                       |
| <code>--restore</code>             | —        | Restore session from --session                                                        |                       |
| <code>--restore-disable</code>     | —        | Do not write restore file                                                             |                       |

| Flag                                   | Argument | What it does                                                                                         | When you would use it |
|----------------------------------------|----------|------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--restore-file-path</code>       | —        | File   Specific path to restore file   <code>--restore-file-path=x.restore</code>                    |                       |
| <code>-o</code>                        | —        | File   Define outfile for recovered hash   <code>-o outfile.txt</code>                               |                       |
| <code>--outfile</code>                 | —        | File   Define outfile for recovered hash   <code>-o outfile.txt</code>                               |                       |
| <code>--outfile-format</code>          | —        | Str   Outfile format to use, separated with commas   <code>--outfile-format=1,3</code>               |                       |
| <code>--outfile-autohex-disable</code> | —        | Disable the use of \$HEX[] in output plains                                                          |                       |
| <code>--outfile-check-timer</code>     | —        | Num   Sets seconds between outfile checks to X   <code>--outfile-check-timer=30</code>               |                       |
| <code>-p</code>                        | —        | Char   Separator char for hashlists and outfile   <code>-p :</code>                                  |                       |
| <code>--separator</code>               | —        | Char   Separator char for hashlists and outfile   <code>-p :</code>                                  |                       |
| <code>--stdout</code>                  | —        | Do not crack a hash, instead print candidates only                                                   |                       |
| <code>--show</code>                    | —        | Compare hashlist with potfile; show cracked hashes                                                   |                       |
| <code>--left</code>                    | —        | Compare hashlist with potfile; show uncracked hashes                                                 |                       |
| <code>--username</code>                | —        | Enable ignoring of usernames in hashfile                                                             |                       |
| <code>--remove</code>                  | —        | Enable removal of hashes once they are cracked                                                       |                       |
| <code>--remove-timer</code>            | —        | Num   Update input hash file each X seconds   <code>--remove-timer=30</code>                         |                       |
| <code>--potfile-disable</code>         | —        | Do not write potfile                                                                                 |                       |
| <code>--potfile-path</code>            | —        | File   Specific path to potfile   <code>--potfile-path=my.pot</code>                                 |                       |
| <code>--encoding-from</code>           | —        | Code   Force internal wordlist encoding from X   <code>--encoding-from=iso-8859-15</code>            |                       |
| <code>--encoding-to</code>             | —        | Code   Force internal wordlist encoding to X   <code>--encoding-to=utf-32le</code>                   |                       |
| <code>--debug-mode</code>              | —        | Num   Defines the debug mode (hybrid only by using rules)   <code>--debug-mode=4</code>              |                       |
| <code>--debug-file</code>              | —        | File   Output file for debugging rules   <code>--debug-file=good.log</code>                          |                       |
| <code>--induction-dir</code>           | —        | Dir   Specify the induction directory to use for loopback   <code>--induction=inducts</code>         |                       |
| <code>--outfile-check-dir</code>       | —        | Dir   Specify the outfile directory to monitor for plains   <code>--outfile-check-dir=x</code>       |                       |
| <code>--logfile-disable</code>         | —        | Disable the logfile                                                                                  |                       |
| <code>--hccapx-message-pair</code>     | —        | Num   Load only message pairs from hccapx matching X   <code>--hccapx-message-pair=2</code>          |                       |
| <code>--nonce-error-corrections</code> | —        | Num   The BF size range to replace AP's nonce last bytes   <code>--nonce-error-corrections=16</code> |                       |
| <code>--keyboard-layout-mapping</code> | —        | File   Keyboard layout mapping table for special hash-modes   <code>--keyb=german.hckmap</code>      |                       |

| Flag                                 | Argument | What it does                                                                                  | When you would use it |
|--------------------------------------|----------|-----------------------------------------------------------------------------------------------|-----------------------|
| <code>--truecrypt-keyfiles</code>    | —        | File   Keyfiles to use, separated with commas   <code>--truecrypt-keyf=x.png</code>           |                       |
| <code>--veracrypt-keyfiles</code>    | —        | File   Keyfiles to use, separated with commas   <code>--veracrypt-keyf=x.txt</code>           |                       |
| <code>--veracrypt-pim-start</code>   | —        | Num   VeraCrypt personal iterations multiplier start   <code>--veracrypt-pim-start=450</code> |                       |
| <code>--veracrypt-pim-stop</code>    | —        | Num   VeraCrypt personal iterations multiplier stop   <code>--veracrypt-pim-stop=500</code>   |                       |
| <code>-b</code>                      | —        | Run benchmark of selected hash-modes                                                          |                       |
| <code>--benchmark</code>             | —        | Run benchmark of selected hash-modes                                                          |                       |
| <code>--benchmark-all</code>         | —        | Run benchmark of all hash-modes (requires -b)                                                 |                       |
| <code>--speed-only</code>            | —        | Return expected speed of the attack, then quit                                                |                       |
| <code>--progress-only</code>         | —        | Return ideal progress step size and time to process                                           |                       |
| <code>-c</code>                      | —        | Num   Sets size in MB to cache from the wordfile to X   <code>-c 32</code>                    |                       |
| <code>--segment-size</code>          | —        | Num   Sets size in MB to cache from the wordfile to X   <code>-c 32</code>                    |                       |
| <code>--bitmap-min</code>            | —        | Num   Sets minimum bits allowed for bitmaps to X   <code>--bitmap-min=24</code>               |                       |
| <code>--bitmap-max</code>            | —        | Num   Sets maximum bits allowed for bitmaps to X   <code>--bitmap-max=24</code>               |                       |
| <code>--cpu-affinity</code>          | —        | Str   Locks to CPU devices, separated with commas   <code>--cpu-affinity=1,2,3</code>         |                       |
| <code>--hook-threads</code>          | —        | Num   Sets number of threads for a hook (per compute unit)   <code>--hook-threads=8</code>    |                       |
| <code>--hash-info</code>             | —        | Show information for each hash-mode                                                           |                       |
| <code>--example-hashes</code>        | —        | Alias of <code>--hash-info</code>                                                             |                       |
| <code>--backend-ignore-cuda</code>   | —        | Do not try to open CUDA interface on startup                                                  |                       |
| <code>--backend-ignore-hip</code>    | —        | Do not try to open HIP interface on startup                                                   |                       |
| <code>--backend-ignore-metal</code>  | —        | Do not try to open Metal interface on startup                                                 |                       |
| <code>--backend-ignore-opengl</code> | —        | Do not try to open OpenGL interface on startup                                                |                       |
| <code>-I</code>                      | —        | Show system/environment/backend API info   <code>-I</code> or <code>-II</code>                |                       |
| <code>--backend-info</code>          | —        | Show system/environment/backend API info   <code>-I</code> or <code>-II</code>                |                       |
| <code>-d</code>                      | —        | Str   Backend devices to use, separated with commas   <code>-d 1</code>                       |                       |
| <code>--backend-devices</code>       | —        | Str   Backend devices to use, separated with commas   <code>-d 1</code>                       |                       |
| <code>-D</code>                      | —        | Str   OpenCL device-types to use, separated with commas   <code>-D 1</code>                   |                       |

| Flag                                   | Argument | What it does                                                                     | When you would use it |
|----------------------------------------|----------|----------------------------------------------------------------------------------|-----------------------|
| <code>--opencl-device-types</code>     | —        | Str   OpenCL device-types to use, separated with commas   -D 1                   |                       |
| <code>-O</code>                        | —        | Enable optimized kernels (limits password length)                                |                       |
| <code>--optimized-kernel-enable</code> | —        | Enable optimized kernels (limits password length)                                |                       |
| <code>-M</code>                        | —        | Disable multiply kernel-accel with processor count                               |                       |
| <code>--multiply-accel-disable</code>  | —        | Disable multiply kernel-accel with processor count                               |                       |
| <code>-w</code>                        | —        | Num   Enable a specific workload profile, see pool below   -w 3                  |                       |
| <code>--workload-profile</code>        | —        | Num   Enable a specific workload profile, see pool below   -w 3                  |                       |
| <code>-n</code>                        | —        | Num   Manual workload tuning, set outerloop step size to X   -n 64               |                       |
| <code>--kernel-accel</code>            | —        | Num   Manual workload tuning, set outerloop step size to X   -n 64               |                       |
| <code>-u</code>                        | —        | Num   Manual workload tuning, set innerloop step size to X   -u 256              |                       |
| <code>--kernel-loops</code>            | —        | Num   Manual workload tuning, set innerloop step size to X   -u 256              |                       |
| <code>-T</code>                        | —        | Num   Manual workload tuning, set thread count to X   -T 64                      |                       |
| <code>--kernel-threads</code>          | —        | Num   Manual workload tuning, set thread count to X   -T 64                      |                       |
| <code>--backend-vector-width</code>    | —        | Num   Manually override backend vector-width to X   --backend-vector=4           |                       |
| <code>--spin-damp</code>               | —        | Num   Use CPU for device synchronization, in percent   --spin-damp=10            |                       |
| <code>--hwmon-disable</code>           | —        | Disable temperature and fanspeed reads and triggers                              |                       |
| <code>--hwmon-temp-abort</code>        | —        | Num   Abort if temperature reaches X degrees Celsius   --hwmon-temp-abort=100    |                       |
| <code>--scrypt-tmto</code>             | —        | Num   Manually override TMTO value for scrypt to X   --scrypt-tmto=3             |                       |
| <code>-s</code>                        | —        | Num   Skip X words from the start   -s 1000000                                   |                       |
| <code>--skip</code>                    | —        | Num   Skip X words from the start   -s 1000000                                   |                       |
| <code>-l</code>                        | —        | Num   Limit X words from the start + skipped words   -l 1000000                  |                       |
| <code>--limit</code>                   | —        | Num   Limit X words from the start + skipped words   -l 1000000                  |                       |
| <code>--keyspace</code>                | —        | Show keyspace base:mod values and quit                                           |                       |
| <code>-j</code>                        | —        | Rule   Single rule applied to each word from left wordlist   -j 'c'              |                       |
| <code>--rule-left</code>               | —        | Rule   Single rule applied to each word from left wordlist   -j 'c'              |                       |
| <code>-k</code>                        | —        | Rule   Single rule applied to each word from right wordlist   -k '^_'            |                       |
| <code>--rule-right</code>              | —        | Rule   Single rule applied to each word from right wordlist   -k '^_'            |                       |
| <code>-r</code>                        | —        | File   Multiple rules applied to each word from wordlists   -r rules/best64.rule |                       |

| Flag                                   | Argument | What it does                                                                                | When you would use it |
|----------------------------------------|----------|---------------------------------------------------------------------------------------------|-----------------------|
| <code>--rules-file</code>              | —        | File   Multiple rules applied to each word from wordlists   -r rules/best64.rule            |                       |
| <code>-g</code>                        | —        | Num   Generate X random rules   -g 10000                                                    |                       |
| <code>--generate-rules</code>          | —        | Num   Generate X random rules   -g 10000                                                    |                       |
| <code>--generate-rules-func-min</code> | —        | Num   Force min X functions per rule                                                        |                       |
| <code>--generate-rules-func-max</code> | —        | Num   Force max X functions per rule                                                        |                       |
| <code>--generate-rules-func-sel</code> | —        | Str   Pool of rule operators valid for random rule engine   --generate-rules-func-sel=ioTlc |                       |
| <code>--generate-rules-seed</code>     | —        | Num   Force RNG seed set to X                                                               |                       |
| <code>-1</code>                        | —        | CS   User-defined charset ?1   -1 ?!d?u                                                     |                       |
| <code>--custom-charset1</code>         | —        | CS   User-defined charset ?1   -1 ?!d?u                                                     |                       |
| <code>-2</code>                        | —        | CS   User-defined charset ?2   -2 ?!d?s                                                     |                       |
| <code>--custom-charset2</code>         | —        | CS   User-defined charset ?2   -2 ?!d?s                                                     |                       |
| <code>-3</code>                        | —        | CS   User-defined charset ?3                                                                |                       |
| <code>--custom-charset3</code>         | —        | CS   User-defined charset ?3                                                                |                       |
| <code>-4</code>                        | —        | CS   User-defined charset ?4                                                                |                       |
| <code>--custom-charset4</code>         | —        | CS   User-defined charset ?4                                                                |                       |
| <code>--identify</code>                | —        | Shows all supported algorithms for input hashes   --identify my.hash                        |                       |
| <code>-i</code>                        | —        | Enable mask increment mode                                                                  |                       |
| <code>--increment</code>               | —        | Enable mask increment mode                                                                  |                       |
| <code>--increment-min</code>           | —        | Num   Start mask incrementing at X   --increment-min=4                                      |                       |
| <code>--increment-max</code>           | —        | Num   Stop mask incrementing at X   --increment-max=8                                       |                       |
| <code>-S</code>                        | —        | Enable slower (but advanced) candidate generators                                           |                       |
| <code>--slow-candidates</code>         | —        | Enable slower (but advanced) candidate generators                                           |                       |
| <code>--brain-server</code>            | —        | Enable brain server                                                                         |                       |
| <code>--brain-server-timer</code>      | —        | Num   Update the brain server dump each X seconds (min:60)   --brain-server-timer=300       |                       |
| <code>-z</code>                        | —        | Enable brain client, activates -S                                                           |                       |
| <code>--brain-client</code>            | —        | Enable brain client, activates -S                                                           |                       |
| <code>--brain-client-features</code>   | —        | Num   Define brain client features, see below   --brain-client-features=3                   |                       |
| <code>--brain-host</code>              | —        | Str   Brain server host (IP or domain)   --brain-host=127.0.0.1                             |                       |



| Flag                                   | Argument | What it does                                                                                               | When you would use it |
|----------------------------------------|----------|------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--brain-port</code>              | —        | Port   Brain server port   <code>--brain-port=13743</code>                                                 |                       |
| <code>--brain-password</code>          | —        | Str   Brain server authentication password   <code>--brain-password=bZfhCvGUSjRq</code>                    |                       |
| <code>--brain-session</code>           | —        | Hex   Overrides automatically calculated brain session   <code>--brain-session=0x2ae611db</code>           |                       |
| <code>--brain-session-whitelist</code> | —        | Hex   Allow given sessions only, separated with commas   <code>--brain-session-whitelist=0x2ae611db</code> |                       |

## Gotchas

*TODO: operational traps.*

## See also

`john`, `hydra`

# hydra

---

**Kit:** Kali Linux · SIFT Workstation **Capability:** Crack passwords and hashes **Version:** Hydra v9.4 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

*TODO: one-line imperative purpose.*

## Common invocations

---

```
# from cyberlab 11-offensive-kali
hydra -h 2>&1 | head -n 1
# from cyberlab 11-offensive-kali
hydra -U ssh 2>&1 | head -n 5
```

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

[hashcat](#) , [john](#)

# john

**Kit:** Kali Linux **Capability:** Crack passwords and hashes **Version:** stat: version: No such file or directory **Captured:** cyberlab-aio via (no args) on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

John the Ripper password cracker, version 1.9.0

## Synopsis

```
john [OPTIONS] [PASSWORD-FILES]
```

## Common invocations

```
# from cyberlab 11-offensive-kali
john --list=build-info 2>&1 | head -n 1
# from cyberlab 11-offensive-kali
john --format=Raw-MD5 --wordlist=/usr/share/wordlists/rockyou.txt exercise/lab_hash.txt
# from cyberlab 11-offensive-kali
john --show --format=Raw-MD5 exercise/lab_hash.txt
# from cyberlab 40-password-cracking
john --version 2>&1 | head -n 1
# from cyberlab 40-password-cracking
john --list=formats | tr ',' '\n' | grep -i -m 5 md5
# from cyberlab 40-password-cracking
john --format=raw-md5 --wordlist=exercise/wordlist.txt exercise/hash.txt
# from cyberlab 40-password-cracking
john --format=raw-md5 --show exercise/hash.txt
```

## Options

All 21 options parsed from the captured help text. The final column is filled in by review.

| Flag                    | Argument | What it does                                 | When you would use it |
|-------------------------|----------|----------------------------------------------|-----------------------|
| <code>--single</code>   | —        | "single crack" mode                          |                       |
| <code>--wordlist</code> | FILE     | wordlist mode, read words from FILE or stdin |                       |
| <code>--stdin</code>    | —        | wordlist mode, read words from FILE or stdin |                       |

| Flag                        | Argument           | What it does                                                                         | When you would use it |
|-----------------------------|--------------------|--------------------------------------------------------------------------------------|-----------------------|
| <code>--rules</code>        | —                  | enable word mangling rules for wordlist mode                                         |                       |
| <code>--incremental</code>  | MODE (optional)    | "incremental" mode [using section MODE]                                              |                       |
| <code>--external</code>     | MODE               | external mode or word filter                                                         |                       |
| <code>--stdout</code>       | LENGTH (optional)  | just output candidate passwords [cut at LENGTH]                                      |                       |
| <code>--restore</code>      | NAME (optional)    | restore an interrupted session [called NAME]                                         |                       |
| <code>--session</code>      | NAME               | give a new session the NAME                                                          |                       |
| <code>--status</code>       | NAME (optional)    | print status of a session [called NAME]                                              |                       |
| <code>--make-charset</code> | FILE               | make a charset, FILE will be overwritten                                             |                       |
| <code>--show</code>         | —                  | show cracked passwords                                                               |                       |
| <code>--test</code>         | TIME (optional)    | run tests and benchmarks for TIME seconds each                                       |                       |
| <code>--users</code>        | [-]LOGIN UID[,...] | [do not] load this (these) user(s) only                                              |                       |
| <code>--groups</code>       | [-]GID[,...]       | load users [not] of this (these) group(s) only                                       |                       |
| <code>--shells</code>       | [-]SHELL[,...]     | load users with[out] this (these) shell(s) only                                      |                       |
| <code>--salts</code>        | [-]N               | load salts with[out] at least N passwords only                                       |                       |
| <code>--save-memory</code>  | LEVEL              | enable memory saving, at LEVEL 1..3                                                  |                       |
| <code>--node</code>         | MIN[-MAX]          | this node's number range out of TOTAL count                                          |                       |
| <code>--fork</code>         | N                  | fork N processes                                                                     |                       |
| <code>--format</code>       | NAME               | force hash type NAME: descrypt/bsdicrypt/md5crypt/bcrypt/LM/AFS/tripcode/dummy/crypt |                       |

## Gotchas

*TODO: operational traps.*

## See also

[hashcat](#) , [hydra](#)

# openssl

---

**Kit:** Base OS — present on every Linux image **Capability:** Decode, decrypt or transform encoded data **Version:** Invalid command '--version'; type "help" for a list. **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Standard commands

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`cyberchef`, `base64dump.py`, `rax2`, `xxd`, `numbers-to-string.py`

# rax2

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Decode, decrypt or transform encoded data **Version:** rax2 6.1.9 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.radare.org/n/radare2.html>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

## Synopsis

```
rax2 [-h|...] [- | expr ...] # convert between numeric bases
int      ->  hex      ; rax2 10
hex      ->  int      ; rax2 0xa
-int     ->  hex      ; rax2 -77
-hex     ->  int      ; rax2 0xffffffffb3
int      ->  bin      ; rax2 b30
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 30 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                           | When you would use it |
|------|----------|--------------------------------------------------------|-----------------------|
| -a   | —        | show ascii table ; rax2 -a                             |                       |
| -b   | base     | output in ; rax2 -b 10 0x46                            |                       |
| -c   | —        | output in C string ; rax2 -c 0x1234 # \x34\x12\x00\x00 |                       |
| -C   | —        | dump as C byte array ; rax2 -C < bytes                 |                       |
| -d   | —        | force integer ; rax2 -d 3 -> 3 instead of 0x3          |                       |
| -e   | —        | swap endianness ; rax2 -e 0x33                         |                       |
| -D   | —        | base64 decode ; rax2 -D "aGVsbG8="                     |                       |
| -E   | —        | base64 encode ; rax2 -E "hello"                        |                       |
| -f   | —        | floating point ; rax2 -f 6.3+2.1                       |                       |

| Flag            | Argument | What it does                                                    | When you would use it |
|-----------------|----------|-----------------------------------------------------------------|-----------------------|
| <code>-F</code> | —        | stdin slurp code hex ; rax2 -F < shellcode.[c/py/js]            |                       |
| <code>-h</code> | —        | help ; rax2 -h                                                  |                       |
| <code>-H</code> | —        | hash string ; rax2 -H linux osx                                 |                       |
| <code>-i</code> | —        | IP address <-> LONG ; rax2 -i 3530468537                        |                       |
| <code>-j</code> | —        | json format output ; rax2 -j 0x1234 # same as r2 -c '?j 0x1234' |                       |
| <code>-k</code> | —        | keep base ; rax2 -k 33+3 -> 36                                  |                       |
| <code>-K</code> | —        | randomart ; rax2 -K 0x34 1020304050                             |                       |
| <code>-n</code> | —        | newline ; append newline to output (for -E/-D/-r/..)            |                       |
| <code>-o</code> | —        | octalstr -> raw ; rax2 -o \162 \62 # r2                         |                       |
| <code>-q</code> | —        | quiet mode ; rax2 -qC < /etc/hosts # be quiet                   |                       |
| <code>-r</code> | —        | r2 style output ; rax2 -r 0x1234 # same as r2 -c '? 0x1234'     |                       |
| <code>-s</code> | —        | hexstr -> raw ; rax2 -s 43 4a 50                                |                       |
| <code>-S</code> | —        | raw -> hexstr ; rax2 -S < /bin/lS > lS.hex                      |                       |
| <code>-t</code> | —        | tstamp -> str ; rax2 -t 1234567890                              |                       |
| <code>-u</code> | —        | units ; rax2 -u 389289238 # 317.0M                              |                       |
| <code>-v</code> | —        | version ; rax2 -v                                               |                       |
| <code>-w</code> | —        | signed word ; rax2 -w 0xffff 0xffff_ffff '0xff&0xffff'          |                       |
| <code>-x</code> | —        | output in hexpairs ; rax2 -x 0x1234 # 34120000                  |                       |
| <code>-X</code> | —        | bin -> hex(bignum) ; rax2 -X 111111111 # 0x1ff                  |                       |
| <code>-z</code> | —        | str -> bin ; rax2 -z hello                                      |                       |
| <code>-Z</code> | —        | bin -> str ; rax2 -Z 01000101 01110110                          |                       |

## Gotchas

*TODO: operational traps.*

## See also

[cyberchef](#) , [base64dump.py](#) , [xxd](#) , [openssl](#) , [numbers-to-string.py](#)

# binwalk

**Kit:** REMnux · Kali Linux **Capability:** Carve files out of unstructured data; Detect and reverse packing; Find hidden data **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://github.com/ReFirmLabs/binwalk>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Find and extract embedded files and filesystems inside a binary blob or firmware image.

## Synopsis

```
binwalk [OPTIONS] [FILE1] [FILE2] [FILE3] ...
```

## Common invocations

```
# from cyberlab 33-binwalk-firmware
binwalk --help | head -n 3
# from cyberlab 33-binwalk-firmware
binwalk firmware.bin
# from cyberlab 33-binwalk-firmware
binwalk -e firmware.bin
# from cyberlab 33-binwalk-firmware
binwalk -E firmware.bin
```

## Options

All 102 options parsed from the captured help text; 8 reviewed with usage guidance.

| Flag        | Argument | What it does                                                | When you would use it                                 |
|-------------|----------|-------------------------------------------------------------|-------------------------------------------------------|
| -B          | —        | Scan target file(s) for common file signatures              |                                                       |
| --signature | —        | Scan target file(s) for common file signatures              |                                                       |
| -R          | str      | Scan target file(s) for the specified sequence of bytes     |                                                       |
| --raw       | str      | Scan target file(s) for the specified sequence of bytes     |                                                       |
| -A          | —        | Scan target file(s) for common executable opcode signatures | Scan for executable opcodes to identify architecture. |



| Flag                      | Argument         | What it does                                                                          | When you would use it                                            |
|---------------------------|------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <code>--opcodes</code>    | —                | Scan target file(s) for common executable opcode signatures                           |                                                                  |
| <code>-m</code>           | file             | Specify a custom magic file to use                                                    |                                                                  |
| <code>--magic</code>      | file             | Specify a custom magic file to use                                                    |                                                                  |
| <code>-b</code>           | —                | Disable smart signature keywords                                                      |                                                                  |
| <code>--dumb</code>       | —                | Disable smart signature keywords                                                      |                                                                  |
| <code>-I</code>           | —                | Show results marked as invalid                                                        |                                                                  |
| <code>--invalid</code>    | —                | Show results marked as invalid                                                        |                                                                  |
| <code>-x</code>           | str              | Exclude results that match                                                            | Exclude signatures matching this string, to cut false hits.      |
| <code>--exclude</code>    | str              | Exclude results that match                                                            |                                                                  |
| <code>-y</code>           | str              | Only show results that match                                                          | Only report signatures matching this string.                     |
| <code>--include</code>    | str              | Only show results that match                                                          |                                                                  |
| <code>-e</code>           | —                | Automatically extract known file types                                                | Extract what is found, rather than only listing it.              |
| <code>--extract</code>    | —                | Automatically extract known file types                                                |                                                                  |
| <code>-D</code>           | type[:ext[:cmd]] | Extract signatures (regular expression), give the files an extension of , and execute |                                                                  |
| <code>--dd</code>         | type[:ext[:cmd]] | Extract signatures (regular expression), give the files an extension of , and execute |                                                                  |
| <code>-M</code>           | —                | Recursively scan extracted files                                                      | Recurse into extracted files (matryoshka) — for nested firmware. |
| <code>--matryoshka</code> | —                | Recursively scan extracted files                                                      |                                                                  |
| <code>-d</code>           | int              | Limit matryoshka recursion depth (default: 8 levels deep)                             | Limit recursion depth; unbounded <code>-M</code> can explode.    |
| <code>--depth</code>      | int              | Limit matryoshka recursion depth (default: 8 levels deep)                             |                                                                  |
| <code>-C</code>           | str              | Extract files/folders to a custom directory (default: current working directory)      | Choose the output directory for extractions.                     |
| <code>--directory</code>  | str              | Extract files/folders to a custom directory (default: current working directory)      |                                                                  |
| <code>-j</code>           | int              | Limit the size of each extracted file                                                 |                                                                  |
| <code>--size</code>       | int              | Limit the size of each extracted file                                                 |                                                                  |
| <code>-n</code>           | int              | Limit the number of extracted files                                                   |                                                                  |
| <code>--count</code>      | int              | Limit the number of extracted files                                                   |                                                                  |
| <code>-0</code>           | str              | Execute external extraction utilities with the specified user's privileges            |                                                                  |

| Flag                             | Argument | What it does                                                                               | When you would use it                                              |
|----------------------------------|----------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <code>--run-as</code>            | str      | Execute external extraction utilities with the specified user's privileges                 |                                                                    |
| <code>-1</code>                  | —        | Do not sanitize extracted symlinks that point outside the extraction directory (dangerous) |                                                                    |
| <code>--preserve-symlinks</code> | —        | Do not sanitize extracted symlinks that point outside the extraction directory (dangerous) |                                                                    |
| <code>-r</code>                  | —        | Delete carved files after extraction                                                       |                                                                    |
| <code>--rm</code>                | —        | Delete carved files after extraction                                                       |                                                                    |
| <code>-z</code>                  | —        | Carve data from files, but don't execute extraction utilities                              |                                                                    |
| <code>--carve</code>             | —        | Carve data from files, but don't execute extraction utilities                              |                                                                    |
| <code>-V</code>                  | —        | Extract into sub-directories named by the offset                                           |                                                                    |
| <code>--subdirs</code>           | —        | Extract into sub-directories named by the offset                                           |                                                                    |
| <code>-E</code>                  | —        | Calculate file entropy                                                                     | Entropy analysis — the fast way to spot encryption or compression. |
| <code>--entropy</code>           | —        | Calculate file entropy                                                                     |                                                                    |
| <code>-F</code>                  | —        | Use faster, but less detailed, entropy analysis                                            |                                                                    |
| <code>--fast</code>              | —        | Use faster, but less detailed, entropy analysis                                            |                                                                    |
| <code>-J</code>                  | —        | Save plot as a PNG                                                                         |                                                                    |
| <code>--save</code>              | —        | Save plot as a PNG                                                                         |                                                                    |
| <code>-Q</code>                  | —        | Omit the legend from the entropy plot graph                                                |                                                                    |
| <code>--nlegend</code>           | —        | Omit the legend from the entropy plot graph                                                |                                                                    |
| <code>-N</code>                  | —        | Do not generate an entropy plot graph                                                      |                                                                    |
| <code>--nplot</code>             | —        | Do not generate an entropy plot graph                                                      |                                                                    |
| <code>-H</code>                  | float    | Set the rising edge entropy trigger threshold (default: 0.95)                              |                                                                    |
| <code>--high</code>              | float    | Set the rising edge entropy trigger threshold (default: 0.95)                              |                                                                    |
| <code>-L</code>                  | float    | Set the falling edge entropy trigger threshold (default: 0.85)                             |                                                                    |
| <code>--low</code>               | float    | Set the falling edge entropy trigger threshold (default: 0.85)                             |                                                                    |
| <code>-W</code>                  | —        | Perform a hexdump / diff of a file or files                                                |                                                                    |
| <code>--hexdump</code>           | —        | Perform a hexdump / diff of a file or files                                                |                                                                    |
| <code>-G</code>                  | —        | Only show lines containing bytes that are the same among all files                         |                                                                    |

| Flag                   | Argument | What it does                                                         | When you would use it |
|------------------------|----------|----------------------------------------------------------------------|-----------------------|
| <code>--green</code>   | —        | Only show lines containing bytes that are the same among all files   |                       |
| <code>-i</code>        | —        | Only show lines containing bytes that are different among all files  |                       |
| <code>--red</code>     | —        | Only show lines containing bytes that are different among all files  |                       |
| <code>-U</code>        | —        | Only show lines containing bytes that are different among some files |                       |
| <code>--blue</code>    | —        | Only show lines containing bytes that are different among some files |                       |
| <code>-u</code>        | —        | Only display lines that are the same between all files               |                       |
| <code>--similar</code> | —        | Only display lines that are the same between all files               |                       |
| <code>-w</code>        | —        | Diff all files, but only display a hex dump of the first file        |                       |
| <code>--terse</code>   | —        | Diff all files, but only display a hex dump of the first file        |                       |
| <code>-X</code>        | —        | Scan for raw deflate compression streams                             |                       |
| <code>--deflate</code> | —        | Scan for raw deflate compression streams                             |                       |
| <code>-Z</code>        | —        | Scan for raw LZMA compression streams                                |                       |
| <code>--lzma</code>    | —        | Scan for raw LZMA compression streams                                |                       |
| <code>-P</code>        | —        | Perform a superficial, but faster, scan                              |                       |
| <code>--partial</code> | —        | Perform a superficial, but faster, scan                              |                       |
| <code>-S</code>        | —        | Stop after the first result                                          |                       |
| <code>--stop</code>    | —        | Stop after the first result                                          |                       |
| <code>-l</code>        | int      | Number of bytes to scan                                              |                       |
| <code>--length</code>  | int      | Number of bytes to scan                                              |                       |
| <code>-o</code>        | int      | Start scan at this file offset                                       |                       |
| <code>--offset</code>  | int      | Start scan at this file offset                                       |                       |
| <code>-O</code>        | int      | Add a base address to all printed offsets                            |                       |
| <code>--base</code>    | int      | Add a base address to all printed offsets                            |                       |
| <code>-K</code>        | int      | Set file block size                                                  |                       |
| <code>--block</code>   | int      | Set file block size                                                  |                       |
| <code>-g</code>        | int      | Reverse every n bytes before scanning                                |                       |
| <code>--swap</code>    | int      | Reverse every n bytes before scanning                                |                       |
| <code>-f</code>        | file     | Log results to file                                                  |                       |

| Flag                    | Argument | What it does                                   | When you would use it |
|-------------------------|----------|------------------------------------------------|-----------------------|
| <code>--log</code>      | file     | Log results to file                            |                       |
| <code>-c</code>         | —        | Log results to file in CSV format              |                       |
| <code>--csv</code>      | —        | Log results to file in CSV format              |                       |
| <code>-t</code>         | —        | Format output to fit the terminal window       |                       |
| <code>--term</code>     | —        | Format output to fit the terminal window       |                       |
| <code>-q</code>         | —        | Suppress output to stdout                      |                       |
| <code>--quiet</code>    | —        | Suppress output to stdout                      |                       |
| <code>-v</code>         | —        | Enable verbose output                          |                       |
| <code>--verbose</code>  | —        | Enable verbose output                          |                       |
| <code>-h</code>         | —        | Show help output                               |                       |
| <code>--help</code>     | —        | Show help output                               |                       |
| <code>-a</code>         | str      | Only scan files whose names match this regex   |                       |
| <code>--finclude</code> | str      | Only scan files whose names match this regex   |                       |
| <code>-p</code>         | str      | Do not scan files whose names match this regex |                       |
| <code>--fexclude</code> | str      | Do not scan files whose names match this regex |                       |
| <code>-s</code>         | int      | Enable the status server on the specified port |                       |
| <code>--status</code>   | int      | Enable the status server on the specified port |                       |

## Gotchas

- **Always timeout-guard binwalk in a harness.** Signature scanning on a large or synthetic image can run effectively forever; this project has been bitten by exactly that.
- `-e -M` on an unknown blob can produce an enormous directory tree. Set `-d` and carve into a scratch filesystem, not your case directory.
- A signature hit is a guess based on magic bytes. Confirm with `file` or entropy before reporting an embedded filesystem as fact.

## See also

`foremost`, `scalpel`, `bulk_extractor`, `tcpextract`, `upx`, `die`, `diec`, `7za`

# blkls

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Extract filesystem blocks — by default the unallocated ones, which is the input a carver wants.

## Synopsis

```
blkls [-aAelvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_vo
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 13 options parsed from the captured help text; 12 reviewed with usage guidance.

| Flag | Argument        | What it does                                                     | When you would use it                                                                                                                                                              |
|------|-----------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -e   | —               | every block (including file system metadata blocks)              | Every block, including filesystem metadata.                                                                                                                                        |
| -l   | —               | print details in time machine list format                        | List block details rather than emitting their contents.                                                                                                                            |
| -a   | —               | Display allocated blocks                                         | Allocated blocks only — the inverse, when isolating live data.                                                                                                                     |
| -A   | —               | Display unallocated blocks                                       | Unallocated blocks. The default and the usual intent: pipe this into <code>foremost</code> or <code>scalpel</code> so the carver reads only free space instead of the whole image. |
| -f   | fstype          | File system type (use '-f list' for supported types)             | Force the filesystem type ( <code>-f list</code> shows the options).                                                                                                               |
| -i   | imgtype         | The format of the image file (use '-i list' for supported types) | Image format for non-raw evidence such as E01 or AFF.                                                                                                                              |
| -b   | dev_sector_size | The size (in bytes) of the device sectors                        | Device sector size; required on 4Kn drives.                                                                                                                                        |

| Flag            | Argument          | What it does                                            | When you would use it                                                                                                                                                             |
|-----------------|-------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-o</code> | imgoffset         | The offset of the file system in the image (in sectors) | Offset, in sectors, from <code>mm1s</code> .                                                                                                                                      |
| <code>-P</code> | pooltype          | Pool container type (use '-P list' for supported types) | Pool type, for APFS or LVM containers.                                                                                                                                            |
| <code>-B</code> | pool_volume_block | Starting block (for pool volumes only)                  | Starting block within a pool volume.                                                                                                                                              |
| <code>-s</code> | —                 | print slack space only (other flags are ignored)        | Slack space only: the tail of the last block of each file, where fragments of previous contents survive. A distinct hunt from carving free space, and it ignores the other flags. |
| <code>-v</code> | —                 | verbose to stderr                                       | Verbose diagnostics to stderr.                                                                                                                                                    |
| <code>-V</code> | —                 | print version                                           |                                                                                                                                                                                   |

## Gotchas

- Output goes to stdout and is the size of the free space — redirect it to a file on a volume that can hold it, not into a pager.
- Block offsets in the extracted stream do **not** match offsets in the original image, because only unallocated blocks were written. Use `-l` if you need to map a hit back to its real location.

## See also

`tsk_recover`, `icat`, `photorec`, `testdisk`

# bulk\_extractor

---

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data; Search raw data for a pattern; Recover encryption keys from memory **Version:** bulk\_extractor 2.2.0 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** [https://github.com/simsong/bulk\\_extractor/](https://github.com/simsong/bulk_extractor/)

← [Capability index](#) · [Kit tool list](#)

## Purpose

---

Extract interesting strings from binary files.

## Synopsis

---

```
bulk_extractor [OPTION...] image_name
```

## Common invocations

---

```
# from cyberlab 02-memory-forensics
bulk_extractor -V
# from cyberlab 02-memory-forensics
bulk_extractor -o be_out sample.mem
# from cyberlab 05-file-carving
bulk_extractor -o /tmp/bulk_out exercise/sample.dd
# from cyberlab 05-file-carving
bulk_extractor -o /tmp/ak_bulk exercise/sample.dd
# from cyberlab 20-volatility-deep
bulk_extractor -o be_out exercise/memdump.raw
# from cyberlab 47-ransomware-memory-case
bulk_extractor -E aes -E zip -o be_out memory.raw
# from cyberlab 47-ransomware-memory-case
bulk_extractor -E aes -o exercise/be_out exercise/memory.raw
# from cyberlab 51-linux-triage-workflow
bulk_extractor -o be_out disk.raw
```

## Options

---

All 69 options parsed from the captured help text. The final column is filled in by review.

| Flag                                | Argument | What it does                                                               | When you would use it |
|-------------------------------------|----------|----------------------------------------------------------------------------|-----------------------|
| <code>-A</code>                     | arg      | Offset added (in bytes) to feature locations (default: 0)                  |                       |
| <code>--offset_add</code>           | arg      | Offset added (in bytes) to feature locations (default: 0)                  |                       |
| <code>-b</code>                     | arg      | Path of file whose contents are prepended to top of all feature files      |                       |
| <code>--banner_file</code>          | arg      | Path of file whose contents are prepended to top of all feature files      |                       |
| <code>-C</code>                     | arg      | Size of context window reported in bytes (default: 16)                     |                       |
| <code>--context_window</code>       | arg      | Size of context window reported in bytes (default: 16)                     |                       |
| <code>-d</code>                     | —        | enable debug-level diagnostic logging                                      |                       |
| <code>--debug</code>                | —        | enable debug-level diagnostic logging                                      |                       |
| <code>-E</code>                     | arg      | disable all scanners except the one specified. Same as -x all -E scanner.  |                       |
| <code>--enable_exclusive</code>     | arg      | disable all scanners except the one specified. Same as -x all -E scanner.  |                       |
| <code>-e</code>                     | arg      | enable a scanner (can be repeated)                                         |                       |
| <code>--enable</code>               | arg      | enable a scanner (can be repeated)                                         |                       |
| <code>-x</code>                     | arg      | disable a scanner (can be repeated)                                        |                       |
| <code>--disable</code>              | arg      | disable a scanner (can be repeated)                                        |                       |
| <code>-f</code>                     | arg      | search for a pattern (can be repeated)                                     |                       |
| <code>--find</code>                 | arg      | search for a pattern (can be repeated)                                     |                       |
| <code>-F</code>                     | arg      | read patterns to search from a file (can be repeated)                      |                       |
| <code>--find_file</code>            | arg      | read patterns to search from a file (can be repeated)                      |                       |
| <code>--find-case-sensitive</code>  | —        | make -f and -F patterns case-sensitive                                     |                       |
| <code>-G</code>                     | arg      | page size in bytes (default: 16777216)                                     |                       |
| <code>--pagesize</code>             | arg      | page size in bytes (default: 16777216)                                     |                       |
| <code>-g</code>                     | arg      | margin size in bytes (default: 4194304)                                    |                       |
| <code>--margin_size</code>          | arg      | margin size in bytes (default: 4194304)                                    |                       |
| <code>-j</code>                     | arg      | number of threads (default: 12)                                            |                       |
| <code>--threads</code>              | arg      | number of threads (default: 12)                                            |                       |
| <code>-J</code>                     | —        | read and process data in the primary thread                                |                       |
| <code>--no_threads</code>           | —        | read and process data in the primary thread                                |                       |
| <code>-M</code>                     | arg      | max recursion depth (default: 12)                                          |                       |
| <code>--max_depth</code>            | arg      | max recursion depth (default: 12)                                          |                       |
| <code>--max_bad_alloc_errors</code> | arg      | max bad allocation errors (default: 3)                                     |                       |
| <code>--max_minute_wait</code>      | arg      | maximum number of minutes to wait until all data are read (default: 60)    |                       |
| <code>--log-level</code>            | arg      | diagnostic log level: trace, debug, info, warning, error, critical, or off |                       |



| Flag                              | Argument | What it does                                                                                                                                                                                            | When you would use it |
|-----------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--log-file</code>           | arg      | diagnostic log file (default: /bulk_extractor.log)                                                                                                                                                      |                       |
| <code>--notify_main_thread</code> | —        | Display notifications in the main thread after phase1 completes. Useful for running with ThreadSanitizer                                                                                                |                       |
| <code>--notify_async</code>       | —        | Display notificaitons asynchronously (default)                                                                                                                                                          |                       |
| <code>-o</code>                   | arg      | output directory [REQUIRED]                                                                                                                                                                             |                       |
| <code>--outdir</code>             | arg      | output directory [REQUIRED]                                                                                                                                                                             |                       |
| <code>-P</code>                   | arg      | directories for scanner shared libraries (can be repeated). Default directories include /usr/local/lib/bulk_extractor, /usr/lib/bulk_extractor and any directories specified in the BE_PATH environment |                       |
| <code>--scanner_dir</code>        | arg      | directories for scanner shared libraries (can be repeated). Default directories include /usr/local/lib/bulk_extractor, /usr/lib/bulk_extractor and any directories specified in the BE_PATH environment |                       |
| <code>-p</code>                   | arg      | print the value of [:length][h]/[r] with optional length, hex output, or raw output.                                                                                                                    |                       |
| <code>--path</code>               | arg      | print the value of [:length][h]/[r] with optional length, hex output, or raw output.                                                                                                                    |                       |
| <code>-q</code>                   | —        | no status or performance output                                                                                                                                                                         |                       |
| <code>--quit</code>               | —        | no status or performance output                                                                                                                                                                         |                       |
| <code>-r</code>                   | arg      | file to read alert list from                                                                                                                                                                            |                       |
| <code>--alert_list</code>         | arg      | file to read alert list from                                                                                                                                                                            |                       |
| <code>-R</code>                   | —        | treat image file as a directory to recursively explore                                                                                                                                                  |                       |
| <code>--recurse</code>            | —        | treat image file as a directory to recursively explore                                                                                                                                                  |                       |
| <code>-S</code>                   | arg      | set a name=value option (can be repeated)                                                                                                                                                               |                       |
| <code>--set</code>                | arg      | set a name=value option (can be repeated)                                                                                                                                                               |                       |
| <code>-s</code>                   | arg      | random sampling parameter frac[:passes]                                                                                                                                                                 |                       |
| <code>--sampling</code>           | arg      | random sampling parameter frac[:passes]                                                                                                                                                                 |                       |
| <code>-V</code>                   | —        | Display PACKAGE_VERSION (currently) 2.2.0-DEVELOP                                                                                                                                                       |                       |
| <code>--version</code>            | —        | Display PACKAGE_VERSION (currently) 2.2.0-DEVELOP                                                                                                                                                       |                       |
| <code>-w</code>                   | arg      | file to read stop list from                                                                                                                                                                             |                       |
| <code>--stop_list</code>          | arg      | file to read stop list from                                                                                                                                                                             |                       |
| <code>-Y</code>                   | arg      | specify [-end] of area on disk to scan                                                                                                                                                                  |                       |
| <code>--scan</code>               | arg      | specify [-end] of area on disk to scan                                                                                                                                                                  |                       |
| <code>-z</code>                   | arg      | specify a starting page number                                                                                                                                                                          |                       |
| <code>--page_start</code>         | arg      | specify a starting page number                                                                                                                                                                          |                       |
| <code>-Z</code>                   | —        | wipe the output directory (recursively) before starting                                                                                                                                                 |                       |

| Flag                         | Argument | What it does                                            | When you would use it |
|------------------------------|----------|---------------------------------------------------------|-----------------------|
| <code>--zap</code>           | —        | wipe the output directory (recursively) before starting |                       |
| <code>-0</code>              | —        | disable real-time notification                          |                       |
| <code>--no_notify</code>     | —        | disable real-time notification                          |                       |
| <code>-1</code>              | —        | version 1.0 notification (console-output)               |                       |
| <code>--version1</code>      | —        | version 1.0 notification (console-output)               |                       |
| <code>-H</code>              | —        | report information about each scanner                   |                       |
| <code>--info_scanners</code> | —        | report information about each scanner                   |                       |
| <code>-h</code>              | —        | print help screen                                       |                       |
| <code>--help</code>          | —        | print help screen                                       |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`foremost`, `scalpel`, `binwalk`, `tcpextract`, `rafind2`, `strings`, `grep`, `xxd`

# ffind

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw Docs:](#) <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Find the file name that points at a given inode — the reverse of a directory lookup.

## Synopsis

```
ffind [-aduvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_vol]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 11 options parsed from the captured help text; 10 reviewed with usage guidance.

| Flag            | Argument        | What it does                                                     | When you would use it                                                                                      |
|-----------------|-----------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>-a</code> | —               | Find all occurrences                                             | Show every name for the inode. Hard-linked files have more than one, and stopping at the first hides that. |
| <code>-d</code> | —               | Find deleted entries ONLY                                        | Deleted names only. The direct answer to "what was this inode called before it was removed?"               |
| <code>-u</code> | —               | Find undeleted entries ONLY                                      | Undeleted names only, when a recycled inode is returning stale hits.                                       |
| <code>-f</code> | fstype          | Image file system type (use '-f list' for supported types)       | Force the filesystem type ( <code>-f list</code> shows the options).                                       |
| <code>-i</code> | imgtype         | The format of the image file (use '-i list' for supported types) | Image format for non-raw evidence such as E01 or AFF.                                                      |
| <code>-b</code> | dev_sector_size | The size (in bytes) of the device sectors                        | Device sector size; required on 4Kn drives.                                                                |
| <code>-o</code> | imgoffset       | The offset of the file system in the image (in sectors)          | Offset, in sectors, from <code>mmls</code> .                                                               |

| Flag            | Argument          | What it does                                            | When you would use it                  |
|-----------------|-------------------|---------------------------------------------------------|----------------------------------------|
| <code>-P</code> | pooltype          | Pool container type (use '-p list' for supported types) | Pool type, for APFS or LVM containers. |
| <code>-B</code> | pool_volume_block | Starting block (for pool volumes only)                  | Starting block within a pool volume.   |
| <code>-v</code> | —                 | Verbose output to stderr                                | Verbose diagnostics to stderr.         |
| <code>-V</code> | —                 | Print version                                           |                                        |

## Gotchas

---

- Inodes are reused. A name returned for a deleted inode may belong to whatever claimed it next, not to the file you are chasing — corroborate with `istat` timestamps before naming it in a report.
- This is the tool for the question `icat` provokes: you carved data out by inode and now need to say what it was called.

## See also

---

`fls`, `ils`, `tsk_recover`

# file

**Kit:** REMnux · FLARE-VM · SIFT Workstation **Capability:** Inspect metadata for one file or inode; Identify what a file actually is **Version:** file-5.44 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw Docs:](https://github.com/file/file) <https://github.com/file/file>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Identify file type using "magic" numbers.

## Synopsis

```
file [OPTION...] [FILE...]
```

## Common invocations

```
# from cyberlab 33-binwalk-firmware
file foremost_out/jpg/*.jpg 2>/dev/null || file foremost_out/**/*.jpg
# from cyberlab 35-radare2-intro
file exercise/hello
# from cyberlab 38-network-emulation
file /tmp/beacon_reply.bin # -> data / HTML (INetSim default object)
```

## Options

All 54 options parsed from the captured help text. The final column is filled in by review.

| Flag                      | Argument | What it does                                             | When you would use it |
|---------------------------|----------|----------------------------------------------------------|-----------------------|
| <code>--help</code>       | —        | display this help and exit                               |                       |
| <code>-v</code>           | —        | output version information and exit                      |                       |
| <code>--version</code>    | —        | output version information and exit                      |                       |
| <code>-m</code>           | LIST     | use LIST as a colon-separated list of magic number files |                       |
| <code>--magic-file</code> | LIST     | use LIST as a colon-separated list of magic number files |                       |
| <code>-z</code>           | —        | try to look inside compressed files                      |                       |
| <code>--uncompress</code> | —        | try to look inside compressed files                      |                       |

| Flag                               | Argument | What it does                                                                                                                                                   | When you would use it |
|------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-Z</code>                    | —        | only print the contents of compressed files                                                                                                                    |                       |
| <code>--uncompress-noreport</code> | —        | only print the contents of compressed files                                                                                                                    |                       |
| <code>-b</code>                    | —        | do not prepend filenames to output lines                                                                                                                       |                       |
| <code>--brief</code>               | —        | do not prepend filenames to output lines                                                                                                                       |                       |
| <code>-c</code>                    | —        | print the parsed form of the magic file, use in conjunction with <code>-m</code> to debug a new magic file before installing it                                |                       |
| <code>--checking-printout</code>   | —        | print the parsed form of the magic file, use in conjunction with <code>-m</code> to debug a new magic file before installing it                                |                       |
| <code>-e</code>                    | TEST     | exclude TEST from the list of test to be performed for file. Valid tests are: apptype, ascii, cdf, compress, csv, elf, encoding, soft, tar, json, text, tokens |                       |
| <code>--exclude</code>             | TEST     | exclude TEST from the list of test to be performed for file. Valid tests are: apptype, ascii, cdf, compress, csv, elf, encoding, soft, tar, json, text, tokens |                       |
| <code>--exclude-quiet</code>       | TEST     | like exclude, but ignore unknown tests                                                                                                                         |                       |
| <code>-f</code>                    | FILE     | read the filenames to be examined from FILE                                                                                                                    |                       |
| <code>--files-from</code>          | FILE     | read the filenames to be examined from FILE                                                                                                                    |                       |
| <code>-F</code>                    | STRING   | use string as separator instead of `:'                                                                                                                         |                       |
| <code>--separator</code>           | STRING   | use string as separator instead of `:'                                                                                                                         |                       |
| <code>-i</code>                    | —        | output MIME type strings ( <code>--mime-type</code> and <code>--mime-encoding</code> )                                                                         |                       |
| <code>--mime</code>                | —        | output MIME type strings ( <code>--mime-type</code> and <code>--mime-encoding</code> )                                                                         |                       |
| <code>--apple</code>               | —        | output the Apple CREATOR/TYPE                                                                                                                                  |                       |
| <code>--extension</code>           | —        | output a slash-separated list of extensions                                                                                                                    |                       |
| <code>--mime-type</code>           | —        | output the MIME type                                                                                                                                           |                       |
| <code>--mime-encoding</code>       | —        | output the MIME encoding                                                                                                                                       |                       |
| <code>-k</code>                    | —        | don't stop at the first match                                                                                                                                  |                       |
| <code>--keep-going</code>          | —        | don't stop at the first match                                                                                                                                  |                       |
| <code>-l</code>                    | —        | list magic strength                                                                                                                                            |                       |
| <code>--list</code>                | —        | list magic strength                                                                                                                                            |                       |
| <code>-L</code>                    | —        | follow symlinks (default if POSIXLY_CORRECT is set)                                                                                                            |                       |
| <code>--dereference</code>         | —        | follow symlinks (default if POSIXLY_CORRECT is set)                                                                                                            |                       |
| <code>-h</code>                    | —        | don't follow symlinks (default if POSIXLY_CORRECT is not set) (default)                                                                                        |                       |
| <code>--no-dereference</code>      | —        | don't follow symlinks (default if POSIXLY_CORRECT is not set) (default)                                                                                        |                       |
| <code>-n</code>                    | —        | do not buffer output                                                                                                                                           |                       |

| Flag                         | Argument | What it does                                                                                                                                                                                             | When you would use it |
|------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--no-buffer</code>     | —        | do not buffer output                                                                                                                                                                                     |                       |
| <code>-N</code>              | —        | do not pad output                                                                                                                                                                                        |                       |
| <code>--no-pad</code>        | —        | do not pad output                                                                                                                                                                                        |                       |
| <code>-0</code>              | —        | terminate filenames with ASCII NUL                                                                                                                                                                       |                       |
| <code>--print0</code>        | —        | terminate filenames with ASCII NUL                                                                                                                                                                       |                       |
| <code>-p</code>              | —        | preserve access times on files                                                                                                                                                                           |                       |
| <code>--preserve-date</code> | —        | preserve access times on files                                                                                                                                                                           |                       |
| <code>-P</code>              | —        | set file engine parameter limits bytes 7340032 max bytes to look inside file elf_notes 256 max ELF notes processed elf_phnum 2048 max ELF prog sections processed elf_shnum 32768 max ELF sections proce |                       |
| <code>--parameter</code>     | —        | set file engine parameter limits bytes 7340032 max bytes to look inside file elf_notes 256 max ELF notes processed elf_phnum 2048 max ELF prog sections processed elf_shnum 32768 max ELF sections proce |                       |
| <code>-r</code>              | —        | don't translate unprintable chars to \ooo                                                                                                                                                                |                       |
| <code>--raw</code>           | —        | don't translate unprintable chars to \ooo                                                                                                                                                                |                       |
| <code>-s</code>              | —        | treat special (block/char devices) files as ordinary ones                                                                                                                                                |                       |
| <code>--special-files</code> | —        | treat special (block/char devices) files as ordinary ones                                                                                                                                                |                       |
| <code>-S</code>              | —        | disable system call sandboxing                                                                                                                                                                           |                       |
| <code>--no-sandbox</code>    | —        | disable system call sandboxing                                                                                                                                                                           |                       |
| <code>-C</code>              | —        | compile file specified by -m                                                                                                                                                                             |                       |
| <code>--compile</code>       | —        | compile file specified by -m                                                                                                                                                                             |                       |
| <code>-d</code>              | —        | print debugging messages                                                                                                                                                                                 |                       |
| <code>--debug</code>         | —        | print debugging messages                                                                                                                                                                                 |                       |

## Gotchas

*TODO: operational traps.*

## See also

`istat`, `ils`, `stat`, `die`, `diec`

# fls

---

**Kit:** SIFT Workstation · REMnux · Kali ( `sleuthkit` ) **Category:** Filesystem analysis **Version:** The Sleuth Kit 4.11.1  
**Docs:** <https://sleuthkit.org/sleuthkit/man/fls.html> **Verified:** 2026-07-29 from `cyberlab-aio:v1` — raw output in `capture/cyberlab-aio/help/fls.help.txt`

**Worked example of the format.** Every option below was read off the real binary, not written from memory.  
See [docs/FORMAT.md](#).

---

## Purpose

List file and directory names in a disk image, including deleted entries.

---

## Synopsis

```
fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/]  
    [-o imgoffset] [-z ZONE] [-s seconds] image [images] [inode]
```

If `[inode]` is not given, the root directory is used.

**Note:** the usage line above advertises only `[-adDFlhpruvV]`, but the tool also accepts `-P`, `-B`, `-S` and `-k`. They are documented below.

---

## Common invocations



```
# List the root directory of an image
fls {{path/to/image.dd}}

# List a partition that starts at a sector offset (the usual case for a full disk)
fls -o {{2048}} {{path/to/image.dd}}

# Show deleted entries only – the fast answer to "what was removed?"
fls -d -o {{2048}} {{path/to/image.dd}}

# Recurse the whole filesystem showing full paths, deleted files included
fls -rp -o {{2048}} {{path/to/image.dd}}

# Recurse showing ONLY deleted files with full paths
fls -rpd -o {{2048}} {{path/to/image.dd}}

# Long listing (sizes, MAC times) like ls -l, in a named time zone
fls -l -z {{EST5EDT}} -o {{2048}} {{path/to/image.dd}}

# Emit a body file for timelining, then build the timeline with mactime
fls -r -m {{/}} -o {{2048}} {{path/to/image.dd}} > {{path/to/body.txt}}

# List the contents of one directory by inode number
fls -o {{2048}} {{path/to/image.dd}} {{inode}}
```

## Options — complete

| Flag            | Argument | What it does                                       | When you would use it                                              |
|-----------------|----------|----------------------------------------------------|--------------------------------------------------------------------|
| <code>-a</code> | —        | Display <code>.</code> and <code>..</code> entries | Rarely; they are hidden by default as noise                        |
| <code>-d</code> | —        | Display deleted entries only                       | Recovery hunts — isolates what was removed                         |
| <code>-D</code> | —        | Display only directories                           | Mapping structure without file noise                               |
| <code>-F</code> | —        | Display only files                                 | Excluding directories from a listing                               |
| <code>-l</code> | —        | Display long version (like <code>ls -l</code> )    | When you need sizes, UID/GID and MAC times                         |
| <code>-p</code> | —        | Display full path for each file                    | Almost always with <code>-r</code> ; otherwise output is ambiguous |
| <code>-r</code> | —        | Recurse on directory entries                       | Walking a whole filesystem rather than one directory               |
| <code>-u</code> | —        | Display undeleted entries only                     | The inverse of <code>-d</code> ; filtering live files              |

| Flag            | Argument                       | What it does                                                                      | When you would use it                                                             |
|-----------------|--------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <code>-m</code> | <code>dir/</code>              | Output in mactime input format, with <code>dir/</code> as the mount point         | Building a super-timeline; feeds <code>mactime</code>                             |
| <code>-h</code> | —                              | Include MD5 checksum hash in mactime output                                       | Timeline work where you also need file hashes                                     |
| <code>-f</code> | <code>fstype</code>            | Filesystem type ( <code>-f list</code> for supported types)                       | When auto-detection fails or is wrong                                             |
| <code>-i</code> | <code>imgtype</code>           | Image format ( <code>-i list</code> for supported types)                          | Non-raw images: E01, AFF, etc.                                                    |
| <code>-b</code> | <code>dev_sector_size</code>   | Device sector size in bytes                                                       | 4Kn drives, where the 512-byte assumption breaks                                  |
| <code>-o</code> | <code>imgoffset</code>         | Offset into the image, in sectors                                                 | <b>The most common flag</b> — targeting a partition in a full-disk image          |
| <code>-P</code> | <code>pooltype</code>          | Pool container type ( <code>-P list</code> for supported types)                   | APFS / logical volume containers                                                  |
| <code>-B</code> | <code>pool_volume_block</code> | Starting block, pool volumes only                                                 | Selecting a volume inside a pool                                                  |
| <code>-S</code> | <code>snap_id</code>           | Snapshot ID (APFS only)                                                           | Examining a specific APFS snapshot                                                |
| <code>-k</code> | <code>password</code>          | Decryption password for encrypted volumes                                         | BitLocker/FileVault-style encrypted volumes                                       |
| <code>-z</code> | <code>ZONE</code>              | Time zone of the original machine (e.g. <code>EST5EDT</code> , <code>GMT</code> ) | Only meaningful with <code>-l</code> ; wrong TZ silently skews every timestamp    |
| <code>-s</code> | <code>seconds</code>           | Time skew of the original machine, in seconds                                     | Only with <code>-l</code> / <code>-m</code> ; correcting a known-bad system clock |
| <code>-v</code> | —                              | Verbose output to stderr                                                          | Debugging why an image will not parse                                             |
| <code>-V</code> | —                              | Print version                                                                     | Recording tool provenance in your notes                                           |

## Gotchas

- `-o` **is in sectors, not bytes**. Get it from `mmfs` first. A wrong offset usually yields "Cannot determine file system type", not an obvious error.
- `-z` **only affects `-l` output**. Setting it without `-l` changes nothing, which makes a wrong-timezone timeline easy to produce and hard to notice.
- **Deleted entries may have unrecoverable content**. `fls -d` listing a name does not mean `icat` will return its data — the blocks may be reallocated.
- `-r` **on a large image is slow and noisy**. Pipe to a file, then grep it, rather than scrolling.

## See also

- `mmls` — partition table; where you get the `-o` value
- `icat` — extract the content of an inode `fls` found
- `istat` — detailed metadata for one inode
- `mactime` — consumes the `-m` body file to build the timeline
- `tsk_recover` — bulk export of allocated/deleted files

# foremost

**Kit:** Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data; Extract files and payloads from traffic **Version:** 1.5.7 **Captured:** cyberlab-aio via -h on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Carve files out of an image or raw data by header and footer signatures.

## Common invocations

```
# from cyberlab 05-file-carving
foremost -V
# from cyberlab 05-file-carving
foremost -t jpg,pdf -i exercise/sample.dd -o /tmp/foremost_out
# from cyberlab 05-file-carving
foremost -t jpg,pdf -i exercise/sample.dd -o /tmp/ak_foremost
# from cyberlab 33-binwalk-firmware
foremost -i firmware.bin -o foremost_out
```

## Options

All 11 options parsed from the captured help text; 8 reviewed with usage guidance.

| Flag | Argument | What it does                                                             | When you would use it                                                                                                           |
|------|----------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| -V   | —        | - display copyright information and exit                                 |                                                                                                                                 |
| -t   | —        | - specify file type. (-t jpeg,pdf ...)                                   | Restrict carving to given types ( jpg , pdf , all ). Narrowing this is the difference between a usable result and 40,000 files. |
| -d   | —        | - turn on indirect block detection (for UNIX file-systems)               |                                                                                                                                 |
| -i   | —        | - specify input file (default is stdin)                                  | Input file or device to carve from.                                                                                             |
| -a   | —        | - Write all headers, perform no error detection (corrupted files)        | Write all headers found, even without a valid footer.                                                                           |
| -w   | —        | - Only write the audit file, do not write any detected files to the disk | Write the audit file only, carving nothing — a cheap dry run.                                                                   |
| -o   | —        | - set output directory (defaults to output)                              | Output directory — must be empty or foremost refuses to run.                                                                    |

| Flag            | Argument | What it does                                                       | When you would use it                                                        |
|-----------------|----------|--------------------------------------------------------------------|------------------------------------------------------------------------------|
| <code>-c</code> | —        | - set configuration file to use (defaults to foremost.conf)        | Use a custom configuration file to add signatures.                           |
| <code>-q</code> | —        | - enables quick mode. Search are performed on 512 byte boundaries. | Quick mode: scan only sector boundaries. Much faster, misses embedded files. |
| <code>-Q</code> | —        | - enables quiet mode. Suppress output messages.                    |                                                                              |
| <code>-v</code> | —        | - verbose mode. Logs all messages to screen                        | Verbose output.                                                              |

## Gotchas

- Carving recovers content but **not filenames or timestamps** — those live in filesystem metadata that carving bypasses. Use `tsk_recover` when the filesystem is intact and carve only what it cannot reach.
- The output directory must be empty; foremost aborts otherwise. This trips scripted reruns constantly.

## See also

`scalpel`, `binwalk`, `bulk_extractor`, `tcpextract`, `tcpflow`

# fsstat

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Report a filesystem's layout and parameters: type, block size, inode range, and the geometry every other TSK tool needs.

## Synopsis

```
fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
```

## Common invocations

```
# from cyberlab 01-disk-forensics
fsstat -V
# from cyberlab 01-disk-forensics
fsstat -o 2048 exercise/sample.dd
# from cyberlab 01-disk-forensics
fsstat -o 0 exercise/sample.dd | grep -Ei "file system type|sector size|cluster size"
# from cyberlab 22-sleuthkit-mastery
fsstat -o 2048 exercise/practice.dd
# from cyberlab 51-linux-triage-workflow
fsstat -o 2048 disk.raw
```

## Options

All 10 options parsed from the captured help text; 9 reviewed with usage guidance.

| Flag            | Argument        | What it does                                                     | When you would use it                                                                                      |
|-----------------|-----------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>-t</code> | —               | display type only                                                | Print only the filesystem type. The scriptable form when all you need is a yes/no on what this volume is.  |
| <code>-i</code> | imgtype         | The format of the image file (use '-i list' for supported types) | Set the image format for non-raw evidence such as E01 or AFF.                                              |
| <code>-b</code> | dev_sector_size | The size (in bytes) of the device sectors                        | Device sector size. Needed on 4Kn drives, where the 512-byte default silently computes every offset wrong. |

| Flag            | Argument          | What it does                                            | When you would use it                                                                                                                               |
|-----------------|-------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-f</code> | fstype            | File system type (use '-f list' for supported types)    | Force the filesystem type when detection is wrong or the superblock is damaged ( <code>-f list</code> shows the options).                           |
| <code>-o</code> | imgoffset         | The offset of the file system in the image (in sectors) | Offset, in <b>sectors</b> , of the filesystem inside the image. Take it from <code>mmls</code> ; this is the flag that ties the two tools together. |
| <code>-P</code> | pooltype          | Pool container type (use '-P list' for supported types) | Pool type, for APFS or LVM containers that hold the volume.                                                                                         |
| <code>-B</code> | pool_volume_block | Starting block (for pool volumes only)                  | Starting block within a pool volume.                                                                                                                |
| <code>-v</code> | —                 | verbose output to stderr                                | Verbose diagnostics to stderr, useful when detection fails.                                                                                         |
| <code>-V</code> | —                 | Print version                                           |                                                                                                                                                     |
| <code>-k</code> | password          | Decryption password for encrypted volumes               | Password for an encrypted volume.                                                                                                                   |

## Gotchas

- Run this first. Block size and inode range from `fsstat` are what make `blkls` , `icat` and `ils` output interpretable — starting anywhere else means guessing at the numbers they print.
- If it reports the wrong type or refuses the volume, the `-o` offset is wrong far more often than the image is corrupt.

## See also

`mmls` , `img_stat` , `testdisk`

# grep

**Kit:** Base OS — present on every Linux image **Capability:** Search raw data for a pattern **Version:** grep (GNU grep) 3.8 **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Search for PATTERNS in each FILE.

## Synopsis

```
grep [OPTION]... PATTERNS [FILE]...
```

## Common invocations

```
# from cyberlab 02-memory-forensics
grep -i "benign.lab.local" be_out/url.txt
# from cyberlab 02-memory-forensics
grep -i "analyst@lab.local" be_out/email.txt
# from cyberlab 03-timeline-analysis
grep filestat /tmp/timeline.csv | sort -t',' -k1,2 | head -n 1
# from cyberlab 04-registry-analysis
grep -i "ComputerName" /tmp/system_dump.txt | head
# from cyberlab 05-file-carving
grep -v '^#' /etc/scalpel/scalpel.conf | grep -qi jpg || printf '\njpg y 200000000 \\xff\\xd8\\xff \\xf
# from cyberlab 05-file-carving
grep -E 'jpg:|pdf:' /tmp/ak_foremost/audit.txt
# from cyberlab 06-windows-artifact-libs
grep -c "Record number" /tmp/security_events.txt
# from cyberlab 06-windows-artifact-libs
grep "Event Identifier" /tmp/security_events.txt | awk -F: '{print $2}' | sort | uniq -c | sort -rn |
```

## Options

All 82 options parsed from the captured help text. The final column is filled in by review.



| Flag                           | Argument | What it does                                  | When you would use it |
|--------------------------------|----------|-----------------------------------------------|-----------------------|
| <code>-E</code>                | —        | PATTERNS are extended regular expressions     |                       |
| <code>--extended-regexp</code> | —        | PATTERNS are extended regular expressions     |                       |
| <code>-F</code>                | —        | PATTERNS are strings                          |                       |
| <code>--fixed-strings</code>   | —        | PATTERNS are strings                          |                       |
| <code>-G</code>                | —        | PATTERNS are basic regular expressions        |                       |
| <code>--basic-regexp</code>    | —        | PATTERNS are basic regular expressions        |                       |
| <code>-P</code>                | —        | PATTERNS are Perl regular expressions         |                       |
| <code>--perl-regexp</code>     | —        | PATTERNS are Perl regular expressions         |                       |
| <code>-e</code>                | PATTERNS | use PATTERNS for matching                     |                       |
| <code>--regexp</code>          | PATTERNS | use PATTERNS for matching                     |                       |
| <code>-f</code>                | FILE     | take PATTERNS from FILE                       |                       |
| <code>--file</code>            | FILE     | take PATTERNS from FILE                       |                       |
| <code>-i</code>                | —        | ignore case distinctions in patterns and data |                       |
| <code>--ignore-case</code>     | —        | ignore case distinctions in patterns and data |                       |
| <code>--no-ignore-case</code>  | —        | do not ignore case distinctions (default)     |                       |
| <code>-w</code>                | —        | match only whole words                        |                       |
| <code>--word-regexp</code>     | —        | match only whole words                        |                       |
| <code>-x</code>                | —        | match only whole lines                        |                       |
| <code>--line-regexp</code>     | —        | match only whole lines                        |                       |
| <code>-z</code>                | —        | a data line ends in 0 byte, not newline       |                       |
| <code>--null-data</code>       | —        | a data line ends in 0 byte, not newline       |                       |
| <code>-s</code>                | —        | suppress error messages                       |                       |
| <code>--no-messages</code>     | —        | suppress error messages                       |                       |
| <code>-v</code>                | —        | select non-matching lines                     |                       |
| <code>--invert-match</code>    | —        | select non-matching lines                     |                       |
| <code>-V</code>                | —        | display version information and exit          |                       |
| <code>--version</code>         | —        | display version information and exit          |                       |
| <code>--help</code>            | —        | display this help text and exit               |                       |
| <code>-m</code>                | NUM      | stop after NUM selected lines                 |                       |
| <code>--max-count</code>       | NUM      | stop after NUM selected lines                 |                       |
| <code>-b</code>                | —        | print the byte offset with output lines       |                       |
| <code>--byte-offset</code>     | —        | print the byte offset with output lines       |                       |
| <code>-n</code>                | —        | print line number with output lines           |                       |
| <code>--line-number</code>     | —        | print line number with output lines           |                       |

| Flag                                 | Argument | What it does                                                                    | When you would use it |
|--------------------------------------|----------|---------------------------------------------------------------------------------|-----------------------|
| <code>--line-buffered</code>         | —        | flush output on every line                                                      |                       |
| <code>-H</code>                      | —        | print file name with output lines                                               |                       |
| <code>--with-filename</code>         | —        | print file name with output lines                                               |                       |
| <code>-h</code>                      | —        | suppress the file name prefix on output                                         |                       |
| <code>--no-filename</code>           | —        | suppress the file name prefix on output                                         |                       |
| <code>--label</code>                 | LABEL    | use LABEL as the standard input file name prefix                                |                       |
| <code>-o</code>                      | —        | show only nonempty parts of lines that match                                    |                       |
| <code>--only-matching</code>         | —        | show only nonempty parts of lines that match                                    |                       |
| <code>-q</code>                      | —        | suppress all normal output                                                      |                       |
| <code>--quiet</code>                 | —        | suppress all normal output                                                      |                       |
| <code>--silent</code>                | —        | suppress all normal output                                                      |                       |
| <code>--binary-files</code>          | TYPE     | assume that binary files are TYPE; TYPE is 'binary', 'text', or 'without-match' |                       |
| <code>-a</code>                      | —        | equivalent to <code>--binary-files=text</code>                                  |                       |
| <code>--text</code>                  | —        | equivalent to <code>--binary-files=text</code>                                  |                       |
| <code>-I</code>                      | —        | equivalent to <code>--binary-files=without-match</code>                         |                       |
| <code>-d</code>                      | ACTION   | how to handle directories; ACTION is 'read', 'recurse', or 'skip'               |                       |
| <code>--directories</code>           | ACTION   | how to handle directories; ACTION is 'read', 'recurse', or 'skip'               |                       |
| <code>-D</code>                      | ACTION   | how to handle devices, FIFOs and sockets; ACTION is 'read' or 'skip'            |                       |
| <code>--devices</code>               | ACTION   | how to handle devices, FIFOs and sockets; ACTION is 'read' or 'skip'            |                       |
| <code>-r</code>                      | —        | like <code>--directories=recurse</code>                                         |                       |
| <code>--recursive</code>             | —        | like <code>--directories=recurse</code>                                         |                       |
| <code>-R</code>                      | —        | likewise, but follow all symlinks                                               |                       |
| <code>--dereference-recursive</code> | —        | likewise, but follow all symlinks                                               |                       |
| <code>--include</code>               | GLOB     | search only files that match GLOB (a file pattern)                              |                       |
| <code>--exclude</code>               | GLOB     | skip files that match GLOB                                                      |                       |
| <code>--exclude-from</code>          | FILE     | skip files that match any file pattern from FILE                                |                       |
| <code>--exclude-dir</code>           | GLOB     | skip directories that match GLOB                                                |                       |
| <code>-L</code>                      | —        | print only names of FILEs with no selected lines                                |                       |
| <code>--files-without-match</code>   | —        | print only names of FILEs with no selected lines                                |                       |
| <code>-l</code>                      | —        | print only names of FILEs with selected lines                                   |                       |

| Flag                              | Argument        | What it does                                                                        | When you would use it |
|-----------------------------------|-----------------|-------------------------------------------------------------------------------------|-----------------------|
| <code>--files-with-matches</code> | —               | print only names of FILES with selected lines                                       |                       |
| <code>-c</code>                   | —               | print only a count of selected lines per FILE                                       |                       |
| <code>--count</code>              | —               | print only a count of selected lines per FILE                                       |                       |
| <code>-T</code>                   | —               | make tabs line up (if needed)                                                       |                       |
| <code>--initial-tab</code>        | —               | make tabs line up (if needed)                                                       |                       |
| <code>-Z</code>                   | —               | print 0 byte after FILE name                                                        |                       |
| <code>--null</code>               | —               | print 0 byte after FILE name                                                        |                       |
| <code>-B</code>                   | NUM             | print NUM lines of leading context                                                  |                       |
| <code>--before-context</code>     | NUM             | print NUM lines of leading context                                                  |                       |
| <code>-A</code>                   | NUM             | print NUM lines of trailing context                                                 |                       |
| <code>--after-context</code>      | NUM             | print NUM lines of trailing context                                                 |                       |
| <code>-C</code>                   | NUM             | print NUM lines of output context                                                   |                       |
| <code>--context</code>            | NUM             | print NUM lines of output context                                                   |                       |
| <code>--group-separator</code>    | SEP             | print SEP on line between matches with context                                      |                       |
| <code>--no-group-separator</code> | —               | do not print separator for matches with context                                     |                       |
| <code>--colour</code>             | WHEN (optional) | use markers to highlight the matching strings; WHEN is 'always', 'never', or 'auto' |                       |
| <code>-U</code>                   | —               | do not strip CR characters at EOL (MSDOS/Windows)                                   |                       |
| <code>--binary</code>             | —               | do not strip CR characters at EOL (MSDOS/Windows)                                   |                       |

## Gotchas

*TODO: operational traps.*

## See also

`rafind2`, `strings`, `xxd`, `bulk_extractor`

# icat

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Extract the contents of a file by inode, including deleted files.

## Synopsis

```
icat [-hrRsvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image [images] inum[-typ[-
```

## Common invocations

```
# from cyberlab 01-disk-forensics
icat -o 2048 exercise/sample.dd 5 > /tmp/recovered_file.bin
# from cyberlab 01-disk-forensics
icat -o 0 exercise/sample.dd 5 > /tmp/recovered.txt
# from cyberlab 22-sleuthkit-mastery
icat -o 2048 exercise/practice.dd 4 | head
# from cyberlab 22-sleuthkit-mastery
icat -o 2048 exercise/practice.dd 6
# from cyberlab 51-linux-triage-workflow
icat -o 2048 disk.raw 5 > recovered_file.bin
# from cyberlab 51-linux-triage-workflow
icat exercise/triage_sample.raw $(find -p exercise/triage_sample.raw | awk '/eicar.com/{gsub(/:/,"",$2)
```

## Options

All 14 options parsed from the captured help text; 8 reviewed with usage guidance.

| Flag | Argument | What it does                         | When you would use it                                                       |
|------|----------|--------------------------------------|-----------------------------------------------------------------------------|
| -h   | —        | Do not display holes in sparse files | Skip holes in sparse files so the output is not padded with zeroes.         |
| -r   | —        | Recover deleted file                 | Attempt recovery of a deleted file — the reason you usually reach for icat. |

| Flag            | Argument          | What it does                                                     | When you would use it                                                              |
|-----------------|-------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <code>-R</code> | —                 | Recover deleted file and suppress recovery errors                | Recover with slack space included, when you want everything the blocks still hold. |
| <code>-S</code> | —                 | Display slack space at end of file                               | Include slack space in the output.                                                 |
| <code>-i</code> | imgtype           | The format of the image file (use '-i list' for supported types) | Image format for non-raw evidence.                                                 |
| <code>-b</code> | dev_sector_size   | The size (in bytes) of the device sectors                        |                                                                                    |
| <code>-f</code> | fstype            | File system type (use '-f list' for supported types)             | Force the filesystem type when detection is wrong.                                 |
| <code>-o</code> | imgoffset         | The offset of the file system in the image (in sectors)          | Partition offset in sectors, from <code>mm1s</code> .                              |
| <code>-P</code> | pooltype          | Pool container type (use '-P list' for supported types)          |                                                                                    |
| <code>-B</code> | pool_volume_block | Starting block (for pool volumes only)                           |                                                                                    |
| <code>-S</code> | snap_id           | Snapshot ID (for APFS only)                                      |                                                                                    |
| <code>-v</code> | —                 | verbose to stderr                                                |                                                                                    |
| <code>-V</code> | —                 | Print version                                                    |                                                                                    |
| <code>-k</code> | password          | Decryption password for encrypted volumes                        | Supply a decryption password for an encrypted volume.                              |

## Gotchas

- Always redirect to a file ( `icat ... > out.bin` ). Binary content dumped to a terminal will corrupt your session.
- A deleted inode listed by `fls -d` may return nothing or garbage: the metadata survived but the blocks were reallocated. Empty output is evidence, not a tool failure.

## See also

`tsk_recover`, `photorec`, `testdisk`, `blkls`

# ils

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones; Inspect metadata for one file or inode **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

List inode metadata, including inodes that no longer have a name pointing at them.

## Synopsis

```
ils [-emOpvV] [-aAlLzZ] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 19 options parsed from the captured help text; 18 reviewed with usage guidance.

| Flag            | Argument | What it does                                                            | When you would use it                                                                                                                                        |
|-----------------|----------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-e</code> | —        | Display all inodes                                                      | Every inode, allocated or not.                                                                                                                               |
| <code>-m</code> | —        | Display output in the mactime format                                    | mactime format — the form <code>mactime</code> consumes to build a timeline. This is how deleted-file metadata reaches the timeline at all.                  |
| <code>-O</code> | —        | Display inodes that are unallocated, but were sill open (UFS/ExtX only) | Unallocated inodes that were still open at the time of imaging (UFS/ExtX). The same trick, caught mid-deletion.                                              |
| <code>-p</code> | —        | Display orphan inodes (unallocated with no file name)                   | Orphan inodes — allocated content with no directory entry. Files that were unlinked while still open, and a standard hiding place worth checking explicitly. |
| <code>-s</code> | seconds  | Time skew of original machine (in seconds)                              | Correct for a known clock skew on the source machine, in seconds, so times line up with other evidence.                                                      |
| <code>-a</code> | —        | Allocated inodes                                                        | Allocated inodes only.                                                                                                                                       |

| Flag            | Argument          | What it does                                                     | When you would use it                                                                       |
|-----------------|-------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <code>-A</code> | —                 | Unallocated inodes                                               | Unallocated inodes only — deleted file metadata that often survives after the name is gone. |
| <code>-l</code> | —                 | Linked inodes                                                    | Linked inodes (a name still points at them).                                                |
| <code>-L</code> | —                 | Unlinked inodes                                                  | Unlinked inodes (nothing does).                                                             |
| <code>-z</code> | —                 | Unused inodes                                                    | Unused inodes.                                                                              |
| <code>-Z</code> | —                 | Used inodes                                                      | Used inodes.                                                                                |
| <code>-i</code> | imgtype           | The format of the image file (use '-i list' for supported types) | Image format for non-raw evidence such as E01 or AFF.                                       |
| <code>-b</code> | dev_sector_size   | The size (in bytes) of the device sectors                        | Device sector size; required on 4Kn drives.                                                 |
| <code>-f</code> | fstype            | File system type (use '-f list' for supported types)             | Force the filesystem type ( <code>-f list</code> shows the options).                        |
| <code>-o</code> | imgoffset         | The offset of the file system in the image (in sectors)          | Offset, in sectors, from <code>mmls</code> .                                                |
| <code>-p</code> | pooltype          | Pool container type (use '-p list' for supported types)          | Pool type, for APFS or LVM containers.                                                      |
| <code>-B</code> | pool_volume_block | Starting block (for pool volumes only)                           | Starting block within a pool volume.                                                        |
| <code>-v</code> | —                 | verbose output to stderr                                         | Verbose diagnostics to stderr.                                                              |
| <code>-V</code> | —                 | Display version number                                           |                                                                                             |

## Gotchas

- `ils` finds metadata with no name; `ffind` turns an inode back into a name; `icat` extracts its content. Deleted-file work is usually all three in sequence.
- An inode surviving does not mean its data did. The blocks it points at may already be reallocated, so `icat` can return another file's contents entirely.

## See also

`fls`, `ffind`, `tsk_recover`, `istat`, `file`, `stat`

# istat

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect metadata for one file or inode **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Show the full metadata for one inode: times, size, and the blocks it occupies.

## Synopsis

```
istat [-N num] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_vol]
```

## Common invocations

```
# from cyberlab 22-sleuthkit-mastery
istat -o 2048 exercise/practice.dd 4
```

## Options

All 14 options parsed from the captured help text; 5 reviewed with usage guidance.

| Flag | Argument        | What it does                                                     | When you would use it                                        |
|------|-----------------|------------------------------------------------------------------|--------------------------------------------------------------|
| -N   | num             | force the display of NUM address of block pointers               | Limit how many block addresses are printed for a large file. |
| -r   | —               | display run list instead of list of block addresses              | Include recovery information for a deleted inode.            |
| -z   | zone            | time zone of original machine (i.e. EST5EDT or GMT)              | Set the time zone for the displayed timestamps.              |
| -s   | seconds         | Time skew of original machine (in seconds)                       | Apply a clock skew in seconds for a known-bad system clock.  |
| -i   | imgtype         | The format of the image file (use '-i list' for supported types) |                                                              |
| -b   | dev_sector_size | The size (in bytes) of the device sectors                        |                                                              |
| -f   | fstype          | File system type (use '-f list' for supported types)             |                                                              |



| Flag            | Argument          | What it does                                            | When you would use it                                 |
|-----------------|-------------------|---------------------------------------------------------|-------------------------------------------------------|
| <code>-o</code> | imgoffset         | The offset of the file system in the image (in sectors) | Partition offset in sectors, from <code>mm1s</code> . |
| <code>-P</code> | pooltype          | Pool container type (use '-p list' for supported types) |                                                       |
| <code>-B</code> | pool_volume_block | Starting block (for pool volumes only)                  |                                                       |
| <code>-S</code> | snap_id           | Snapshot ID (for APFS only)                             |                                                       |
| <code>-v</code> | —                 | verbose output to stderr                                |                                                       |
| <code>-V</code> | —                 | print version                                           |                                                       |
| <code>-k</code> | password          | Decryption password for encrypted volumes               |                                                       |

## Gotchas

- The block list is what lets you prove whether a deleted file is still recoverable — cross-check it before promising a recovery.

## See also

`ils`, `file`, `stat`

# mmls

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Display the partition layout of a disk image, including unallocated gaps.

## Common invocations

```
# from cyberlab 01-disk-forensics
mmls -V
# from cyberlab 01-disk-forensics
mmls exercise/sample.dd
# from cyberlab 22-sleuthkit-mastery
mmls exercise/practice.dd
# from cyberlab 51-linux-triage-workflow
mmls disk.raw
# from cyberlab 57-forensic-acquisition
sudo mmls /mnt/ewf/ewf1
```

## Options

All 12 options parsed from the captured help text; 9 reviewed with usage guidance.

| Flag | Argument        | What it does                                                                      | When you would use it                                                                                     |
|------|-----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| -t   | vstype          | The type of volume system (use '-t list' for list of supported types)             | Force the volume-system type when auto-detection guesses wrong ( <code>-t list</code> shows the options). |
| -i   | imgtype         | The format of the image file (use '-i list' for list supported types)             | Set the image format for non-raw evidence such as E01 or AFF.                                             |
| -b   | dev_sector_size | The size (in bytes) of the device sectors                                         | Set the device sector size; required on 4Kn drives where the 512-byte default is wrong.                   |
| -o   | imgoffset       | Offset to the start of the volume that contains the partition system (in sectors) | Read a volume system nested at an offset — rare, but needed for nested containers.                        |
| -B   | —               | print the rounded length in bytes                                                 | Print volume sizes in bytes rather than sectors, when reporting.                                          |
| -r   | —               | recurse and look for other partition tables in partitions (DOS Only)              | Recurse into nested volume systems (e.g. an extended partition).                                          |

| Flag            | Argument | What it does             | When you would use it                                                            |
|-----------------|----------|--------------------------|----------------------------------------------------------------------------------|
| <code>-v</code> | —        | verbose output           | Verbose diagnostics to stderr when an image will not parse.                      |
| <code>-V</code> | —        | print the version        |                                                                                  |
| <code>-a</code> | —        | Show allocated volumes   | Show allocated volumes only, when the gap entries are noise.                     |
| <code>-A</code> | —        | Show unallocated volumes | Show unallocated space only — where a hidden or deleted partition would show up. |
| <code>-m</code> | —        | Show metadata volumes    |                                                                                  |
| <code>-M</code> | —        | Hide metadata volumes    |                                                                                  |

## Gotchas

- The **Start** column is in sectors. That value is what every other TSK tool wants for `-o`. Multiplying by the sector size here is the single most common mistake in a TSK workflow.
- If `mmfs` reports no partition table, the image may be a single volume rather than a whole disk — try `fsstat` on it directly at offset 0 before assuming the image is corrupt.

## See also

`fsstat`, `img_stat`, `testdisk`

# photorec

**Kit:** SIFT Workstation **Capability:** Recover deleted or lost files **Version:** PhotoRec 7.1 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

PhotoRec 7.1, Data Recovery Utility, July 2019

## Synopsis

```
photorec [/log] [/debug] [/d recup_dir] [file.dd|file.e01|device]
photorec /version
```

## Common invocations

```
# from cyberlab 01-disk-forensics
photorec /version
# from cyberlab 01-disk-forensics
photorec /log /d /tmp/carved /cmd exercise/sample.dd partition_none,options,mode_ext2,fileopt,everythi
```

## Options

All 2 options parsed from the captured help text. The final column is filled in by review.

| Flag   | Argument | What it does               | When you would use it |
|--------|----------|----------------------------|-----------------------|
| /log   | —        | create a photorec.log file |                       |
| /debug | —        | add debug information      |                       |

## Gotchas

*TODO: operational traps.*

## See also

[tsk\\_recover](#), [icat](#), [testdisk](#), [blkls](#)

# rafind2

---

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Search raw data for a pattern **Version:** rafind2 6.1.9

**Captured:** `cyberlab-aio` via `--help` on 2026-08-01 **raw Docs:** <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

## Synopsis

---

```
rafind2 [-mBXnzZhqv] [-a align] [-b sz] [-f/t from/to] [-[e|s|S] str] [-x hex] [-R str] [-I str] [-g]
```

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`strings`, `grep`, `xxd`, `bulk_extractor`

# scalpel

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data **Version:** Scalpel version 1.60 **Captured:** cyberlab-aio via -h on 2026-08-01 **raw Docs:** <https://github.com/sleuthkit/scalpel>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Carve contents out of binary files, such as partitions.

## Synopsis

```
scalpel [-b] [-c <config file>] [-d] [-h|V] [-i <file>]
[-m blocksize] [-n] [-o <outputdir>] [-O num] [-q clustersize]
[-r] [-s num] [-t <blockmap file>] [-u] [-v]
<imgfile> [<imgfile>] ...
```

## Common invocations

```
# from cyberlab 05-file-carving
scalpel -V
# from cyberlab 05-file-carving
scalpel -c /etc/scalpel/scalpel.conf -o /tmp/scalpel_out exercise/sample.dd
# from cyberlab 05-file-carving
scalpel -c /etc/scalpel/scalpel.conf -o /tmp/ak_scalpel exercise/sample.dd
```

## Options

All 17 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                                                                                                     | When you would use it |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -b   | —        | Carve files even if defined footers aren't discovered within maximum carve size for file type [foremost 0.69 compat mode].                                                       |                       |
| -c   | —        | Choose configuration file.                                                                                                                                                       |                       |
| -d   | —        | Generate header/footer database; will bypass certain optimizations and discover all footers, so performance suffers. Doesn't affect the set of files carved. <b>EXPERIMENTAL</b> |                       |
| -h   | —        | Print this help message and exit.                                                                                                                                                |                       |

| Flag            | Argument | What it does                                                                                                                                                                                             | When you would use it |
|-----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-i</code> | —        | Read names of disk images from specified file.                                                                                                                                                           |                       |
| <code>-m</code> | —        | Generate/update carve coverage blockmap file. The first 32bit unsigned int in the file identifies the block size. Thereafter each 32bit unsigned int entry in the blockmap file corresponds to one block |                       |
| <code>-n</code> | —        | Don't add extensions to extracted files.                                                                                                                                                                 |                       |
| <code>-o</code> | —        | Set output directory for carved files.                                                                                                                                                                   |                       |
| <code>-O</code> | —        | Don't organize carved files by type. Default is to organize carved files into subdirectories.                                                                                                            |                       |
| <code>-p</code> | —        | Perform image file preview; audit log indicates which files would have been carved, but no files are actually carved.                                                                                    |                       |
| <code>-q</code> | —        | Carve only when header is cluster-aligned.                                                                                                                                                               |                       |
| <code>-r</code> | —        | Find only first of overlapping headers/footers [foremost 0.69 compat mode].                                                                                                                              |                       |
| <code>-s</code> | —        | Skip n bytes in each disk image before carving.                                                                                                                                                          |                       |
| <code>-t</code> | —        | Set directory for coverage blockmap. <b>EXPERIMENTAL</b>                                                                                                                                                 |                       |
| <code>-u</code> | —        | Use carve coverage blockmap when carving. Carve only sections of the image whose entries in the blockmap are 0. These areas are treated as contiguous regions. <b>EXPERIMENTAL</b>                       |                       |
| <code>-V</code> | —        | Print copyright information and exit.                                                                                                                                                                    |                       |
| <code>-v</code> | —        | Verbose mode.                                                                                                                                                                                            |                       |

## Gotchas

*TODO: operational traps.*

## See also

`foremost`, `binwalk`, `bulk_extractor`, `tcpxtract`

# stat

**Kit:** Base OS — present on every Linux image **Capability:** Inspect metadata for one file or inode **Version:** stat (GNU coreutils) 9.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Display file or file system status.

## Synopsis

```
stat [OPTION]... FILE...
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 12 options parsed from the captured help text. The final column is filled in by review.

| Flag          | Argument | What it does                                                                                                                                | When you would use it |
|---------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -L            | —        | follow links                                                                                                                                |                       |
| --dereference | —        | follow links                                                                                                                                |                       |
| -f            | —        | display file system status instead of file status                                                                                           |                       |
| --file-system | —        | display file system status instead of file status                                                                                           |                       |
| --cached      | MODE     | specify how to use cached attributes; useful on remote file systems. See MODE below                                                         |                       |
| -c            | —        | use the specified FORMAT instead of the default; output a newline after each use of FORMAT                                                  |                       |
| --format      | FORMAT   | use the specified FORMAT instead of the default; output a newline after each use of FORMAT                                                  |                       |
| --printf      | FORMAT   | like --format, but interpret backslash escapes, and do not output a mandatory trailing newline; if you want a newline, include \n in FORMAT |                       |
| -t            | —        | print the information in terse form                                                                                                         |                       |



| Flag                   | Argument | What it does                        | When you would use it |
|------------------------|----------|-------------------------------------|-----------------------|
| <code>--terse</code>   | —        | print the information in terse form |                       |
| <code>--help</code>    | —        | display this help and exit          |                       |
| <code>--version</code> | —        | output version information and exit |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`istat`, `ils`, `file`

# strings

**Kit:** Base OS — present on every Linux image **Capability:** Search raw data for a pattern; Extract strings, including obfuscated ones **Version:** GNU strings (GNU Binutils for Debian) 2.40 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Print sequences of printable characters found in a binary file.

## Synopsis

```
strings [option(s)] [file(s)]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 14 options parsed from the captured help text; 3 reviewed with usage guidance.

| Flag              | Argument | What it does                                         | When you would use it                                            |
|-------------------|----------|------------------------------------------------------|------------------------------------------------------------------|
| -d                | —        | Only scan the data sections in the file              |                                                                  |
| --data            | —        | Only scan the data sections in the file              |                                                                  |
| -f                | —        | Print the name of the file before each string        |                                                                  |
| --print-file-name | —        | Print the name of the file before each string        |                                                                  |
| -n                | number   | Locate & print any sequence of at least              | Minimum length. Default 4 is noisy; 8–10 cuts most false hits.   |
| --bytes           | number   | displayable characters. (The default is 4).          |                                                                  |
| -t                | o,d,x    | Print the location of the string in base 8, 10 or 16 |                                                                  |
| --radix           | o,d,x    | Print the location of the string in base 8, 10 or 16 |                                                                  |
| -o                | —        | An alias for --radix=o                               | Print the byte offset of each string — lets you seek back to it. |

| Flag                  | Argument    | What it does                                          | When you would use it                                              |
|-----------------------|-------------|-------------------------------------------------------|--------------------------------------------------------------------|
| <code>-T</code>       | —           | Specify the binary file format                        |                                                                    |
| <code>--target</code> | BFDNAME     | Specify the binary file format                        |                                                                    |
| <code>-U</code>       | d s i x e h | Specify how to treat UTF-8 encoded unicode characters | Control how unicode is handled; needed for UTF-16 Windows strings. |
| <code>-h</code>       | —           | Display this information                              |                                                                    |
| <code>--help</code>   | —           | Display this information                              |                                                                    |

## Gotchas

- GNU `strings` reads only initialised, loaded sections by default. Use `-a` to scan the whole file — malware routinely hides outside them.
- It finds ASCII by default and will miss UTF-16LE strings that Windows binaries are full of. When a PE looks empty, that is usually why — reach for `floss` instead.

## See also

`rafind2`, `grep`, `xxd`, `bulk_extractor`, `floss`, `base64dump.py`, `numbers-to-string.py`

# tcpxtract

**Kit:** REMnux · SIFT Workstation **Capability:** Carve files out of unstructured data; Extract files and payloads from traffic **Version:** tcpxtract v1.0.1 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw Docs:](http://tcpxtract.sourceforge.net/) <http://tcpxtract.sourceforge.net/>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Extract files from network traffic.

## Synopsis

```
tcpxtract [OPTIONS] [[-d <DEVICE>] [-f <FILE>]]
```

## Common invocations

```
# from cyberlab 05-file-carving
tcpxtract --version 2>&1 | head -n 1
# from cyberlab 05-file-carving
tcpxtract -f exercise/sample.pcap -o /tmp/tcpxtract_out
# from cyberlab 05-file-carving
tcpxtract -f exercise/sample.pcap -o /tmp/ak_tcp
```

## Options

All 12 options parsed from the captured help text. The final column is filled in by review.

| Flag                   | Argument  | What it does                                         | When you would use it |
|------------------------|-----------|------------------------------------------------------|-----------------------|
| <code>--file</code>    | FILE      | to specify an input capture file instead of a device |                       |
| <code>-f</code>        | FILE      | to specify an input capture file instead of a device |                       |
| <code>--device</code>  | DEVICE    | to specify an input device (i.e. eth0)               |                       |
| <code>-d</code>        | DEVICE    | to specify an input device (i.e. eth0)               |                       |
| <code>--config</code>  | FILE      | use FILE as the config file                          |                       |
| <code>-c</code>        | FILE      | use FILE as the config file                          |                       |
| <code>--output</code>  | DIRECTORY | dump files to DIRECTORY instead of current directory |                       |
| <code>-o</code>        | DIRECTORY | dump files to DIRECTORY instead of current directory |                       |
| <code>--version</code> | —         | display the version number of this program           |                       |

| Flag                | Argument | What it does                               | When you would use it |
|---------------------|----------|--------------------------------------------|-----------------------|
| <code>-v</code>     | —        | display the version number of this program |                       |
| <code>--help</code> | —        | display this lovely screen                 |                       |
| <code>-h</code>     | —        | display this lovely screen                 |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`foremost`, `scalpel`, `binwalk`, `bulk_extractor`, `tcpflow`

# testdisk

**Kit:** SIFT Workstation **Capability:** See the partition and volume layout; Recover deleted or lost files **Version:** TestDisk 7.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

TestDisk 7.1, Data Recovery Utility, July 2019

## Synopsis

```
testdisk [/log] [/debug] [file.dd|file.e01|device]
testdisk /list  [/log]  [file.dd|file.e01|device]
testdisk /version
```

## Common invocations

```
# from cyberlab 01-disk-forensics
testdisk /version
```

## Options

All 3 options parsed from the captured help text. The final column is filled in by review.

| Flag   | Argument | What it does               | When you would use it |
|--------|----------|----------------------------|-----------------------|
| /log   | —        | create a testdisk.log file |                       |
| /debug | —        | add debug information      |                       |
| /list  | —        | display current partitions |                       |

## Gotchas

*TODO: operational traps.*

## See also

[mmls](#) , [fsstat](#) , [img\\_stat](#) , [tsk\\_recover](#) , [icat](#) , [photorec](#) , [blkls](#)

# tsk\_recover

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones; Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)  
**Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Bulk-export files from an image to a directory.

## Synopsis

```
tsk_recover [-vVae] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o sector_offset] [-P pooltype] [-B
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 11 options parsed from the captured help text; 6 reviewed with usage guidance.

| Flag | Argument        | What it does                                                     | When you would use it                                         |
|------|-----------------|------------------------------------------------------------------|---------------------------------------------------------------|
| -i   | imgtype         | The format of the image file (use '-i list' for supported types) | Image format for non-raw evidence.                            |
| -b   | dev_sector_size | The size (in bytes) of the device sectors                        |                                                               |
| -f   | fstype          | The file system type (use '-f list' for supported types)         | Force the filesystem type.                                    |
| -v   | —               | verbose output to stderr                                         |                                                               |
| -V   | —               | Print version                                                    |                                                               |
| -a   | —               | Recover allocated files only                                     | Recover allocated (live) files only.                          |
| -e   | —               | Recover all files (allocated and unallocated)                    | Recover every file, allocated and deleted — the usual choice. |

| Flag            | Argument          | What it does                                                                                                          | When you would use it                                          |
|-----------------|-------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <code>-o</code> | sector_offset     | sector offset for a volume to recover (recovers only that volume)                                                     | Partition offset in sectors, from <code>mmfs</code> .          |
| <code>-P</code> | pooltype          | Pool container type (use '-P list' for supported types)                                                               |                                                                |
| <code>-B</code> | pool_volume_block | Starting block (for pool volumes only)                                                                                |                                                                |
| <code>-d</code> | dir_inum          | Directory inum to recover from (must also specify a specific partition using -o or there must not be a volume system) | Recover from a specified directory inode rather than the root. |

## Gotchas

- Default behaviour recovers only *deleted* files, which surprises people expecting a full export. Use `-e` for everything.
- This preserves paths and names, unlike carving. Prefer it whenever the filesystem metadata is intact, and carve only what it cannot reach.

## See also

`fls`, `ffind`, `ils`, `icat`, `photorec`, `testdisk`, `blkls`



# xxd

**Kit:** Base OS — present on every Linux image **Capability:** Search raw data for a pattern; Decode, decrypt or transform encoded data; Inspect files by hand **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

xxd [options] [infile [outfile]]

## Synopsis

```
xxd [options] [infile [outfile]]
or
xxd -r [-s [-]offset] [-c cols] [-ps] [infile [outfile]]
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 16 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                    | When you would use it |
|------|----------|-----------------------------------------------------------------|-----------------------|
| -a   | —        | toggle autoskip: A single '*' replaces nul-lines. Default off.  |                       |
| -b   | —        | binary digit dump (incompatible with -ps,-i,-r). Default hex.   |                       |
| -C   | —        | capitalize variable names in C include file style (-i).         |                       |
| -c   | cols     | format octets per line. Default 16 (-i: 12, -ps: 30).           |                       |
| -E   | —        | show characters in EBCDIC. Default ASCII.                       |                       |
| -e   | —        | little-endian dump (incompatible with -ps,-i,-r).               |                       |
| -g   | bytes    | number of octets per group in normal output. Default 2 (-e: 4). |                       |
| -h   | —        | print this summary.                                             |                       |
| -i   | —        | output in C include file style.                                 |                       |
| -l   | len      | stop after octets.                                              |                       |
| -n   | name     | set the variable name used in C include output (-i).            |                       |
| -o   | off      | add to the displayed file position.                             |                       |

| Flag            | Argument | What it does                                               | When you would use it |
|-----------------|----------|------------------------------------------------------------|-----------------------|
| <code>-r</code> | —        | reverse operation: convert (or patch) hexdump into binary. |                       |
| <code>-d</code> | —        | show offset in decimal instead of hex.                     |                       |
| <code>-u</code> | —        | use upper case hex letters.                                |                       |
| <code>-v</code> | —        | show version: "xxd 2022-01-14 by Juergen Weigert et al."   |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`rafind2`, `strings`, `grep`, `bulk_extractor`, `cyberchef`, `base64dump.py`, `rax2`, `openssl`

# mraptor

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** MacroRaptor 0.56.2 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
mraptor [options] <filename> [filename2 ...]
```

## Common invocations

```
# from cyberlab 36-oletools-deep
mraptor --version
# from cyberlab 36-oletools-deep
mraptor sample_macro.doc
# from cyberlab 36-oletools-deep
mraptor sample_macro.doc # automated verdict
```

## Options

All 11 options parsed from the captured help text. The final column is filled in by review.

| Flag                | Argument     | What it does                                                                                                    | When you would use it |
|---------------------|--------------|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-h</code>     | —            | show this help message and exit                                                                                 |                       |
| <code>--help</code> | —            | show this help message and exit                                                                                 |                       |
| <code>-r</code>     | —            | find files recursively in subdirectories.                                                                       |                       |
| <code>-z</code>     | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)        |                       |
| <code>--zip</code>  | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)        |                       |
| <code>-f</code>     | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |

| Flag                    | Argument  | What it does                                                                                                    | When you would use it |
|-------------------------|-----------|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--zipfname</code> | ZIP_FNAME | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |
| <code>-l</code>         | LOGLEVEL  | logging level debug/info/warning/error/critical (default=warning)                                               |                       |
| <code>--loglevel</code> | LOGLEVEL  | logging level debug/info/warning/error/critical (default=warning)                                               |                       |
| <code>-m</code>         | —         | Show matched strings.                                                                                           |                       |
| <code>--matches</code>  | —         | Show matched strings.                                                                                           |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

# msodde

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
msodde [-h] [-j] [--nounquote] [-l LOGLEVEL] [-p PASSWORD] [-d] [-f]
[-a]
FILE
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 15 options parsed from the captured help text. The final column is filled in by review.

| Flag        | Argument | What it does                                                                                   | When you would use it |
|-------------|----------|------------------------------------------------------------------------------------------------|-----------------------|
| -h          | —        | show this help message and exit                                                                |                       |
| --help      | —        | show this help message and exit                                                                |                       |
| -j          | —        | Output in json format. Do not use with -ldebug                                                 |                       |
| --json      | —        | Output in json format. Do not use with -ldebug                                                 |                       |
| --nounquote | —        | don't unquote values                                                                           |                       |
| -l          | LOGLEVEL | logging level debug/info/warning/error/critical (default=warning)                              |                       |
| --loglevel  | LOGLEVEL | logging level debug/info/warning/error/critical (default=warning)                              |                       |
| -p          | PASSWORD | if encrypted office files are encountered, try decryption with this password. May be repeated. |                       |
| --password  | PASSWORD | if encrypted office files are encountered, try decryption with this password. May be repeated. |                       |
| -d          | —        | Return only DDE and DDEAUTO fields                                                             |                       |
| --dde-only  | —        | Return only DDE and DDEAUTO fields                                                             |                       |

| Flag                      | Argument | What it does                                      | When you would use it |
|---------------------------|----------|---------------------------------------------------|-----------------------|
| <code>-f</code>           | —        | Return all fields except harmless ones            |                       |
| <code>--filter</code>     | —        | Return all fields except harmless ones            |                       |
| <code>-a</code>           | —        | Return all fields, irrespective of their contents |                       |
| <code>--all-fields</code> | —        | Return all fields, irrespective of their contents |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`olevba`, `oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`

# msoffcrypto-tool

**Kit:** REMnux **Capability:** Analyse other container formats **Captured:** cyberlab-aio via --help on 2026-08-01  
**raw Docs:** <https://github.com/nolze/msoffcrypto-tool>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Decrypt a Microsoft Office file with password, intermediate key, or private key which generated its escrow key.

## Synopsis

```
msoffcrypto-tool [-h] (-p [PASSWORD] | -t) [-e] [-v] [infile] [outfile]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 8 options parsed from the captured help text. The final column is filled in by review.

| Flag       | Argument | What it does                       | When you would use it |
|------------|----------|------------------------------------|-----------------------|
| -h         | —        | show this help message and exit    |                       |
| --help     | —        | show this help message and exit    |                       |
| -p         | PASSWORD | password text                      |                       |
| --password | PASSWORD | password text                      |                       |
| -t         | —        | test if the file is encrypted      |                       |
| --test     | —        | test if the file is encrypted      |                       |
| -e         | —        | encryption mode (default is false) |                       |
| -v         | —        | print verbose information          |                       |

## Gotchas

*TODO: operational traps.*

## See also

7za, unzip



# olebrowse

---

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Microsoft Office OLE2 compound documents.

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`olevba`, `oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olefile`

# oledir

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** oledir 0.54 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
oledir [options] <filename> [filename2 ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 7 options parsed from the captured help text. The final column is filled in by review.

| Flag                    | Argument     | What it does                                                                                                    | When you would use it |
|-------------------------|--------------|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-h</code>         | —            | show this help message and exit                                                                                 |                       |
| <code>--help</code>     | —            | show this help message and exit                                                                                 |                       |
| <code>-r</code>         | —            | find files recursively in subdirectories.                                                                       |                       |
| <code>-z</code>         | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)        |                       |
| <code>--zip</code>      | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)        |                       |
| <code>-f</code>         | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |
| <code>--zipfname</code> | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |

## Gotchas

*TODO: operational traps.*

## See also

---

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [olebrowse](#), [olefile](#)

# oledump.py

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** oledump.py 0.0.85 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
oledump.py [options] [file]
```

## Common invocations

```
# from cyberlab 10-malicious-documents
oledump.py -h | head -n 1
# from cyberlab 10-malicious-documents
oledump.py exercise/sample.doc
# from cyberlab 10-malicious-documents
oledump.py -s 3 -v exercise/sample.doc
# from cyberlab 36-oletools-deep
oledump.py --help | head -n 3
# from cyberlab 36-oletools-deep
oledump.py sample_macro.doc
# from cyberlab 36-oletools-deep
oledump.py -s 3 -v sample_macro.doc
# from cyberlab 36-oletools-deep
oledump.py sample_macro.doc # locate the 'M' stream (3)
# from cyberlab 36-oletools-deep
oledump.py -s 3 -v sample_macro.doc # dump VBA source
```

## Options

All 64 options parsed from the captured help text. The final column is filled in by review.

| Flag                   | Argument | What it does                           | When you would use it |
|------------------------|----------|----------------------------------------|-----------------------|
| <code>--version</code> | —        | show program's version number and exit |                       |

| Flag                                       | Argument  | What it does                                                        | When you would use it |
|--------------------------------------------|-----------|---------------------------------------------------------------------|-----------------------|
| <code>-h</code>                            | —         | show this help message and exit                                     |                       |
| <code>--help</code>                        | —         | show this help message and exit                                     |                       |
| <code>-m</code>                            | —         | Print manual                                                        |                       |
| <code>--man</code>                         | —         | Print manual                                                        |                       |
| <code>-s</code>                            | SELECT    | select item nr for dumping (a for all)                              |                       |
| <code>--select</code>                      | SELECT    | select item nr for dumping (a for all)                              |                       |
| <code>-d</code>                            | —         | perform dump                                                        |                       |
| <code>--dump</code>                        | —         | perform dump                                                        |                       |
| <code>-x</code>                            | —         | perform hex dump                                                    |                       |
| <code>--hexdump</code>                     | —         | perform hex dump                                                    |                       |
| <code>-a</code>                            | —         | perform ascii dump                                                  |                       |
| <code>--asciidump</code>                   | —         | perform ascii dump                                                  |                       |
| <code>-A</code>                            | —         | perform ascii dump with RLE                                         |                       |
| <code>--asciidumprle</code>                | —         | perform ascii dump with RLE                                         |                       |
| <code>-S</code>                            | —         | perform strings dump                                                |                       |
| <code>--strings</code>                     | —         | perform strings dump                                                |                       |
| <code>-T</code>                            | —         | do head & tail                                                      |                       |
| <code>--headtail</code>                    | —         | do head & tail                                                      |                       |
| <code>-v</code>                            | —         | VBA decompression                                                   |                       |
| <code>--vbadecompress</code>               | —         | VBA decompression                                                   |                       |
| <code>--vbadecompressskipattributes</code> | —         | VBA decompression, skipping initial attributes                      |                       |
| <code>--vbadecompresscorrupt</code>        | —         | VBA decompression, display beginning if corrupted                   |                       |
| <code>-r</code>                            | —         | read raw file (use with options -v or -p                            |                       |
| <code>--raw</code>                         | —         | read raw file (use with options -v or -p                            |                       |
| <code>-t</code>                            | TRANSLATE | string translation, like utf16 or .decode("utf8")                   |                       |
| <code>--translate</code>                   | TRANSLATE | string translation, like utf16 or .decode("utf8")                   |                       |
| <code>-e</code>                            | —         | extract OLE embedded file                                           |                       |
| <code>--extract</code>                     | —         | extract OLE embedded file                                           |                       |
| <code>-i</code>                            | —         | print extra info for selected item                                  |                       |
| <code>--info</code>                        | —         | print extra info for selected item                                  |                       |
| <code>-p</code>                            | PLUGINS   | plugins to load (separate plugins with a comma , ; @file supported) |                       |
| <code>--plugins</code>                     | PLUGINS   | plugins to load (separate plugins with a comma , ; @file supported) |                       |

| Flag                          | Argument       | What it does                                                                                         | When you would use it |
|-------------------------------|----------------|------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--pluginoptions</code>  | PLUGINOPTIONS  | options for the plugin                                                                               |                       |
| <code>--plugindir</code>      | PLUGINDIR      | directory for the plugin                                                                             |                       |
| <code>-q</code>               | —              | only print output from plugins                                                                       |                       |
| <code>--quiet</code>          | —              | only print output from plugins                                                                       |                       |
| <code>-y</code>               | YARA           | YARA rule-file, @file, directory or #rule to check streams (YARA search doesn't work with -s option) |                       |
| <code>--yara</code>           | YARA           | YARA rule-file, @file, directory or #rule to check streams (YARA search doesn't work with -s option) |                       |
| <code>-D</code>               | DECODERS       | decoders to load (separate decoders with a comma , ; @file supported)                                |                       |
| <code>--decoders</code>       | DECODERS       | decoders to load (separate decoders with a comma , ; @file supported)                                |                       |
| <code>--decoderoptions</code> | DECODEROPTIONS | options for the decoder                                                                              |                       |
| <code>--decoderdir</code>     | DECODERDIR     | directory for the decoder                                                                            |                       |
| <code>--yarastrings</code>    | —              | Print YARA strings                                                                                   |                       |
| <code>-M</code>               | —              | Print metadata                                                                                       |                       |
| <code>--metadata</code>       | —              | Print metadata                                                                                       |                       |
| <code>-c</code>               | —              | Add extra calculated data to output, like hashes                                                     |                       |
| <code>--calc</code>           | —              | Add extra calculated data to output, like hashes                                                     |                       |
| <code>--decompress</code>     | —              | Search for compressed data in the stream and decompress it                                           |                       |
| <code>-V</code>               | —              | verbose output with decoder errors and YARA rules                                                    |                       |
| <code>--verbose</code>        | —              | verbose output with decoder errors and YARA rules                                                    |                       |
| <code>-C</code>               | CUT            | cut data                                                                                             |                       |
| <code>--cut</code>            | CUT            | cut data                                                                                             |                       |
| <code>-E</code>               | EXTRA          | add extra info (environment variable: OLEDUMP_EXTRA)                                                 |                       |
| <code>--extra</code>          | EXTRA          | add extra info (environment variable: OLEDUMP_EXTRA)                                                 |                       |
| <code>--storages</code>       | —              | Include storages in report                                                                           |                       |
| <code>-f</code>               | FIND           | Find D0CF11E0 MAGIC sequence (use l for listing, number for selecting)                               |                       |
| <code>--find</code>           | FIND           | Find D0CF11E0 MAGIC sequence (use l for listing, number for selecting)                               |                       |
| <code>-j</code>               | —              | produce json output                                                                                  |                       |
| <code>--jsonoutput</code>     | —              | produce json output                                                                                  |                       |
| <code>-u</code>               | —              | Include unused data after end of stream                                                              |                       |

| Flag                      | Argument | What it does                                   | When you would use it |
|---------------------------|----------|------------------------------------------------|-----------------------|
| <code>--unuseddata</code> | —        | Include unused data after end of stream        |                       |
| <code>--password</code>   | PASSWORD | The ZIP password to be used (default infected) |                       |
| <code>--trimnull</code>   | —        | Trim starting at first null byte               |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`olevba`, `oleid`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`, `olefile`

# olefile

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://github.com/decalage2/olefile>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Python package to parse, read and write MS OLE2 files.

## Synopsis

```
olefile [options] <filename> [filename2 ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 7 options parsed from the captured help text. The final column is filled in by review.

| Flag       | Argument | What it does                                                                                 | When you would use it |
|------------|----------|----------------------------------------------------------------------------------------------|-----------------------|
| -h         | —        | show this help message and exit                                                              |                       |
| --help     | —        | show this help message and exit                                                              |                       |
| -c         | —        | check all streams (for debugging purposes)                                                   |                       |
| -p         | —        | extract all user-defined propertires                                                         |                       |
| -d         | —        | debug mode, shortcut for -l debug (displays a lot of debug information, for developers only) |                       |
| -l         | LOGLEVEL | logging level debug/info/warning/error/critical (default=warning)                            |                       |
| --loglevel | LOGLEVEL | logging level debug/info/warning/error/critical (default=warning)                            |                       |

## Gotchas

*TODO: operational traps.*



## See also

---

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

# oleid

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** oleid 0.60.1 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
oleid [-h] [FILE ...]
```

## Common invocations

```
# from cyberlab 10-malicious-documents
oleid exercise/sample.doc
# from cyberlab 36-oletools-deep
oleid -h | head -n 3
# from cyberlab 36-oletools-deep
oleid gives a quick risk indicator table (VBA macros: Yes/No, encryption, etc.)
# from cyberlab 36-oletools-deep
oleid sample_macro.doc
# from cyberlab 48-phishing-doc-case
oleid --version
# from cyberlab 48-phishing-doc-case
oleid exercise/invoice_sample.doc
```

## Options

All 2 options parsed from the captured help text. The final column is filled in by review.

| Flag                | Argument | What it does                    | When you would use it |
|---------------------|----------|---------------------------------|-----------------------|
| <code>-h</code>     | —        | show this help message and exit |                       |
| <code>--help</code> | —        | show this help message and exit |                       |

## Gotchas

*TODO: operational traps.*

## See also

---

[olevba](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

# olemap

---

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Microsoft Office OLE2 compound documents.

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`olevba`, `oleid`, `oledump.py`, `olemeta`, `oletimes`, `oledir`, `olebrowse`, `olefile`

# olemeta

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** olemeta 0.54 **Captured:** cyberlab-aio  
via `--help` on 2026-08-01 **raw Docs:** <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
olemeta [options] <filename> [filename2 ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 7 options parsed from the captured help text. The final column is filled in by review.

| Flag                    | Argument     | What it does                                                                                                    | When you would use it |
|-------------------------|--------------|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-h</code>         | —            | show this help message and exit                                                                                 |                       |
| <code>--help</code>     | —            | show this help message and exit                                                                                 |                       |
| <code>-r</code>         | —            | find files recursively in subdirectories.                                                                       |                       |
| <code>-z</code>         | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)        |                       |
| <code>--zip</code>      | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)        |                       |
| <code>-f</code>         | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |
| <code>--zipfname</code> | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |

## Gotchas

*TODO: operational traps.*

## See also

---

[olevba](#), [oleid](#), [oledump.py](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

# oleobj

**Kit:** REMnux **Capability:** Analyse Office documents and macros; Analyse RTF documents and embedded objects

**Version:** oleobj 0.60.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:**

<https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
usage: oleobj [options] <filename> [filename2 ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 14 options parsed from the captured help text. The final column is filled in by review.

| Flag       | Argument     | What it does                                                                                                    | When you would use it |
|------------|--------------|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| -h         | —            | show this help message and exit                                                                                 |                       |
| --help     | —            | show this help message and exit                                                                                 |                       |
| -r         | —            | find files recursively in subdirectories.                                                                       |                       |
| -d         | OUTPUT_DIR   | use specified directory to output files.                                                                        |                       |
| -z         | ZIP_PASSWORD | if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)       |                       |
| --zip      | ZIP_PASSWORD | if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)       |                       |
| -f         | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |
| --zipfname | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |
| -l         | LOGLEVEL     | logging level debug/info/warning/error/critical (default=warning)                                               |                       |
| --loglevel | LOGLEVEL     | logging level debug/info/warning/error/critical (default=warning)                                               |                       |

| Flag                      | Argument | What it does                                            | When you would use it |
|---------------------------|----------|---------------------------------------------------------|-----------------------|
| <code>-i</code>           | FILE     | Additional file to parse (same as positional arguments) |                       |
| <code>--more-input</code> | FILE     | Additional file to parse (same as positional arguments) |                       |
| <code>-v</code>           | —        | verbose mode, set logging to DEBUG (overwrites -l)      |                       |
| <code>--verbose</code>    | —        | verbose mode, set logging to DEBUG (overwrites -l)      |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`olevba`, `oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`



# oletimes

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** oletimes 0.54 **Captured:** cyberlab-aio  
via `--help` on 2026-08-01 **raw Docs:** <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
oletimes [options] <filename> [filename2 ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 7 options parsed from the captured help text. The final column is filled in by review.

| Flag                    | Argument     | What it does                                                                                                    | When you would use it |
|-------------------------|--------------|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-h</code>         | —            | show this help message and exit                                                                                 |                       |
| <code>--help</code>     | —            | show this help message and exit                                                                                 |                       |
| <code>-r</code>         | —            | find files recursively in subdirectories.                                                                       |                       |
| <code>-z</code>         | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)        |                       |
| <code>--zip</code>      | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)        |                       |
| <code>-f</code>         | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |
| <code>--zipfname</code> | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |

## Gotchas

*TODO: operational traps.*

## See also

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[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

# olevba

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** olevba 0.60.2 **Captured:** cyberlab-aio  
via `--help` on 2026-08-01 [raw Docs](#): <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Extract and analyse VBA macros from Office documents.

## Synopsis

```
usage: olevba [options] <filename> [filename2 ...]
```

## Common invocations

```
# from cyberlab 10-malicious-documents
olevba --version
# from cyberlab 10-malicious-documents
olevba --decode exercise/sample.doc
# from cyberlab 10-malicious-documents
olevba exercise/sample.doc | grep -E "AutoExec|AutoOpen|Suspicious"
# from cyberlab 36-oletools-deep
olevba sample_macro.doc
# from cyberlab 36-oletools-deep
olevba sample_macro.doc # keyword analysis table
# from cyberlab 48-phishing-doc-case
olevba --decode exercise/invoice_sample.doc
```

## Options

All 29 options parsed from the captured help text; 10 reviewed with usage guidance.

| Flag                | Argument     | What it does                                                                       | When you would use it                     |
|---------------------|--------------|------------------------------------------------------------------------------------|-------------------------------------------|
| <code>-h</code>     | —            | show this help message and exit                                                    |                                           |
| <code>--help</code> | —            | show this help message and exit                                                    |                                           |
| <code>-r</code>     | —            | find files recursively in subdirectories.                                          |                                           |
| <code>-z</code>     | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password. | Read documents from inside a zip archive. |

| Flag                      | Argument     | What it does                                                                                                     | When you would use it                                                                                    |
|---------------------------|--------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <code>--zip</code>        | ZIP_PASSWORD | if the file is a zip archive, open all files from it, using the provided password.                               |                                                                                                          |
| <code>-p</code>           | PASSWORD     | if encrypted office files are encountered, try decryption with this password. May be repeated.                   | Supply a password for an encrypted document.                                                             |
| <code>--password</code>   | PASSWORD     | if encrypted office files are encountered, try decryption with this password. May be repeated.                   |                                                                                                          |
| <code>-f</code>           | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default: *) |                                                                                                          |
| <code>--zipfname</code>   | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default: *) |                                                                                                          |
| <code>-a</code>           | —            | display only analysis results, not the macro source code                                                         | Show only the analysis results, skipping the macro source.                                               |
| <code>--analysis</code>   | —            | display only analysis results, not the macro source code                                                         |                                                                                                          |
| <code>-c</code>           | —            | display only VBA source code, do not analyze it                                                                  | Show only the macro source code.                                                                         |
| <code>--code</code>       | —            | display only VBA source code, do not analyze it                                                                  |                                                                                                          |
| <code>--decode</code>     | —            | display all the obfuscated strings with their decoded content (Hex, Base64, StrReverse, Dridex, VBA).            | Display all the decoded strings the analyser recovered.                                                  |
| <code>--attr</code>       | —            | display the attribute lines at the beginning of VBA source code                                                  |                                                                                                          |
| <code>--reveal</code>     | —            | display the macro source code after replacing all the obfuscated strings by their decoded content.               | Substitute deobfuscated values back into the source, which is the most readable view of a hostile macro. |
| <code>-l</code>           | LOGLEVEL     | logging level debug/info/warning/error/critical (default=warning)                                                | Set log level when diagnosing a parse failure.                                                           |
| <code>--loglevel</code>   | LOGLEVEL     | logging level debug/info/warning/error/critical (default=warning)                                                |                                                                                                          |
| <code>--deobf</code>      | —            | Attempt to deobfuscate VBA expressions (slow)                                                                    |                                                                                                          |
| <code>--relaxed</code>    | —            | Do not raise errors if opening of substream fails (this option is now deprecated, enabled by default)            |                                                                                                          |
| <code>--show-pcode</code> | —            | Show disassembled P-code (using pcodedmp)                                                                        | Disassemble the stored p-code — catches macros where the VBA source and compiled p-code disagree.        |
| <code>--no-pcode</code>   | —            | Disable extraction and analysis of pcode                                                                         |                                                                                                          |
| <code>--no-xlm</code>     | —            | Do not extract XLM Excel macros. This may speed up analysis of large files.                                      |                                                                                                          |
| <code>-t</code>           | —            | triage mode, display results as a summary table (default for multiple files)                                     | Triage mode — one compact line per file, ideal for a batch.                                              |
| <code>--triage</code>     | —            | triage mode, display results as a summary table (default for multiple files)                                     |                                                                                                          |

| Flag                    | Argument | What it does                                                  | When you would use it               |
|-------------------------|----------|---------------------------------------------------------------|-------------------------------------|
| <code>-d</code>         | —        | detailed mode, display full results (default for single file) | Show the deobfuscated macro source. |
| <code>--detailed</code> | —        | detailed mode, display full results (default for single file) |                                     |
| <code>-j</code>         | —        | json mode, detailed in json format (never default)            |                                     |
| <code>--json</code>     | —        | json mode, detailed in json format (never default)            |                                     |

## Gotchas

- **VBA source and p-code can differ.** Malware abuses this so the readable source looks benign while the p-code executes. If a document is suspicious but the source is clean, check `--show-pcode`.
- Run this on a copy, in an isolated VM. Nothing here executes the macro, but the sample itself is still live malware.

## See also

`oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`, `olefile`

# pcodedmp

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** pcodedmp version 1.2.6 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://github.com/bontchev/pcodedmp>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Disassemble VBA p-code.

## Synopsis

```
pcodedmp [-h] [-v] [-n] [-d] [-b] [-o OUTPUTFILE]
fileOrDir [fileOrDir ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 12 options parsed from the captured help text. The final column is filled in by review.

| Flag                      | Argument   | What it does                           | When you would use it |
|---------------------------|------------|----------------------------------------|-----------------------|
| <code>-h</code>           | —          | show this help message and exit        |                       |
| <code>--help</code>       | —          | show this help message and exit        |                       |
| <code>-v</code>           | —          | show program's version number and exit |                       |
| <code>--version</code>    | —          | show program's version number and exit |                       |
| <code>-n</code>           | —          | Don't recurse into directories         |                       |
| <code>--norecurse</code>  | —          | Don't recurse into directories         |                       |
| <code>-d</code>           | —          | Only disassemble, no stream dumps      |                       |
| <code>--disasmonly</code> | —          | Only disassemble, no stream dumps      |                       |
| <code>-b</code>           | —          | Dump the stream contents               |                       |
| <code>--verbose</code>    | —          | Dump the stream contents               |                       |
| <code>-o</code>           | OUTPUTFILE | Output file name                       |                       |
| <code>--output</code>     | OUTPUTFILE | Output file name                       |                       |

## Gotchas

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*TODO: operational traps.*

## See also

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[olevba](#) , [oleid](#) , [oledump.py](#) , [olemeta](#) , [oletimes](#) , [olemap](#) , [oledir](#) , [olebrowse](#)

# pdf-parser

**Kit:** Kali Linux **Capability:** Analyse PDFs **Version:** pdf-parser.py 0.7.14 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

pdf-parser, use it to parse a PDF document

## Synopsis

```
pdf-parser.py [options] pdf-file|zip-file|url
```

## Common invocations

```
# from cyberlab 37-pdf-analysis
pdf-parser --version
# from cyberlab 37-pdf-analysis
pdf-parser --search JavaScript exercise/sample.pdf
# from cyberlab 37-pdf-analysis
pdf-parser --object 4 --filter --raw exercise/sample.pdf
```

## Options

All 54 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                                          | When you would use it |
|-----------|----------|-------------------------------------------------------|-----------------------|
| --version | —        | show program's version number and exit                |                       |
| -h        | —        | show this help message and exit                       |                       |
| --help    | —        | show this help message and exit                       |                       |
| -m        | —        | Print manual                                          |                       |
| --man     | —        | Print manual                                          |                       |
| -s        | SEARCH   | string to search in indirect objects (except streams) |                       |
| --search  | SEARCH   | string to search in indirect objects (except streams) |                       |



| Flag                                 | Argument  | What it does                                                                                                        | When you would use it |
|--------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-f</code>                      | —         | pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only) |                       |
| <code>--filter</code>                | —         | pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only) |                       |
| <code>-o</code>                      | OBJECT    | id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)                          |                       |
| <code>--object</code>                | OBJECT    | id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)                          |                       |
| <code>-r</code>                      | REFERENCE | id of indirect object being referenced (version independent)                                                        |                       |
| <code>--reference</code>             | REFERENCE | id of indirect object being referenced (version independent)                                                        |                       |
| <code>-e</code>                      | ELEMENTS  | type of elements to select (ctxsi)                                                                                  |                       |
| <code>--elements</code>              | ELEMENTS  | type of elements to select (ctxsi)                                                                                  |                       |
| <code>-w</code>                      | —         | raw output for data and filters                                                                                     |                       |
| <code>--raw</code>                   | —         | raw output for data and filters                                                                                     |                       |
| <code>-a</code>                      | —         | display stats for pdf document                                                                                      |                       |
| <code>--stats</code>                 | —         | display stats for pdf document                                                                                      |                       |
| <code>-t</code>                      | TYPE      | type of indirect object to select                                                                                   |                       |
| <code>--type</code>                  | TYPE      | type of indirect object to select                                                                                   |                       |
| <code>-0</code>                      | —         | parse stream of /ObjStm objects                                                                                     |                       |
| <code>--objstm</code>                | —         | parse stream of /ObjStm objects                                                                                     |                       |
| <code>-v</code>                      | —         | display malformed PDF elements                                                                                      |                       |
| <code>--verbose</code>               | —         | display malformed PDF elements                                                                                      |                       |
| <code>-x</code>                      | EXTRACT   | filename to extract malformed content to                                                                            |                       |
| <code>--extract</code>               | EXTRACT   | filename to extract malformed content to                                                                            |                       |
| <code>-H</code>                      | —         | display hash of objects                                                                                             |                       |
| <code>--hash</code>                  | —         | display hash of objects                                                                                             |                       |
| <code>-n</code>                      | —         | do not canonicalize the output                                                                                      |                       |
| <code>--nocanonicalizedoutput</code> | —         | do not canonicalize the output                                                                                      |                       |
| <code>-d</code>                      | DUMP      | filename to dump stream content to                                                                                  |                       |
| <code>--dump</code>                  | DUMP      | filename to dump stream content to                                                                                  |                       |
| <code>-D</code>                      | —         | display debug info                                                                                                  |                       |
| <code>--debug</code>                 | —         | display debug info                                                                                                  |                       |

| Flag                             | Argument          | What it does                                                                              | When you would use it |
|----------------------------------|-------------------|-------------------------------------------------------------------------------------------|-----------------------|
| <code>-c</code>                  | —                 | display the content for objects without streams or with streams without filters           |                       |
| <code>--content</code>           | —                 | display the content for objects without streams or with streams without filters           |                       |
| <code>--searchstream</code>      | SEARCHSTREAM      | string to search in streams                                                               |                       |
| <code>--unfiltered</code>        | —                 | search in unfiltered streams                                                              |                       |
| <code>--casesensitive</code>     | —                 | case sensitive search in streams                                                          |                       |
| <code>--regex</code>             | —                 | use regex to search in streams                                                            |                       |
| <code>--overridingfilters</code> | OVERRIDINGFILTERS | override filters with given filters (use raw for the raw stream content)                  |                       |
| <code>-g</code>                  | —                 | generate a Python program that creates the parsed PDF file                                |                       |
| <code>--generate</code>          | —                 | generate a Python program that creates the parsed PDF file                                |                       |
| <code>--generateembedded</code>  | GENERATEEMBEDDED  | generate a Python program that embeds the selected indirect object as a file              |                       |
| <code>-y</code>                  | YARA              | YARA rule (or directory or @file) to check streams (can be used with option --unfiltered) |                       |
| <code>--yara</code>              | YARA              | YARA rule (or directory or @file) to check streams (can be used with option --unfiltered) |                       |
| <code>--yarastrings</code>       | —                 | Print YARA strings                                                                        |                       |
| <code>--decoders</code>          | DECODERS          | decoders to load (separate decoders with a comma , ; @file supported)                     |                       |
| <code>--decoderoptions</code>    | DECODEROPTIONS    | options for the decoder                                                                   |                       |
| <code>-k</code>                  | KEY               | key to search in dictionaries                                                             |                       |
| <code>--key</code>               | KEY               | key to search in dictionaries                                                             |                       |
| <code>-j</code>                  | —                 | produce json output                                                                       |                       |
| <code>--jsonoutput</code>        | —                 | produce json output                                                                       |                       |

## Gotchas

*TODO: operational traps.*

## See also

`pdfid` , `pdfid.py` , `pdf-parser.py`

# pdf-parser.py

**Kit:** REMnux · Kali Linux **Capability:** Analyse PDFs **Version:** pdf-parser.py 0.7.14 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://blog.didierstevens.com/programs/pdf-tools/>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Examine elements of the PDF file.

## Synopsis

```
pdf-parser.py [options] pdf-file|zip-file|url
```

## Common invocations

```
# from cyberlab 10-malicious-documents
pdf-parser.py --version
# from cyberlab 10-malicious-documents
pdf-parser.py --search JavaScript exercise/sample.pdf
# from cyberlab 48-phishing-doc-case
pdf-parser.py --search OpenAction exercise/resume_sample.pdf
# from cyberlab 48-phishing-doc-case
pdf-parser.py --type /JavaScript exercise/resume_sample.pdf
```

## Options

All 54 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                                          | When you would use it |
|-----------|----------|-------------------------------------------------------|-----------------------|
| --version | —        | show program's version number and exit                |                       |
| -h        | —        | show this help message and exit                       |                       |
| --help    | —        | show this help message and exit                       |                       |
| -m        | —        | Print manual                                          |                       |
| --man     | —        | Print manual                                          |                       |
| -s        | SEARCH   | string to search in indirect objects (except streams) |                       |
| --search  | SEARCH   | string to search in indirect objects (except streams) |                       |

| Flag                                 | Argument  | What it does                                                                                                        | When you would use it |
|--------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-f</code>                      | —         | pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only) |                       |
| <code>--filter</code>                | —         | pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only) |                       |
| <code>-o</code>                      | OBJECT    | id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)                          |                       |
| <code>--object</code>                | OBJECT    | id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)                          |                       |
| <code>-r</code>                      | REFERENCE | id of indirect object being referenced (version independent)                                                        |                       |
| <code>--reference</code>             | REFERENCE | id of indirect object being referenced (version independent)                                                        |                       |
| <code>-e</code>                      | ELEMENTS  | type of elements to select (ctxsi)                                                                                  |                       |
| <code>--elements</code>              | ELEMENTS  | type of elements to select (ctxsi)                                                                                  |                       |
| <code>-w</code>                      | —         | raw output for data and filters                                                                                     |                       |
| <code>--raw</code>                   | —         | raw output for data and filters                                                                                     |                       |
| <code>-a</code>                      | —         | display stats for pdf document                                                                                      |                       |
| <code>--stats</code>                 | —         | display stats for pdf document                                                                                      |                       |
| <code>-t</code>                      | TYPE      | type of indirect object to select                                                                                   |                       |
| <code>--type</code>                  | TYPE      | type of indirect object to select                                                                                   |                       |
| <code>-0</code>                      | —         | parse stream of /ObjStm objects                                                                                     |                       |
| <code>--objstm</code>                | —         | parse stream of /ObjStm objects                                                                                     |                       |
| <code>-v</code>                      | —         | display malformed PDF elements                                                                                      |                       |
| <code>--verbose</code>               | —         | display malformed PDF elements                                                                                      |                       |
| <code>-x</code>                      | EXTRACT   | filename to extract malformed content to                                                                            |                       |
| <code>--extract</code>               | EXTRACT   | filename to extract malformed content to                                                                            |                       |
| <code>-H</code>                      | —         | display hash of objects                                                                                             |                       |
| <code>--hash</code>                  | —         | display hash of objects                                                                                             |                       |
| <code>-n</code>                      | —         | do not canonicalize the output                                                                                      |                       |
| <code>--nocanonicalizedoutput</code> | —         | do not canonicalize the output                                                                                      |                       |
| <code>-d</code>                      | DUMP      | filename to dump stream content to                                                                                  |                       |
| <code>--dump</code>                  | DUMP      | filename to dump stream content to                                                                                  |                       |
| <code>-D</code>                      | —         | display debug info                                                                                                  |                       |
| <code>--debug</code>                 | —         | display debug info                                                                                                  |                       |

| Flag                             | Argument          | What it does                                                                              | When you would use it |
|----------------------------------|-------------------|-------------------------------------------------------------------------------------------|-----------------------|
| <code>-c</code>                  | —                 | display the content for objects without streams or with streams without filters           |                       |
| <code>--content</code>           | —                 | display the content for objects without streams or with streams without filters           |                       |
| <code>--searchstream</code>      | SEARCHSTREAM      | string to search in streams                                                               |                       |
| <code>--unfiltered</code>        | —                 | search in unfiltered streams                                                              |                       |
| <code>--casesensitive</code>     | —                 | case sensitive search in streams                                                          |                       |
| <code>--regex</code>             | —                 | use regex to search in streams                                                            |                       |
| <code>--overridingfilters</code> | OVERRIDINGFILTERS | override filters with given filters (use raw for the raw stream content)                  |                       |
| <code>-g</code>                  | —                 | generate a Python program that creates the parsed PDF file                                |                       |
| <code>--generate</code>          | —                 | generate a Python program that creates the parsed PDF file                                |                       |
| <code>--generateembedded</code>  | GENERATEEMBEDDED  | generate a Python program that embeds the selected indirect object as a file              |                       |
| <code>-y</code>                  | YARA              | YARA rule (or directory or @file) to check streams (can be used with option --unfiltered) |                       |
| <code>--yara</code>              | YARA              | YARA rule (or directory or @file) to check streams (can be used with option --unfiltered) |                       |
| <code>--yarastrings</code>       | —                 | Print YARA strings                                                                        |                       |
| <code>--decoders</code>          | DECODERS          | decoders to load (separate decoders with a comma , ; @file supported)                     |                       |
| <code>--decoderoptions</code>    | DECODEROPTIONS    | options for the decoder                                                                   |                       |
| <code>-k</code>                  | KEY               | key to search in dictionaries                                                             |                       |
| <code>--key</code>               | KEY               | key to search in dictionaries                                                             |                       |
| <code>-j</code>                  | —                 | produce json output                                                                       |                       |
| <code>--jsonoutput</code>        | —                 | produce json output                                                                       |                       |

## Gotchas

*TODO: operational traps.*

## See also

`pdfid` , `pdfid.py` , `pdf-parser`

# pdfid

**Kit:** Kali Linux **Capability:** Analyse PDFs **Version:** pdfid.py 0.2.10 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Triage a PDF by counting the structural keywords that indicate active content.

## Synopsis

```
pdfid.py [options] [pdf-file|zip-file|url|@file] ...
```

## Common invocations

```
# from cyberlab 37-pdf-analysis
pdfid --version
# from cyberlab 37-pdf-analysis
pdfid exercise/sample.pdf
```

## Options

All 31 options parsed from the captured help text; 7 reviewed with usage guidance.

| Flag      | Argument | What it does                                                | When you would use it                               |
|-----------|----------|-------------------------------------------------------------|-----------------------------------------------------|
| --version | —        | show program's version number and exit                      |                                                     |
| -h        | —        | show this help message and exit                             |                                                     |
| --help    | —        | show this help message and exit                             |                                                     |
| -s        | —        | scan the given directory                                    | Scan a directory of PDFs rather than a single file. |
| --scan    | —        | scan the given directory                                    |                                                     |
| -a        | —        | display all the names                                       | Display all keyword counts, including zero entries. |
| --all     | —        | display all the names                                       |                                                     |
| -e        | —        | display extra data, like dates                              | Include extra keywords beyond the default set.      |
| --extra   | —        | display extra data, like dates                              |                                                     |
| -f        | —        | force the scan of the file, even without proper %PDF header |                                                     |

| Flag                            | Argument      | What it does                                                        | When you would use it                                                                                        |
|---------------------------------|---------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>--force</code>            | —             | force the scan of the file, even without proper %PDF header         |                                                                                                              |
| <code>-d</code>                 | —             | disable JavaScript and auto launch                                  | Disarm the PDF — neutralise /JS and /JavaScript. Produces a safe-to-open copy; never overwrite the original. |
| <code>--disarm</code>           | —             | disable JavaScript and auto launch                                  |                                                                                                              |
| <code>-p</code>                 | PLUGINS       | plugins to load (separate plugins with a comma , ; @file supported) |                                                                                                              |
| <code>--plugins</code>          | PLUGINS       | plugins to load (separate plugins with a comma , ; @file supported) |                                                                                                              |
| <code>-c</code>                 | —             | output csv data when using plugins                                  | CSV output for batch triage.                                                                                 |
| <code>--csv</code>              | —             | output csv data when using plugins                                  |                                                                                                              |
| <code>-m</code>                 | MINIMUMSCORE  | minimum score for plugin results output                             | Only report files scoring above a minimum, for a large sweep.                                                |
| <code>--minimumscore</code>     | MINIMUMSCORE  | minimum score for plugin results output                             |                                                                                                              |
| <code>-v</code>                 | —             | verbose (will also raise caught exceptions)                         |                                                                                                              |
| <code>--verbose</code>          | —             | verbose (will also raise caught exceptions)                         |                                                                                                              |
| <code>-S</code>                 | SELECT        | selection expression                                                |                                                                                                              |
| <code>--select</code>           | SELECT        | selection expression                                                |                                                                                                              |
| <code>-n</code>                 | —             | supress output for counts equal to zero                             | Hide zero-count keywords to shorten the output.                                                              |
| <code>--nozero</code>           | —             | supress output for counts equal to zero                             |                                                                                                              |
| <code>-o</code>                 | OUTPUT        | output to log file                                                  |                                                                                                              |
| <code>--output</code>           | OUTPUT        | output to log file                                                  |                                                                                                              |
| <code>--pluginoptions</code>    | PLUGINOPTIONS | options for the plugin                                              |                                                                                                              |
| <code>-l</code>                 | —             | take filenames literally, no wildcard matching                      |                                                                                                              |
| <code>--literalfilenames</code> | —             | take filenames literally, no wildcard matching                      |                                                                                                              |
| <code>--recursemdir</code>      | —             | Recurse directories (wildcards and here files (@...) allowed)       |                                                                                                              |

## Gotchas

---

- `pdfid` **counts keywords; it does not parse**. A high `/JS` count is a reason to look, not proof of malice — and a crafted PDF can hide content from it. Follow up with `pdf-parser`.
- `/OpenAction` plus `/JS` is the classic auto-execute pair worth prioritising in triage.

## See also

---

`pdfid.py` , `pdf-parser` , `pdf-parser.py`



# pdfid.py

**Kit:** REMnux · Kali Linux **Capability:** Analyse PDFs **Version:** pdfid.py 0.2.10 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://blog.didierstevens.com/programs/pdf-tools/>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Identify suspicious elements of the PDF file.

## Synopsis

```
pdfid.py [options] [pdf-file|zip-file|url|@file] ...
```

## Common invocations

```
# from cyberlab 10-malicious-documents
pdfid.py --version
# from cyberlab 10-malicious-documents
pdfid.py exercise/sample.pdf
```

## Options

All 31 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                                                | When you would use it |
|-----------|----------|-------------------------------------------------------------|-----------------------|
| --version | —        | show program's version number and exit                      |                       |
| -h        | —        | show this help message and exit                             |                       |
| --help    | —        | show this help message and exit                             |                       |
| -s        | —        | scan the given directory                                    |                       |
| --scan    | —        | scan the given directory                                    |                       |
| -a        | —        | display all the names                                       |                       |
| --all     | —        | display all the names                                       |                       |
| -e        | —        | display extra data, like dates                              |                       |
| --extra   | —        | display extra data, like dates                              |                       |
| -f        | —        | force the scan of the file, even without proper %PDF header |                       |

| Flag                            | Argument      | What it does                                                        | When you would use it |
|---------------------------------|---------------|---------------------------------------------------------------------|-----------------------|
| <code>--force</code>            | —             | force the scan of the file, even without proper %PDF header         |                       |
| <code>-d</code>                 | —             | disable JavaScript and auto launch                                  |                       |
| <code>--disarm</code>           | —             | disable JavaScript and auto launch                                  |                       |
| <code>-p</code>                 | PLUGINS       | plugins to load (separate plugins with a comma , ; @file supported) |                       |
| <code>--plugins</code>          | PLUGINS       | plugins to load (separate plugins with a comma , ; @file supported) |                       |
| <code>-c</code>                 | —             | output csv data when using plugins                                  |                       |
| <code>--csv</code>              | —             | output csv data when using plugins                                  |                       |
| <code>-m</code>                 | MINIMUMSCORE  | minimum score for plugin results output                             |                       |
| <code>--minimumscore</code>     | MINIMUMSCORE  | minimum score for plugin results output                             |                       |
| <code>-v</code>                 | —             | verbose (will also raise caught exceptions)                         |                       |
| <code>--verbose</code>          | —             | verbose (will also raise caught exceptions)                         |                       |
| <code>-S</code>                 | SELECT        | selection expression                                                |                       |
| <code>--select</code>           | SELECT        | selection expression                                                |                       |
| <code>-n</code>                 | —             | supress output for counts equal to zero                             |                       |
| <code>--nozero</code>           | —             | supress output for counts equal to zero                             |                       |
| <code>-o</code>                 | OUTPUT        | output to log file                                                  |                       |
| <code>--output</code>           | OUTPUT        | output to log file                                                  |                       |
| <code>--pluginoptions</code>    | PLUGINOPTIONS | options for the plugin                                              |                       |
| <code>-l</code>                 | —             | take filenames literally, no wildcard matching                      |                       |
| <code>--literalfilenames</code> | —             | take filenames literally, no wildcard matching                      |                       |
| <code>--recursedir</code>       | —             | Recurse directories (wildcards and here files (@...) allowed)       |                       |

## Gotchas

*TODO: operational traps.*

## See also

`pdfid`, `pdf-parser`, `pdf-parser.py`

# pyxswf

**Kit:** REMnux **Capability:** Analyse Office documents and macros **Version:** pyxswf 0.54 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
pyxswf.py
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 20 options parsed from the captured help text. The final column is filled in by review.

| Flag                   | Argument | What it does                                                                                              | When you would use it |
|------------------------|----------|-----------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-h</code>        | —        | show this help message and exit                                                                           |                       |
| <code>--help</code>    | —        | show this help message and exit                                                                           |                       |
| <code>-x</code>        | —        | Extracts the embedded SWF(s), names it MD5HASH.swf & saves it in the working dir. No addition args needed |                       |
| <code>--extract</code> | —        | Extracts the embedded SWF(s), names it MD5HASH.swf & saves it in the working dir. No addition args needed |                       |
| <code>-y</code>        | —        | Scans the SWF(s) with yara. If the SWF(s) is compressed it will be deflated. No addition args needed      |                       |
| <code>--yara</code>    | —        | Scans the SWF(s) with yara. If the SWF(s) is compressed it will be deflated. No addition args needed      |                       |
| <code>-s</code>        | —        | Scans the SWF(s) for MD5 signatures. Please see func checkMD5 to define hashes. No addition args needed   |                       |
| <code>--md5scan</code> | —        | Scans the SWF(s) for MD5 signatures. Please see func checkMD5 to define hashes. No addition args needed   |                       |
| <code>-H</code>        | —        | Displays the SWFs file header. No addition args needed                                                    |                       |

| Flag                      | Argument | What it does                                                                               | When you would use it |
|---------------------------|----------|--------------------------------------------------------------------------------------------|-----------------------|
| <code>--header</code>     | —        | Displays the SWFs file header. No addition args needed                                     |                       |
| <code>-d</code>           | —        | Deflates compressed SWFS(s)                                                                |                       |
| <code>--decompress</code> | —        | Deflates compressed SWFS(s)                                                                |                       |
| <code>-r</code>           | PATH     | Will recursively scan a directory for files that contain SWFs. Must provide path in quotes |                       |
| <code>--recdir</code>     | PATH     | Will recursively scan a directory for files that contain SWFs. Must provide path in quotes |                       |
| <code>-c</code>           | —        | Compresses the SWF using Zlib                                                              |                       |
| <code>--compress</code>   | —        | Compresses the SWF using Zlib                                                              |                       |
| <code>-o</code>           | —        | Parse an OLE file (e.g. Word, Excel) to look for SWF in each stream                        |                       |
| <code>--ole</code>        | —        | Parse an OLE file (e.g. Word, Excel) to look for SWF in each stream                        |                       |
| <code>-f</code>           | —        | Parse an RTF file to look for SWF in each embedded object                                  |                       |
| <code>--rtf</code>        | —        | Parse an RTF file to look for SWF in each embedded object                                  |                       |

## Gotchas

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*TODO: operational traps.*

## See also

---

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

# rtfobj

**Kit:** REMnux **Capability:** Analyse RTF documents and embedded objects **Version:** rtfobj 0.60.1 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Microsoft Office OLE2 compound documents.

## Synopsis

```
rtfobj [options] <filename> [filename2 ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 12 options parsed from the captured help text. The final column is filled in by review.

| Flag                    | Argument     | What it does                                                                                                    | When you would use it |
|-------------------------|--------------|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-h</code>         | —            | show this help message and exit                                                                                 |                       |
| <code>--help</code>     | —            | show this help message and exit                                                                                 |                       |
| <code>-r</code>         | —            | find files recursively in subdirectories.                                                                       |                       |
| <code>-z</code>         | ZIP_PASSWORD | if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)       |                       |
| <code>--zip</code>      | ZIP_PASSWORD | if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)       |                       |
| <code>-f</code>         | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |
| <code>--zipfname</code> | ZIP_FNAME    | if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*) |                       |
| <code>-l</code>         | LOGLEVEL     | logging level debug/info/warning/error/critical (default=warning)                                               |                       |
| <code>--loglevel</code> | LOGLEVEL     | logging level debug/info/warning/error/critical (default=warning)                                               |                       |

| Flag                | Argument    | What it does                                                                                                                  | When you would use it |
|---------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-s</code>     | SAVE_OBJECT | Save the object corresponding to the provided number to a file, for example "-s 2". Use "-s all" to save all objects at once. |                       |
| <code>--save</code> | SAVE_OBJECT | Save the object corresponding to the provided number to a file, for example "-s 2". Use "-s all" to save all objects at once. |                       |
| <code>-d</code>     | OUTPUT_DIR  | use specified directory to save output files.                                                                                 |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`oleobj`

# xlmdeobfuscator

**Kit:** REMnux **Capability:** Analyse Office documents and macros; Break simple obfuscation **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://github.com/DissectMalware/XLMMacroDeobfuscator>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Deobfuscate XLM macros (also known as Excel 4.0 macros) from Microsoft Office files.

## Synopsis

```
xlmdeobfuscator [-h] [-c FILE_PATH] [-f FILE_PATH] [-n] [-x]
[--sort-formulas] [--defined-names] [-2]
[--with-ms-excel] [-s] [-d DAY]
[--output-formula-format OUTPUT_FORMULA_FORMAT]
[--extract-formula-format EXTRACT_FORMULA_FORMAT]
[--no-indent] [--silent] [--export-json FILE_PATH]
```

## Common invocations

```
# from cyberlab 10-malicious-documents
xlmdeobfuscator --help | head -n 1
# from cyberlab 10-malicious-documents
xlmdeobfuscator --file exercise/sample.xls
```

## Options

All 30 options parsed from the captured help text. The final column is filled in by review.

| Flag                       | Argument  | What it does                                      | When you would use it |
|----------------------------|-----------|---------------------------------------------------|-----------------------|
| <code>-h</code>            | —         | show this help message and exit                   |                       |
| <code>--help</code>        | —         | show this help message and exit                   |                       |
| <code>-c</code>            | FILE_PATH | Specify a config file (must be a valid JSON file) |                       |
| <code>--config-file</code> | FILE_PATH | Specify a config file (must be a valid JSON file) |                       |
| <code>-f</code>            | FILE_PATH | The path of a XLSM file                           |                       |
| <code>--file</code>        | FILE_PATH | The path of a XLSM file                           |                       |

| Flag                                  | Argument               | What it does                                                                                                                     | When you would use it |
|---------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-n</code>                       | —                      | Disable interactive shell                                                                                                        |                       |
| <code>--noninteractive</code>         | —                      | Disable interactive shell                                                                                                        |                       |
| <code>-x</code>                       | —                      | Only extract cells without any emulation                                                                                         |                       |
| <code>--extract-only</code>           | —                      | Only extract cells without any emulation                                                                                         |                       |
| <code>--sort-formulas</code>          | —                      | Sort extracted formulas based on their cell address (requires -x)                                                                |                       |
| <code>--defined-names</code>          | —                      | Extract all defined names                                                                                                        |                       |
| <code>-2</code>                       | —                      | [Deprecated] Do not use MS Excel to process XLS files                                                                            |                       |
| <code>--no-ms-excel</code>            | —                      | [Deprecated] Do not use MS Excel to process XLS files                                                                            |                       |
| <code>--with-ms-excel</code>          | —                      | Use MS Excel to process XLS files                                                                                                |                       |
| <code>-s</code>                       | —                      | Open an XLM shell before interpreting the macros in the input                                                                    |                       |
| <code>--start-with-shell</code>       | —                      | Open an XLM shell before interpreting the macros in the input                                                                    |                       |
| <code>-d</code>                       | DAY                    | Specify the day of month                                                                                                         |                       |
| <code>--day</code>                    | DAY                    | Specify the day of month                                                                                                         |                       |
| <code>--output-formula-format</code>  | OUTPUT_FORMULA_FORMAT  | Specify the format for output formulas ([[CELL-ADDR]], [[INT-FORMULA]], and [[STATUS]])                                          |                       |
| <code>--extract-formula-format</code> | EXTRACT_FORMULA_FORMAT | Specify the format for extracted formulas ([[CELL-ADDR]], [[CELL-FORMULA]], and [[CELL-VALUE]])                                  |                       |
| <code>--no-indent</code>              | —                      | Do not show indent before formulas                                                                                               |                       |
| <code>--silent</code>                 | —                      | Do not print output                                                                                                              |                       |
| <code>--export-json</code>            | FILE_PATH              | Export the output to JSON                                                                                                        |                       |
| <code>--start-point</code>            | CELL_ADDR              | Start interpretation from a specific cell address                                                                                |                       |
| <code>-p</code>                       | PASSWORD               | Password to decrypt the protected document                                                                                       |                       |
| <code>--password</code>               | PASSWORD               | Password to decrypt the protected document                                                                                       |                       |
| <code>-o</code>                       | OUTPUT_LEVEL           | Set the level of details to be shown (0:all commands, 1: commands no jump 2:important commands 3:strings in important commands). |                       |
| <code>--output-level</code>           | OUTPUT_LEVEL           | Set the level of details to be shown (0:all commands, 1: commands no jump 2:important commands 3:strings in important commands). |                       |
| <code>--timeout</code>                | N                      | stop emulation after N seconds (0: not interruption N>0: stop emulation after N seconds)                                         |                       |

## Gotchas

*TODO: operational traps.*



## See also

---

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

# 7za

**Kit:** REMnux **Capability:** Detect and reverse packing; Analyse other container formats **Captured:** cyberlab-aio  
via `--help` on 2026-08-01 [raw](#) **Docs:** <https://www.7-zip.org>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Compress and decompress files using a variety of algorithms.

## Synopsis

```
7za <command> [<switches>...] <archive_name> [<file_names>...] [@listfile]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 1 options parsed from the captured help text. The final column is filled in by review.

| Flag            | Argument | What it does              | When you would use it |
|-----------------|----------|---------------------------|-----------------------|
| <code>-y</code> | —        | assume Yes on all queries |                       |

## Gotchas

*TODO: operational traps.*

## See also

`upx`, `die`, `diec`, `binwalk`, `unzip`, `msoffcrypto-tool`

# base64dump.py

**Kit:** REMnux **Capability:** Extract strings, including obfuscated ones; Decode, decrypt or transform encoded data

**Version:** base64dump.py 0.0.30 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw Docs:](#)

<https://blog.didierstevens.com/2020/07/03/update-base64dump-py-version-0-0-12/>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Locate and decode strings encoded in Base64 and other common encodings.

## Synopsis

```
base64dump.py [options] [file]
```

## Common invocations

```
# from cyberlab 09-deobfuscation
base64dump.py --version 2>&1 | head -n 1
# from cyberlab 09-deobfuscation
base64dump.py exercise/encoded_payload.bin
# from cyberlab 09-deobfuscation
base64dump.py -s 1 -d exercise/encoded_payload.bin | head -c 200
# from cyberlab 25-cyberchef-recipes
base64dump.py --version
# from cyberlab 25-cyberchef-recipes
base64dump.py encoded_payload.ps1.txt
# from cyberlab 25-cyberchef-recipes
base64dump.py -s 1 -d encoded_payload.ps1.txt | xxd | head -n 2
```

## Options

All 54 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                           | When you would use it |
|-----------|----------|----------------------------------------|-----------------------|
| --version | —        | show program's version number and exit |                       |
| -h        | —        | show this help message and exit        |                       |
| --help    | —        | show this help message and exit        |                       |

| Flag                            | Argument  | What it does                                                            | When you would use it |
|---------------------------------|-----------|-------------------------------------------------------------------------|-----------------------|
| <code>-m</code>                 | —         | Print manual                                                            |                       |
| <code>--man</code>              | —         | Print manual                                                            |                       |
| <code>-e</code>                 | ENCODING  | select encoding to use, use "all" to try all encodings (default base64) |                       |
| <code>--encoding</code>         | ENCODING  | select encoding to use, use "all" to try all encodings (default base64) |                       |
| <code>-s</code>                 | SELECT    | select item nr for dumping (a for all, L for largest)                   |                       |
| <code>--select</code>           | SELECT    | select item nr for dumping (a for all, L for largest)                   |                       |
| <code>-d</code>                 | —         | perform dump                                                            |                       |
| <code>--dump</code>             | —         | perform dump                                                            |                       |
| <code>-x</code>                 | —         | perform hex dump                                                        |                       |
| <code>--hexdump</code>          | —         | perform hex dump                                                        |                       |
| <code>-a</code>                 | —         | perform ascii dump                                                      |                       |
| <code>--asciidump</code>        | —         | perform ascii dump                                                      |                       |
| <code>-A</code>                 | —         | perform ascii dump with RLE                                             |                       |
| <code>--asciidumprle</code>     | —         | perform ascii dump with RLE                                             |                       |
| <code>-S</code>                 | —         | perform strings dump                                                    |                       |
| <code>--strings</code>          | —         | perform strings dump                                                    |                       |
| <code>-t</code>                 | TRANSLATE | string translation, like utf16 or .decode("utf8")                       |                       |
| <code>--translate</code>        | TRANSLATE | string translation, like utf16 or .decode("utf8")                       |                       |
| <code>-n</code>                 | NUMBER    | minimum number of bytes in decoded data                                 |                       |
| <code>--number</code>           | NUMBER    | minimum number of bytes in decoded data                                 |                       |
| <code>-c</code>                 | CUT       | cut data                                                                |                       |
| <code>--cut</code>              | CUT       | cut data                                                                |                       |
| <code>-w</code>                 | —         | ignore whitespace                                                       |                       |
| <code>--ignorewhitespace</code> | —         | ignore whitespace                                                       |                       |
| <code>-u</code>                 | —         | do not repeat identical decoded data                                    |                       |
| <code>--unique</code>           | —         | do not repeat identical decoded data                                    |                       |
| <code>-z</code>                 | —         | ignore null (zero) bytes                                                |                       |
| <code>--ignorenullbytes</code>  | —         | ignore null (zero) bytes                                                |                       |
| <code>-i</code>                 | IGNORE    | characters to ignore                                                    |                       |
| <code>--ignore</code>           | IGNORE    | characters to ignore                                                    |                       |
| <code>-I</code>                 | IGNOREHEX | characters to ignore (hexadecimal)                                      |                       |

| Flag                          | Argument       | What it does                                                                                      | When you would use it |
|-------------------------------|----------------|---------------------------------------------------------------------------------------------------|-----------------------|
| <code>--ignorehex</code>      | IGNOREHEX      | characters to ignore (hexadecimal)                                                                |                       |
| <code>-y</code>               | YARA           | YARA rule-file, @file, directory or #rule to check data (YARA search doesn't work with -s option) |                       |
| <code>--yara</code>           | YARA           | YARA rule-file, @file, directory or #rule to check data (YARA search doesn't work with -s option) |                       |
| <code>-D</code>               | DECODERS       | decoders to load (separate decoders with a comma , ; @file supported)                             |                       |
| <code>--decoders</code>       | DECODERS       | decoders to load (separate decoders with a comma , ; @file supported)                             |                       |
| <code>--decoderoptions</code> | DECODEROPTIONS | options for the decoder                                                                           |                       |
| <code>--decoderdir</code>     | DECODERDIR     | directory for the decoder                                                                         |                       |
| <code>--yarastrings</code>    | —              | Print YARA strings                                                                                |                       |
| <code>-V</code>               | —              | verbose output with decoder errors and YARA rules                                                 |                       |
| <code>--verbose</code>        | —              | verbose output with decoder errors and YARA rules                                                 |                       |
| <code>--jsonoutput</code>     | —              | produce json output                                                                               |                       |
| <code>--jsoninput</code>      | —              | consume json input                                                                                |                       |
| <code>-T</code>               | —              | do head & tail                                                                                    |                       |
| <code>--headtail</code>       | —              | do head & tail                                                                                    |                       |
| <code>-p</code>               | PREPROCESS     | Python function to process encodings prior to decoding (like L4, lambda bytes: bytes[:-1], ...)   |                       |
| <code>--preprocess</code>     | PREPROCESS     | Python function to process encodings prior to decoding (like L4, lambda bytes: bytes[:-1], ...)   |                       |
| <code>-P</code>               | POSTPROCESS    | Python function to post-process decodings                                                         |                       |
| <code>--postprocess</code>    | POSTPROCESS    | Python function to post-process decodings                                                         |                       |
| <code>--sort</code>           | SORT           | Sort output                                                                                       |                       |
| <code>--stats</code>          | —              | provide statistics for encoded data                                                               |                       |

## Gotchas

*TODO: operational traps.*

## See also

`floss`, `strings`, `numbers-to-string.py`, `cyberchef`, `rax2`, `xxd`, `openssl`

# capa

**Kit:** REMnux · FLARE-VM **Capability:** Identify capabilities in a binary **Version:** capa 9.4.0 **Captured:** cyberlab-aio  
via `--help` on 2026-08-01 **raw Docs:** <https://github.com/mandiant/capa>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Identify the capabilities of a binary by matching rules against its disassembly, mapped to MITRE ATT&CK.

## Synopsis

```
capa [-h] [--version] [-v] [-vv] [-d] [-q]
[--color {auto,always,never}]
[-f {auto,pe,dotnet,elf,sc32,sc64,cape,drakvuf,vmray,freeze,binexport2,binja_database}]
[-b {auto,vivisect,ida,pefile,binja,dotnet,binexport2,ghidra,freeze,cape,drakvuf,vmray}]
[--restrict-to-functions RESTRICT_TO_FUNCTIONS]
[--restrict-to-processes RESTRICT_TO_PROCESSES]
```

## Common invocations

```
# from cyberlab 08-malware-static-triage
capa --version
# from cyberlab 08-malware-static-triage
capa exercise/sample.bin | head -n 40
# from cyberlab 21-yara-authoring
capa -q /bin/ls | head -n 30
# from cyberlab 12-static-re
capa exercise\sample_static.exe
# from cyberlab 12-static-re
capa -v exercise\sample_static.exe
# from cyberlab 42-floss-strings
capa .\exercise\sample.exe
# from cyberlab 42-floss-strings
capa -v .\exercise\sample.exe | Select-String -Pattern "XOR|encode|T1027"
# from cyberlab 46-cutter-windows
capa -v .\exercise\sample.exe
```

## Options

All 21 options parsed from the captured help text; 8 reviewed with usage guidance.

| Flag                                 | Argument                 | What it does                                                                                               | When you would use it                                                                 |
|--------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <code>-h</code>                      | —                        | show this help message and exit                                                                            |                                                                                       |
| <code>--help</code>                  | —                        | show this help message and exit                                                                            |                                                                                       |
| <code>--version</code>               | —                        | show program's version number and exit                                                                     |                                                                                       |
| <code>-v</code>                      | —                        | enable verbose result document (no effect with <code>--json</code> )                                       | Verbose: show which rules matched and where.                                          |
| <code>--verbose</code>               | —                        | enable verbose result document (no effect with <code>--json</code> )                                       |                                                                                       |
| <code>-d</code>                      | —                        | enable debugging output on STDERR                                                                          |                                                                                       |
| <code>--debug</code>                 | —                        | enable debugging output on STDERR                                                                          |                                                                                       |
| <code>-q</code>                      | —                        | disable all output but errors                                                                              | Quiet output for scripted use.                                                        |
| <code>--quiet</code>                 | —                        | disable all output but errors                                                                              |                                                                                       |
| <code>--color</code>                 | auto,always,never        | enable ANSI color codes in results, default: only during interactive session                               |                                                                                       |
| <code>--restrict-to-functions</code> | RESTRICT_TO_FUNCTIONS    | provide a list of comma-separated function virtual addresses to analyze (static analysis).                 | Analyse only the named functions — useful on a large binary you have already triaged. |
| <code>--restrict-to-processes</code> | RESTRICT_TO_PROCESSES    | provide a list of comma-separated process IDs to analyze (dynamic analysis).                               |                                                                                       |
| <code>--os</code>                    | auto,linux,macos,windows | select sample OS: auto (detect OS automatically - default), linux, macos, windows                          | Force the target OS when auto-detection is wrong.                                     |
| <code>-r</code>                      | RULES                    | path to rule file or directory, use embedded rules by default                                              | Use an alternate rules directory.                                                     |
| <code>--rules</code>                 | RULES                    | path to rule file or directory, use embedded rules by default                                              |                                                                                       |
| <code>-s</code>                      | SIGNATURES               | path to .sig/.pat file or directory used to identify library functions, use embedded signatures by default | Point at the library-identification signatures.                                       |
| <code>--signatures</code>            | SIGNATURES               | path to .sig/.pat file or directory used to identify library functions, use embedded signatures by default |                                                                                       |
| <code>-t</code>                      | TAG                      | filter on rule meta field values                                                                           | Only run rules carrying a given tag.                                                  |
| <code>--tag</code>                   | TAG                      | filter on rule meta field values                                                                           |                                                                                       |
| <code>-j</code>                      | —                        | emit JSON instead of text                                                                                  | JSON output, for feeding a triage pipeline.                                           |
| <code>--json</code>                  | —                        | emit JSON instead of text                                                                                  |                                                                                       |

- capa reports *capabilities*, not verdicts. 'Can encrypt data' describes a backup tool as readily as ransomware — it narrows where to look.
- Packed samples yield almost nothing useful. Unpack first, or capa will just describe the packer stub.

## See also

---

floss



# clamscan

**Kit:** REMnux · SIFT Workstation **Capability:** Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#) **Docs:** <https://www.clamav.net>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Scan files and directories with the ClamAV signature engine.

## Common invocations

```
# from cyberlab 34-clamav-scanning
clamscan --version
# from cyberlab 34-clamav-scanning
clamscan -r -i /tmp/samples
# from cyberlab 34-clamav-scanning
clamscan /tmp/eicar.com
# from cyberlab 34-clamav-scanning
clamscan exercise/sample.txt
# from cyberlab 34-clamav-scanning
clamscan --database=exercise/lab34.yar exercise/sample.txt
# from cyberlab 51-linux-triage-workflow
clamscan -r --infected --stdout be_out recovered_file.bin
# from cyberlab 51-linux-triage-workflow
clamscan --infected --stdout exercise/extract/eicar.com
```

## Options

All 53 options parsed from the captured help text; 11 reviewed with usage guidance.

| Flag                           | Argument | What it does                           | When you would use it |
|--------------------------------|----------|----------------------------------------|-----------------------|
| <code>--help</code>            | —        | Show this help                         |                       |
| <code>-h</code>                | —        | Show this help                         |                       |
| <code>--version</code>         | —        | Print version number                   |                       |
| <code>-V</code>                | —        | Print version number                   |                       |
| <code>--verbose</code>         | —        | Be verbose                             |                       |
| <code>-v</code>                | —        | Be verbose                             |                       |
| <code>--archive-verbose</code> | —        | Show filenames inside scanned archives |                       |

| Flag                                  | Argument  | What it does                                                                      | When you would use it                                                                        |
|---------------------------------------|-----------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <code>-a</code>                       | —         | Show filenames inside scanned archives                                            |                                                                                              |
| <code>--debug</code>                  | —         | Enable libclamav's debug messages                                                 |                                                                                              |
| <code>--quiet</code>                  | —         | Only output error messages                                                        | Only report errors, for scripted runs.                                                       |
| <code>--stdout</code>                 | —         | Write to stdout instead of stderr. Does not affect 'debug' messages.              |                                                                                              |
| <code>--no-summary</code>             | —         | Disable summary at end of scanning                                                | Suppress the trailing summary block when parsing output.                                     |
| <code>--infected</code>               | —         | Only print infected files                                                         |                                                                                              |
| <code>-i</code>                       | —         | Only print infected files                                                         | Print infected files only — the flag that makes output usable.                               |
| <code>--bell</code>                   | —         | Sound bell on virus detection                                                     | Audible alert on detection; useful during a long live scan.                                  |
| <code>--tempdir</code>                | DIRECTORY | Create temporary files in DIRECTORY                                               | Redirect temp extraction; important when scanning large archives on a small root filesystem. |
| <code>--database</code>               | FILE      | -d FILE/DIR Load virus database from FILE or load all supported db files from DIR |                                                                                              |
| <code>--fail-if-cvd-older-than</code> | days      | Return with a nonzero error code if virus database outdated.                      |                                                                                              |
| <code>--log</code>                    | FILE      | -l FILE Save scan report to FILE                                                  | Write results to a log file for the case record.                                             |
| <code>--file-list</code>              | FILE      | -f FILE Scan files from FILE                                                      | Scan exactly the paths listed in a file.                                                     |
| <code>--move</code>                   | DIRECTORY | Move infected files into DIRECTORY                                                | Quarantine detections into a directory. Never point this at evidence you must preserve.      |
| <code>--copy</code>                   | DIRECTORY | Copy infected files into DIRECTORY                                                | Copy rather than move detections, preserving the original.                                   |
| <code>--exclude</code>                | REGEX     | Don't scan file names matching REGEX                                              | Skip paths matching a regex — noisy or irrelevant trees.                                     |
| <code>--exclude-dir</code>            | REGEX     | Don't scan directories matching REGEX                                             |                                                                                              |
| <code>--include</code>                | REGEX     | Only scan file names matching REGEX                                               | Restrict the scan to paths matching a regex.                                                 |
| <code>--include-dir</code>            | REGEX     | Only scan directories matching REGEX                                              |                                                                                              |
| <code>--bytecode-timeout</code>       | N         | Set bytecode timeout (in milliseconds)                                            |                                                                                              |
| <code>--exclude-pua</code>            | CAT       | Skip PUA sigs of category CAT                                                     |                                                                                              |
| <code>--include-pua</code>            | CAT       | Load PUA sigs of category CAT                                                     |                                                                                              |
| <code>--structured-ssn-format</code>  | X         | SSN format (0=normal,1=stripped,2=both)                                           |                                                                                              |

| Flag                                | Argument | What it does                                                                | When you would use it |
|-------------------------------------|----------|-----------------------------------------------------------------------------|-----------------------|
| <code>--structured-ssn-count</code> | N        | Min SSN count to generate a detect                                          |                       |
| <code>--structured-cc-count</code>  | N        | Min CC count to generate a detect                                           |                       |
| <code>--structured-cc-mode</code>   | X        | CC mode (0=credit debit and private label, 1=credit cards only)             |                       |
| <code>--nocerts</code>              | —        | Disable authenticode certificate chain verification in PE files             |                       |
| <code>--dumpcerts</code>            | —        | Dump authenticode certificate chain in PE files                             |                       |
| <code>--max-scantime</code>         | #n       | Scan time longer than this will be skipped and assumed clean (milliseconds) |                       |
| <code>--max-filesize</code>         | #n       | Files larger than this will be skipped and assumed clean                    |                       |
| <code>--max-scansize</code>         | #n       | The maximum amount of data to scan for each container file (**)             |                       |
| <code>--max-files</code>            | #n       | The maximum number of files to scan for each container file (**)            |                       |
| <code>--max-recursion</code>        | #n       | Maximum archive recursion level for container file (**)                     |                       |
| <code>--max-dir-recursion</code>    | #n       | Maximum directory recursion level                                           |                       |
| <code>--max-embeddedpe</code>       | #n       | Maximum size file to check for embedded PE                                  |                       |
| <code>--max-htmlnormalize</code>    | #n       | Maximum size of HTML file to normalize                                      |                       |
| <code>--max-htmlnotags</code>       | #n       | Maximum size of normalized HTML file to scan                                |                       |
| <code>--max-scriptnormalize</code>  | #n       | Maximum size of script file to normalize                                    |                       |
| <code>--max-ziptypercg</code>       | #n       | Maximum size zip to type reanalyze                                          |                       |
| <code>--max-partitions</code>       | #n       | Maximum number of partitions in disk image to be scanned                    |                       |
| <code>--max-iconspe</code>          | #n       | Maximum number of icons in PE file to be scanned                            |                       |
| <code>--max-rechwp3</code>          | #n       | Maximum recursive calls to HWP3 parsing function                            |                       |
| <code>--pcre-match-limit</code>     | #n       | Maximum calls to the PCRE match function.                                   |                       |
| <code>--pcre-recmatch-limit</code>  | #n       | Maximum recursive calls to the PCRE match function.                         |                       |
| <code>--pcre-max-filesize</code>    | #n       | Maximum size file to perform PCRE subsig matching.                          |                       |

| Flag                         | Argument | What it does                                                     | When you would use it |
|------------------------------|----------|------------------------------------------------------------------|-----------------------|
| <code>--disable-cache</code> | —        | Disable caching and cache checks for hash sums of scanned files. |                       |

## Gotchas

---

- Without `-i` the output lists every clean file too, which buries the hits in a large tree.
- `--move` and `--remove` MUTATE evidence. On an investigation, scan a copy or use `--copy` — a quarantine action on original evidence is not defensible.
- Signatures must be current for the result to mean anything; a stale database yields a comfortable and worthless clean result.

## See also

---

`yara`, `yarac`, `freshclam`, `sigtool`

# die

---

**Kit:** FLARE-VM **Capability:** Identify what a file actually is; Detect and reverse packing **Version:** Detect It Easy v3.10  
**Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

qt.qpa.xcb: could not connect to display

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`file`, `diec`, `upx`, `binwalk`, `7za`, `unzip`

# diec

**Kit:** REMnux · FLARE-VM **Capability:** Identify what a file actually is; Detect and reverse packing **Version:** die 3.10  
**Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://github.com/horsicq/Detect-It-Easy>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Determine types of files and examine file properties.

## Synopsis

```
diec [options] target
```

## Common invocations

```
# from cyberlab 08-malware-static-triage
diec --version
# from cyberlab 08-malware-static-triage
diec -j exercise/sample.bin | head -n 40
# from cyberlab 32-remnux-static-triage
diec sample.exe
# from cyberlab 32-remnux-static-triage
diec exercise/sample.exe | grep -iE "compiler|packer|linker"
```

## Options

All 47 options parsed from the captured help text. The final column is filled in by review.

| Flag            | Argument | What it does                                 | When you would use it |
|-----------------|----------|----------------------------------------------|-----------------------|
| -h              | —        | Displays help on commandline options.        |                       |
| --help          | —        | Displays help on commandline options.        |                       |
| --help-all      | —        | Displays help including Qt specific options. |                       |
| -v              | —        | Displays version information.                |                       |
| --version       | —        | Displays version information.                |                       |
| -r              | —        | Recursive scan.                              |                       |
| --recursivescan | —        | Recursive scan.                              |                       |
| -d              | —        | Deep scan.                                   |                       |

| Flag                          | Argument | What it does                                                    | When you would use it |
|-------------------------------|----------|-----------------------------------------------------------------|-----------------------|
| <code>--deepscan</code>       | —        | Deep scan.                                                      |                       |
| <code>-u</code>               | —        | Heuristic scan.                                                 |                       |
| <code>--heuristicscan</code>  | —        | Heuristic scan.                                                 |                       |
| <code>-b</code>               | —        | Verbose.                                                        |                       |
| <code>--verbose</code>        | —        | Verbose.                                                        |                       |
| <code>-g</code>               | —        | Aggressive scan.                                                |                       |
| <code>--aggressivescan</code> | —        | Aggressive scan.                                                |                       |
| <code>-a</code>               | —        | Scan all types.                                                 |                       |
| <code>--alltypes</code>       | —        | Scan all types.                                                 |                       |
| <code>-l</code>               | —        | Profiling signatures.                                           |                       |
| <code>--profiling</code>      | —        | Profiling signatures.                                           |                       |
| <code>-U</code>               | —        | Hide unknown.                                                   |                       |
| <code>--hideunknown</code>    | —        | Hide unknown.                                                   |                       |
| <code>-e</code>               | —        | Show entropy.                                                   |                       |
| <code>--entropy</code>        | —        | Show entropy.                                                   |                       |
| <code>-i</code>               | —        | Show file info.                                                 |                       |
| <code>--info</code>           | —        | Show file info.                                                 |                       |
| <code>-S</code>               | method   | Special file info for . For example -S "Hash" or -S "Hash#MD5". |                       |
| <code>--special</code>        | method   | Special file info for . For example -S "Hash" or -S "Hash#MD5". |                       |
| <code>-x</code>               | —        | Result as XML.                                                  |                       |
| <code>--xml</code>            | —        | Result as XML.                                                  |                       |
| <code>-j</code>               | —        | Result as JSON.                                                 |                       |
| <code>--json</code>           | —        | Result as JSON.                                                 |                       |
| <code>-c</code>               | —        | Result as CSV.                                                  |                       |
| <code>--csv</code>            | —        | Result as CSV.                                                  |                       |
| <code>-t</code>               | —        | Result as TSV.                                                  |                       |
| <code>--tsv</code>            | —        | Result as TSV.                                                  |                       |
| <code>-p</code>               | —        | Result as Plain Text.                                           |                       |
| <code>--plaintext</code>      | —        | Result as Plain Text.                                           |                       |
| <code>-D</code>               | path     | Set database.                                                   |                       |
| <code>--database</code>       | path     | Set database.                                                   |                       |
| <code>-E</code>               | path     | Set extra database.                                             |                       |
| <code>--extradatabase</code>  | path     | Set extra database.                                             |                       |
| <code>-C</code>               | path     | Set custom database.                                            |                       |

| Flag                          | Argument | What it does                           | When you would use it |
|-------------------------------|----------|----------------------------------------|-----------------------|
| <code>--customdatabase</code> | path     | Set custom database.                   |                       |
| <code>-s</code>               | —        | Show database.                         |                       |
| <code>--showdatabase</code>   | —        | Show database.                         |                       |
| <code>-m</code>               | —        | Show all special methods for the file. |                       |
| <code>--showmethods</code>    | —        | Show all special methods for the file. |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`file` , `die` , `upx` , `binwalk` , `7za` , `unzip`



# floss

**Kit:** REMnux · FLARE-VM **Capability:** Identify capabilities in a binary; Extract strings, including obfuscated ones; Break simple obfuscation **Version:** floss 3.1.1 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw Docs: https://github.com/mandiant/flare-floss](https://github.com/mandiant/flare-floss)

← [Capability index](#) · [Kit tool list](#)

## Purpose

Extract strings from a binary, including obfuscated strings that `strings` cannot see.

## Synopsis

```
floss [-h] [-H] [-n MIN_LENGTH]
[--no {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]]
[--only {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]]
[-j] [-v] [-d] [-q] [--color {auto,always,never}]
sample
```

## Common invocations

```
# from cyberlab 08-malware-static-triage
floss --version
# from cyberlab 08-malware-static-triage
floss --no-color exercise/sample.bin | head -n 60
# from cyberlab 12-static-re
floss --no-static exercise\sample_static.exe
# from cyberlab 12-static-re
floss exercise\sample_static.exe
# from cyberlab 42-floss-strings
floss --help
# from cyberlab 42-floss-strings
floss .\exercise\sample.exe
# from cyberlab 42-floss-strings
floss --json --only decoded stack .\exercise\sample.exe > .\exercise\floss.json
# from cyberlab 42-floss-strings
floss --only static .\exercise\sample.exe | Select-String "HELLO"
```

## Options

All 14 options parsed from the captured help text; 5 reviewed with usage guidance.

| Flag                          | Argument          | What it does                                                                          | When you would use it                                               |
|-------------------------------|-------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <code>-h</code>               | —                 | show this help message and exit                                                       |                                                                     |
| <code>--help</code>           | —                 | show this help message and exit                                                       |                                                                     |
| <code>-H</code>               | —                 | show advanced options and exit                                                        |                                                                     |
| <code>-n</code>               | MIN_LENGTH        | minimum string length                                                                 | Minimum string length; raise it to cut noise on large binaries.     |
| <code>--minimum-length</code> | MIN_LENGTH        | minimum string length                                                                 |                                                                     |
| <code>-j</code>               | —                 | emit JSON instead of text                                                             | Emit JSON, for feeding results into other tooling.                  |
| <code>--json</code>           | —                 | emit JSON instead of text                                                             |                                                                     |
| <code>-v</code>               | —                 | enable verbose results, e.g. including function offsets (does not affect JSON output) | Verbose progress; FLOSS emulation can be slow and silent otherwise. |
| <code>--verbose</code>        | —                 | enable verbose results, e.g. including function offsets (does not affect JSON output) |                                                                     |
| <code>-d</code>               | —                 | enable debugging output on STDERR, specify multiple times to increase verbosity       |                                                                     |
| <code>--debug</code>          | —                 | enable debugging output on STDERR, specify multiple times to increase verbosity       |                                                                     |
| <code>-q</code>               | —                 | disable all status output on STDOUT except fatal errors                               | Quiet mode — just the strings, for piping.                          |
| <code>--quiet</code>          | —                 | disable all status output on STDOUT except fatal errors                               |                                                                     |
| <code>--color</code>          | auto,always,never | enable ANSI color codes in results, default: only during interactive session          | Control colour output when redirecting to a file.                   |

## Gotchas

- FLOSS emulates code to recover decoded strings, so it is far slower than `strings` and can take minutes on a large sample. Budget for it.
- It is built for Windows PE files. Pointing it at an ELF or a script gives you little more than plain `strings` would.

## See also

`capa`, `strings`, `base64dump.py`, `numbers-to-string.py`, `xortool`, `xlmdeobfuscator`

# freshclam

**Kit:** REMnux · SIFT Workstation **Capability:** Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.clamav.net>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Scan files for malware signatures.

## Common invocations

```
# from cyberlab 34-clamav-scanning
freshclam --version
# from cyberlab 34-clamav-scanning
sudo freshclam
```

## Options

All 27 options parsed from the captured help text. The final column is filled in by review.

| Flag                         | Argument | What it does                                                         | When you would use it |
|------------------------------|----------|----------------------------------------------------------------------|-----------------------|
| <code>--help</code>          | —        | Show this help                                                       |                       |
| <code>-h</code>              | —        | Show this help                                                       |                       |
| <code>--version</code>       | —        | Print version number and exit                                        |                       |
| <code>-V</code>              | —        | Print version number and exit                                        |                       |
| <code>--verbose</code>       | —        | Be verbose                                                           |                       |
| <code>-v</code>              | —        | Be verbose                                                           |                       |
| <code>--debug</code>         | —        | Enable debug messages                                                |                       |
| <code>--quiet</code>         | —        | Only output error messages                                           |                       |
| <code>--no-warnings</code>   | —        | Don't print and log warnings                                         |                       |
| <code>--stdout</code>        | —        | Write to stdout instead of stderr. Does not affect 'debug' messages. |                       |
| <code>--show-progress</code> | —        | Show download progress percentage                                    |                       |
| <code>--config-file</code>   | FILE     | Read configuration from FILE.                                        |                       |

| Flag                               | Argument  | What it does                                                                                                                                                                                             | When you would use it |
|------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--log</code>                 | FILE      | -l FILE Log into FILE                                                                                                                                                                                    |                       |
| <code>--daemon</code>              | —         | Run in daemon mode                                                                                                                                                                                       |                       |
| <code>-d</code>                    | —         | Run in daemon mode                                                                                                                                                                                       |                       |
| <code>--pid</code>                 | FILE      | -p FILE Write the daemon's pid to FILE                                                                                                                                                                   |                       |
| <code>--foreground</code>          | —         | Don't fork into background (for use in daemon mode).                                                                                                                                                     |                       |
| <code>-F</code>                    | —         | Don't fork into background (for use in daemon mode).                                                                                                                                                     |                       |
| <code>--user</code>                | USER      | -u USER Run as USER                                                                                                                                                                                      |                       |
| <code>--no-dns</code>              | —         | Force old non-DNS verification method                                                                                                                                                                    |                       |
| <code>--checks</code>              | #n        | -c #n Number of checks per day, 1 <= n <= 50                                                                                                                                                             |                       |
| <code>--datadir</code>             | DIRECTORY | Download new databases into DIRECTORY NOTE: DIRECTORY must already exist, be an absolute path, and be writeable by freshclam and readable by clamd/clamscan. --daemon-notify[=/path/clamd.conf] Send REL |                       |
| <code>--local-address</code>       | IP        | -a IP Bind to IP for HTTP downloads                                                                                                                                                                      |                       |
| <code>--on-update-execute</code>   | COMMAND   | Execute COMMAND after successful update. Use EXIT_1 to return 1 after successful database update.                                                                                                        |                       |
| <code>--on-error-execute</code>    | COMMAND   | Execute COMMAND if errors occurred                                                                                                                                                                       |                       |
| <code>--on-outdated-execute</code> | COMMAND   | Execute COMMAND when software is outdated                                                                                                                                                                |                       |
| <code>--update-db</code>           | DBNAME    | Only update database DBNAME                                                                                                                                                                              |                       |

## Gotchas

*TODO: operational traps.*

## See also

[yara](#), [yarak](#), [clamscan](#), [sigtool](#)

# numbers-to-string.py

**Kit:** REMnux **Capability:** Extract strings, including obfuscated ones; Decode, decrypt or transform encoded data

**Version:** numbers-to-string.py 0.0.11 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw Docs:](#)

<https://blog.didierstevens.com/2020/12/12/update-numbers-to-string-py-version-0-0-11/>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Translate number sequences into ASCII characters.

## Synopsis

```
numbers-to-string.py [options] [expression [[@]file ...]]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 31 options parsed from the captured help text. The final column is filled in by review.

| Flag                   | Argument | What it does                                          | When you would use it |
|------------------------|----------|-------------------------------------------------------|-----------------------|
| <code>--version</code> | —        | show program's version number and exit                |                       |
| <code>-h</code>        | —        | show this help message and exit                       |                       |
| <code>--help</code>    | —        | show this help message and exit                       |                       |
| <code>-m</code>        | —        | Print manual                                          |                       |
| <code>--man</code>     | —        | Print manual                                          |                       |
| <code>-o</code>        | OUTPUT   | Output to file                                        |                       |
| <code>--output</code>  | OUTPUT   | Output to file                                        |                       |
| <code>-e</code>        | —        | Generate error when error occurs in Python expression |                       |
| <code>--error</code>   | —        | Generate error when error occurs in Python expression |                       |
| <code>-i</code>        | —        | Ignore numbers greater than 255                       |                       |
| <code>--ignore</code>  | —        | Ignore numbers greater than 255                       |                       |
| <code>-n</code>        | NUMBER   | Minimum number of numbers (3 by default)              |                       |
| <code>--number</code>  | NUMBER   | Minimum number of numbers (3 by default)              |                       |

| Flag                       | Argument   | What it does                  | When you would use it |
|----------------------------|------------|-------------------------------|-----------------------|
| <code>-v</code>            | —          | Verbose                       |                       |
| <code>--verbose</code>     | —          | Verbose                       |                       |
| <code>-j</code>            | —          | Join output                   |                       |
| <code>--join</code>        | —          | Join output                   |                       |
| <code>-S</code>            | —          | Generate statistics           |                       |
| <code>--statistics</code>  | —          | Generate statistics           |                       |
| <code>-t</code>            | TABLE      | Translation table             |                       |
| <code>--table</code>       | TABLE      | Translation table             |                       |
| <code>-T</code>            | BEGINTABLE | Begin translation table       |                       |
| <code>--begintable</code>  | BEGINTABLE | Begin translation table       |                       |
| <code>-b</code>            | —          | Produce binary output         |                       |
| <code>--binary</code>      | —          | Produce binary output         |                       |
| <code>-l</code>            | LINE       | Select line with given number |                       |
| <code>--line</code>        | LINE       | Select line with given number |                       |
| <code>--grep</code>        | GREP       | Grep expression               |                       |
| <code>--grepoptions</code> | GREOPTIONS | Grep options                  |                       |
| <code>--begin</code>       | BEGIN      | Begin substring               |                       |
| <code>--end</code>         | END        | End substring                 |                       |

## Gotchas

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*TODO: operational traps.*

## See also

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`floss`, `strings`, `base64dump.py`, `cyberchef`, `rax2`, `xxd`, `openssl`

# objdump

**Kit:** REMnux **Capability:** Inspect PE / ELF structure; Disassemble and explore a binary **Version:** GNU objdump (GNU Binutils for Debian) 2.40 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw Docs:](https://en.wikipedia.org/wiki/Objdump) <https://en.wikipedia.org/wiki/Objdump>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Disassemble binary files.

## Synopsis

```
objdump <option(s)> <file(s)>
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 78 options parsed from the captured help text. The final column is filled in by review.

| Flag                           | Argument | What it does                                        | When you would use it |
|--------------------------------|----------|-----------------------------------------------------|-----------------------|
| <code>-a</code>                | —        | Display archive header information                  |                       |
| <code>--archive-headers</code> | —        | Display archive header information                  |                       |
| <code>-f</code>                | —        | Display the contents of the overall file header     |                       |
| <code>--file-headers</code>    | —        | Display the contents of the overall file header     |                       |
| <code>-p</code>                | —        | Display object format specific file header contents |                       |
| <code>--private-headers</code> | —        | Display object format specific file header contents |                       |
| <code>-x</code>                | —        | Display the contents of all headers                 |                       |
| <code>--all-headers</code>     | —        | Display the contents of all headers                 |                       |
| <code>-d</code>                | —        | Display assembler contents of executable sections   |                       |
| <code>--disassemble</code>     | —        | Display assembler contents of executable sections   |                       |
| <code>-D</code>                | —        | Display assembler contents of all sections          |                       |

| Flag                           | Argument           | What it does                                                                                            | When you would use it |
|--------------------------------|--------------------|---------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--disassemble-all</code> | —                  | Display assembler contents of all sections                                                              |                       |
| <code>-S</code>                | —                  | Intermix source code with disassembly <code>--source-comment[=]</code> Prefix lines of source code with |                       |
| <code>--source</code>          | —                  | Intermix source code with disassembly <code>--source-comment[=]</code> Prefix lines of source code with |                       |
| <code>-s</code>                | —                  | Display the full contents of all sections requested                                                     |                       |
| <code>--full-contents</code>   | —                  | Display the full contents of all sections requested                                                     |                       |
| <code>-g</code>                | —                  | Display debug information in object file                                                                |                       |
| <code>--debugging</code>       | —                  | Display debug information in object file                                                                |                       |
| <code>-e</code>                | —                  | Display debug information using ctags style                                                             |                       |
| <code>--debugging-tags</code>  | —                  | Display debug information using ctags style                                                             |                       |
| <code>-G</code>                | —                  | Display (in raw form) any STABS info in the file                                                        |                       |
| <code>--stabs</code>           | —                  | Display (in raw form) any STABS info in the file                                                        |                       |
| <code>-L</code>                | —                  | Display the contents of non-debug sections in separate debuginfo files. (Implies <code>-WK</code> )     |                       |
| <code>--process-links</code>   | —                  | Display the contents of non-debug sections in separate debuginfo files. (Implies <code>-WK</code> )     |                       |
| <code>--ctf</code>             | SECTION (optional) | Display CTF info from SECTION, (default <code>`.ctf'</code> )                                           |                       |
| <code>--sframe</code>          | SECTION (optional) | Display SFrame info from SECTION, (default <code>`.sframe'</code> )                                     |                       |
| <code>-t</code>                | —                  | Display the contents of the symbol table(s)                                                             |                       |
| <code>--syms</code>            | —                  | Display the contents of the symbol table(s)                                                             |                       |
| <code>-T</code>                | —                  | Display the contents of the dynamic symbol table                                                        |                       |
| <code>--dynamic-syms</code>    | —                  | Display the contents of the dynamic symbol table                                                        |                       |
| <code>-r</code>                | —                  | Display the relocation entries in the file                                                              |                       |
| <code>--reloc</code>           | —                  | Display the relocation entries in the file                                                              |                       |
| <code>-R</code>                | —                  | Display the dynamic relocation entries in the file                                                      |                       |
| <code>--dynamic-reloc</code>   | —                  | Display the dynamic relocation entries in the file                                                      |                       |
| <code>-v</code>                | —                  | Display this program's version number                                                                   |                       |
| <code>--version</code>         | —                  | Display this program's version number                                                                   |                       |
| <code>-i</code>                | —                  | List object formats and architectures supported                                                         |                       |
| <code>--info</code>            | —                  | List object formats and architectures supported                                                         |                       |
| <code>-H</code>                | —                  | Display this information                                                                                |                       |
| <code>--help</code>            | —                  | Display this information                                                                                |                       |



| Flag                              | Argument         | What it does                                                                                                 | When you would use it |
|-----------------------------------|------------------|--------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-b</code>                   | BFDNAME          | Specify the target object format as BFDNAME                                                                  |                       |
| <code>--target</code>             | BFDNAME          | Specify the target object format as BFDNAME                                                                  |                       |
| <code>-m</code>                   | MACHINE          | Specify the target architecture as MACHINE                                                                   |                       |
| <code>--architecture</code>       | MACHINE          | Specify the target architecture as MACHINE                                                                   |                       |
| <code>-j</code>                   | NAME             | Only display information for section NAME                                                                    |                       |
| <code>--section</code>            | NAME             | Only display information for section NAME                                                                    |                       |
| <code>--file-start-context</code> | —                | Include context from start of file (with -S)                                                                 |                       |
| <code>-I</code>                   | DIR              | Add DIR to search list for source files                                                                      |                       |
| <code>--include</code>            | DIR              | Add DIR to search list for source files                                                                      |                       |
| <code>-l</code>                   | —                | Include line numbers and filenames in output                                                                 |                       |
| <code>--line-numbers</code>       | —                | Include line numbers and filenames in output                                                                 |                       |
| <code>-F</code>                   | —                | Include file offsets when displaying information                                                             |                       |
| <code>--file-offsets</code>       | —                | Include file offsets when displaying information                                                             |                       |
| <code>-C</code>                   | STYLE (optional) | Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust" |                       |
| <code>--demangle</code>           | STYLE (optional) | Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust" |                       |
| <code>--recurse-limit</code>      | —                | Enable a limit on recursion whilst demangling (default)                                                      |                       |
| <code>--no-recurse-limit</code>   | —                | Disable a limit on recursion whilst demangling                                                               |                       |
| <code>-w</code>                   | —                | Format output for more than 80 columns                                                                       |                       |
| <code>--wide</code>               | —                | Format output for more than 80 columns                                                                       |                       |
| <code>-z</code>                   | —                | Do not skip blocks of zeroes when disassembling                                                              |                       |
| <code>--disassemble-zeroes</code> | —                | Do not skip blocks of zeroes when disassembling                                                              |                       |
| <code>--start-address</code>      | ADDR             | Only process data whose address is >= ADDR                                                                   |                       |
| <code>--stop-address</code>       | ADDR             | Only process data whose address is < ADDR                                                                    |                       |
| <code>--no-addresses</code>       | —                | Do not print address alongside disassembly                                                                   |                       |
| <code>--prefix-addresses</code>   | —                | Print complete address alongside disassembly                                                                 |                       |
| <code>--insn-width</code>         | WIDTH            | Display WIDTH bytes on a single line for -d                                                                  |                       |
| <code>--adjust-vma</code>         | OFFSET           | Add OFFSET to all displayed section addresses                                                                |                       |
| <code>--show-all-symbols</code>   | —                | When disassembling, display all symbols at a given address                                                   |                       |
| <code>--special-syms</code>       | —                | Include special symbols in symbol dumps                                                                      |                       |

| Flag                              | Argument | What it does                                  | When you would use it |
|-----------------------------------|----------|-----------------------------------------------|-----------------------|
| <code>--inlines</code>            | —        | Print all inlines for source line (with -l)   |                       |
| <code>--prefix</code>             | PREFIX   | Add PREFIX to absolute paths for -S           |                       |
| <code>--prefix-strip</code>       | LEVEL    | Strip initial directory names for -S          |                       |
| <code>--dwarf-depth</code>        | N        | Do not display DIEs at depth N or greater     |                       |
| <code>--dwarf-start</code>        | N        | Display DIEs starting at offset N             |                       |
| <code>--dwarf-check</code>        | —        | Make additional dwarf consistency checks.     |                       |
| <code>--ctf-parent</code>         | NAME     | Use CTF archive member NAME as the CTF parent |                       |
| <code>--visualize-jumps</code>    | —        | Visualize jumps by drawing ASCII art lines    |                       |
| <code>--disassembler-color</code> | off      | Disable disassembler color output. (default)  |                       |

## Gotchas

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*TODO: operational traps.*

## See also

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[rabin2](#), [readelf](#), [readelf.py](#), [r2](#), [rasm2](#), [vivbin](#), [vdbbin](#)

# rabin2

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect PE / ELF structure; Disassemble and explore a binary  
**Version:** rabin2 6.1.9 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw Docs:](#)  
<https://www.radare.org/n/radare2.html>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

## Synopsis

```
rabin2 [-AcdeEghHiIjJlLMqrRsSuvVxzZ] [-@ at] [-a arch] [-b bits] [-B addr]
[-C F:C:D] [-f str] [-m addr] [-n str] [-N m:M] [-P[-P] pdb]
[-o str] [-O help] [-k query] [-D lang mangledsymbol] file
```

## Common invocations

```
# from cyberlab 35-radare2-intro
rabin2 (shipped with radare2) lists strings from data sections; r2 disassembles main
# from cyberlab 35-radare2-intro
rabin2 -z exercise/hello
# from cyberlab 35-radare2-intro
rabin2 -z exercise/hello | grep r2rules
```

## Options

All 47 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                | When you would use it |
|------|----------|---------------------------------------------|-----------------------|
| -@   | addr     | show section, symbol or import at addr      |                       |
| -A   | —        | list sub-binaries and their arch-bits pairs |                       |
| -a   | arch     | set arch (x86, arm, .. or _)                |                       |
| -b   | bits     | set bits (32, 64 ...)                       |                       |
| -B   | addr     | override base address (pie bins)            |                       |
| -c   | —        | list classes                                |                       |

| Flag      | Argument  | What it does                                               | When you would use it |
|-----------|-----------|------------------------------------------------------------|-----------------------|
| <b>-C</b> | fmt:C:D   | create [elf,mach0,pe] with Code and Data hexpairs (see -a) |                       |
| <b>-d</b> | —         | show debug/dwarf information                               |                       |
| <b>-e</b> | —         | program entrypoint                                         |                       |
| <b>-E</b> | —         | globally exportable symbols                                |                       |
| <b>-f</b> | str       | select sub-bin named str                                   |                       |
| <b>-F</b> | binfmt    | force to use that bin plugin (ignore header check)         |                       |
| <b>-g</b> | —         | same as -SMZIHVResizcld -SS -SSS -ee (show all info)       |                       |
| <b>-G</b> | addr      | load address . address to header                           |                       |
| <b>-h</b> | —         | this help message                                          |                       |
| <b>-H</b> | —         | header fields                                              |                       |
| <b>-i</b> | —         | imports (symbols imported from libraries)                  |                       |
| <b>-I</b> | —         | binary info                                                |                       |
| <b>-j</b> | —         | output in json                                             |                       |
| <b>-k</b> | sdb-query | run sdb query. for example: '*'                            |                       |
| <b>-K</b> | algo      | calculate checksums (md5, sha1, ..)                        |                       |
| <b>-l</b> | —         | linked libraries                                           |                       |
| <b>-L</b> | plugin    | list supported bin plugins or plugin details               |                       |
| <b>-m</b> | addr      | show source line at addr                                   |                       |
| <b>-M</b> | —         | main (show address of main symbol)                         |                       |
| <b>-n</b> | str       | show section, symbol or import named str                   |                       |
| <b>-N</b> | min:max   | force min:max number of chars per string (see -z and -zz)  |                       |
| <b>-o</b> | str       | output file or extraction directory for write operations   |                       |
| <b>-O</b> | str       | write/extract operations (-O help)                         |                       |
| <b>-p</b> | —         | show always physical addresses                             |                       |
| <b>-P</b> | —         | show debug/pdb information                                 |                       |
| <b>-q</b> | —         | be quiet, just show fewer data                             |                       |
| <b>-Q</b> | —         | show load address used by dlopen (non-aslr libs)           |                       |
| <b>-r</b> | —         | radare output                                              |                       |
| <b>-R</b> | —         | relocations                                                |                       |
| <b>-s</b> | —         | symbols                                                    |                       |
| <b>-S</b> | —         | sections                                                   |                       |
| <b>-t</b> | —         | display file hashes                                        |                       |
| <b>-T</b> | —         | display file signature                                     |                       |
| <b>-u</b> | —         | resources                                                  |                       |

| Flag            | Argument | What it does                                                      | When you would use it |
|-----------------|----------|-------------------------------------------------------------------|-----------------------|
| <code>-v</code> | —        | display version and quit                                          |                       |
| <code>-V</code> | —        | show binary version information                                   |                       |
| <code>-w</code> | —        | display try/catch blocks                                          |                       |
| <code>-x</code> | —        | extract sub-binaries (-xS sections, -xSS segments, -xu resources) |                       |
| <code>-y</code> | —        | show types (structs, enums, function signatures)                  |                       |
| <code>-z</code> | —        | strings (from data section)                                       |                       |
| <code>-Z</code> | —        | guess size of binary program                                      |                       |

## Gotchas

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*TODO: operational traps.*

## See also

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[readelf](#), [readelf.py](#), [objdump](#), [r2](#), [rasm2](#), [vivbin](#), [vdbbin](#)

# radiff2

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Compare or cluster samples; Diff two binaries **Version:** radiff2 6.1.9 **Captured:** cyberlab-aio via -h on 2026-08-01 **raw Docs:** <https://www.radare.org/n/radare2.html>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

## Synopsis

```
radiff2 [-options] [-A[A]] [-B #] [-g sym] [-m graph_mode][-t %] [file] [file]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 29 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                 | When you would use it |
|------|----------|--------------------------------------------------------------|-----------------------|
| -a   | arch     | specify architecture plugin to use (x86, arm, ..)            |                       |
| -A   | -A       | run aaa or aaaa after loading each binary (see -C)           |                       |
| -b   | bits     | specify register size for arch (16 (thumb), 32, 64, ..)      |                       |
| -c   | cmd      | run given command on every RCore instance                    |                       |
| -C   | —        | graphdiff code (columns: off-A, match-ratio, off-B) (see -A) |                       |
| -d   | —        | use delta diffing                                            |                       |
| -D   | —        | show disasm instead of hexpairs                              |                       |
| -e   | k=v      | set eval config var value for all RCore instances            |                       |
| -f   | help     | select output format (see '-f help' for details)             |                       |
| -g   | arg      | graph diff of [sym] or functions in [off1,off2]              |                       |
| -i   | help     | compare bin information (symbols, strings, classes, ..)      |                       |
| -j   | —        | output in json format (see -f json)                          |                       |
| -l   | addr     | specify final offset (length) for diffing, same format as -o |                       |

| Flag            | Argument | What it does                                                                  | When you would use it |
|-----------------|----------|-------------------------------------------------------------------------------|-----------------------|
| <code>-m</code> | mode     | choose the graph output mode (aditsjJ)                                        |                       |
| <code>-n</code> | —        | count of changes                                                              |                       |
| <code>-o</code> | addr     | specify initial offset for diffing, can use A:B format                        |                       |
| <code>-O</code> | —        | code diffing with opcode bytes only                                           |                       |
| <code>-p</code> | —        | use physical addressing (io.va=false) (only for radiff2 -AC)                  |                       |
| <code>-q</code> | —        | quiet mode (disable colors, reduce output)                                    |                       |
| <code>-r</code> | —        | output in radare commands                                                     |                       |
| <code>-s</code> | —        | compute edit distance (no substitution, Eugene W. Myers O(ND) diff algorithm) |                       |
| <code>-S</code> | name     | sort code diff (name, namelen, addr, size, type, dist) (only for -C or -g)    |                       |
| <code>-T</code> | —        | analyze files in threads (EXPERIMENTAL, 30% faster and crashy)                |                       |
| <code>-x</code> | —        | show two column hexdump diffing                                               |                       |
| <code>-X</code> | —        | use xpatch format for the diffing output                                      |                       |
| <code>-u</code> | —        | unified output (---+++)                                                       |                       |
| <code>-U</code> | —        | unified output using system 'diff'                                            |                       |
| <code>-v</code> | —        | show version information                                                      |                       |
| <code>-V</code> | —        | be verbose (current only for -s)                                              |                       |

## Gotchas

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*TODO: operational traps.*

## See also

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[ssdeep](#)

# readelf

**Kit:** Base OS — present on every Linux image **Capability:** Inspect PE / ELF structure **Version:** GNU readelf (GNU Binutils for Debian) 2.40 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Display information about the contents of ELF format files

## Synopsis

```
readelf <option(s)> elf-file(s)
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 68 options parsed from the captured help text. The final column is filled in by review.

| Flag              | Argument | What it does                              | When you would use it |
|-------------------|----------|-------------------------------------------|-----------------------|
| -a                | —        | Equivalent to: -h -l -S -s -r -d -V -A -l |                       |
| --all             | —        | Equivalent to: -h -l -S -s -r -d -V -A -l |                       |
| -h                | —        | Display the ELF file header               |                       |
| --file-header     | —        | Display the ELF file header               |                       |
| -l                | —        | Display the program headers               |                       |
| --program-headers | —        | Display the program headers               |                       |
| --segments        | —        | An alias for --program-headers            |                       |
| -S                | —        | Display the sections' header              |                       |
| --section-headers | —        | Display the sections' header              |                       |
| --sections        | —        | An alias for --section-headers            |                       |
| -g                | —        | Display the section groups                |                       |



| Flag                            | Argument         | What it does                                                                                                 | When you would use it |
|---------------------------------|------------------|--------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--section-groups</code>   | —                | Display the section groups                                                                                   |                       |
| <code>-t</code>                 | —                | Display the section details                                                                                  |                       |
| <code>--section-details</code>  | —                | Display the section details                                                                                  |                       |
| <code>-e</code>                 | —                | Equivalent to: <code>-h -l -S</code>                                                                         |                       |
| <code>--headers</code>          | —                | Equivalent to: <code>-h -l -S</code>                                                                         |                       |
| <code>-s</code>                 | —                | Display the symbol table                                                                                     |                       |
| <code>--syms</code>             | —                | Display the symbol table                                                                                     |                       |
| <code>--symbols</code>          | —                | An alias for <code>--syms</code>                                                                             |                       |
| <code>--dyn-syms</code>         | —                | Display the dynamic symbol table                                                                             |                       |
| <code>--lto-syms</code>         | —                | Display LTO symbol tables                                                                                    |                       |
| <code>--sym-base</code>         | 0 8 10 16        | Force base for symbol sizes. The options are mixed (the default), octal, decimal, hexadecimal.               |                       |
| <code>-C</code>                 | —                | Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust" |                       |
| <code>--demangle</code>         | STYLE (optional) | Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust" |                       |
| <code>--no-demangle</code>      | —                | Do not demangle low-level symbol names. (default)                                                            |                       |
| <code>--recurse-limit</code>    | —                | Enable a demangling recursion limit. (default)                                                               |                       |
| <code>--no-recurse-limit</code> | —                | Disable a demangling recursion limit                                                                         |                       |
| <code>-n</code>                 | —                | Display the core notes (if present)                                                                          |                       |
| <code>--notes</code>            | —                | Display the core notes (if present)                                                                          |                       |
| <code>-r</code>                 | —                | Display the relocations (if present)                                                                         |                       |
| <code>--relocs</code>           | —                | Display the relocations (if present)                                                                         |                       |
| <code>-u</code>                 | —                | Display the unwind info (if present)                                                                         |                       |
| <code>--unwind</code>           | —                | Display the unwind info (if present)                                                                         |                       |
| <code>-d</code>                 | —                | Display the dynamic section (if present)                                                                     |                       |
| <code>--dynamic</code>          | —                | Display the dynamic section (if present)                                                                     |                       |
| <code>-V</code>                 | —                | Display the version sections (if present)                                                                    |                       |
| <code>--version-info</code>     | —                | Display the version sections (if present)                                                                    |                       |
| <code>-A</code>                 | —                | Display architecture specific information (if any)                                                           |                       |
| <code>--arch-specific</code>    | —                | Display architecture specific information (if any)                                                           |                       |

| Flag                          | Argument        | What it does                                                                          | When you would use it |
|-------------------------------|-----------------|---------------------------------------------------------------------------------------|-----------------------|
| <code>-c</code>               | —               | Display the symbol/file index in an archive                                           |                       |
| <code>--archive-index</code>  | —               | Display the symbol/file index in an archive                                           |                       |
| <code>-D</code>               | —               | Use the dynamic section info when displaying symbols                                  |                       |
| <code>--use-dynamic</code>    | —               | Use the dynamic section info when displaying symbols                                  |                       |
| <code>-x</code>               | —               | Dump the contents of section as bytes                                                 |                       |
| <code>--hex-dump</code>       | number name     | Dump the contents of section as bytes                                                 |                       |
| <code>-p</code>               | —               | Dump the contents of section as strings                                               |                       |
| <code>--string-dump</code>    | number name     | Dump the contents of section as strings                                               |                       |
| <code>-R</code>               | —               | Dump the relocated contents of section                                                |                       |
| <code>--relocated-dump</code> | number name     | Dump the relocated contents of section                                                |                       |
| <code>-z</code>               | —               | Decompress section before dumping it                                                  |                       |
| <code>--decompress</code>     | —               | Decompress section before dumping it                                                  |                       |
| <code>-P</code>               | —               | Display the contents of non-debug sections in separate debuginfo files. (Implies -wK) |                       |
| <code>--process-links</code>  | —               | Display the contents of non-debug sections in separate debuginfo files. (Implies -wK) |                       |
| <code>--dwarf-depth</code>    | N               | Do not display DIEs at depth N or greater                                             |                       |
| <code>--dwarf-start</code>    | N               | Display DIEs starting at offset N                                                     |                       |
| <code>--ctf</code>            | number name     | Display CTF info from section                                                         |                       |
| <code>--ctf-parent</code>     | name            | Use CTF archive member as the CTF parent                                              |                       |
| <code>--ctf-symbols</code>    | number name     | Use section as the CTF external symtab                                                |                       |
| <code>--ctf-strings</code>    | number name     | Use section as the CTF external strtab                                                |                       |
| <code>--sframe</code>         | NAME (optional) | Display SFrame info from section NAME, (default '.sframe')                            |                       |
| <code>-I</code>               | —               | Display histogram of bucket list lengths                                              |                       |
| <code>--histogram</code>      | —               | Display histogram of bucket list lengths                                              |                       |
| <code>-W</code>               | —               | Allow output width to exceed 80 characters                                            |                       |
| <code>--wide</code>           | —               | Allow output width to exceed 80 characters                                            |                       |
| <code>-H</code>               | —               | Display this information                                                              |                       |
| <code>--help</code>           | —               | Display this information                                                              |                       |
| <code>-v</code>               | —               | Display the version number of readelf                                                 |                       |
| <code>--version</code>        | —               | Display the version number of readelf                                                 |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

[rabin2](#), [readelf.py](#), [objdump](#)

# readelf.py

**Kit:** REMnux **Capability:** Inspect PE / ELF structure **Version:** based on pyelftools 0.33 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://github.com/eliben/pyelftools>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Python library for parsing and analyzing ELF files and DWARF debugging information.

## Synopsis

```
usage: readelf.py [options] <elf-file>
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 33 options parsed from the captured help text. The final column is filled in by review.

| Flag                           | Argument | What it does                           | When you would use it |
|--------------------------------|----------|----------------------------------------|-----------------------|
| <code>-v</code>                | —        | show program's version number and exit |                       |
| <code>--version</code>         | —        | show program's version number and exit |                       |
| <code>-d</code>                | —        | Display the dynamic section            |                       |
| <code>--dynamic</code>         | —        | Display the dynamic section            |                       |
| <code>-H</code>                | —        | Display this information               |                       |
| <code>--help</code>            | —        | Display this information               |                       |
| <code>-h</code>                | —        | Display the ELF file header            |                       |
| <code>--file-header</code>     | —        | Display the ELF file header            |                       |
| <code>-l</code>                | —        | Display the program headers            |                       |
| <code>--program-headers</code> | —        | Display the program headers            |                       |
| <code>--segments</code>        | —        | Display the program headers            |                       |
| <code>-S</code>                | —        | Display the sections' headers          |                       |

| Flag                           | Argument    | What it does                                                                                                                           | When you would use it |
|--------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--section-headers</code> | —           | Display the sections' headers                                                                                                          |                       |
| <code>--sections</code>        | —           | Display the sections' headers                                                                                                          |                       |
| <code>-e</code>                | —           | Equivalent to: <code>-h -l -S</code>                                                                                                   |                       |
| <code>--headers</code>         | —           | Equivalent to: <code>-h -l -S</code>                                                                                                   |                       |
| <code>-s</code>                | —           | Display the symbol table                                                                                                               |                       |
| <code>--symbols</code>         | —           | Display the symbol table                                                                                                               |                       |
| <code>--syms</code>            | —           | Display the symbol table                                                                                                               |                       |
| <code>-n</code>                | —           | Display the core notes (if present)                                                                                                    |                       |
| <code>--notes</code>           | —           | Display the core notes (if present)                                                                                                    |                       |
| <code>-r</code>                | —           | Display the relocations (if present)                                                                                                   |                       |
| <code>--relocs</code>          | —           | Display the relocations (if present)                                                                                                   |                       |
| <code>-x</code>                | number name | Dump the contents of section as bytes                                                                                                  |                       |
| <code>--hex-dump</code>        | number name | Dump the contents of section as bytes                                                                                                  |                       |
| <code>-p</code>                | number name | Dump the contents of section as strings                                                                                                |                       |
| <code>--string-dump</code>     | number name | Dump the contents of section as strings                                                                                                |                       |
| <code>-V</code>                | —           | Display the version sections (if present)                                                                                              |                       |
| <code>--version-info</code>    | —           | Display the version sections (if present)                                                                                              |                       |
| <code>-A</code>                | —           | Display the architecture-specific information (if present)                                                                             |                       |
| <code>--arch-specific</code>   | —           | Display the architecture-specific information (if present)                                                                             |                       |
| <code>--debug-dump</code>      | what        | Display the contents of DWARF debug sections. can one of {info,decodedline,frames,frames- interp,aranges,pubtypes,pubnames,loc,Ranges} |                       |
| <code>--traceback</code>       | —           | Dump the Python traceback on ELFError exceptions from elftools                                                                         |                       |

## Gotchas

*TODO: operational traps.*

## See also

`rabin2`, `readelf`, `objdump`

# unzip

**Kit:** REMnux **Capability:** Detect and reverse packing; Analyse other container formats **Version:** UnZip 6.00  
**Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <http://infozip.sourceforge.net>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Compress and decompress files using the zip algorithm.

## Synopsis

```
unzip [-Z] [-opts[modifiers]] file[.zip] [list] [-x xlist] [-d exdir]
Default action is to extract files in list, except those in xlist, to exdir;
file[.zip] may be a wildcard. -Z => ZipInfo mode ("unzip -Z" for usage).
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 12 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                        | When you would use it |
|------|----------|---------------------------------------------------------------------|-----------------------|
| -p   | —        | extract files to pipe, no messages -l list files (short format)     |                       |
| -f   | —        | freshen existing files, create none -t test compressed archive data |                       |
| -u   | —        | update files, create if necessary -z display archive comment only   |                       |
| -v   | —        | list verbosely/show version info -T timestamp archive to latest     |                       |
| -x   | —        | exclude files that follow (in xlist) -d extract files into exdir    |                       |
| -n   | —        | never overwrite existing files -q quiet mode (-qq => quieter)       |                       |
| -o   | —        | overwrite files WITHOUT prompting -a auto-convert any text files    |                       |
| -j   | —        | junk paths (do not make directories) -aa treat ALL files as text    |                       |
| -U   | —        | use escapes for all non-ASCII Unicode -UU ignore any Unicode fields |                       |
| -C   | —        | match filenames case-insensitively -L make (some) names lowercase   |                       |
| -X   | —        | restore UID/GID info -V retain VMS version numbers                  |                       |
| -K   | —        | keep setuid/setgid/tacky permissions -M pipe through "more" pager   |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

[upx](#), [die](#), [diec](#), [binwalk](#), [7za](#), [msoffcrypto-tool](#)

# upx

**Kit:** REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Detect and reverse packing **Version:** upx 4.2.4  
**Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://upx.github.io>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Pack and unpack executables with the UPX compressor.

## Synopsis

```
upx [-123456789dlthVL] [-qvfk] [-o file] file..
```

## Common invocations

```
# from cyberlab 28-x64dbg-workflow
upx --best .\exercise\packed_hello.exe
# from cyberlab 28-x64dbg-workflow
upx -d .\exercise\packed_hello.exe -o .\exercise\unpacked_ref.exe
# from cyberlab 43-pe-bear-deep
upx --best -o C:\work\pe-lab\exercise\packed_hello.exe C:\work\pe-lab\exercise\hello.exe
# from cyberlab 52-unpacking-case
upx --best .\exercise\sample_packed.exe
```

## Options

All 37 options parsed from the captured help text; 5 reviewed with usage guidance.

| Flag | Argument | What it does                                   | When you would use it                                     |
|------|----------|------------------------------------------------|-----------------------------------------------------------|
| -1   | —        | compress faster -9 compress better             |                                                           |
| -d   | —        | decompress -l list compressed file             | Decompress — the only flag you normally want in analysis. |
| -t   | —        | test compressed file -V display version number | Test the file's integrity without writing anything.       |
| -h   | —        | give this help -L display software license     |                                                           |
| -q   | —        | be quiet -v be verbose                         | Quiet output.                                             |
| -f   | —        | force compression of suspicious files          | Force the operation on a file UPX considers questionable. |



| Flag                              | Argument | What it does                                           | When you would use it               |
|-----------------------------------|----------|--------------------------------------------------------|-------------------------------------|
| <code>--no-color</code>           | —        | change look                                            |                                     |
| <code>--mono</code>               | —        | change look                                            |                                     |
| <code>--color</code>              | —        | change look                                            |                                     |
| <code>--no-progress</code>        | —        | change look                                            |                                     |
| <code>--lzma</code>               | —        | try LZMA [slower but tighter than NRV]                 |                                     |
| <code>--brute</code>              | —        | try all available compression methods & filters [slow] |                                     |
| <code>--ultra-brute</code>        | —        | try even more compression variants [very slow]         |                                     |
| <code>-k</code>                   | —        | keep backup files                                      | Keep a backup of the original file. |
| <code>--backup</code>             | —        | keep backup files                                      |                                     |
| <code>--no-backup</code>          | —        | no backup files [default]                              |                                     |
| <code>--overlay</code>            | copy     | copy any extra data attached to the file [default]     |                                     |
| <code>--force-overwrite</code>    | —        | force overwrite of output files                        |                                     |
| <code>--link</code>               | —        | preserve hard links (Unix only) [USE WITH CARE]        |                                     |
| <code>--no-link</code>            | —        | do not preserve hard links but rename files [default]  |                                     |
| <code>--no-mode</code>            | —        | do not preserve file mode (aka permissions)            |                                     |
| <code>--no-owner</code>           | —        | do not preserve file ownership                         |                                     |
| <code>--no-time</code>            | —        | do not preserve file timestamp                         |                                     |
| <code>--coff</code>               | —        | produce COFF output [default: EXE]                     |                                     |
| <code>--8086</code>               | —        | make compressed com work on any 8086                   |                                     |
| <code>--no-reloc</code>           | —        | put no relocations in to the exe header                |                                     |
| <code>--8-bit</code>              | —        | uses 8 bit size compression [default: 32 bit]          |                                     |
| <code>--8mib-ram</code>           | —        | 8 megabyte memory limit [default: 2 MiB]               |                                     |
| <code>--boot-only</code>          | —        | disables client/host transfer compatibility            |                                     |
| <code>--no-align</code>           | —        | don't align to 2048 bytes [enables: --console-run]     |                                     |
| <code>--le</code>                 | —        | produce LE output [default: EXE]                       |                                     |
| <code>--compress-exports</code>   | 0        | do not compress the export section                     |                                     |
| <code>--compress-icons</code>     | 0        | do not compress any icons                              |                                     |
| <code>--compress-resources</code> | 0        | do not compress any resources at all                   |                                     |

| Flag                             | Argument | What it does                                 | When you would use it |
|----------------------------------|----------|----------------------------------------------|-----------------------|
| <code>--keep-resource</code>     | list     | do not compress resources specified by list  |                       |
| <code>--strip-relocs</code>      | 0        | do not strip relocations                     |                       |
| <code>--preserve-build-id</code> | —        | copy .gnu.note.build-id to compressed output |                       |

## Gotchas

---

- **Work on a copy.** `upx -d` rewrites the file in place, mutating your sample and invalidating its hash.
- Malware routinely uses a modified UPX header so stock `upx -d` refuses. Failure to unpack is itself an indicator, not a dead end.

## See also

---

`die`, `diec`, `binwalk`, `7za`, `unzip`

# yara

**Kit:** REMnux · Kali Linux · FLARE-VM **Capability:** Scan with signatures for known-bad **Version:** 4.2.3 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#) **Docs:** <https://virustotal.github.io/yara/>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Scan files, directories or live processes against YARA rules.

## Synopsis

```
yara [OPTION]... [NAMESPACE:]RULES_FILE... FILE | DIR | PID
```

## Common invocations

```
# from cyberlab 08-malware-static-triage
yara --version
# from cyberlab 08-malware-static-triage
yara -r exercise/triage.yar exercise/sample.bin
# from cyberlab 21-yara-authoring
yara --help | head -n 20
# from cyberlab 21-yara-authoring
yara -w /tmp/hunt_eicar.yar /tmp/hunt_eicar.yar
# from cyberlab 21-yara-authoring
yara -r -s /tmp/hunt_eicar.yar /tmp
# from cyberlab 21-yara-authoring
yara -r -s /tmp/hunt_eicar.yar exercise/
# from cyberlab 21-yara-authoring
yara your_rule.yar exercise/eicar_sample.txt
# from cyberlab 34-clamav-scanning
yara -s exercise/lab34.yar exercise/sample.txt
```

## Options

All 55 options parsed from the captured help text; 12 reviewed with usage guidance.

| Flag                              | Argument | What it does                               | When you would use it |
|-----------------------------------|----------|--------------------------------------------|-----------------------|
| <code>--atom-quality-table</code> | FILE     | path to a file with the atom quality table |                       |

| Flag                                    | Argument    | What it does                                                             | When you would use it                                                         |
|-----------------------------------------|-------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <code>-C</code>                         | —           | load compiled rules                                                      | Load a pre-compiled ruleset (from <code>yarac</code> ) instead of source.     |
| <code>--compiled-rules</code>           | —           | load compiled rules                                                      |                                                                               |
| <code>-c</code>                         | —           | print only number of matches                                             | Print only the count of matches per file, for a quick sweep.                  |
| <code>--count</code>                    | —           | print only number of matches                                             |                                                                               |
| <code>-d</code>                         | VAR=VALUE   | define external variable                                                 | Define an external variable a rule expects, e.g. <code>-d filename=x</code> . |
| <code>--define</code>                   | VAR=VALUE   | define external variable                                                 |                                                                               |
| <code>--fail-on-warnings</code>         | —           | fail on warnings                                                         |                                                                               |
| <code>-f</code>                         | —           | fast matching mode                                                       | Fast matching mode; faster but can miss some overlapping matches.             |
| <code>--fast-scan</code>                | —           | fast matching mode                                                       |                                                                               |
| <code>-h</code>                         | —           | show this help and exit                                                  |                                                                               |
| <code>--help</code>                     | —           | show this help and exit                                                  |                                                                               |
| <code>-i</code>                         | IDENTIFIER  | print only rules named IDENTIFIER                                        | Scan only the rule with the given identifier.                                 |
| <code>--identifier</code>               | IDENTIFIER  | print only rules named IDENTIFIER                                        |                                                                               |
| <code>--max-process-memory-chunk</code> | NUMBER      | set maximum chunk size while reading process memory (default=1073741824) |                                                                               |
| <code>-l</code>                         | NUMBER      | abort scanning after matching a NUMBER of rules                          |                                                                               |
| <code>--max-rules</code>                | NUMBER      | abort scanning after matching a NUMBER of rules                          |                                                                               |
| <code>--max-strings-per-rule</code>     | NUMBER      | set maximum number of strings per rule (default=10000)                   |                                                                               |
| <code>-x</code>                         | MODULE=FILE | pass FILE's content as extra data to MODULE                              |                                                                               |
| <code>--module-data</code>              | MODULE=FILE | pass FILE's content as extra data to MODULE                              |                                                                               |
| <code>-n</code>                         | —           | print only not satisfied rules (negate)                                  | Print files that do NOT match — useful for finding the odd one out.           |
| <code>--negate</code>                   | —           | print only not satisfied rules (negate)                                  |                                                                               |
| <code>-N</code>                         | —           | do not follow symlinks when scanning                                     |                                                                               |
| <code>--no-follow-symlinks</code>       | —           | do not follow symlinks when scanning                                     |                                                                               |
| <code>-w</code>                         | —           | disable warnings                                                         | Suppress warnings, which are noisy on large third-party rulesets.             |

| Flag                               | Argument | What it does                                                    | When you would use it                                                                                                   |
|------------------------------------|----------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <code>--no-warnings</code>         | —        | disable warnings                                                |                                                                                                                         |
| <code>-m</code>                    | —        | print metadata                                                  | Print rule metadata, which usually carries author and reference.                                                        |
| <code>--print-meta</code>          | —        | print metadata                                                  |                                                                                                                         |
| <code>-D</code>                    | —        | print module data                                               |                                                                                                                         |
| <code>--print-module-data</code>   | —        | print module data                                               |                                                                                                                         |
| <code>-e</code>                    | —        | print rules' namespace                                          |                                                                                                                         |
| <code>--print-namespace</code>     | —        | print rules' namespace                                          |                                                                                                                         |
| <code>-S</code>                    | —        | print rules' statistics                                         |                                                                                                                         |
| <code>--print-stats</code>         | —        | print rules' statistics                                         |                                                                                                                         |
| <code>-s</code>                    | —        | print matching strings                                          | Print the matching strings and their offsets. Essential for verifying a hit is real rather than trusting the rule name. |
| <code>--print-strings</code>       | —        | print matching strings                                          |                                                                                                                         |
| <code>-L</code>                    | —        | print length of matched strings                                 |                                                                                                                         |
| <code>--print-string-length</code> | —        | print length of matched strings                                 |                                                                                                                         |
| <code>-g</code>                    | —        | print tags                                                      |                                                                                                                         |
| <code>--print-tags</code>          | —        | print tags                                                      |                                                                                                                         |
| <code>-r</code>                    | —        | recursively search directories                                  | Recurse into directories — the usual mode when triaging a tree.                                                         |
| <code>--recursive</code>           | —        | recursively search directories                                  |                                                                                                                         |
| <code>--scan-list</code>           | —        | scan files listed in FILE, one per line                         |                                                                                                                         |
| <code>-z</code>                    | NUMBER   | skip files larger than the given size when scanning a directory |                                                                                                                         |
| <code>--skip-larger</code>         | NUMBER   | skip files larger than the given size when scanning a directory |                                                                                                                         |
| <code>-k</code>                    | SLOTS    | set maximum stack size (default=16384)                          |                                                                                                                         |
| <code>--stack-size</code>          | SLOTS    | set maximum stack size (default=16384)                          |                                                                                                                         |
| <code>-t</code>                    | TAG      | print only rules tagged as TAG                                  | Scan only rules carrying a given tag.                                                                                   |
| <code>--tag</code>                 | TAG      | print only rules tagged as TAG                                  |                                                                                                                         |
| <code>-p</code>                    | NUMBER   | use the specified NUMBER of threads to scan a directory         | Use N threads to speed up a large scan.                                                                                 |
| <code>--threads</code>             | NUMBER   | use the specified NUMBER of threads to scan a directory         |                                                                                                                         |

| Flag                   | Argument | What it does                                     | When you would use it |
|------------------------|----------|--------------------------------------------------|-----------------------|
| <code>-a</code>        | SECONDS  | abort scanning after the given number of SECONDS |                       |
| <code>--timeout</code> | SECONDS  | abort scanning after the given number of SECONDS |                       |
| <code>-v</code>        | —        | show version information                         |                       |
| <code>--version</code> | —        | show version information                         |                       |

## Gotchas

- A rule that references an external variable fails to compile unless you supply it with `-d`. This is the most common 'the rule is broken' false alarm.
- Scanning a live process needs a PID rather than a path, and root.
- Compile large rulesets once with `yarac` and scan with `-C`; recompiling thousands of rules per scan dominates runtime.

## See also

`yarac`, `clamscan`, `freshclam`, `sigtool`

# yarac

**Kit:** REMnux · Kali Linux · FLARE-VM **Capability:** Scan with signatures for known-bad **Version:** 4.2.3 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#) **Docs:** <https://virustotal.github.io/yara/>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Identify and classify malware samples using Yara rules.

## Synopsis

```
yarac [OPTION]... [NAMESPACE:]SOURCE_FILE... OUTPUT_FILE
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 11 options parsed from the captured help text. The final column is filled in by review.

| Flag                                | Argument  | What it does                                           | When you would use it |
|-------------------------------------|-----------|--------------------------------------------------------|-----------------------|
| <code>--atom-quality-table</code>   | FILE      | path to a file with the atom quality table             |                       |
| <code>-d</code>                     | VAR=VALUE | define external variable                               |                       |
| <code>--define</code>               | VAR=VALUE | define external variable                               |                       |
| <code>--fail-on-warnings</code>     | —         | fail on warnings                                       |                       |
| <code>-h</code>                     | —         | show this help and exit                                |                       |
| <code>--help</code>                 | —         | show this help and exit                                |                       |
| <code>--max-strings-per-rule</code> | NUMBER    | set maximum number of strings per rule (default=10000) |                       |
| <code>-w</code>                     | —         | disable warnings                                       |                       |
| <code>--no-warnings</code>          | —         | disable warnings                                       |                       |
| <code>-v</code>                     | —         | show version information                               |                       |
| <code>--version</code>              | —         | show version information                               |                       |

## Gotchas

*TODO: operational traps.*

## See also

---

`yara`, `clamscan`, `freshclam`, `sigtool`



# aeskeyfind

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Recover encryption keys from memory **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://citp.princeton.edu/our-work/memory/>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Find 128-bit and 256-bit AES keys in a memory image.

## Synopsis

```
aeskeyfind [OPTION]... MEMORY-IMAGE
```

## Common invocations

```
# from cyberlab 02-memory-forensics
aeskeyfind 2>&1 | head -n 1
# from cyberlab 02-memory-forensics
aeskeyfind sample.mem
```

## Options

All 3 options parsed from the captured help text. The final column is filled in by review.

| Flag            | Argument | What it does                                                                                   | When you would use it |
|-----------------|----------|------------------------------------------------------------------------------------------------|-----------------------|
| <code>-v</code> | —        | verbose output -- prints the extended keys and the constraints on the rows of the key schedule |                       |
| <code>-q</code> | —        | don't display a progress bar                                                                   |                       |
| <code>-h</code> | —        | displays this help message                                                                     |                       |

## Gotchas

*TODO: operational traps.*

## See also

```
rsakeyfind, bulk_extractor
```

# rsakeyfind

---

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Recover encryption keys from memory **Captured:** cyberlab-aio via (no args) on 2026-08-01 **raw Docs:** <https://citp.princeton.edu/our-work/memory/>

← [Capability index](#) · [Kit tool list](#)

## Purpose

---

Find BER-encoded RSA private keys in a memory image.

## Synopsis

---

```
rsakeyfind MEMORY-IMAGE [MODULUS-FILE]
```

## Common invocations

---

```
# from cyberlab 02-memory-forensics
rsakeyfind 2>&1 | head -n 1
# from cyberlab 02-memory-forensics
rsakeyfind sample.mem
```

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

[aeskeyfind](#) , [bulk\\_extractor](#)

# vol

**Kit:** SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Extract and analyse artifacts from a memory image using Volatility 3 plugins.

## Synopsis

```
vol [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]]
[-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
[-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
[--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL] [--filters FILTERS]
[--hide-columns [HIDE_COLUMNS ...]] [-r RENDERER]
```

## Common invocations

```
# from cyberlab 02-memory-forensics
vol --help | head -n 3
# from cyberlab 02-memory-forensics
vol -h | grep -i -E "pslist|netscan|windows.info" | head -n 10
# from cyberlab 02-memory-forensics
vol -f $IMAGE windows.pslist | head -n 20
# from cyberlab 20-volatility-deep
vol --version
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.info
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.pslist
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.pstree
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.psscan
```

## Options

All 35 options parsed from the captured help text; 9 reviewed with usage guidance.

| Flag                       | Argument                | What it does                                                                  | When you would use it                                                                                  |
|----------------------------|-------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>-h</code>            | —                       | Show this help message and exit, for specific plugin options use 'vol --help' |                                                                                                        |
| <code>--help</code>        | —                       | Show this help message and exit, for specific plugin options use 'vol --help' |                                                                                                        |
| <code>-c</code>            | CONFIG                  | Load the configuration from a json file                                       | Load saved configuration from a JSON file.                                                             |
| <code>--config</code>      | CONFIG                  | Load the configuration from a json file                                       |                                                                                                        |
| <code>--parallelism</code> | {processes,threads,off} | Enables parallelism (defaults to off if no argument given)                    |                                                                                                        |
| <code>-e</code>            | EXTEND                  | Extend the configuration with a new (or changed) setting                      |                                                                                                        |
| <code>--extend</code>      | EXTEND                  | Extend the configuration with a new (or changed) setting                      |                                                                                                        |
| <code>-p</code>            | PLUGIN_DIRS             | Semi-colon separated list of paths to find plugins                            | Add a directory of custom plugins.                                                                     |
| <code>--plugin-dirs</code> | PLUGIN_DIRS             | Semi-colon separated list of paths to find plugins                            |                                                                                                        |
| <code>-s</code>            | SYMBOL_DIRS             | Semi-colon separated list of paths to find symbols                            | Point at a local symbol-table directory — the fix for an air-gapped host that cannot download symbols. |
| <code>--symbol-dirs</code> | SYMBOL_DIRS             | Semi-colon separated list of paths to find symbols                            |                                                                                                        |
| <code>-v</code>            | —                       | Increase output verbosity                                                     | Increase verbosity while diagnosing a symbol-table failure.                                            |
| <code>--verbosity</code>   | —                       | Increase output verbosity                                                     |                                                                                                        |
| <code>-l</code>            | LOG                     | Log output to a file as well as the console                                   |                                                                                                        |
| <code>--log</code>         | LOG                     | Log output to a file as well as the console                                   |                                                                                                        |
| <code>-o</code>            | OUTPUT_DIR              | Directory in which to output any generated files                              | Directory for files the plugin dumps (processes, DLLs, files).                                         |
| <code>--output-dir</code>  | OUTPUT_DIR              | Directory in which to output any generated files                              |                                                                                                        |
| <code>-q</code>            | —                       | Remove progress feedback                                                      | Quiet — suppress the progress and INFO banner when scripting.                                          |
| <code>--quiet</code>       | —                       | Remove progress feedback                                                      |                                                                                                        |
| <code>-f</code>            | FILE                    | Shorthand for --single-location=file:// if single- location is not defined    | The memory image to analyse — required for almost every plugin.                                        |
| <code>--file</code>        | FILE                    | Shorthand for --single-location=file:// if single- location is not defined    |                                                                                                        |

| Flag                                 | Argument                  | What it does                                                                                                   | When you would use it                                                                                          |
|--------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>--write-config</code>          | —                         | Write configuration JSON file out to config.json                                                               |                                                                                                                |
| <code>--save-config</code>           | SAVE_CONFIG               | Save configuration JSON file to a file                                                                         |                                                                                                                |
| <code>--clear-cache</code>           | —                         | Clears out all short-term cached items                                                                         |                                                                                                                |
| <code>--cache-path</code>            | CACHE_PATH                | Change the default path (/root/.cache/volatility3) used to store the cache                                     |                                                                                                                |
| <code>--offline</code>               | —                         | Do not search online for additional JSON files                                                                 | Never attempt a network fetch for symbols. Use this on evidence networks so a scan cannot stall on a download. |
| <code>-u</code>                      | URL                       | Search online for ISF json files                                                                               |                                                                                                                |
| <code>--remote-isf-url</code>        | URL                       | Search online for ISF json files                                                                               |                                                                                                                |
| <code>--filters</code>               | FILTERS                   | List of filters to apply to the output (in the form of [+ -]columnname,pattern[!])                             |                                                                                                                |
| <code>--hide-columns</code>          | HIDE_COLUMNS ...          | Case-insensitive space separated list of prefixes to determine which columns to hide in the output if provided |                                                                                                                |
| <code>-r</code>                      | RENDERER                  | Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)                    | Choose the renderer: <code>csv</code> / <code>json</code> when feeding another tool.                           |
| <code>--renderer</code>              | RENDERER                  | Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)                    |                                                                                                                |
| <code>--single-location</code>       | SINGLE_LOCATION           | Specifies a base location on which to stack                                                                    |                                                                                                                |
| <code>--stackers</code>              | STACKERS ...              | List of stackers                                                                                               |                                                                                                                |
| <code>--single-swap-locations</code> | SINGLE_SWAP_LOCATIONS ... | Specifies a list of swap layer URIs for use with single-location                                               |                                                                                                                |

## Gotchas

- **Timeout-guard** `vol` **in any harness.** On a synthetic or truncated memory image it has pinned a core at 99% indefinitely on this project. Never run it unbounded in an automated pass.
- Volatility 3 takes plugin names, not v2 syntax: `windows.pslist`, not `--profile=... pslist`. Most 'plugin not found' errors are pasted v2 commands.
- The first run against an unfamiliar image downloads symbol tables. On an isolated network that hangs — pre-stage symbols and use `--offline`.

# See also

---

`volatility3`, `volshell`

# volatility3

**Kit:** SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

An open-source memory forensics framework

## Synopsis

```
volatility3 [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]]
[-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
[-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
[--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL]
[--filters FILTERS] [--hide-columns [HIDE_COLUMNS ...]]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 35 options parsed from the captured help text. The final column is filled in by review.

| Flag                       | Argument                | What it does                                                                          | When you would use it |
|----------------------------|-------------------------|---------------------------------------------------------------------------------------|-----------------------|
| <code>-h</code>            | —                       | Show this help message and exit, for specific plugin options use 'volatility3 --help' |                       |
| <code>--help</code>        | —                       | Show this help message and exit, for specific plugin options use 'volatility3 --help' |                       |
| <code>-c</code>            | CONFIG                  | Load the configuration from a json file                                               |                       |
| <code>--config</code>      | CONFIG                  | Load the configuration from a json file                                               |                       |
| <code>--parallelism</code> | {processes,threads,off} | Enables parallelism (defaults to off if no argument given)                            |                       |
| <code>-e</code>            | EXTEND                  | Extend the configuration with a new (or changed) setting                              |                       |
| <code>--extend</code>      | EXTEND                  | Extend the configuration with a new (or changed) setting                              |                       |
| <code>-p</code>            | PLUGIN_DIRS             | Semi-colon separated list of paths to find plugins                                    |                       |



| Flag                                 | Argument                  | What it does                                                                                                   | When you would use it |
|--------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--plugin-dirs</code>           | PLUGIN_DIRS               | Semi-colon separated list of paths to find plugins                                                             |                       |
| <code>-s</code>                      | SYMBOL_DIRS               | Semi-colon separated list of paths to find symbols                                                             |                       |
| <code>--symbol-dirs</code>           | SYMBOL_DIRS               | Semi-colon separated list of paths to find symbols                                                             |                       |
| <code>-v</code>                      | —                         | Increase output verbosity                                                                                      |                       |
| <code>--verbosity</code>             | —                         | Increase output verbosity                                                                                      |                       |
| <code>-l</code>                      | LOG                       | Log output to a file as well as the console                                                                    |                       |
| <code>--log</code>                   | LOG                       | Log output to a file as well as the console                                                                    |                       |
| <code>-o</code>                      | OUTPUT_DIR                | Directory in which to output any generated files                                                               |                       |
| <code>--output-dir</code>            | OUTPUT_DIR                | Directory in which to output any generated files                                                               |                       |
| <code>-q</code>                      | —                         | Remove progress feedback                                                                                       |                       |
| <code>--quiet</code>                 | —                         | Remove progress feedback                                                                                       |                       |
| <code>-f</code>                      | FILE                      | Shorthand for <code>--single-location=file://</code> if single- location is not defined                        |                       |
| <code>--file</code>                  | FILE                      | Shorthand for <code>--single-location=file://</code> if single- location is not defined                        |                       |
| <code>--write-config</code>          | —                         | Write configuration JSON file out to config.json                                                               |                       |
| <code>--save-config</code>           | SAVE_CONFIG               | Save configuration JSON file to a file                                                                         |                       |
| <code>--clear-cache</code>           | —                         | Clears out all short-term cached items                                                                         |                       |
| <code>--cache-path</code>            | CACHE_PATH                | Change the default path ( <code>/root/.cache/volatility3</code> ) used to store the cache                      |                       |
| <code>--offline</code>               | —                         | Do not search online for additional JSON files                                                                 |                       |
| <code>-u</code>                      | URL                       | Search online for ISF json files                                                                               |                       |
| <code>--remote-isf-url</code>        | URL                       | Search online for ISF json files                                                                               |                       |
| <code>--filters</code>               | FILTERS                   | List of filters to apply to the output (in the form of <code>[+ -]columnname,pattern[!]</code> )               |                       |
| <code>--hide-columns</code>          | HIDE_COLUMNS ...          | Case-insensitive space separated list of prefixes to determine which columns to hide in the output if provided |                       |
| <code>-r</code>                      | RENDERER                  | Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)                    |                       |
| <code>--renderer</code>              | RENDERER                  | Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)                    |                       |
| <code>--single-location</code>       | SINGLE_LOCATION           | Specifies a base location on which to stack                                                                    |                       |
| <code>--stackers</code>              | STACKERS ...              | List of stackers                                                                                               |                       |
| <code>--single-swap-locations</code> | SINGLE_SWAP_LOCATIONS ... | Specifies a list of swap layer URIs for use with single-location                                               |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`vol`, `volshell`

# volshell

**Kit:** SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Volshell (Volatility 3 Framework) 2.28.0

## Synopsis

```
volshell [-h] [-c CONFIG] [-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS]
[-v] [-o OUTPUT_DIR] [-q] [--log LOG] [-f FILE]
[--write-config] [--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL] [-w | -l | -m]
[--single-location SINGLE_LOCATION]
[--stackers [STACKERS ...]]
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 38 options parsed from the captured help text. The final column is filled in by review.

| Flag          | Argument    | What it does                                             | When you would use it |
|---------------|-------------|----------------------------------------------------------|-----------------------|
| -h            | —           | show this help message and exit                          |                       |
| --help        | —           | show this help message and exit                          |                       |
| -c            | CONFIG      | Load the configuration from a json file                  |                       |
| --config      | CONFIG      | Load the configuration from a json file                  |                       |
| -e            | EXTEND      | Extend the configuration with a new (or changed) setting |                       |
| --extend      | EXTEND      | Extend the configuration with a new (or changed) setting |                       |
| -p            | PLUGIN_DIRS | Semi-colon separated list of paths to find plugins       |                       |
| --plugin-dirs | PLUGIN_DIRS | Semi-colon separated list of paths to find plugins       |                       |

| Flag                                 | Argument                  | What it does                                                                              | When you would use it |
|--------------------------------------|---------------------------|-------------------------------------------------------------------------------------------|-----------------------|
| <code>-s</code>                      | SYMBOL_DIRS               | Semi-colon separated list of paths to find symbols                                        |                       |
| <code>--symbol-dirs</code>           | SYMBOL_DIRS               | Semi-colon separated list of paths to find symbols                                        |                       |
| <code>-v</code>                      | —                         | Increase output verbosity                                                                 |                       |
| <code>--verbosity</code>             | —                         | Increase output verbosity                                                                 |                       |
| <code>-o</code>                      | OUTPUT_DIR                | Directory in which to output any generated files                                          |                       |
| <code>--output-dir</code>            | OUTPUT_DIR                | Directory in which to output any generated files                                          |                       |
| <code>-q</code>                      | —                         | Remove progress feedback                                                                  |                       |
| <code>--quiet</code>                 | —                         | Remove progress feedback                                                                  |                       |
| <code>--log</code>                   | LOG                       | Log output to a file as well as the console                                               |                       |
| <code>-f</code>                      | FILE                      | Shorthand for <code>--single-location=file://</code> if single-location is not defined    |                       |
| <code>--file</code>                  | FILE                      | Shorthand for <code>--single-location=file://</code> if single-location is not defined    |                       |
| <code>--write-config</code>          | —                         | Write configuration JSON file out to config.json                                          |                       |
| <code>--save-config</code>           | SAVE_CONFIG               | Save configuration JSON file to a file                                                    |                       |
| <code>--clear-cache</code>           | —                         | Clears out all short-term cached items                                                    |                       |
| <code>--cache-path</code>            | CACHE_PATH                | Change the default path ( <code>/root/.cache/volatility3</code> ) used to store the cache |                       |
| <code>--offline</code>               | —                         | Do not search online for additional JSON files                                            |                       |
| <code>-u</code>                      | URL                       | Search online for ISF json files                                                          |                       |
| <code>--remote-isf-url</code>        | URL                       | Search online for ISF json files                                                          |                       |
| <code>-w</code>                      | —                         | Run a Windows volshell                                                                    |                       |
| <code>--windows</code>               | —                         | Run a Windows volshell                                                                    |                       |
| <code>-l</code>                      | —                         | Run a Linux volshell                                                                      |                       |
| <code>--linux</code>                 | —                         | Run a Linux volshell                                                                      |                       |
| <code>-m</code>                      | —                         | Run a Mac volshell                                                                        |                       |
| <code>--mac</code>                   | —                         | Run a Mac volshell                                                                        |                       |
| <code>--single-location</code>       | SINGLE_LOCATION           | Specifies a base location on which to stack                                               |                       |
| <code>--stackers</code>              | STACKERS ...              | List of stackers                                                                          |                       |
| <code>--single-swap-locations</code> | SINGLE_SWAP_LOCATIONS ... | Specifies a list of swap layer URIs for use with single-location                          |                       |
| <code>--script</code>                | SCRIPT                    | File to load and execute at start                                                         |                       |
| <code>--script-only</code>           | —                         | Exit volshell after the script specified in <code>--script</code> completes               |                       |
| <code>--pid</code>                   | PID                       | Process ID                                                                                |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`vol`, `volatility3`

# arp-scan

---

**Kit:** SIFT Workstation **Capability:** Probe or scan hosts and services **Version:** arp-scan 1.10.0 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Target hosts must be specified on the command line unless the --file or

## Synopsis

---

```
arp-scan [options] [hosts...]
```

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

[nmap](#), [nping](#)

# capinfos

**Kit:** REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Read and filter packet captures **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Capture and analyze network traffic with this sniffer.

## Synopsis

```
capinfos [options] <infile> ...
```

## Common invocations

```
# from cyberlab 24-wireshark-deep
capinfos exercise/sample.pcap
```

## Options

All 4 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                  | When you would use it |
|-----------|----------|-------------------------------|-----------------------|
| -h        | —        | display this help and exit    |                       |
| --help    | —        | display this help and exit    |                       |
| -v        | —        | display version info and exit |                       |
| --version | —        | display version info and exit |                       |

## Gotchas

*TODO: operational traps.*

## See also

[tshark](#), [ngrep](#), [tcpflow](#)

# editcap

**Kit:** REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Capture and analyze network traffic with this sniffer.

## Synopsis

```
editcap [options] ... <infile> <outfile> [ <packet#>[<-<packet#>] ... ]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 25 options parsed from the captured help text. The final column is filled in by review.

| Flag     | Argument        | What it does                                                                                                                                                                       | When you would use it |
|----------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -r       | —               | keep the selected packets; default is to delete them.                                                                                                                              |                       |
| -A       | start time      | only read packets whose timestamp is after (or equal to) the given time.                                                                                                           |                       |
| -B       | stop time       | only read packets whose timestamp is before the given time. Time format for -A/-B options is YYYY-MM-DDThh:mm:ss[.nnnnnnnnn][Z + -hh:mm] Unix epoch timestamps are also supported. |                       |
| --novlan | —               | remove vlan info from packets before checking for duplicates.                                                                                                                      |                       |
| -d       | —               | remove packet if duplicate (window == 5).                                                                                                                                          |                       |
| -D       | dup window      | remove packet if duplicate; configurable . Valid values are 0 to 1000000. NOTE: A of 0 with -V (verbose option) is useful to print MD5 hashes.                                     |                       |
| -w       | dup time window | remove packet if duplicate packet is found EQUAL TO OR LESS THAN prior to current packet. A is specified in relative seconds (e.g. 0.000001). NOTE: The use of the '               |                       |
| -s       | snaplen         | truncate each packet to max. bytes of data.                                                                                                                                        |                       |
| -L       | —               | adjust the frame (i.e. reported) length when chopping and/or snapping.                                                                                                             |                       |



| Flag                                   | Argument         | What it does                                                                                                                                                                                             | When you would use it |
|----------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-t</code>                        | time adjustment  | adjust the timestamp of each packet. is in relative seconds (e.g. -0.5).                                                                                                                                 |                       |
| <code>-o</code>                        | change offset    | When used in conjunction with -E, skip some bytes from the beginning of the packet. This allows one to preserve some bytes, in order to have some headers untouched.                                     |                       |
| <code>--seed</code>                    | seed             | When used in conjunction with -E, set the seed to use for the pseudo-random number generator. This allows one to repeat a particular sequence of errors.                                                 |                       |
| <code>-I</code>                        | bytes to ignore  | ignore the specified number of bytes at the beginning of the frame during MD5 hash calculation, unless the frame is too short, then the full frame is used. Useful to remove duplicated packets taken on |                       |
| <code>-c</code>                        | packets per file | split the packet output to different files based on uniform packet counts with a maximum of each.                                                                                                        |                       |
| <code>-i</code>                        | seconds per file | split the packet output to different files based on uniform time intervals with a maximum of each.                                                                                                       |                       |
| <code>-F</code>                        | capture type     | set the output file type; default is pcapng. An empty "-F" option will list the file types.                                                                                                              |                       |
| <code>-T</code>                        | encap type       | set the output file encapsulation type; default is the same as the input file. An empty "-T" option will list the encapsulation types.                                                                   |                       |
| <code>--discard-all-secrets</code>     | —                | Discard all decryption secrets from the input file when writing the output file. Does not discard secrets added by "--inject-secrets" in the same command line.                                          |                       |
| <code>--capture-comment</code>         | comment          | Add a capture file comment, if supported.                                                                                                                                                                |                       |
| <code>--discard-capture-comment</code> | —                | Discard capture file comments from the input file when writing the output file. Does not discard comments added by "--capture-comment" in the same command line.                                         |                       |
| <code>-h</code>                        | —                | display this help and exit.                                                                                                                                                                              |                       |
| <code>--help</code>                    | —                | display this help and exit.                                                                                                                                                                              |                       |
| <code>-V</code>                        | —                | verbose output. If -V is used with any of the 'Duplicate Packet Removal' options (-d, -D or -w) then Packet lengths and MD5 hashes are printed to standard-error.                                        |                       |
| <code>-v</code>                        | —                | print version information and exit.                                                                                                                                                                      |                       |
| <code>--version</code>                 | —                | print version information and exit.                                                                                                                                                                      |                       |

## Gotchas

*TODO: operational traps.*

## See also

`mergcap`, `reordercap`

# inetsim

**Kit:** REMnux · Kali Linux **Capability:** Simulate network services for detonation **Version:** INetSim 1.3.2 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw](#) **Docs:** <https://www.inetsim.org/>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Emulate common network services and interact with malware.

## Synopsis

```
/usr/bin/inetsim [options]
```

## Common invocations

```
# from cyberlab 38-network-emulation
inetsim --version 2>&1 | head -n 3
# from cyberlab 38-network-emulation
sudo inetsim --config ~/lab38/inetsim.conf --data-dir /var/lib/inetsim --log-dir ~/lab38/log &
```

## Options

All 14 options parsed from the captured help text. The final column is filled in by review.

| Flag                        | Argument   | What it does                                                                                                | When you would use it |
|-----------------------------|------------|-------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--help</code>         | —          | Print this help message.                                                                                    |                       |
| <code>--version</code>      | —          | Show version information.                                                                                   |                       |
| <code>--config</code>       | filename   | Configuration file to use.                                                                                  |                       |
| <code>--log-dir</code>      | directory  | Directory logfiles are written to.                                                                          |                       |
| <code>--data-dir</code>     | directory  | Directory containing service data.                                                                          |                       |
| <code>--report-dir</code>   | directory  | Directory reports are written to.                                                                           |                       |
| <code>--bind-address</code> | IP address | Default IP address to bind services to. Overrides configuration option 'default_bind_address'.              |                       |
| <code>--max-childs</code>   | num        | Default maximum number of child processes per service. Overrides configuration option 'default_max_childs'. |                       |

| Flag                               | Argument | What it does                                                                                              | When you would use it |
|------------------------------------|----------|-----------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--user</code>                | username | Default user to run services. Overrides configuration option 'default_run_as_user'.                       |                       |
| <code>--faketime-init-delta</code> | secs     | Initial faketime delta (seconds). Overrides configuration option 'faketime_init_delta'.                   |                       |
| <code>--faketime-auto-delay</code> | secs     | Delay for auto incrementing faketime (seconds). Overrides configuration option 'faketime_auto_delay'.     |                       |
| <code>--faketime-auto-incr</code>  | secs     | Delta for auto incrementing faketime (seconds). Overrides configuration option 'faketime_auto_increment'. |                       |
| <code>--session</code>             | id       | Session id to use. Defaults to main process id.                                                           |                       |
| <code>--pidfile</code>             | filename | Pid file to use. Defaults to '/var/run/inetsim.pid'.                                                      |                       |

## Gotchas

---

*TODO: operational traps.*

# mergecap

**Kit:** REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.wireshark.org>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Capture and analyze network traffic with this sniffer.

## Synopsis

```
mergecap [options] -w <outfile>|- <infile> [<infile> ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 7 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                                                                        | When you would use it |
|-----------|----------|-------------------------------------------------------------------------------------|-----------------------|
| -a        | —        | concatenate rather than merge files. default is to merge based on frame timestamps. |                       |
| -s        | snaplen  | truncate packets to bytes of data.                                                  |                       |
| -h        | —        | display this help and exit.                                                         |                       |
| --help    | —        | display this help and exit.                                                         |                       |
| -V        | —        | verbose output.                                                                     |                       |
| -v        | —        | print version information and exit.                                                 |                       |
| --version | —        | print version information and exit.                                                 |                       |

## Gotchas

*TODO: operational traps.*

## See also

---

`editcap`, `reordercap`

# ngrep

**Kit:** REMnux · SIFT Workstation **Capability:** Read and filter packet captures **Version:** V1.47.1 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://github.com/jpr5/ngrep/>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Look for patterns in network traffic.

## Synopsis

```
ngrep <-hNXViqpevxLDtTRM> <-IO pcap_dump> <-n num> <-d dev> <-A num>
<-s snaplen> <-S limitlen> <-W normal|byline|single|none> <-c cols>
<-P char> <-F file> <-K count>
<match expression> <bpf filter>
```

## Common invocations

```
# from cyberlab 07-network-pcap
ngrep -V 2>&1 | head -n 1
# from cyberlab 07-network-pcap
ngrep -I exercise/sample.pcap -q -W byline 'User-Agent'
# from cyberlab 07-network-pcap
ngrep -I exercise/sample.pcap -q -W byline 'User-Agent' | grep -i 'User-Agent'
# from cyberlab 24-wireshark-deep
ngrep -I exercise/sample.pcap -q -W byline "User-Agent"
# from cyberlab 24-wireshark-deep
ngrep -I exercise/sample.pcap -q -W byline "User-Agent" | grep -i "User-Agent"
```

## Options

All 29 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does           | When you would use it |
|------|----------|------------------------|-----------------------|
| -h   | —        | is help/usage          |                       |
| -V   | —        | is version information |                       |

| Flag            | Argument | What it does                                                                                     | When you would use it |
|-----------------|----------|--------------------------------------------------------------------------------------------------|-----------------------|
| <code>-q</code> | —        | is be quiet (don't print packet reception hash marks)                                            |                       |
| <code>-e</code> | —        | is show empty packets                                                                            |                       |
| <code>-i</code> | —        | is ignore case                                                                                   |                       |
| <code>-v</code> | —        | is invert match                                                                                  |                       |
| <code>-R</code> | —        | is don't do privilege revocation logic                                                           |                       |
| <code>-x</code> | —        | is print in alternate hexdump format                                                             |                       |
| <code>-X</code> | —        | is interpret match expression as hexadecimal                                                     |                       |
| <code>-w</code> | —        | is word-regex (expression must match as a word)                                                  |                       |
| <code>-p</code> | —        | is don't go into promiscuous mode                                                                |                       |
| <code>-l</code> | —        | is make stdout line buffered                                                                     |                       |
| <code>-D</code> | —        | is replay pcap_dumps with their recorded time intervals                                          |                       |
| <code>-t</code> | —        | is print timestamp every time a packet is matched                                                |                       |
| <code>-T</code> | —        | is print delta timestamp every time a packet is matched specify twice for delta from first match |                       |
| <code>-M</code> | —        | is don't do multi-line match (do single-line match instead)                                      |                       |
| <code>-I</code> | —        | is read packet stream from pcap format file pcap_dump                                            |                       |
| <code>-O</code> | —        | is dump matched packets in pcap format to pcap_dump                                              |                       |
| <code>-n</code> | —        | is look at only num packets                                                                      |                       |
| <code>-A</code> | —        | is dump num packets after a match                                                                |                       |
| <code>-s</code> | —        | is set the bpf caplen                                                                            |                       |
| <code>-S</code> | —        | is set the limitlen on matched packets                                                           |                       |

| Flag            | Argument | What it does                                               | When you would use it |
|-----------------|----------|------------------------------------------------------------|-----------------------|
| <code>-W</code> | —        | is set the dump format (normal, byline, single, none)      |                       |
| <code>-C</code> | —        | is force the column width to the specified size            |                       |
| <code>-P</code> | —        | is set the non-printable display char to what is specified |                       |
| <code>-F</code> | —        | is read the bpf filter from the specified file             |                       |
| <code>-N</code> | —        | is show sub protocol number                                |                       |
| <code>-d</code> | —        | is use specified device instead of the pcap default        |                       |
| <code>-K</code> | —        | is send N packets to kill observed connections             |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`tshark`, `capinfos`, `tcpflow`



# nmap

**Kit:** Kali Linux · FLARE-VM **Capability:** Probe or scan hosts and services **Version:** Nmap version 7.93 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

← [Capability index](#) · [Kit tool list](#)

## Purpose

Nmap 7.93 ( <https://nmap.org> )

## Synopsis

```
nmap [Scan Type(s)] [Options] {target specification}
```

## Common invocations

```
# from cyberlab 11-offensive-kali
nmap --version
# from cyberlab 11-offensive-kali
nmap -sT -Pn -p 1-1024 127.0.0.1
# from cyberlab 26-metasploit-workflow
nmap -sV -Pn -p 1-1000 -oN scan.txt "$TARGET"
# from cyberlab 26-metasploit-workflow
nmap -sV -Pn -oX scan.xml "$TARGET"
# from cyberlab 41-web-app-testing
nmap -sV -p 80,443,8080 "$TARGET"
# from cyberlab 41-web-app-testing
nmap -p 80,443 --script http-title,http-headers,http-server-header "$TARGET"
# from cyberlab 41-web-app-testing
nmap -p 8000 --script http-title 127.0.0.1
```

## Options

All 66 options parsed from the captured help text. The final column is filled in by review.

| Flag          | Argument                  | What it does           | When you would use it |
|---------------|---------------------------|------------------------|-----------------------|
| --exclude     | host1[,host2][,host3],... | Exclude hosts/networks |                       |
| --excludefile | exclude_file              | Exclude list from file |                       |

| Flag                             | Argument          | What it does                                                                             | When you would use it |
|----------------------------------|-------------------|------------------------------------------------------------------------------------------|-----------------------|
| <code>--dns-servers</code>       | serv1[,serv2],... | Specify custom DNS servers                                                               |                       |
| <code>--system-dns</code>        | —                 | Use OS's DNS resolver                                                                    |                       |
| <code>--traceroute</code>        | —                 | Trace hop path to each host                                                              |                       |
| <code>--scanflags</code>         | flags             | Customize TCP scan flags                                                                 |                       |
| <code>-b</code>                  | FTP relay host    | FTP bounce scan                                                                          |                       |
| <code>-p</code>                  | port ranges       | Only scan specified ports Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9   |                       |
| <code>--exclude-ports</code>     | port ranges       | Exclude the specified ports from scanning                                                |                       |
| <code>-F</code>                  | —                 | Fast mode - Scan fewer ports than the default scan                                       |                       |
| <code>-r</code>                  | —                 | Scan ports sequentially - don't randomize                                                |                       |
| <code>--top-ports</code>         | number            | Scan most common ports                                                                   |                       |
| <code>--port-ratio</code>        | ratio             | Scan ports more common than                                                              |                       |
| <code>--version-intensity</code> | level             | Set from 0 (light) to 9 (try all probes)                                                 |                       |
| <code>--version-light</code>     | —                 | Limit to most likely probes (intensity 2)                                                |                       |
| <code>--version-all</code>       | —                 | Try every single probe (intensity 9)                                                     |                       |
| <code>--version-trace</code>     | —                 | Show detailed version scan activity (for debugging)                                      |                       |
| <code>--script</code>            | Lua scripts       | is a comma separated list of directories, script-files or script-categories              |                       |
| <code>--script-args</code>       | n1=v1,[n2=v2,...] | provide arguments to scripts                                                             |                       |
| <code>--script-args-file</code>  | filename          | provide NSE script args in a file                                                        |                       |
| <code>--script-trace</code>      | —                 | Show all data sent and received                                                          |                       |
| <code>--script-updatedb</code>   | —                 | Update the script database.                                                              |                       |
| <code>--script-help</code>       | Lua scripts       | Show help about scripts. is a comma-separated list of script-files or script-categories. |                       |
| <code>-O</code>                  | —                 | Enable OS detection                                                                      |                       |
| <code>--osscan-limit</code>      | —                 | Limit OS detection to promising targets                                                  |                       |
| <code>--osscan-guess</code>      | —                 | Guess OS more aggressively                                                               |                       |
| <code>-T</code>                  | 0-5               | Set timing template (higher is faster)                                                   |                       |
| <code>--min-hostgroup</code>     | —                 | Parallel host scan group sizes                                                           |                       |
| <code>--min-parallelism</code>   | —                 | Probe parallelization                                                                    |                       |
| <code>--min-rtt-timeout</code>   | —                 | Specifies probe round trip time.                                                         |                       |

| Flag                          | Argument               | What it does                                                            | When you would use it |
|-------------------------------|------------------------|-------------------------------------------------------------------------|-----------------------|
| <code>--max-retries</code>    | tries                  | Caps number of port scan probe retransmissions.                         |                       |
| <code>--host-timeout</code>   | time                   | Give up on target after this long                                       |                       |
| <code>--min-rate</code>       | number                 | Send packets no slower than per second                                  |                       |
| <code>--max-rate</code>       | number                 | Send packets no faster than per second                                  |                       |
| <code>-f</code>               | —                      | fragment packets (optionally w/given MTU)                               |                       |
| <code>--mtu</code>            | val                    | fragment packets (optionally w/given MTU)                               |                       |
| <code>-D</code>               | decoy1,decoy2[,ME],... | Cloak a scan with decoys                                                |                       |
| <code>-S</code>               | IP_Address             | Spoof source address                                                    |                       |
| <code>-e</code>               | iface                  | Use specified interface                                                 |                       |
| <code>--proxies</code>        | url1,[url2],...        | Relay connections through HTTP/SOCKS4 proxies                           |                       |
| <code>--data</code>           | hex string             | Append a custom payload to sent packets                                 |                       |
| <code>--data-string</code>    | string                 | Append a custom ASCII string to sent packets                            |                       |
| <code>--data-length</code>    | num                    | Append random data to sent packets                                      |                       |
| <code>--ip-options</code>     | options                | Send packets with specified ip options                                  |                       |
| <code>--ttl</code>            | val                    | Set IP time-to-live field                                               |                       |
| <code>--spoof-mac</code>      | mac address            | Spoof your MAC address                                                  |                       |
| <code>--badsum</code>         | —                      | Send packets with a bogus TCP/UDP/SCTP checksum                         |                       |
| <code>-v</code>               | —                      | Increase verbosity level (use -vv or more for greater effect)           |                       |
| <code>-d</code>               | —                      | Increase debugging level (use -dd or more for greater effect)           |                       |
| <code>--reason</code>         | —                      | Display the reason a port is in a particular state                      |                       |
| <code>--open</code>           | —                      | Only show open (or possibly open) ports                                 |                       |
| <code>--packet-trace</code>   | —                      | Show all packets sent and received                                      |                       |
| <code>--iflist</code>         | —                      | Print host interfaces and routes (for debugging)                        |                       |
| <code>--append-output</code>  | —                      | Append to rather than clobber specified output files                    |                       |
| <code>--resume</code>         | filename               | Resume an aborted scan                                                  |                       |
| <code>--noninteractive</code> | —                      | Disable runtime interactions via keyboard                               |                       |
| <code>--stylesheet</code>     | path                   | XSL stylesheet to transform XML output to HTML                          |                       |
| <code>--webxml</code>         | —                      | Reference stylesheet from Nmap.Org for more portable XML                |                       |
| <code>--no-stylesheet</code>  | —                      | Prevent associating of XSL stylesheet w/XML output                      |                       |
| <code>-6</code>               | —                      | Enable IPv6 scanning                                                    |                       |
| <code>-A</code>               | —                      | Enable OS detection, version detection, script scanning, and traceroute |                       |
| <code>--datadir</code>        | dirname                | Specify custom Nmap data file location                                  |                       |
| <code>--privileged</code>     | —                      | Assume that the user is fully privileged                                |                       |

| Flag                        | Argument | What it does                                | When you would use it |
|-----------------------------|----------|---------------------------------------------|-----------------------|
| <code>--unprivileged</code> | —        | Assume the user lacks raw socket privileges |                       |
| <code>-V</code>             | —        | Print version number                        |                       |
| <code>-h</code>             | —        | Print this help summary page.               |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`nping`, `arp-scan`

# nping

**Kit:** Kali Linux · FLARE-VM **Capability:** Probe or scan hosts and services **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Nping 0.7.93 ( <https://nmap.org/nping> )

## Synopsis

```
nping [Probe mode] [Options] {target specification}
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 85 options parsed from the captured help text. The final column is filled in by review.

| Flag                       | Argument   | What it does                                                | When you would use it |
|----------------------------|------------|-------------------------------------------------------------|-----------------------|
| <code>--tcp-connect</code> | —          | Unprivileged TCP connect probe mode.                        |                       |
| <code>--tcp</code>         | —          | TCP probe mode.                                             |                       |
| <code>--udp</code>         | —          | UDP probe mode.                                             |                       |
| <code>--icmp</code>        | —          | ICMP probe mode.                                            |                       |
| <code>--arp</code>         | —          | ARP/RARP probe mode.                                        |                       |
| <code>--tr</code>          | —          | Traceroute mode (can only be used with TCP/UDP/ICMP modes). |                       |
| <code>--traceroute</code>  | —          | Traceroute mode (can only be used with TCP/UDP/ICMP modes). |                       |
| <code>-p</code>            | port spec  | Set destination port(s).                                    |                       |
| <code>--dest-port</code>   | port spec  | Set destination port(s).                                    |                       |
| <code>-g</code>            | portnumber | Try to use a custom source port.                            |                       |
| <code>--source-port</code> | portnumber | Try to use a custom source port.                            |                       |
| <code>--seq</code>         | seqnumber  | Set sequence number.                                        |                       |

| Flag                                | Argument  | What it does                                                                    | When you would use it |
|-------------------------------------|-----------|---------------------------------------------------------------------------------|-----------------------|
| <code>--flags</code>                | flag list | Set TCP flags (ACK,PSH,RST,SYN,FIN...)                                          |                       |
| <code>--ack</code>                  | acknumber | Set ACK number.                                                                 |                       |
| <code>--win</code>                  | size      | Set window size.                                                                |                       |
| <code>--badsum</code>               | —         | Use a random invalid checksum.                                                  |                       |
| <code>--icmp-type</code>            | type      | ICMP type.                                                                      |                       |
| <code>--icmp-code</code>            | code      | ICMP code.                                                                      |                       |
| <code>--icmp-id</code>              | id        | Set identifier.                                                                 |                       |
| <code>--icmp-seq</code>             | n         | Set sequence number.                                                            |                       |
| <code>--icmp-redirect-addr</code>   | addr      | Set redirect address.                                                           |                       |
| <code>--icmp-param-pointer</code>   | pnt       | Set parameter problem pointer.                                                  |                       |
| <code>--icmp-advert-lifetime</code> | time      | Set router advertisement lifetime.                                              |                       |
| <code>--icmp-advert-entry</code>    | IP,pref   | Add router advertisement entry.                                                 |                       |
| <code>--icmp-orig-time</code>       | —         | : Set originate timestamp.                                                      |                       |
| <code>--icmp-recv-time</code>       | —         | : Set receive timestamp.                                                        |                       |
| <code>--icmp-trans-time</code>      | timestamp | Set transmit timestamp.                                                         |                       |
| <code>--arp-type</code>             | type      | Type: ARP, ARP-reply, RARP, RARP-reply.                                         |                       |
| <code>--arp-sender-mac</code>       | mac       | Set sender MAC address.                                                         |                       |
| <code>--arp-sender-ip</code>        | —         | : Set sender IP address.                                                        |                       |
| <code>--arp-target-mac</code>       | mac       | Set target MAC address.                                                         |                       |
| <code>--arp-target-ip</code>        | —         | : Set target IP address.                                                        |                       |
| <code>-S</code>                     | —         | Set source IP address.                                                          |                       |
| <code>--source-ip</code>            | —         | Set source IP address.                                                          |                       |
| <code>--dest-ip</code>              | addr      | Set destination IP address (used as an alternative to {target specification} ). |                       |
| <code>--tos</code>                  | tos       | Set type of service field (8bits).                                              |                       |
| <code>--id</code>                   | —         | : Set identification field (16 bits).                                           |                       |
| <code>--df</code>                   | —         | Set Don't Fragment flag.                                                        |                       |
| <code>--mf</code>                   | —         | Set More Fragments flag.                                                        |                       |
| <code>--evil</code>                 | —         | Set Reserved / Evil flag.                                                       |                       |
| <code>--ttl</code>                  | hops      | Set time to live [0-255].                                                       |                       |
| <code>--badsum-ip</code>            | —         | Use a random invalid checksum.                                                  |                       |

| Flag                         | Argument                         | What it does                                            | When you would use it |
|------------------------------|----------------------------------|---------------------------------------------------------|-----------------------|
| <code>--ip-options</code>    | S R [route]]L<br>[route]]T U ... | Set IP options                                          |                       |
| <code>--mtu</code>           | size                             | Set MTU. Packets get fragmented if MTU is small enough. |                       |
| <code>-6</code>              | —                                | Use IP version 6.                                       |                       |
| <code>--IPv6</code>          | —                                | Use IP version 6.                                       |                       |
| <code>--hop-limit</code>     | —                                | Set hop limit (same as IPv4 TTL).                       |                       |
| <code>--traffic-class</code> | class                            | : Set traffic class.                                    |                       |
| <code>--flow</code>          | label                            | Set flow label.                                         |                       |
| <code>--dest-mac</code>      | mac                              | Set destination mac address. (Disables ARP resolution)  |                       |
| <code>--source-mac</code>    | mac                              | Set source MAC address.                                 |                       |
| <code>--ether-type</code>    | type                             | Set EtherType value.                                    |                       |
| <code>--data</code>          | hex string                       | Include a custom payload.                               |                       |
| <code>--data-string</code>   | text                             | Include a custom ASCII text.                            |                       |
| <code>--data-length</code>   | len                              | Include len random bytes as payload.                    |                       |
| <code>--echo-client</code>   | passphrase                       | Run Nping in client mode.                               |                       |
| <code>--echo-server</code>   | passphrase                       | Run Nping in server mode.                               |                       |
| <code>--echo-port</code>     | port                             | Use custom to listen or connect.                        |                       |
| <code>--no-crypto</code>     | —                                | Disable encryption and authentication.                  |                       |
| <code>--once</code>          | —                                | Stop the server after one connection.                   |                       |
| <code>--safe-payloads</code> | —                                | Erase application data in echoed packets.               |                       |
| <code>--delay</code>         | time                             | Adjust delay between probes.                            |                       |
| <code>--rate</code>          | —                                | : Send num packets per second.                          |                       |
| <code>-h</code>              | —                                | Display help information.                               |                       |
| <code>--help</code>          | —                                | Display help information.                               |                       |
| <code>-V</code>              | —                                | Display current version number.                         |                       |
| <code>--version</code>       | —                                | Display current version number.                         |                       |
| <code>-c</code>              | n                                | Stop after rounds.                                      |                       |
| <code>--count</code>         | n                                | Stop after rounds.                                      |                       |
| <code>-e</code>              | name                             | Use supplied network interface.                         |                       |
| <code>--interface</code>     | name                             | Use supplied network interface.                         |                       |
| <code>-H</code>              | —                                | Do not display sent packets.                            |                       |
| <code>--hide-sent</code>     | —                                | Do not display sent packets.                            |                       |
| <code>-N</code>              | —                                | Do not try to capture replies.                          |                       |
| <code>--no-capture</code>    | —                                | Do not try to capture replies.                          |                       |

| Flag                        | Argument    | What it does                              | When you would use it |
|-----------------------------|-------------|-------------------------------------------|-----------------------|
| <code>--privileged</code>   | —           | Assume user is fully privileged.          |                       |
| <code>--unprivileged</code> | —           | Assume user lacks raw socket privileges.  |                       |
| <code>--send-eth</code>     | —           | Send packets at the raw Ethernet layer.   |                       |
| <code>--send-ip</code>      | —           | Send packets using raw IP sockets.        |                       |
| <code>--bpf-filter</code>   | filter spec | Specify custom BPF filter.                |                       |
| <code>-v</code>             | —           | Increment verbosity level by one.         |                       |
| <code>-d</code>             | —           | Increment debugging level by one.         |                       |
| <code>-q</code>             | —           | Decrease verbosity level by one.          |                       |
| <code>--quiet</code>        | —           | Set verbosity and debug level to minimum. |                       |
| <code>--debug</code>        | —           | Set verbosity and debug to the max level. |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`nmap`, `arp-scan`



# reordercap

**Kit:** REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-01 **raw Docs:** <https://www.wireshark.org>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Capture and analyze network traffic with this sniffer.

## Synopsis

```
reordercap [options] <infile> <outfile>
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 3 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                             | When you would use it |
|------|----------|----------------------------------------------------------|-----------------------|
| -n   | —        | don't write to output file if the input file is ordered. |                       |
| -h   | —        | display this help and exit.                              |                       |
| -v   | —        | print version information and exit.                      |                       |

## Gotchas

*TODO: operational traps.*

## See also

[editcap](#), [mergecap](#)

# tcpflow

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Read and filter packet captures; Extract files and payloads from traffic **Version:** TCPFLOW 1.6.1 **Captured:** cyberlab-aio via `--help` on 2026-08-01 [raw Docs:](#) <https://downloads.digitalcorpora.org/downloads/tcpflow/>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Analyze the flow of network traffic.

## Synopsis

```
tcpflow [-aBcCDhIpsvVZ] [-b max_bytes] [-d debug_level]
[-[eE] scanner] [-f max_fds] [-F[ctTXMkmg]] [-h|--help] [-i iface]
[-l files...] [-L semlock] [-m min_bytes] [-o outdir] [-r file] [-R file]
[-S name=value] [-T template] [-U|--relinquish-privileges user] [-v|--verbose]
[-w file] [-x scanner] [-X xmlfile] [-z|--chroot dir] [expression]
```

## Common invocations

```
# from cyberlab 07-network-pcap
tcpflow --version | head -n 1
```

## Options

All 33 options parsed from the captured help text. The final column is filled in by review.

| Flag            | Argument                 | What it does                                                                | When you would use it |
|-----------------|--------------------------|-----------------------------------------------------------------------------|-----------------------|
| <code>-a</code> | —                        | do ALL post-processing.                                                     |                       |
| <code>-b</code> | <code>max_bytes</code>   | max number of bytes per flow to save                                        |                       |
| <code>-d</code> | <code>debug_level</code> | debug level; default is 1                                                   |                       |
| <code>-f</code> | —                        | maximum number of file descriptors to use                                   |                       |
| <code>-h</code> | —                        | print this help message (-hh for more help)                                 |                       |
| <code>-H</code> | —                        | print detailed information about each scanner                               |                       |
| <code>-i</code> | —                        | network interface on which to listen                                        |                       |
| <code>-I</code> | —                        | write for each flow another file *.findx to provide byte-indexed timestamps |                       |
| <code>-g</code> | —                        | output each flow in alternating colors (note change!)                       |                       |

| Flag            | Argument | What it does                                                          | When you would use it |
|-----------------|----------|-----------------------------------------------------------------------|-----------------------|
| <code>-l</code> | —        | treat non-flag arguments as input files rather than a pcap expression |                       |
| <code>-L</code> | —        | semlock - specifies that writes are locked using a named semaphore    |                       |
| <code>-p</code> | —        | don't use promiscuous mode                                            |                       |
| <code>-q</code> | —        | quiet mode - do not print warnings                                    |                       |
| <code>-r</code> | file     | read packets from tcpdump pcap file (may be repeated)                 |                       |
| <code>-R</code> | file     | read packets from tcpdump pcap file TO FINISH CONNECTIONS             |                       |
| <code>-v</code> | —        | verbose operation equivalent to -d 10                                 |                       |
| <code>-V</code> | —        | print version number and exit                                         |                       |
| <code>-w</code> | —        | file : write packets not processed to file                            |                       |
| <code>-o</code> | —        | outdir : specify output directory (default '.')                       |                       |
| <code>-X</code> | —        | filename : DFXML output to filename                                   |                       |
| <code>-m</code> | —        | bytes : specifies skip that starts a new stream (default 16777216).   |                       |
| <code>-Z</code> | —        | do not decompress gzip-compressed HTTP transactions                   |                       |
| <code>-K</code> | —        | output keep pcap flow structure.                                      |                       |
| <code>-U</code> | user     | relinquish privileges and become user (if running as root)            |                       |
| <code>-z</code> | dir      | chroot to dir (requires that -U be used).                             |                       |
| <code>-E</code> | scanner  | - turn off all scanners except scanner                                |                       |
| <code>-B</code> | —        | binary output, even with -c or -C (normally -c or -C turn it off)     |                       |
| <code>-c</code> | —        | console print only (don't create files)                               |                       |
| <code>-C</code> | —        | console print only, but without the display of source/dest header     |                       |
| <code>-0</code> | —        | don't print newlines after packets when printing to console           |                       |
| <code>-s</code> | —        | strip non-printable characters (change to '.')                        |                       |
| <code>-J</code> | —        | output json format.                                                   |                       |
| <code>-D</code> | —        | output in hex (useful to combine with -c or -C)                       |                       |

## Gotchas

*TODO: operational traps.*

## See also

`tshark`, `capinfos`, `ngrep`, `tcpxtract`, `foremost`

# curl

**Kit:** REMnux **Capability:** Fetch and verify external references **Version:** curl 7.88.1 **Captured:** cyberlab-ai0 via --help on 2026-08-01 **raw Docs:** <https://curl.se>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool.

## Synopsis

```
curl [options...] <url>
```

## Common invocations

```
# from cyberlab 38-network-emulation
curl -s http://127.0.0.1/malware.bin -o /tmp/served.bin && file /tmp/served.bin
# from cyberlab 38-network-emulation
curl -s "http://$TARGET_DOMAIN/gate.php?id=203.0.113.10" -o /tmp/beacon_reply.bin
```

## Options

All 22 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                                    | When you would use it |
|-----------|----------|-------------------------------------------------|-----------------------|
| -d        | data     | HTTP POST data                                  |                       |
| --data    | data     | HTTP POST data                                  |                       |
| -f        | —        | Fail fast with no output on HTTP errors         |                       |
| --fail    | —        | Fail fast with no output on HTTP errors         |                       |
| -h        | category | Get help for commands                           |                       |
| --help    | category | Get help for commands                           |                       |
| -i        | —        | Include protocol response headers in the output |                       |
| --include | —        | Include protocol response headers in the output |                       |
| -o        | file     | Write to file instead of stdout                 |                       |
| --output  | file     | Write to file instead of stdout                 |                       |
| -O        | —        | Write output to a file named as the remote file |                       |

| Flag                       | Argument | What it does                                    | When you would use it |
|----------------------------|----------|-------------------------------------------------|-----------------------|
| <code>--remote-name</code> | —        | Write output to a file named as the remote file |                       |
| <code>-s</code>            | —        | Silent mode                                     |                       |
| <code>--silent</code>      | —        | Silent mode                                     |                       |
| <code>-T</code>            | file     | Transfer local FILE to destination              |                       |
| <code>--upload-file</code> | file     | Transfer local FILE to destination              |                       |
| <code>-A</code>            | name     | Send User-Agent to server                       |                       |
| <code>--user-agent</code>  | name     | Send User-Agent to server                       |                       |
| <code>-v</code>            | —        | Make the operation more talkative               |                       |
| <code>--verbose</code>     | —        | Make the operation more talkative               |                       |
| <code>-V</code>            | —        | Show version number and quit                    |                       |
| <code>--version</code>     | —        | Show version number and quit                    |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

[wget](#)

# ezhexviewer

---

**Kit:** REMnux **Capability:** Inspect files by hand **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

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## Purpose

---

Microsoft Office OLE2 compound documents.

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`xxd`, `less`



| Flag                    | Argument                            | What it does                                                                 | When you would use it |
|-------------------------|-------------------------------------|------------------------------------------------------------------------------|-----------------------|
| <code>-I</code>         | —                                   | ..... --IGNORE-CASE Ignore case in all searches.                             |                       |
| <code>-j</code>         | <code>_ON</code>                    | ... --jump-target=[_ON] Screen position of target lines.                     |                       |
| <code>-J</code>         | —                                   | ..... --status-column Display a status column at left edge of screen.        |                       |
| <code>-k</code>         | <code>□/□i_□l_□e</code>             | . --lesskey-file=[□/□i_□l_□e] Use a lesskey file.                            |                       |
| <code>-K</code>         | —                                   | ..... --quit-on-intr Exit less in response to ctrl-C.                        |                       |
| <code>-L</code>         | —                                   | ..... --no-lessopen Ignore the LESSOPEN environment variable.                |                       |
| <code>-m</code>         | —                                   | ... --long-prompt --LONG-PROMPT Set prompt style.                            |                       |
| <code>-M</code>         | —                                   | ... --long-prompt --LONG-PROMPT Set prompt style.                            |                       |
| <code>-n</code>         | —                                   | ... --line-numbers --LINE-NUMBERS Don't use line numbers.                    |                       |
| <code>-N</code>         | —                                   | ... --line-numbers --LINE-NUMBERS Don't use line numbers.                    |                       |
| <code>-o</code>         | <code>□/□i_□l_□e</code>             | . --log-file=[□/□i_□l_□e] Copy to log file (standard input only).            |                       |
| <code>-O</code>         | <code>□/□i_□l_□e</code>             | . --LOG-FILE=[□/□i_□l_□e] Copy to log file (unconditionally overwrite).      |                       |
| <code>-p</code>         | <code>□p□a_□t_□t_□e_□r_□n</code>    | Start at pattern (from command line).                                        |                       |
| <code>--pattern</code>  | <code>□p□a_□t_□t_□e_□r_□n</code>    | Start at pattern (from command line).                                        |                       |
| <code>-P</code>         | <code>□p□r_□o_□m_□p_□t</code>       | Define new prompt.                                                           |                       |
| <code>--prompt</code>   | <code>□p□r_□o_□m_□p_□t</code>       | Define new prompt.                                                           |                       |
| <code>-q</code>         | —                                   | ... --quiet --QUIET --silent --SILENT Quiet the terminal bell.               |                       |
| <code>-Q</code>         | —                                   | ... --quiet --QUIET --silent --SILENT Quiet the terminal bell.               |                       |
| <code>-r</code>         | —                                   | ... --raw-control-chars --RAW-CONTROL-CHARS Output "raw" control characters. |                       |
| <code>-R</code>         | —                                   | ... --raw-control-chars --RAW-CONTROL-CHARS Output "raw" control characters. |                       |
| <code>-s</code>         | —                                   | ..... --squeeze-blank-lines Squeeze multiple blank lines.                    |                       |
| <code>-S</code>         | —                                   | ..... --chop-long-lines Chop (truncate) long lines rather than wrapping.     |                       |
| <code>-t</code>         | <code>□t□a_□g</code>                | .. --tag=[□t□a_□g] Find a tag.                                               |                       |
| <code>-T</code>         | <code>□t□a_□g_□s_□f_□i_□l_□e</code> | Use an alternate tags file.                                                  |                       |
| <code>--tag-file</code> | <code>□t□a_□g_□s_□f_□i_□l_□e</code> | Use an alternate tags file.                                                  |                       |
| <code>-u</code>         | —                                   | ... --underline-special --UNDERLINE-SPECIAL Change handling of backspaces.   |                       |
| <code>-U</code>         | —                                   | ... --underline-special --UNDERLINE-SPECIAL Change handling of backspaces.   |                       |
| <code>-V</code>         | —                                   | ..... --version Display the version number of "less".                        |                       |



| Flag                            | Argument         | What it does                                                                               | When you would use it |
|---------------------------------|------------------|--------------------------------------------------------------------------------------------|-----------------------|
| <code>-w</code>                 | —                | ..... --hilite-unread Highlight first new line after forward-screen.                       |                       |
| <code>-W</code>                 | —                | ..... --HILITE-UNREAD Highlight first new line after any forward movement.                 |                       |
| <code>-X</code>                 | —                | ..... --no-init Don't use termcap init/deinit strings.                                     |                       |
| <code>-y</code>                 | <code>_ON</code> | ... --max-forw-scroll=[ <code>_ON</code> ] Forward scroll limit.                           |                       |
| <code>-z</code>                 | <code>_ON</code> | ... --window=[ <code>_ON</code> ] Set size of window.                                      |                       |
| <code>-#</code>                 | <code>_ON</code> | ... --shift=[ <code>_ON</code> ] Set horizontal scroll amount (0 = one half screen width). |                       |
| <code>--file-size</code>        | —                | Automatically determine the size of the input file.                                        |                       |
| <code>--follow-name</code>      | —                | The F command changes files if the input file is renamed.                                  |                       |
| <code>--incsearch</code>        | —                | Search file as each pattern character is typed in.                                         |                       |
| <code>--line-num-width</code>   | <code>N</code>   | Set the width of the -N line number field to N characters.                                 |                       |
| <code>--mouse</code>            | —                | Enable mouse input.                                                                        |                       |
| <code>--no-keypad</code>        | —                | Don't send termcap keypad init/deinit strings.                                             |                       |
| <code>--no-histdups</code>      | —                | Remove duplicates from command history.                                                    |                       |
| <code>--rscroll</code>          | <code>C</code>   | Set the character used to mark truncated lines.                                            |                       |
| <code>--save-marks</code>       | —                | Retain marks across invocations of less.                                                   |                       |
| <code>--status-col-width</code> | <code>N</code>   | Set the width of the -J status column to N characters.                                     |                       |
| <code>--use-backslash</code>    | —                | Subsequent options use backslash as escape char.                                           |                       |
| <code>--use-color</code>        | —                | Enables colored text.                                                                      |                       |
| <code>--wheel-lines</code>      | <code>N</code>   | Each click of the mouse wheel moves N lines.                                               |                       |

## Gotchas

*TODO: operational traps.*

## See also

`xxd`, `ezhexviewer`

# wget

**Kit:** REMnux **Capability:** Fetch and verify external references **Version:** GNU Wget 1.21.3 **Captured:** cyberlab-aio  
via `--help` on 2026-08-01 **raw Docs:** <https://www.gnu.org/software/wget/>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Interact with servers via HTTP, HTTPS, FTP, and FTPS using this command-line tool.

## Synopsis

```
wget [OPTION]... [URL]...
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 185 options parsed from the captured help text. The final column is filled in by review.

| Flag                         | Argument | What it does                         | When you would use it |
|------------------------------|----------|--------------------------------------|-----------------------|
| <code>-V</code>              | —        | display the version of Wget and exit |                       |
| <code>--version</code>       | —        | display the version of Wget and exit |                       |
| <code>-h</code>              | —        | print this help                      |                       |
| <code>--help</code>          | —        | print this help                      |                       |
| <code>-b</code>              | —        | go to background after startup       |                       |
| <code>--background</code>    | —        | go to background after startup       |                       |
| <code>-e</code>              | COMMAND  | execute a '.wgetrc'-style command    |                       |
| <code>--execute</code>       | COMMAND  | execute a '.wgetrc'-style command    |                       |
| <code>-o</code>              | FILE     | log messages to FILE                 |                       |
| <code>--output-file</code>   | FILE     | log messages to FILE                 |                       |
| <code>-a</code>              | FILE     | append messages to FILE              |                       |
| <code>--append-output</code> | FILE     | append messages to FILE              |                       |
| <code>-d</code>              | —        | print lots of debugging information  |                       |

| Flag                                | Argument | What it does                                                              | When you would use it |
|-------------------------------------|----------|---------------------------------------------------------------------------|-----------------------|
| <code>--debug</code>                | —        | print lots of debugging information                                       |                       |
| <code>-q</code>                     | —        | quiet (no output)                                                         |                       |
| <code>--quiet</code>                | —        | quiet (no output)                                                         |                       |
| <code>-v</code>                     | —        | be verbose (this is the default)                                          |                       |
| <code>--verbose</code>              | —        | be verbose (this is the default)                                          |                       |
| <code>--report-speed</code>         | TYPE     | output bandwidth as TYPE. TYPE can be bits                                |                       |
| <code>-i</code>                     | FILE     | download URLs found in local or external FILE                             |                       |
| <code>--input-file</code>           | FILE     | download URLs found in local or external FILE                             |                       |
| <code>-F</code>                     | —        | treat input file as HTML                                                  |                       |
| <code>--force-html</code>           | —        | treat input file as HTML                                                  |                       |
| <code>-B</code>                     | URL      | resolves HTML input-file links (-i -F) relative to URL                    |                       |
| <code>--base</code>                 | URL      | resolves HTML input-file links (-i -F) relative to URL                    |                       |
| <code>--config</code>               | FILE     | specify config file to use                                                |                       |
| <code>--no-config</code>            | —        | do not read any config file                                               |                       |
| <code>--rejected-log</code>         | FILE     | log reasons for URL rejection to FILE                                     |                       |
| <code>-t</code>                     | NUMBER   | set number of retries to NUMBER (0 unlimits)                              |                       |
| <code>--tries</code>                | NUMBER   | set number of retries to NUMBER (0 unlimits)                              |                       |
| <code>--retry-connrefused</code>    | —        | retry even if connection is refused                                       |                       |
| <code>--retry-on-http-error</code>  | ERRORS   | comma-separated list of HTTP errors to retry                              |                       |
| <code>-O</code>                     | FILE     | write documents to FILE                                                   |                       |
| <code>--output-document</code>      | FILE     | write documents to FILE                                                   |                       |
| <code>--no-netrc</code>             | —        | don't try to obtain credentials from .netrc                               |                       |
| <code>-c</code>                     | —        | resume getting a partially-downloaded file                                |                       |
| <code>--continue</code>             | —        | resume getting a partially-downloaded file                                |                       |
| <code>--start-pos</code>            | OFFSET   | start downloading from zero-based position OFFSET                         |                       |
| <code>--progress</code>             | TYPE     | select progress gauge type                                                |                       |
| <code>--show-progress</code>        | —        | display the progress bar in any verbosity mode                            |                       |
| <code>-N</code>                     | —        | don't re-retrieve files unless newer than local                           |                       |
| <code>--timestamping</code>         | —        | don't re-retrieve files unless newer than local                           |                       |
| <code>--no-if-modified-since</code> | —        | don't use conditional if-modified-since get requests in timestamping mode |                       |

| Flag                                    | Argument | What it does                                                                                        | When you would use it |
|-----------------------------------------|----------|-----------------------------------------------------------------------------------------------------|-----------------------|
| <code>--no-use-server-timestamps</code> | —        | don't set the local file's timestamp by the one on the server                                       |                       |
| <code>-S</code>                         | —        | print server response                                                                               |                       |
| <code>--server-response</code>          | —        | print server response                                                                               |                       |
| <code>--spider</code>                   | —        | don't download anything                                                                             |                       |
| <code>-T</code>                         | SECONDS  | set all timeout values to SECONDS                                                                   |                       |
| <code>--timeout</code>                  | SECONDS  | set all timeout values to SECONDS                                                                   |                       |
| <code>--dns-timeout</code>              | SECS     | set the DNS lookup timeout to SECS                                                                  |                       |
| <code>--connect-timeout</code>          | SECS     | set the connect timeout to SECS                                                                     |                       |
| <code>--read-timeout</code>             | SECS     | set the read timeout to SECS                                                                        |                       |
| <code>-w</code>                         | SECONDS  | wait SECONDS between retrievals (applies if more than 1 URL is to be retrieved)                     |                       |
| <code>--wait</code>                     | SECONDS  | wait SECONDS between retrievals (applies if more than 1 URL is to be retrieved)                     |                       |
| <code>--waitretry</code>                | SECONDS  | wait 1..SECONDS between retries of a retrieval (applies if more than 1 URL is to be retrieved)      |                       |
| <code>--random-wait</code>              | —        | wait from 0.5WAIT...1.5WAIT secs between retrievals (applies if more than 1 URL is to be retrieved) |                       |
| <code>--no-proxy</code>                 | —        | explicitly turn off proxy                                                                           |                       |
| <code>-Q</code>                         | NUMBER   | set retrieval quota to NUMBER                                                                       |                       |
| <code>--quota</code>                    | NUMBER   | set retrieval quota to NUMBER                                                                       |                       |
| <code>--bind-address</code>             | ADDRESS  | bind to ADDRESS (hostname or IP) on local host                                                      |                       |
| <code>--limit-rate</code>               | RATE     | limit download rate to RATE                                                                         |                       |
| <code>--no-dns-cache</code>             | —        | disable caching DNS lookups                                                                         |                       |
| <code>--restrict-file-names</code>      | OS       | restrict chars in file names to ones OS allows                                                      |                       |
| <code>--ignore-case</code>              | —        | ignore case when matching files/directories                                                         |                       |
| <code>-4</code>                         | —        | connect only to IPv4 addresses                                                                      |                       |
| <code>--inet4-only</code>               | —        | connect only to IPv4 addresses                                                                      |                       |
| <code>-6</code>                         | —        | connect only to IPv6 addresses                                                                      |                       |
| <code>--inet6-only</code>               | —        | connect only to IPv6 addresses                                                                      |                       |
| <code>--prefer-family</code>            | FAMILY   | connect first to addresses of specified family, one of IPv6, IPv4, or none                          |                       |
| <code>--user</code>                     | USER     | set both ftp and http user to USER                                                                  |                       |
| <code>--password</code>                 | PASS     | set both ftp and http password to PASS                                                              |                       |

| Flag                                | Argument | What it does                                                                                                                                                  | When you would use it |
|-------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--ask-password</code>         | —        | prompt for passwords                                                                                                                                          |                       |
| <code>--use-askpass</code>          | COMMAND  | specify credential handler for requesting username and password. If no COMMAND is specified the WGET_ASKPASS or the SSH_ASKPASS environment variable is used. |                       |
| <code>--no-iri</code>               | —        | turn off IRI support                                                                                                                                          |                       |
| <code>--local-encoding</code>       | ENC      | use ENC as the local encoding for IRIs                                                                                                                        |                       |
| <code>--remote-encoding</code>      | ENC      | use ENC as the default remote encoding                                                                                                                        |                       |
| <code>--unlink</code>               | —        | remove file before clobber                                                                                                                                    |                       |
| <code>--xattr</code>                | —        | turn on storage of metadata in extended file attributes                                                                                                       |                       |
| <code>-x</code>                     | —        | force creation of directories                                                                                                                                 |                       |
| <code>--force-directories</code>    | —        | force creation of directories                                                                                                                                 |                       |
| <code>--protocol-directories</code> | —        | use protocol name in directories                                                                                                                              |                       |
| <code>-P</code>                     | PREFIX   | save files to PREFIX/..                                                                                                                                       |                       |
| <code>--directory-prefix</code>     | PREFIX   | save files to PREFIX/..                                                                                                                                       |                       |
| <code>--cut-dirs</code>             | NUMBER   | ignore NUMBER remote directory components                                                                                                                     |                       |
| <code>--http-user</code>            | USER     | set http user to USER                                                                                                                                         |                       |
| <code>--http-password</code>        | PASS     | set http password to PASS                                                                                                                                     |                       |
| <code>--no-cache</code>             | —        | disallow server-cached data                                                                                                                                   |                       |
| <code>--default-page</code>         | NAME     | change the default page name (normally this is 'index.html').                                                                                                 |                       |
| <code>-E</code>                     | —        | save HTML/CSS documents with proper extensions                                                                                                                |                       |
| <code>--adjust-extension</code>     | —        | save HTML/CSS documents with proper extensions                                                                                                                |                       |
| <code>--ignore-length</code>        | —        | ignore 'Content-Length' header field                                                                                                                          |                       |
| <code>--header</code>               | STRING   | insert STRING among the headers                                                                                                                               |                       |
| <code>--compression</code>          | TYPE     | choose compression, one of auto, gzip and none. (default: none)                                                                                               |                       |
| <code>--max-redirect</code>         | —        | maximum redirections allowed per page                                                                                                                         |                       |
| <code>--proxy-user</code>           | USER     | set USER as proxy username                                                                                                                                    |                       |
| <code>--proxy-password</code>       | PASS     | set PASS as proxy password                                                                                                                                    |                       |
| <code>--referer</code>              | URL      | include 'Referer: URL' header in HTTP request                                                                                                                 |                       |
| <code>--save-headers</code>         | —        | save the HTTP headers to file                                                                                                                                 |                       |
| <code>-U</code>                     | AGENT    | identify as AGENT instead of Wget/VERSION                                                                                                                     |                       |

| Flag                                | Argument   | What it does                                                                                                                                | When you would use it |
|-------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--user-agent</code>           | AGENT      | identify as AGENT instead of Wget/VERSION                                                                                                   |                       |
| <code>--no-http-keep-alive</code>   | —          | disable HTTP keep-alive (persistent connections)                                                                                            |                       |
| <code>--no-cookies</code>           | —          | don't use cookies                                                                                                                           |                       |
| <code>--load-cookies</code>         | FILE       | load cookies from FILE before session                                                                                                       |                       |
| <code>--save-cookies</code>         | FILE       | save cookies to FILE after session                                                                                                          |                       |
| <code>--keep-session-cookies</code> | —          | load and save session (non-permanent) cookies                                                                                               |                       |
| <code>--post-data</code>            | STRING     | use the POST method; send STRING as the data                                                                                                |                       |
| <code>--post-file</code>            | FILE       | use the POST method; send contents of FILE                                                                                                  |                       |
| <code>--method</code>               | HTTPMethod | use method "HTTPMethod" in the request                                                                                                      |                       |
| <code>--body-data</code>            | STRING     | send STRING as data. --method MUST be set                                                                                                   |                       |
| <code>--body-file</code>            | FILE       | send contents of FILE. --method MUST be set                                                                                                 |                       |
| <code>--content-disposition</code>  | —          | honor the Content-Disposition header when choosing local file names (EXPERIMENTAL)                                                          |                       |
| <code>--content-on-error</code>     | —          | output the received content on server errors                                                                                                |                       |
| <code>--auth-no-challenge</code>    | —          | send Basic HTTP authentication information without first waiting for the server's challenge                                                 |                       |
| <code>--secure-protocol</code>      | PR         | choose secure protocol, one of auto, SSLv2, SSLv3, TLSv1, TLSv1_1, TLSv1_2, TLSv1_3 and PFS                                                 |                       |
| <code>--https-only</code>           | —          | only follow secure HTTPS links                                                                                                              |                       |
| <code>--no-check-certificate</code> | —          | don't validate the server's certificate                                                                                                     |                       |
| <code>--certificate</code>          | FILE       | client certificate file                                                                                                                     |                       |
| <code>--certificate-type</code>     | TYPE       | client certificate type, PEM or DER                                                                                                         |                       |
| <code>--private-key</code>          | FILE       | private key file                                                                                                                            |                       |
| <code>--private-key-type</code>     | TYPE       | private key type, PEM or DER                                                                                                                |                       |
| <code>--ca-certificate</code>       | FILE       | file with the bundle of CAs                                                                                                                 |                       |
| <code>--ca-directory</code>         | DIR        | directory where hash list of CAs is stored                                                                                                  |                       |
| <code>--crl-file</code>             | FILE       | file with bundle of CRLs                                                                                                                    |                       |
| <code>--pinnedpubkey</code>         | FILE       | Public key (PEM/DER) file, or any number of base64 encoded sha256 hashes preceded by 'sha256/' and separated by ';', to verify peer against |                       |

| Flag                                      | Argument  | What it does                                                                                                                                                                                                           | When you would use it |
|-------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--ciphers</code>                    | STR       | Set the priority string (GnuTLS) or cipher list string (OpenSSL) directly. Use with care. This option overrides <code>--secure-protocol</code> . The format and syntax of this string depend on the specific SSL/TLS e |                       |
| <code>--no-hsts</code>                    | —         | disable HSTS                                                                                                                                                                                                           |                       |
| <code>--hsts-file</code>                  | —         | path of HSTS database (will override default)                                                                                                                                                                          |                       |
| <code>--ftp-user</code>                   | USER      | set ftp user to USER                                                                                                                                                                                                   |                       |
| <code>--ftp-password</code>               | PASS      | set ftp password to PASS                                                                                                                                                                                               |                       |
| <code>--no-remove-listing</code>          | —         | don't remove '.listing' files                                                                                                                                                                                          |                       |
| <code>--no-glob</code>                    | —         | turn off FTP file name globbing                                                                                                                                                                                        |                       |
| <code>--no-passive-ftp</code>             | —         | disable the "passive" transfer mode                                                                                                                                                                                    |                       |
| <code>--preserve-permissions</code>       | —         | preserve remote file permissions                                                                                                                                                                                       |                       |
| <code>--retr-symlinks</code>              | —         | when recursing, get linked-to files (not dir)                                                                                                                                                                          |                       |
| <code>--ftps-implicit</code>              | —         | use implicit FTPS (default port is 990)                                                                                                                                                                                |                       |
| <code>--ftps-resume-ssl</code>            | —         | resume the SSL/TLS session started in the control connection when opening a data connection                                                                                                                            |                       |
| <code>--ftps-clear-data-connection</code> | —         | cipher the control channel only; all the data will be in plaintext                                                                                                                                                     |                       |
| <code>--ftps-fallback-to-ftp</code>       | —         | fall back to FTP if FTPS is not supported in the target server                                                                                                                                                         |                       |
| <code>--warc-file</code>                  | FILENAME  | save request/response data to a .warc.gz file                                                                                                                                                                          |                       |
| <code>--warc-header</code>                | STRING    | insert STRING into the warcinfo record                                                                                                                                                                                 |                       |
| <code>--warc-max-size</code>              | NUMBER    | set maximum size of WARC files to NUMBER                                                                                                                                                                               |                       |
| <code>--warc-cdx</code>                   | —         | write CDX index files                                                                                                                                                                                                  |                       |
| <code>--warc-dedup</code>                 | FILENAME  | do not store records listed in this CDX file                                                                                                                                                                           |                       |
| <code>--no-warc-compression</code>        | —         | do not compress WARC files with GZIP                                                                                                                                                                                   |                       |
| <code>--no-warc-digests</code>            | —         | do not calculate SHA1 digests                                                                                                                                                                                          |                       |
| <code>--no-warc-keep-log</code>           | —         | do not store the log file in a WARC record                                                                                                                                                                             |                       |
| <code>--warc-tempdir</code>               | DIRECTORY | location for temporary files created by the WARC writer                                                                                                                                                                |                       |
| <code>-r</code>                           | —         | specify recursive download                                                                                                                                                                                             |                       |
| <code>--recursive</code>                  | —         | specify recursive download                                                                                                                                                                                             |                       |
| <code>-l</code>                           | NUMBER    | maximum recursion depth (inf or 0 for infinite)                                                                                                                                                                        |                       |

| Flag                             | Argument | What it does                                                           | When you would use it |
|----------------------------------|----------|------------------------------------------------------------------------|-----------------------|
| <code>--level</code>             | NUMBER   | maximum recursion depth (inf or 0 for infinite)                        |                       |
| <code>--delete-after</code>      | —        | delete files locally after downloading them                            |                       |
| <code>-k</code>                  | —        | make links in downloaded HTML or CSS point to local files              |                       |
| <code>--convert-links</code>     | —        | make links in downloaded HTML or CSS point to local files              |                       |
| <code>--convert-file-only</code> | —        | convert the file part of the URLs only (usually known as the basename) |                       |
| <code>--backups</code>           | N        | before writing file X, rotate up to N backup files                     |                       |
| <code>-K</code>                  | —        | before converting file X, back up as X.orig                            |                       |
| <code>--backup-converted</code>  | —        | before converting file X, back up as X.orig                            |                       |
| <code>-m</code>                  | —        | shortcut for <code>-N -r -l inf --no-remove-listing</code>             |                       |
| <code>--mirror</code>            | —        | shortcut for <code>-N -r -l inf --no-remove-listing</code>             |                       |
| <code>-p</code>                  | —        | get all images, etc. needed to display HTML page                       |                       |
| <code>--page-requisites</code>   | —        | get all images, etc. needed to display HTML page                       |                       |
| <code>--strict-comments</code>   | —        | turn on strict (SGML) handling of HTML comments                        |                       |
| <code>-A</code>                  | LIST     | comma-separated list of accepted extensions                            |                       |
| <code>--accept</code>            | LIST     | comma-separated list of accepted extensions                            |                       |
| <code>-R</code>                  | LIST     | comma-separated list of rejected extensions                            |                       |
| <code>--reject</code>            | LIST     | comma-separated list of rejected extensions                            |                       |
| <code>--accept-regex</code>      | REGEX    | regex matching accepted URLs                                           |                       |
| <code>--reject-regex</code>      | REGEX    | regex matching rejected URLs                                           |                       |
| <code>--regex-type</code>        | TYPE     | regex type (posix pcre)                                                |                       |
| <code>-D</code>                  | LIST     | comma-separated list of accepted domains                               |                       |
| <code>--domains</code>           | LIST     | comma-separated list of accepted domains                               |                       |
| <code>--exclude-domains</code>   | LIST     | comma-separated list of rejected domains                               |                       |
| <code>--follow-ftp</code>        | —        | follow FTP links from HTML documents                                   |                       |
| <code>--follow-tags</code>       | LIST     | comma-separated list of followed HTML tags                             |                       |
| <code>--ignore-tags</code>       | LIST     | comma-separated list of ignored HTML tags                              |                       |
| <code>-H</code>                  | —        | go to foreign hosts when recursive                                     |                       |
| <code>--span-hosts</code>        | —        | go to foreign hosts when recursive                                     |                       |
| <code>-L</code>                  | —        | follow relative links only                                             |                       |
| <code>--relative</code>          | —        | follow relative links only                                             |                       |



| Flag                               | Argument | What it does                                                   | When you would use it |
|------------------------------------|----------|----------------------------------------------------------------|-----------------------|
| <code>-I</code>                    | LIST     | list of allowed directories                                    |                       |
| <code>--include-directories</code> | LIST     | list of allowed directories                                    |                       |
| <code>--trust-server-names</code>  | —        | use the name specified by the redirection URL's last component |                       |
| <code>-X</code>                    | LIST     | list of excluded directories                                   |                       |
| <code>--exclude-directories</code> | LIST     | list of excluded directories                                   |                       |

## Gotchas

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*TODO: operational traps.*

## See also

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[curl](#)

# frida

**Kit:** REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Trace the execution of a process to analyze its behavior.

## Synopsis

```
frida [options] target
```

## Common invocations

```
# from cyberlab 39-frida-tracing
frida --version
# from cyberlab 39-frida-tracing
frida -f ./target -l /dev/stdin --no-pause <<'EOF'
```

## Options

All 66 options parsed from the captured help text. The final column is filled in by review.

| Flag     | Argument | What it does                           | When you would use it |
|----------|----------|----------------------------------------|-----------------------|
| -h       | —        | show this help message and exit        |                       |
| --help   | —        | show this help message and exit        |                       |
| -D       | ID       | connect to device with the given ID    |                       |
| --device | ID       | connect to device with the given ID    |                       |
| -U       | —        | connect to USB device                  |                       |
| --usb    | —        | connect to USB device                  |                       |
| -R       | —        | connect to remote frida-server         |                       |
| --remote | —        | connect to remote frida-server         |                       |
| -H       | HOST     | connect to remote frida-server on HOST |                       |
| --host   | HOST     | connect to remote frida-server on HOST |                       |

| Flag                                          | Argument              | What it does                                                                                                                                     | When you would use it |
|-----------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--certificate</code>                    | CERTIFICATE           | speak TLS with HOST, expecting CERTIFICATE                                                                                                       |                       |
| <code>--origin</code>                         | ORIGIN                | connect to remote server with "Origin" header set to ORIGIN                                                                                      |                       |
| <code>--token</code>                          | TOKEN                 | authenticate with HOST using TOKEN                                                                                                               |                       |
| <code>--keepalive-interval</code>             | INTERVAL              | set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)                                            |                       |
| <code>--device-option</code>                  | option                | override a backend-specific option, such as "control- endpoint= (string)localabstract:/my-frida-server" (supported types are: string, bool, int) |                       |
| <code>--p2p</code>                            | —                     | establish a peer-to-peer connection with target                                                                                                  |                       |
| <code>--stun-server</code>                    | ADDRESS               | set STUN server ADDRESS to use with --p2p                                                                                                        |                       |
| <code>-f</code>                               | TARGET                | spawn FILE                                                                                                                                       |                       |
| <code>--file</code>                           | TARGET                | spawn FILE                                                                                                                                       |                       |
| <code>-F</code>                               | —                     | attach to frontmost application                                                                                                                  |                       |
| <code>--attach-frontmost</code>               | —                     | attach to frontmost application                                                                                                                  |                       |
| <code>-n</code>                               | NAME                  | attach to NAME                                                                                                                                   |                       |
| <code>--attach-name</code>                    | NAME                  | attach to NAME                                                                                                                                   |                       |
| <code>-N</code>                               | IDENTIFIER            | attach to IDENTIFIER                                                                                                                             |                       |
| <code>--attach-identifier</code>              | IDENTIFIER            | attach to IDENTIFIER                                                                                                                             |                       |
| <code>-p</code>                               | PID                   | attach to PID                                                                                                                                    |                       |
| <code>--attach-pid</code>                     | PID                   | attach to PID                                                                                                                                    |                       |
| <code>-W</code>                               | PATTERN               | await spawn matching PATTERN                                                                                                                     |                       |
| <code>--await</code>                          | PATTERN               | await spawn matching PATTERN                                                                                                                     |                       |
| <code>--stdio</code>                          | inherit,pipe          | stdio behavior when spawning (defaults to "inherit")                                                                                             |                       |
| <code>--aux</code>                            | option                | set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)                                                     |                       |
| <code>--realm</code>                          | native,emulated       | realm to attach in                                                                                                                               |                       |
| <code>--exceptor</code>                       | full,handler-only,off | configure the exception handling mode                                                                                                            |                       |
| <code>--disable-unwind-broker</code>          | —                     | disable the unwind broker                                                                                                                        |                       |
| <code>--disable-exit-monitor</code>           | —                     | disable the exit monitor                                                                                                                         |                       |
| <code>--disable-thread-suspend-monitor</code> | —                     | disable the thread suspend monitor                                                                                                               |                       |
| <code>--linker-notifier-offset</code>         | OFFSET                | add a linker notifier OFFSET (may be specified multiple times)                                                                                   |                       |
| <code>--runtime</code>                        | qjs,v8                | script runtime to use                                                                                                                            |                       |

| Flag                          | Argument              | What it does                                                                                        | When you would use it |
|-------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------|-----------------------|
| <code>--debug</code>          | —                     | enable the Node.js compatible script debugger                                                       |                       |
| <code>--squelch-crash</code>  | —                     | if enabled, will not dump crash report to console                                                   |                       |
| <code>-O</code>               | FILE                  | text file containing additional command line options                                                |                       |
| <code>--options-file</code>   | FILE                  | text file containing additional command line options                                                |                       |
| <code>--version</code>        | —                     | show program's version number and exit                                                              |                       |
| <code>-l</code>               | SCRIPT                | load SCRIPT                                                                                         |                       |
| <code>--load</code>           | SCRIPT                | load SCRIPT                                                                                         |                       |
| <code>-P</code>               | PARAMETERS_JSON       | parameters as JSON, same as Gadget                                                                  |                       |
| <code>--parameters</code>     | PARAMETERS_JSON       | parameters as JSON, same as Gadget                                                                  |                       |
| <code>-C</code>               | USER_CMODULE          | load CMODULE                                                                                        |                       |
| <code>--cmodule</code>        | USER_CMODULE          | load CMODULE                                                                                        |                       |
| <code>--toolchain</code>      | any,internal,external | CModule toolchain to use when compiling from source code                                            |                       |
| <code>-c</code>               | CODESHARE_URI         | load CODESHARE_URI                                                                                  |                       |
| <code>--codeshare</code>      | CODESHARE_URI         | load CODESHARE_URI                                                                                  |                       |
| <code>-e</code>               | CODE                  | evaluate CODE                                                                                       |                       |
| <code>--eval</code>           | CODE                  | evaluate CODE                                                                                       |                       |
| <code>-q</code>               | —                     | quiet mode (no prompt) and quit after -l and -e                                                     |                       |
| <code>-t</code>               | TIMEOUT               | seconds to wait before terminating in quiet mode (or 'inf' to run forever)                          |                       |
| <code>--timeout</code>        | TIMEOUT               | seconds to wait before terminating in quiet mode (or 'inf' to run forever)                          |                       |
| <code>--pause</code>          | —                     | leave main thread paused after spawning program                                                     |                       |
| <code>-o</code>               | LOGFILE               | output to log file                                                                                  |                       |
| <code>--output</code>         | LOGFILE               | output to log file                                                                                  |                       |
| <code>--eternalize</code>     | —                     | eternalize the script before exit                                                                   |                       |
| <code>--exit-on-error</code>  | —                     | exit with code 1 after encountering any exception in the SCRIPT                                     |                       |
| <code>--kill-on-exit</code>   | —                     | kill the spawned program when Frida exits                                                           |                       |
| <code>--auto-perform</code>   | —                     | wrap entered code with Java.perform                                                                 |                       |
| <code>--auto-reload</code>    | —                     | Enable auto reload of provided scripts and c module (on by default, will be required in the future) |                       |
| <code>--no-auto-reload</code> | —                     | Disable auto reload of provided scripts and c module                                                |                       |

## Gotchas

*TODO: operational traps.*

## See also

---

`frida-trace`, `frida-ps`, `frida-discover`, `frida-kill`, `frida-ls-devices`

# frida-discover

**Kit:** REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Trace the execution of a process to analyze its behavior.

## Synopsis

```
frida-discover [options] target
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 43 options parsed from the captured help text. The final column is filled in by review.

| Flag          | Argument    | What it does                                                | When you would use it |
|---------------|-------------|-------------------------------------------------------------|-----------------------|
| -h            | —           | show this help message and exit                             |                       |
| --help        | —           | show this help message and exit                             |                       |
| -D            | ID          | connect to device with the given ID                         |                       |
| --device      | ID          | connect to device with the given ID                         |                       |
| -U            | —           | connect to USB device                                       |                       |
| --usb         | —           | connect to USB device                                       |                       |
| -R            | —           | connect to remote frida-server                              |                       |
| --remote      | —           | connect to remote frida-server                              |                       |
| -H            | HOST        | connect to remote frida-server on HOST                      |                       |
| --host        | HOST        | connect to remote frida-server on HOST                      |                       |
| --certificate | CERTIFICATE | speak TLS with HOST, expecting CERTIFICATE                  |                       |
| --origin      | ORIGIN      | connect to remote server with "Origin" header set to ORIGIN |                       |
| --token       | TOKEN       | authenticate with HOST using TOKEN                          |                       |

| Flag                                          | Argument              | What it does                                                                                                                                     | When you would use it |
|-----------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--keepalive-interval</code>             | INTERVAL              | set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)                                            |                       |
| <code>--device-option</code>                  | option                | override a backend-specific option, such as "control- endpoint= (string)localabstract:/my-frida-server" (supported types are: string, bool, int) |                       |
| <code>--p2p</code>                            | —                     | establish a peer-to-peer connection with target                                                                                                  |                       |
| <code>--stun-server</code>                    | ADDRESS               | set STUN server ADDRESS to use with --p2p                                                                                                        |                       |
| <code>-f</code>                               | TARGET                | spawn FILE                                                                                                                                       |                       |
| <code>--file</code>                           | TARGET                | spawn FILE                                                                                                                                       |                       |
| <code>-F</code>                               | —                     | attach to frontmost application                                                                                                                  |                       |
| <code>--attach-frontmost</code>               | —                     | attach to frontmost application                                                                                                                  |                       |
| <code>-n</code>                               | NAME                  | attach to NAME                                                                                                                                   |                       |
| <code>--attach-name</code>                    | NAME                  | attach to NAME                                                                                                                                   |                       |
| <code>-N</code>                               | IDENTIFIER            | attach to IDENTIFIER                                                                                                                             |                       |
| <code>--attach-identifier</code>              | IDENTIFIER            | attach to IDENTIFIER                                                                                                                             |                       |
| <code>-p</code>                               | PID                   | attach to PID                                                                                                                                    |                       |
| <code>--attach-pid</code>                     | PID                   | attach to PID                                                                                                                                    |                       |
| <code>-W</code>                               | PATTERN               | await spawn matching PATTERN                                                                                                                     |                       |
| <code>--await</code>                          | PATTERN               | await spawn matching PATTERN                                                                                                                     |                       |
| <code>--stdio</code>                          | inherit,pipe          | stdio behavior when spawning (defaults to "inherit")                                                                                             |                       |
| <code>--aux</code>                            | option                | set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)                                                     |                       |
| <code>--realm</code>                          | native,emulated       | realm to attach in                                                                                                                               |                       |
| <code>--exceptor</code>                       | full,handler-only,off | configure the exception handling mode                                                                                                            |                       |
| <code>--disable-unwind-broker</code>          | —                     | disable the unwind broker                                                                                                                        |                       |
| <code>--disable-exit-monitor</code>           | —                     | disable the exit monitor                                                                                                                         |                       |
| <code>--disable-thread-suspend-monitor</code> | —                     | disable the thread suspend monitor                                                                                                               |                       |
| <code>--linker-notifier-offset</code>         | OFFSET                | add a linker notifier OFFSET (may be specified multiple times)                                                                                   |                       |
| <code>--runtime</code>                        | qjs,v8                | script runtime to use                                                                                                                            |                       |
| <code>--debug</code>                          | —                     | enable the Node.js compatible script debugger                                                                                                    |                       |
| <code>--squelch-crash</code>                  | —                     | if enabled, will not dump crash report to console                                                                                                |                       |

| Flag                        | Argument | What it does                                         | When you would use it |
|-----------------------------|----------|------------------------------------------------------|-----------------------|
| <code>-O</code>             | FILE     | text file containing additional command line options |                       |
| <code>--options-file</code> | FILE     | text file containing additional command line options |                       |
| <code>--version</code>      | —        | show program's version number and exit               |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`frida`, `frida-trace`, `frida-ps`, `frida-kill`, `frida-ls-devices`



# frida-kill

**Kit:** REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://frida.re>

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## Purpose

Trace the execution of a process to analyze its behavior.

## Synopsis

```
frida-kill [options] process
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 20 options parsed from the captured help text. The final column is filled in by review.

| Flag          | Argument    | What it does                                                | When you would use it |
|---------------|-------------|-------------------------------------------------------------|-----------------------|
| -h            | —           | show this help message and exit                             |                       |
| --help        | —           | show this help message and exit                             |                       |
| -D            | ID          | connect to device with the given ID                         |                       |
| --device      | ID          | connect to device with the given ID                         |                       |
| -U            | —           | connect to USB device                                       |                       |
| --usb         | —           | connect to USB device                                       |                       |
| -R            | —           | connect to remote frida-server                              |                       |
| --remote      | —           | connect to remote frida-server                              |                       |
| -H            | HOST        | connect to remote frida-server on HOST                      |                       |
| --host        | HOST        | connect to remote frida-server on HOST                      |                       |
| --certificate | CERTIFICATE | speak TLS with HOST, expecting CERTIFICATE                  |                       |
| --origin      | ORIGIN      | connect to remote server with “Origin” header set to ORIGIN |                       |
| --token       | TOKEN       | authenticate with HOST using TOKEN                          |                       |

| Flag                              | Argument | What it does                                                                                                                                     | When you would use it |
|-----------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--keepalive-interval</code> | INTERVAL | set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)                                            |                       |
| <code>--device-option</code>      | option   | override a backend-specific option, such as "control- endpoint= (string)localabstract:/my-frida-server" (supported types are: string, bool, int) |                       |
| <code>--p2p</code>                | —        | establish a peer-to-peer connection with target                                                                                                  |                       |
| <code>--stun-server</code>        | ADDRESS  | set STUN server ADDRESS to use with --p2p                                                                                                        |                       |
| <code>-O</code>                   | FILE     | text file containing additional command line options                                                                                             |                       |
| <code>--options-file</code>       | FILE     | text file containing additional command line options                                                                                             |                       |
| <code>--version</code>            | —        | show program's version number and exit                                                                                                           |                       |

## Gotchas

*TODO: operational traps.*

## See also

`frida`, `frida-trace`, `frida-ps`, `frida-discover`, `frida-ls-devices`

# frida-ls-devices

**Kit:** REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://frida.re>

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## Purpose

Trace the execution of a process to analyze its behavior.

## Synopsis

```
frida-ls-devices [options]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 5 options parsed from the captured help text. The final column is filled in by review.

| Flag           | Argument | What it does                                         | When you would use it |
|----------------|----------|------------------------------------------------------|-----------------------|
| -h             | —        | show this help message and exit                      |                       |
| --help         | —        | show this help message and exit                      |                       |
| -O             | FILE     | text file containing additional command line options |                       |
| --options-file | FILE     | text file containing additional command line options |                       |
| --version      | —        | show program's version number and exit               |                       |

## Gotchas

*TODO: operational traps.*

## See also

[frida](#), [frida-trace](#), [frida-ps](#), [frida-discover](#), [frida-kill](#)

# frida-ps

**Kit:** REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Trace the execution of a process to analyze its behavior.

## Synopsis

```
frida-ps [options]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 28 options parsed from the captured help text. The final column is filled in by review.

| Flag          | Argument    | What it does                                                | When you would use it |
|---------------|-------------|-------------------------------------------------------------|-----------------------|
| -h            | —           | show this help message and exit                             |                       |
| --help        | —           | show this help message and exit                             |                       |
| -D            | ID          | connect to device with the given ID                         |                       |
| --device      | ID          | connect to device with the given ID                         |                       |
| -U            | —           | connect to USB device                                       |                       |
| --usb         | —           | connect to USB device                                       |                       |
| -R            | —           | connect to remote frida-server                              |                       |
| --remote      | —           | connect to remote frida-server                              |                       |
| -H            | HOST        | connect to remote frida-server on HOST                      |                       |
| --host        | HOST        | connect to remote frida-server on HOST                      |                       |
| --certificate | CERTIFICATE | speak TLS with HOST, expecting CERTIFICATE                  |                       |
| --origin      | ORIGIN      | connect to remote server with “Origin” header set to ORIGIN |                       |
| --token       | TOKEN       | authenticate with HOST using TOKEN                          |                       |

| Flag                              | Argument | What it does                                                                                                                                   | When you would use it |
|-----------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--keepalive-interval</code> | INTERVAL | set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)                                          |                       |
| <code>--device-option</code>      | option   | override a backend-specific option, such as "control-endpoint=(string)localabstract:/my-frida-server" (supported types are: string, bool, int) |                       |
| <code>--p2p</code>                | —        | establish a peer-to-peer connection with target                                                                                                |                       |
| <code>--stun-server</code>        | ADDRESS  | set STUN server ADDRESS to use with --p2p                                                                                                      |                       |
| <code>-O</code>                   | FILE     | text file containing additional command line options                                                                                           |                       |
| <code>--options-file</code>       | FILE     | text file containing additional command line options                                                                                           |                       |
| <code>--version</code>            | —        | show program's version number and exit                                                                                                         |                       |
| <code>-a</code>                   | —        | list only applications                                                                                                                         |                       |
| <code>--applications</code>       | —        | list only applications                                                                                                                         |                       |
| <code>-i</code>                   | —        | include all installed applications                                                                                                             |                       |
| <code>--installed</code>          | —        | include all installed applications                                                                                                             |                       |
| <code>-j</code>                   | —        | output results as JSON                                                                                                                         |                       |
| <code>--json</code>               | —        | output results as JSON                                                                                                                         |                       |
| <code>-e</code>                   | —        | exclude icons in output                                                                                                                        |                       |
| <code>--exclude-icons</code>      | —        | exclude icons in output                                                                                                                        |                       |

## Gotchas

*TODO: operational traps.*

## See also

`frida`, `frida-trace`, `frida-discover`, `frida-kill`, `frida-ls-devices`

# frida-trace

**Kit:** REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#) **Docs:** <https://frida.re>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Trace the execution of a process to analyze its behavior.

## Synopsis

```
frida-trace [options] target
```

## Common invocations

```
# from cyberlab 39-frida-tracing
frida-trace --version
# from cyberlab 39-frida-tracing
timeout 8 frida-trace -f ./target -i getenv
```

## Options

All 82 options parsed from the captured help text. The final column is filled in by review.

| Flag     | Argument | What it does                           | When you would use it |
|----------|----------|----------------------------------------|-----------------------|
| -h       | —        | show this help message and exit        |                       |
| --help   | —        | show this help message and exit        |                       |
| -D       | ID       | connect to device with the given ID    |                       |
| --device | ID       | connect to device with the given ID    |                       |
| -U       | —        | connect to USB device                  |                       |
| --usb    | —        | connect to USB device                  |                       |
| -R       | —        | connect to remote frida-server         |                       |
| --remote | —        | connect to remote frida-server         |                       |
| -H       | HOST     | connect to remote frida-server on HOST |                       |
| --host   | HOST     | connect to remote frida-server on HOST |                       |

| Flag                                          | Argument              | What it does                                                                                                                                     | When you would use it |
|-----------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--certificate</code>                    | CERTIFICATE           | speak TLS with HOST, expecting CERTIFICATE                                                                                                       |                       |
| <code>--origin</code>                         | ORIGIN                | connect to remote server with "Origin" header set to ORIGIN                                                                                      |                       |
| <code>--token</code>                          | TOKEN                 | authenticate with HOST using TOKEN                                                                                                               |                       |
| <code>--keepalive-interval</code>             | INTERVAL              | set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)                                            |                       |
| <code>--device-option</code>                  | option                | override a backend-specific option, such as "control- endpoint= (string)localabstract:/my-frida-server" (supported types are: string, bool, int) |                       |
| <code>--p2p</code>                            | —                     | establish a peer-to-peer connection with target                                                                                                  |                       |
| <code>--stun-server</code>                    | ADDRESS               | set STUN server ADDRESS to use with --p2p                                                                                                        |                       |
| <code>-f</code>                               | TARGET                | spawn FILE                                                                                                                                       |                       |
| <code>--file</code>                           | TARGET                | spawn FILE                                                                                                                                       |                       |
| <code>-F</code>                               | —                     | attach to frontmost application                                                                                                                  |                       |
| <code>--attach-frontmost</code>               | —                     | attach to frontmost application                                                                                                                  |                       |
| <code>-n</code>                               | NAME                  | attach to NAME                                                                                                                                   |                       |
| <code>--attach-name</code>                    | NAME                  | attach to NAME                                                                                                                                   |                       |
| <code>-N</code>                               | IDENTIFIER            | attach to IDENTIFIER                                                                                                                             |                       |
| <code>--attach-identifier</code>              | IDENTIFIER            | attach to IDENTIFIER                                                                                                                             |                       |
| <code>-p</code>                               | PID                   | attach to PID                                                                                                                                    |                       |
| <code>--attach-pid</code>                     | PID                   | attach to PID                                                                                                                                    |                       |
| <code>-W</code>                               | PATTERN               | await spawn matching PATTERN                                                                                                                     |                       |
| <code>--await</code>                          | PATTERN               | await spawn matching PATTERN                                                                                                                     |                       |
| <code>--stdio</code>                          | inherit,pipe          | stdio behavior when spawning (defaults to "inherit")                                                                                             |                       |
| <code>--aux</code>                            | option                | set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)                                                     |                       |
| <code>--realm</code>                          | native,emulated       | realm to attach in                                                                                                                               |                       |
| <code>--exceptor</code>                       | full,handler-only,off | configure the exception handling mode                                                                                                            |                       |
| <code>--disable-unwind-broker</code>          | —                     | disable the unwind broker                                                                                                                        |                       |
| <code>--disable-exit-monitor</code>           | —                     | disable the exit monitor                                                                                                                         |                       |
| <code>--disable-thread-suspend-monitor</code> | —                     | disable the thread suspend monitor                                                                                                               |                       |
| <code>--linker-notifier-offset</code>         | OFFSET                | add a linker notifier OFFSET (may be specified multiple times)                                                                                   |                       |
| <code>--runtime</code>                        | qjs,v8                | script runtime to use                                                                                                                            |                       |

| Flag                                  | Argument        | What it does                                         | When you would use it |
|---------------------------------------|-----------------|------------------------------------------------------|-----------------------|
| <code>--debug</code>                  | —               | enable the Node.js compatible script debugger        |                       |
| <code>--squelch-crash</code>          | —               | if enabled, will not dump crash report to console    |                       |
| <code>-O</code>                       | FILE            | text file containing additional command line options |                       |
| <code>--options-file</code>           | FILE            | text file containing additional command line options |                       |
| <code>--version</code>                | —               | show program's version number and exit               |                       |
| <code>-I</code>                       | MODULE          | include MODULE                                       |                       |
| <code>--include-module</code>         | MODULE          | include MODULE                                       |                       |
| <code>-X</code>                       | MODULE          | exclude MODULE                                       |                       |
| <code>--exclude-module</code>         | MODULE          | exclude MODULE                                       |                       |
| <code>-i</code>                       | FUNCTION        | include [MODULE!]FUNCTION                            |                       |
| <code>--include</code>                | FUNCTION        | include [MODULE!]FUNCTION                            |                       |
| <code>-x</code>                       | FUNCTION        | exclude [MODULE!]FUNCTION                            |                       |
| <code>--exclude</code>                | FUNCTION        | exclude [MODULE!]FUNCTION                            |                       |
| <code>-T</code>                       | INCLUDE_IMPORTS | include program's imports                            |                       |
| <code>--include-imports</code>        | INCLUDE_IMPORTS | include program's imports                            |                       |
| <code>-t</code>                       | MODULE          | include MODULE imports                               |                       |
| <code>--include-module-imports</code> | MODULE          | include MODULE imports                               |                       |
| <code>-m</code>                       | OBJC_METHOD     | include OBJC_METHOD                                  |                       |
| <code>--include-objc-method</code>    | OBJC_METHOD     | include OBJC_METHOD                                  |                       |
| <code>-M</code>                       | OBJC_METHOD     | exclude OBJC_METHOD                                  |                       |
| <code>--exclude-objc-method</code>    | OBJC_METHOD     | exclude OBJC_METHOD                                  |                       |
| <code>-y</code>                       | SWIFT_FUNC      | include SWIFT_FUNC                                   |                       |
| <code>--include-swift-func</code>     | SWIFT_FUNC      | include SWIFT_FUNC                                   |                       |
| <code>-Y</code>                       | SWIFT_FUNC      | exclude SWIFT_FUNC                                   |                       |
| <code>--exclude-swift-func</code>     | SWIFT_FUNC      | exclude SWIFT_FUNC                                   |                       |
| <code>-j</code>                       | JAVA_METHOD     | include JAVA_METHOD                                  |                       |
| <code>--include-java-method</code>    | JAVA_METHOD     | include JAVA_METHOD                                  |                       |
| <code>-J</code>                       | JAVA_METHOD     | exclude JAVA_METHOD                                  |                       |
| <code>--exclude-java-method</code>    | JAVA_METHOD     | exclude JAVA_METHOD                                  |                       |



| Flag                                | Argument        | What it does                                                        | When you would use it |
|-------------------------------------|-----------------|---------------------------------------------------------------------|-----------------------|
| <code>-s</code>                     | DEBUG_SYMBOL    | include DEBUG_SYMBOL                                                |                       |
| <code>--include-debug-symbol</code> | DEBUG_SYMBOL    | include DEBUG_SYMBOL                                                |                       |
| <code>-q</code>                     | —               | do not format output messages                                       |                       |
| <code>--quiet</code>                | —               | do not format output messages                                       |                       |
| <code>-d</code>                     | —               | add module name to generated onEnter log statement                  |                       |
| <code>--decorate</code>             | —               | add module name to generated onEnter log statement                  |                       |
| <code>-S</code>                     | PATH            | path to JavaScript file used to initialize the session              |                       |
| <code>--init-session</code>         | PATH            | path to JavaScript file used to initialize the session              |                       |
| <code>-P</code>                     | PARAMETERS_JSON | parameters as JSON, exposed as a global named 'parameters'          |                       |
| <code>--parameters</code>           | PARAMETERS_JSON | parameters as JSON, exposed as a global named 'parameters'          |                       |
| <code>-o</code>                     | OUTPUT          | dump messages to file                                               |                       |
| <code>--output</code>               | OUTPUT          | dump messages to file                                               |                       |
| <code>--ui-host</code>              | UI_HOST         | the host to serve the UI on (default localhost)                     |                       |
| <code>--ui-port</code>              | UI_PORT         | the TCP port to serve the UI on                                     |                       |
| <code>--ui-allow-origin</code>      | ORIGIN          | allow browser requests from ORIGIN; may be specified multiple times |                       |

## Gotchas

*TODO: operational traps.*

## See also

`frida`, `frida-ps`, `frida-discover`, `frida-kill`, `frida-ls-devices`

# r2

**Kit:** REMnux · Kali Linux · SIFT Workstation **Capability:** Disassemble and explore a binary **Version:** 6.1.9-124 r2  
**Captured:** cyberlab-aio via -h on 2026-08-01 [raw](#) **Docs:** <https://www.radare.org/n/radare2.html>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

## Synopsis

```
r2 [-ACdfjLMnQStuvwzX] [-P patch] [-p prj] [-a arch] [-b bits] [-c cmd]
[-s addr] [-B baddr] [-m maddr] [-i script] [-e k=v] file|pid|-|--|=
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 31 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                           | When you would use it |
|------|----------|--------------------------------------------------------|-----------------------|
| -0   | —        | print \x00 after init and every command                |                       |
| -1   | —        | redirect stderr to stdout                              |                       |
| -2   | —        | close stderr file descriptor (silent warning messages) |                       |
| -a   | arch     | set asm.arch                                           |                       |
| -A   | —        | run 'aaa' command to analyze all referenced code       |                       |
| -b   | bits     | set asm.bits                                           |                       |
| -B   | baddr    | set base address for PIE binaries                      |                       |
| -C   | —        | file is host:port (alias for -c+=http://%s/cmd/)       |                       |
| -d   | —        | debug the executable 'file' or running process 'pid'   |                       |
| -E   | —        | open cfg.editor with the user rc file                  |                       |
| -f   | —        | block size = file size                                 |                       |
| -i   | file     | run rlang program, r2script file or load plugin        |                       |

| Flag            | Argument | What it does                                                | When you would use it |
|-----------------|----------|-------------------------------------------------------------|-----------------------|
| <code>-I</code> | file     | run script file before the file is opened                   |                       |
| <code>-j</code> | —        | use json for -v, -L and maybe others                        |                       |
| <code>-m</code> | addr     | map file at given address (loadaddr)                        |                       |
| <code>-M</code> | —        | do not demangle symbol names                                |                       |
| <code>-N</code> | —        | do not load user settings and scripts                       |                       |
| <code>-q</code> | —        | quiet mode (no prompt) and quit after -i                    |                       |
| <code>-Q</code> | —        | quiet mode (no prompt) and quit faster (quickLeak=true)     |                       |
| <code>-p</code> | prj      | use project, list if no arg, load if no file                |                       |
| <code>-P</code> | file     | apply rapatch file and quit                                 |                       |
| <code>-r</code> | rarun2   | specify rarun2 profile to load (same as -e dbg.profile=X)   |                       |
| <code>-s</code> | addr     | initial seek                                                |                       |
| <code>-S</code> | —        | start r2 in sandbox mode                                    |                       |
| <code>-t</code> | —        | load rabin2 info in thread                                  |                       |
| <code>-u</code> | —        | set bin.filter=false to get raw sym/sec/cls names           |                       |
| <code>-v</code> | —        | show radare2 version (-V show lib versions)                 |                       |
| <code>-V</code> | —        | show radare2 version (-V show lib versions)                 |                       |
| <code>-w</code> | —        | open file in write mode                                     |                       |
| <code>-x</code> | —        | open without exec-flag (asm.emu will not work), See io.exec |                       |
| <code>-X</code> | —        | same as -e bin.usextr=false (useful for dyldcache)          |                       |

## Gotchas

*TODO: operational traps.*

## See also

`rabin2`, `rasm2`, `objdump`, `vivbin`, `vdbbin`

# rasm2

**Kit:** REMnux **Capability:** Disassemble and explore a binary; Analyse shellcode **Version:** rasm2 6.1.9 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

## Synopsis

```
rasm2 [-ACdDehHLBvw] [-a arch] [-b bits] [-s addr] [-S syntax]
[-f file] [-o file] [-F fil:ter] [-i skip] [-l len] 'code'|hex|0101b|-
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 25 options parsed from the captured help text. The final column is filled in by review.

| Flag            | Argument | What it does                                           | When you would use it |
|-----------------|----------|--------------------------------------------------------|-----------------------|
| <code>-a</code> | arch     | set architecture to assemble/disassemble (see -L)      |                       |
| <code>-A</code> | —        | show Analysis information from given hexpairs          |                       |
| <code>-b</code> | bits     | set cpu register size (8, 16, 32, 64) (RASM2_BITS)     |                       |
| <code>-B</code> | —        | binary input/output (-l is mandatory for binary input) |                       |
| <code>-c</code> | cpu      | select specific CPU (depends on arch)                  |                       |
| <code>-C</code> | —        | output in C format                                     |                       |
| <code>-d</code> | —        | disassemble from hexpair bytes (-D show hexpairs)      |                       |
| <code>-D</code> | —        | disassemble from hexpair bytes (-D show hexpairs)      |                       |
| <code>-e</code> | —        | use big endian instead of little endian                |                       |
| <code>-E</code> | —        | display ESIL expression (same input as in -d)          |                       |
| <code>-f</code> | file     | read data from file                                    |                       |
| <code>-F</code> | parser   | specify which parse filter use (see -LL)               |                       |

| Flag            | Argument | What it does                                                       | When you would use it |
|-----------------|----------|--------------------------------------------------------------------|-----------------------|
| <code>-i</code> | len      | ignore/skip N bytes of the input buffer                            |                       |
| <code>-j</code> | —        | output in json format                                              |                       |
| <code>-k</code> | kernel   | select operating system (linux, windows, darwin, android, ios, ..) |                       |
| <code>-l</code> | len      | input/Output length                                                |                       |
| <code>-N</code> | —        | same as r2 -N (or R2_NOPLUGINS) (not load any plugin)              |                       |
| <code>-o</code> | file     | output file name (rasm2 -Bf a.asm -o a)                            |                       |
| <code>-p</code> | —        | run SPP over input for assembly                                    |                       |
| <code>-q</code> | —        | quiet mode                                                         |                       |
| <code>-r</code> | —        | output in radare commands                                          |                       |
| <code>-S</code> | syntax   | select syntax (intel, att)                                         |                       |
| <code>-v</code> | —        | show version information                                           |                       |
| <code>-x</code> | —        | use hex dwords instead of hex pairs when assembling.               |                       |
| <code>-w</code> | —        | what's this instruction for? describe opcode                       |                       |

## Gotchas

*TODO: operational traps.*

## See also

`r2`, `rabin2`, `objdump`, `vivbin`, `vdbbin`, `xortool`

# vdbbin

**Kit:** REMnux **Capability:** Disassemble and explore a binary **Captured:** cyberlab-aio via --help on 2026-08-01  
**raw Docs:** <https://github.com/vivisect/vivisect>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Statically examine and emulate binary files.

## Synopsis

```
vdbbin [options] [platformopt=foo, ...]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 27 options parsed from the captured help text. The final column is filled in by review.

| Flag       | Argument   | What it does                     | When you would use it |
|------------|------------|----------------------------------|-----------------------|
| -h         | —          | show this help message and exit  |                       |
| --help     | —          | show this help message and exit  |                       |
| -c         | COMMAND    | Debug a fired command            |                       |
| --cmd      | COMMAND    | Debug a fired command            |                       |
| -p         | PROCESS    | Attach to process by name or pid |                       |
| --process  | PROCESS    | Attach to process by name or pid |                       |
| -Q         | —          | Run the QT gui                   |                       |
| --qt       | —          | Run the QT gui                   |                       |
| -R         | REMOTEHOST | Attach to remote VDB server      |                       |
| --remote   | REMOTEHOST | Attach to remote VDB server      |                       |
| -r         | —          | Do not stop on attach            |                       |
| --run      | —          | Do not stop on attach            |                       |
| -s         | SNAPSHOT   | Load a vtrace snapshot file      |                       |
| --snapshot | SNAPSHOT   | Load a vtrace snapshot file      |                       |

| Flag                   | Argument | What it does                                  | When you would use it |
|------------------------|----------|-----------------------------------------------|-----------------------|
| <code>-S</code>        | —        | —                                             |                       |
| <code>--server</code>  | —        | —                                             |                       |
| <code>-v</code>        | —        | —                                             |                       |
| <code>--verbose</code> | —        | —                                             |                       |
| <code>-t</code>        | TARGET   | Activate special vdb target ( -t ? for list ) |                       |
| <code>--target</code>  | TARGET   | Activate special vdb target ( -t ? for list ) |                       |
| <code>--android</code> | —        | Debug Android with ADB!                       |                       |
| <code>-e</code>        | EVENTID  | Used for Windows JIT                          |                       |
| <code>--eventid</code> | EVENTID  | Used for Windows JIT                          |                       |
| <code>-w</code>        | WAITFOR  | Wait for process with name                    |                       |
| <code>--waitfor</code> | WAITFOR  | Wait for process with name                    |                       |
| <code>--LI</code>      | —        | Breakpoint on Library initialization          |                       |
| <code>--LL</code>      | —        | Breakpoint on Library load time               |                       |

## Gotchas

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*TODO: operational traps.*

## See also

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[r2](#) , [rabin2](#) , [rasm2](#) , [objdump](#) , [vivbin](#)

# vivbin

**Kit:** REMnux **Capability:** Disassemble and explore a binary **Captured:** cyberlab-aio via --help on 2026-08-01  
**raw Docs:** <https://github.com/vivisect/vivisect>

← [Capability index](#) · [Kit tool list](#)

## Purpose

Statically examine and emulate binary files.

## Synopsis

```
vivbin [options] <workspace|binaries...>
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 34 options parsed from the captured help text. The final column is filled in by review.

| Flag            | Argument    | What it does                                                       | When you would use it |
|-----------------|-------------|--------------------------------------------------------------------|-----------------------|
| -h              | —           | show this help message and exit                                    |                       |
| --help          | —           | show this help message and exit                                    |                       |
| -M              | MODNAME     | run the file listed as an analysis module in non-gui mode and exit |                       |
| --module        | MODNAME     | run the file listed as an analysis module in non-gui mode and exit |                       |
| -A              | —           | Do <i>not</i> do an initial auto-analysis pass                     |                       |
| --skip-analysis | —           | Do <i>not</i> do an initial auto-analysis pass                     |                       |
| -B              | —           | Do <i>not</i> start the gui, just load, analyze and save           |                       |
| --bulk          | —           | Do <i>not</i> start the gui, just load, analyze and save           |                       |
| -C              | —           | Output vivisect performace profiling (cProfile) info               |                       |
| --cprofile      | —           | Output vivisect performace profiling (cProfile) info               |                       |
| -E              | ENTRYPOINTS | Add Entry Point for bulk analysis (can have multiple "-E " args    |                       |
| --entrypoint    | ENTRYPOINTS | Add Entry Point for bulk analysis (can have multiple "-E " args    |                       |
| -O              | OPTION      | .= (optval must be json syntax)                                    |                       |



| Flag                        | Argument     | What it does                                                                                  | When you would use it |
|-----------------------------|--------------|-----------------------------------------------------------------------------------------------|-----------------------|
| <code>--option</code>       | OPTION       | .= (optval must be json syntax)                                                               |                       |
| <code>-p</code>             | PARSEMOD     | Manually specify the parser module (pe/elf/blob/...)                                          |                       |
| <code>--parser</code>       | PARSEMOD     | Manually specify the parser module (pe/elf/blob/...)                                          |                       |
| <code>-s</code>             | STORAGE_NAME | Specify a storage module by name                                                              |                       |
| <code>--storage</code>      | STORAGE_NAME | Specify a storage module by name                                                              |                       |
| <code>-S</code>             | —            | Run Vivisect Server. Last argument should be the VivWorkspaces directory.                     |                       |
| <code>--server</code>       | —            | Run Vivisect Server. Last argument should be the VivWorkspaces directory.                     |                       |
| <code>-P</code>             | SERVER_PORT  | Port to run Vivisect Server on (only meaningful with --server)                                |                       |
| <code>--port</code>         | SERVER_PORT  | Port to run Vivisect Server on (only meaningful with --server)                                |                       |
| <code>-v</code>             | —            | Enable verbose mode (multiples matter: -vvvv)                                                 |                       |
| <code>--verbose</code>      | —            | Enable verbose mode (multiples matter: -vvvv)                                                 |                       |
| <code>-V</code>             | VERSION      | Add file version (if available) to save file name                                             |                       |
| <code>--version</code>      | VERSION      | Add file version (if available) to save file name                                             |                       |
| <code>-c</code>             | CONFIG       | Path to a directory to use for config data                                                    |                       |
| <code>--config</code>       | CONFIG       | Path to a directory to use for config data                                                    |                       |
| <code>-a</code>             | —            | Autosave configuration data                                                                   |                       |
| <code>--autosave</code>     | —            | Autosave configuration data                                                                   |                       |
| <code>-o</code>             | OUTFILE      | Name of VivWorkspace file to create (useful for loading multiple binaries into one workspace) |                       |
| <code>--outfile</code>      | OUTFILE      | Name of VivWorkspace file to create (useful for loading multiple binaries into one workspace) |                       |
| <code>-m</code>             | —            | List Architectures and their version/maturity                                                 |                       |
| <code>--archmaturity</code> | —            | List Architectures and their version/maturity                                                 |                       |

## Gotchas

*TODO: operational traps.*

## See also

`r2`, `rabin2`, `rasm2`, `objdump`, `vdbbin`

# xortool

**Kit:** REMnux **Capability:** Analyse shellcode; Break simple obfuscation **Version:** 1.1.0 **Captured:** cyberlab-aio via `--help` on 2026-08-01 **raw Docs:** <https://github.com/hellman/xortool>

[← Capability index](#) · [Kit tool list](#)

## Purpose

Recover the key length and key of an XOR-encrypted file by frequency analysis.

## Synopsis

```
xortool [-x] [-m MAX-LEN] [-f] [-t CHARSET] [FILE]
xortool [-x] [-l LEN] [-c CHAR | -b | -o] [-f] [-t CHARSET] [-p PLAIN] [-r PERCENT] [FILE]
xortool [-x] [-m MAX-LEN| -l LEN] [-c CHAR | -b | -o] [-f] [-t CHARSET] [-p PLAIN] [-r PERCENT] [FILE]
xortool [-h | --help]
xortool --version
```

## Common invocations

```
# from cyberlab 09-deobfuscation
xortool --version
# from cyberlab 09-deobfuscation
xortool -c 20 exercise/encoded_payload.bin
```

## Options

All 20 options parsed from the captured help text; 4 reviewed with usage guidance.

| Flag                      | Argument | What it does                              | When you would use it                        |
|---------------------------|----------|-------------------------------------------|----------------------------------------------|
| <code>-x</code>           | —        | input is hex-encoded str                  |                                              |
| <code>--hex</code>        | —        | input is hex-encoded str                  |                                              |
| <code>-l</code>           | LEN      | length of the key                         | Fix the key length when you already know it. |
| <code>--key-length</code> | LEN      | length of the key                         |                                              |
| <code>-m</code>           | MAX-LEN  | maximum key length to probe [default: 65] | Maximum key length to consider.              |
| <code>--max-keylen</code> | MAX-LEN  | maximum key length to probe [default: 65] |                                              |

| Flag                           | Argument | What it does                                                | When you would use it                                                                                                                                                         |
|--------------------------------|----------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-c</code>                | CHAR     | most frequent char (one char or hex code)                   | Give the most frequent character of the plaintext — usually <code>20</code> (space) for text, <code>00</code> for binaries. This is the flag that makes or breaks the attack. |
| <code>--char</code>            | CHAR     | most frequent char (one char or hex code)                   |                                                                                                                                                                               |
| <code>-b</code>                | —        | brute force all possible most frequent chars                | Brute-force the most frequent character rather than guessing.                                                                                                                 |
| <code>--brute-chars</code>     | —        | brute force all possible most frequent chars                |                                                                                                                                                                               |
| <code>-o</code>                | —        | same as <code>-b</code> but will only check printable chars |                                                                                                                                                                               |
| <code>--brute-printable</code> | —        | same as <code>-b</code> but will only check printable chars |                                                                                                                                                                               |
| <code>-f</code>                | —        | filter outputs based on the charset                         |                                                                                                                                                                               |
| <code>--filter-output</code>   | —        | filter outputs based on the charset                         |                                                                                                                                                                               |
| <code>-p</code>                | PLAIN    | use known plaintext for decoding                            |                                                                                                                                                                               |
| <code>--known-plaintext</code> | PLAIN    | use known plaintext for decoding                            |                                                                                                                                                                               |
| <code>-r</code>                | PERCENT  | threshold validity percentage [default: 95]                 |                                                                                                                                                                               |
| <code>--threshold</code>       | PERCENT  | threshold validity percentage [default: 95]                 |                                                                                                                                                                               |
| <code>-h</code>                | —        | show this help                                              |                                                                                                                                                                               |
| <code>--help</code>            | —        | show this help                                              |                                                                                                                                                                               |

## Gotchas

- Output lands in an `xortool_out/` directory, not stdout. People routinely think it did nothing.
- `-c 00` is right far more often than the default for packed binaries, because null padding dominates.

## See also

`rasm2`, `floss`, `xlmdeobfuscator`

# AmcacheParser

**Kit:** FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse ESE / SRUM / Amcache databases; Parse execution and persistence artifacts **Version:** 2026.5.0+76dc8354aa98ce1e1c6f942abcfb09f583f411dd **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Description:

## Synopsis

```
AmcacheParser [options]
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 15 options parsed from the captured help text. The final column is filled in by review.

| Flag   | Argument | What it does                                                                                                                                                                                              | When you would use it |
|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -f     | f        | Amcache.hve file to parse                                                                                                                                                                                 |                       |
| -i     | —        | Include file entries for Programs entries                                                                                                                                                                 |                       |
| -w     | w        | Path to file containing SHA-1 hashes to <i>exclude</i> from the results. Blacklisting overrides whitelisting                                                                                              |                       |
| -b     | b        | Path to file containing SHA-1 hashes to <i>include</i> from the results. Blacklisting overrides whitelisting                                                                                              |                       |
| --csv  | csv      | Directory to save CSV formatted results to. Be sure to include the full path in double quotes                                                                                                             |                       |
| --csvf | csvf     | File name to save CSV formatted results to. When present, overrides default name                                                                                                                          |                       |
| --dt   | dt       | The custom date/time format to use when displaying time stamps. See <a href="https://goo.gl/CNVq0k">https://goo.gl/CNVq0k</a> for options. Default is: yyyy-MM-dd HH:mm:ss [default: yyyy-MM-dd HH:mm:ss] |                       |
| --mp   | —        | When true, display higher precision for timestamps                                                                                                                                                        |                       |
| --nl   | —        | When true, ignore transaction log files for dirty hives. Default is FALSE                                                                                                                                 |                       |

| Flag                   | Argument | What it does                             | When you would use it |
|------------------------|----------|------------------------------------------|-----------------------|
| <code>--debug</code>   | —        | Show debug information during processing |                       |
| <code>--trace</code>   | —        | Show trace information during processing |                       |
| <code>-?</code>        | —        | Show help and usage information          |                       |
| <code>-h</code>        | —        | Show help and usage information          |                       |
| <code>--help</code>    | —        | Show help and usage information          |                       |
| <code>--version</code> | —        | Show version information                 |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

[esedbexport](#), [esedbinfo](#), [SrumECmd](#), [PECmd](#), [AppCompatCacheParser](#), [MFTECmd](#)

# AppCompatCacheParser

**Kit:** FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+0cf059f40c2f7b31acdccb142461945402217398 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Description:

## Synopsis

```
AppCompatCacheParser [options]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 13 options parsed from the captured help text. The final column is filled in by review.

| Flag                 | Argument | What it does                                                                                                                                                             | When you would use it |
|----------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-f</code>      | f        | Full path to SYSTEM hive to process. If this option is not specified, the live Registry will be used                                                                     |                       |
| <code>--csv</code>   | csv      | Directory to save CSV formatted results to. Be sure to include the full path in double quotes                                                                            |                       |
| <code>--csvf</code>  | csvf     | File name to save CSV formatted results to. When present, overrides default name                                                                                         |                       |
| <code>-c</code>      | c        | The ControlSet to parse. Default is to extract all control sets [default: -1]                                                                                            |                       |
| <code>-t</code>      | —        | Sorts last modified timestamps in descending order                                                                                                                       |                       |
| <code>--dt</code>    | dt       | The custom date/time format to use when displaying time stamps. See <a href="https://goo.gl/CNVq0k">https://goo.gl/CNVq0k</a> for options [default: yyyy-MM-dd HH:mm:ss] |                       |
| <code>--nl</code>    | —        | When true, ignore transaction log files for dirty hives                                                                                                                  |                       |
| <code>--debug</code> | —        | Show debug information during processing                                                                                                                                 |                       |
| <code>--trace</code> | —        | Show trace information during processing                                                                                                                                 |                       |
| <code>-?</code>      | —        | Show help and usage information                                                                                                                                          |                       |
| <code>-h</code>      | —        | Show help and usage information                                                                                                                                          |                       |

| Flag                   | Argument | What it does                    | When you would use it |
|------------------------|----------|---------------------------------|-----------------------|
| <code>--help</code>    | —        | Show help and usage information |                       |
| <code>--version</code> | —        | Show version information        |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

[PECmd](#), [MFTECmd](#), [AmcacheParser](#)

# chainsaw

**Kit:** SIFT / Security Onion (Sigma-based log hunting) **Capability:** Parse Windows event logs **Version:** chainsaw 2.16.0 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Rapidly work with Forensic Artefacts

## Synopsis

```
chainsaw [OPTIONS] <COMMAND>
```

## Common invocations

```
# from cyberlab 58-eventlog-hunting
chainsaw hunt Security_sample.evtx -s sigma/ --mapping mapping.yml
```

## Options

All 6 options parsed from the captured help text. The final column is filled in by review.

| Flag          | Argument    | What it does                                   | When you would use it |
|---------------|-------------|------------------------------------------------|-----------------------|
| --no-banner   | —           | Hide Chainsaw's banner                         |                       |
| --num-threads | NUM_THREADS | Limit the thread number (default: num of CPUs) |                       |
| -h            | —           | Print help                                     |                       |
| --help        | —           | Print help                                     |                       |
| -V            | —           | Print version                                  |                       |
| --version     | —           | Print version                                  |                       |

## Gotchas

*TODO: operational traps.*

## See also

[evtxexport](#) , [evtxinfo](#) , [EvtxECmd](#) , [hayabusa](#)



# esedbexport

**Kit:** SIFT Workstation (libyal) **Capability:** Parse ESE / SRUM / Amcache databases **Captured:** cyberlab-aio via --help on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: esedbexport

## Synopsis

```
esedbexport [ -c codepage ] [ -l logfile ] [ -m mode ] [ -t target ]
[ -T table_name ] [ -hvV ] source
```

## Common invocations

```
# from cyberlab 06-windows-artifact-libs
esedbexport -V
# from cyberlab 06-windows-artifact-libs
esedbexport -t /tmp/edb_out exercise/Current.edb
```

## Options

All 8 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                                                                                                                             | When you would use it |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -c   | —        | codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254 windows-1255, windows-1256, windows-125 |                       |
| -h   | —        | shows this help                                                                                                                                                                                          |                       |
| -l   | —        | logs information about the exported items                                                                                                                                                                |                       |
| -m   | —        | export mode, option: all, tables (default) 'all' exports all the tables or a single specified table with indexes, 'tables' exports all the tables or a single specified table                            |                       |
| -t   | —        | specify the basename of the target directory to export to (default is the source filename) esedbexport will add the suffix .export to the basename                                                       |                       |

| Flag            | Argument | What it does                  | When you would use it |
|-----------------|----------|-------------------------------|-----------------------|
| <code>-T</code> | —        | exports only a specific table |                       |
| <code>-v</code> | —        | verbose output to stderr      |                       |
| <code>-V</code> | —        | print version                 |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

[esedbinfo](#) , [SrumECmd](#) , [AmcacheParser](#)

# esedbinfo

**Kit:** SIFT Workstation (libyal) **Capability:** Parse ESE / SRUM / Amcache databases **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: esedbinfo

## Synopsis

```
esedbinfo [ -hvV ] source
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 3 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does             | When you would use it |
|------|----------|--------------------------|-----------------------|
| -h   | —        | shows this help          |                       |
| -v   | —        | verbose output to stderr |                       |
| -V   | —        | print version            |                       |

## Gotchas

TODO: operational traps.

## See also

[esedbexport](#), [SrumECmd](#), [AmcacheParser](#)

# EvtxECmd

**Kit:** FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse Windows event logs **Version:** 2026.5.0+bfc7f47ccbf65ffc9a3777cde5498db2fdd94664 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Description:

## Synopsis

```
EvtxECmd [options]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 26 options parsed from the captured help text. The final column is filled in by review.

| Flag                 | Argument | What it does                                                                                                                                                                    | When you would use it |
|----------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-f</code>      | f        | File to process. Either this or -d is required                                                                                                                                  |                       |
| <code>-d</code>      | d        | Directory to recursively process. Either this or -f is required                                                                                                                 |                       |
| <code>--csv</code>   | csv      | Directory to save CSV formatted results to. Be sure to include the full path in double quotes                                                                                   |                       |
| <code>--csvf</code>  | csvf     | File name to save CSV formatted results to. When present, overrides default name                                                                                                |                       |
| <code>--json</code>  | json     | Directory to save JSON formatted results to. Be sure to include the full path in double quotes                                                                                  |                       |
| <code>--jsonf</code> | jsonf    | File name to save JSON formatted results to. When present, overrides default name                                                                                               |                       |
| <code>--xml</code>   | xml      | Directory to save XML formatted results to                                                                                                                                      |                       |
| <code>--xmlf</code>  | xmlf     | File name to save XML formatted results to. When present, overrides default name                                                                                                |                       |
| <code>--dt</code>    | dt       | The custom date/time format to use when displaying time stamps. See <a href="https://goo.gl/CNVq0k">https://goo.gl/CNVq0k</a> for options [default: yyyy-MM-dd HH:mm:ss.ffffff] |                       |
| <code>--inc</code>   | inc      | List of Event IDs to process. All others are ignored. Overrides --exc Format is 4624,4625,5410,5500-5600                                                                        |                       |

| Flag                   | Argument | What it does                                                                                                                                                                                              | When you would use it |
|------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--exc</code>     | exc      | List of Event IDs to IGNORE. All others are included. Format is 4624,4625,5410,5500-5600                                                                                                                  |                       |
| <code>--sd</code>      | sd       | Start date for including events (UTC). Anything OLDER than this is dropped. Format should match --dt                                                                                                      |                       |
| <code>--ed</code>      | ed       | End date for including events (UTC). Anything NEWER than this is dropped. Format should match --dt                                                                                                        |                       |
| <code>--fj</code>      | —        | When true, export all available data when using --json                                                                                                                                                    |                       |
| <code>--tdt</code>     | tdt      | The number of seconds to use for time discrepancy detection. Default is 1 [default: 1]                                                                                                                    |                       |
| <code>--met</code>     | —        | When true, show metrics about processed event log. Default is true                                                                                                                                        |                       |
| <code>--maps</code>    | maps     | The path where event maps are located. Defaults to 'Maps' folder where program was executed [default: /opt/ezttools/EvtxECmd/Maps]                                                                        |                       |
| <code>--vss</code>     | —        | Process all Volume Shadow Copies that exist on drive specified by -f or -d                                                                                                                                |                       |
| <code>--dedupe</code>  | —        | Deduplicate -f or -d & VSCs based on SHA-1. First file found wins                                                                                                                                         |                       |
| <code>--sync</code>    | —        | If true, the latest maps from <a href="https://github.com/EricZimmerman/evtx/tree/master/evtx/Maps">https://github.com/EricZimmerman/evtx/tree/master/evtx/Maps</a> are downloaded and local maps updated |                       |
| <code>--debug</code>   | —        | Show debug information during processing                                                                                                                                                                  |                       |
| <code>--trace</code>   | —        | Show trace information during processing                                                                                                                                                                  |                       |
| <code>-?</code>        | —        | Show help and usage information                                                                                                                                                                           |                       |
| <code>-h</code>        | —        | Show help and usage information                                                                                                                                                                           |                       |
| <code>--help</code>    | —        | Show help and usage information                                                                                                                                                                           |                       |
| <code>--version</code> | —        | Show version information                                                                                                                                                                                  |                       |

## Gotchas

*TODO: operational traps.*

## See also

`evtxexport`, `evtxinfo`, `chainsaw`, `hayabusa`

# evtxexport

**Kit:** SIFT Workstation (libyal) **Capability:** Parse Windows event logs **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: evtxexport

## Synopsis

```
evtxexport [ -c codepage ] [ -f format ] [ -l log_file ]
[ -m mode ] [ -p resource_files_path ]
[ -r registry_files_path ] [ -s system_file ]
[ -S software_file ] [ -t event_log_type ]
[ -hTvV ] source
```

## Common invocations

```
# from cyberlab 06-windows-artifact-libs
evtxexport -V
# from cyberlab 06-windows-artifact-libs
evtxexport -h
# from cyberlab 06-windows-artifact-libs
evtxexport -f text exercise/Security.evtx > /tmp/security_events.txt
```

## Options

All 13 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                                                                                                                             | When you would use it |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -c   | —        | codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255 |                       |
| -f   | —        | output format, options: xml, text (default)                                                                                                                                                              |                       |
| -h   | —        | shows this help                                                                                                                                                                                          |                       |

| Flag            | Argument | What it does                                                                                                                                                                                    | When you would use it |
|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-l</code> | —        | logs information about the exported items                                                                                                                                                       |                       |
| <code>-m</code> | —        | export mode, option: all, items (default), recovered 'all' exports the (allocated) items and recovered items, 'items' exports the (allocated) items and 'recovered' exports the recovered items |                       |
| <code>-p</code> | —        | search PATH for the resource files                                                                                                                                                              |                       |
| <code>-r</code> | —        | name of the directory containing the SOFTWARE and SYSTEM (Windows) Registry file                                                                                                                |                       |
| <code>-s</code> | —        | filename of the SYSTEM (Windows) Registry file. This option overrides the path provided by -r                                                                                                   |                       |
| <code>-S</code> | —        | filename of the SOFTWARE (Windows) Registry file. This option overrides the path provided by -r                                                                                                 |                       |
| <code>-t</code> | —        | event log type, options: application, security, system if not specified the event log type is determined based on the filename.                                                                 |                       |
| <code>-T</code> | —        | use event template definitions to parse the event record data                                                                                                                                   |                       |
| <code>-v</code> | —        | verbose output to stderr                                                                                                                                                                        |                       |
| <code>-V</code> | —        | print version                                                                                                                                                                                   |                       |

## Gotchas

*TODO: operational traps.*

## See also

[evtxinfo](#), [EvtxECmd](#), [chainsaw](#), [hayabusa](#)

# evtxinfo

**Kit:** SIFT Workstation (libyal) **Capability:** Parse Windows event logs **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: evtxinfo

## Synopsis

```
evtxinfo [ -c codepage ] [ -hvV ] source
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 4 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                                                                                                                             | When you would use it |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -c   | —        | codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255 |                       |
| -h   | —        | shows this help                                                                                                                                                                                          |                       |
| -v   | —        | verbose output to stderr                                                                                                                                                                                 |                       |
| -V   | —        | print version                                                                                                                                                                                            |                       |

## Gotchas

TODO: operational traps.

## See also



evtxexport , EvtxECmd , chainsaw , hayabusa

# hayabusa

---

**Kit:** SIFT / Security Onion (Sigma-based log hunting) **Capability:** Parse Windows event logs **Version:** error: unexpected argument '--version' found **Captured:** `cyberlab-aio` via `help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Hayabusa v3.9.0 - Showa Day Release

## Synopsis

---

```
hayabusa.exe <COMMAND> [OPTIONS]
hayabusa.exe help <COMMAND> or hayabusa.exe <COMMAND> -h
```

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

[evtxexport](#), [evtxinfo](#), [EvtxECmd](#), [chainsaw](#)

# hivexsh

---

**Kit:** Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

hivexsh: failed to open hive file: help: No such file or directory

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`rip.pl`, `regripper`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

# MFTECmd

**Kit:** FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+4fd94a6bd12237e8501baff5fc9e5b1b01c53862 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Description:

## Synopsis

```
MFTECmd [options]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 32 options parsed from the captured help text. The final column is filled in by review.

| Flag                 | Argument | What it does                                                                                                                                              | When you would use it |
|----------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-f</code>      | f        | File to process (\$MFT   \$J   \$Boot   \$SDS   \$I30). Required                                                                                          |                       |
| <code>-m</code>      | m        | \$MFT file to use when -f points to a \$J file (Use this to resolve parent path in \$J CSV output)                                                        |                       |
| <code>--csv</code>   | csv      | Directory to save CSV formatted results to. Be sure to include the full path in double quotes. This or --json required unless --de or --body is specified |                       |
| <code>--csvf</code>  | csvf     | File name to save CSV formatted results to. When present, overrides default name                                                                          |                       |
| <code>--json</code>  | json     | Directory to save JSON formatted results to. Be sure to include the full path in double quotes. This or --csv required unless --de or --body is specified |                       |
| <code>--jsonf</code> | jsonf    | File name to save JSON formatted results to. When present, overrides default name                                                                         |                       |
| <code>--body</code>  | body     | Directory to save bodyfile formatted results to. --bdl is also required when using this option                                                            |                       |
| <code>--bodyf</code> | bodyf    | File name to save body formatted results to. When present, overrides default name                                                                         |                       |
| <code>--bdl</code>   | bdl      | Drive letter (C, D, etc.) to use with bodyfile. Only the drive letter itself should be provided                                                           |                       |

| Flag                   | Argument | What it does                                                                                                                                                                    | When you would use it |
|------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--blf</code>     | —        | When true, use LF vs CRLF for newlines                                                                                                                                          |                       |
| <code>--dd</code>      | dd       | Directory to save exported \$MFT FILE record. --do is also required when using this option                                                                                      |                       |
| <code>--do</code>      | do       | Offset of the \$MFT FILE record to dump as decimal or hex. Ex: 5120 or 0x1400 Use --de or --debug to see offsets                                                                |                       |
| <code>--de</code>      | de       | Dump full details for \$MFT entry/sequence #. Format is 'Entry' or 'Entry-Seq' as decimal or hex. Example: 5, 624-5 or 0x270-0x5.                                               |                       |
| <code>--dr</code>      | —        | When true, dump \$MFT resident files to dir specified by --csv or --json, in 'Resident' subdirectory. Files will be named '--_bin'                                              |                       |
| <code>--fls</code>     | —        | When true, displays contents of directory from \$MFT specified by --de. Ignored when --de points to a file                                                                      |                       |
| <code>--ds</code>      | ds       | Dump full details for Security Id from \$SDS as decimal or hex. Example: 624 or 0x270                                                                                           |                       |
| <code>--dt</code>      | dt       | The custom date/time format to use when displaying time stamps. See <a href="https://goo.gl/CNVq0k">https://goo.gl/CNVq0k</a> for options [default: yyyy-MM-dd HH:mm:ss.ffffff] |                       |
| <code>--sn</code>      | —        | Include DOS file name types in \$MFT output                                                                                                                                     |                       |
| <code>--fl</code>      | —        | Generate condensed file listing of parsed \$MFT contents. Requires --csv                                                                                                        |                       |
| <code>--at</code>      | —        | When true, include all timestamps from 0x30 attribute vs only when they differ from 0x10 in the \$MFT                                                                           |                       |
| <code>--rs</code>      | —        | When true, recover slack space from FILE records when processing \$MFT files. This option has no effect for \$I30 files                                                         |                       |
| <code>--vss</code>     | —        | Process all Volume Shadow Copies that exist on drive specified by -f                                                                                                            |                       |
| <code>--dedupe</code>  | —        | Deduplicate -f & VSCs based on SHA-1. First file found wins                                                                                                                     |                       |
| <code>--ir</code>      | —        | Include resident data in JSON/CSV output                                                                                                                                        |                       |
| <code>--re</code>      | re       | Comma-separated list of extensions to include for resident data (e.g., '.txt,.ps1,.bat'). If omitted, includes all                                                              |                       |
| <code>--rm</code>      | rm       | Maximum size in bytes for resident data to include (max: 1024000) [default: 1024]                                                                                               |                       |
| <code>--debug</code>   | —        | Show debug information during processing                                                                                                                                        |                       |
| <code>--trace</code>   | —        | Show trace information during processing                                                                                                                                        |                       |
| <code>-?</code>        | —        | Show help and usage information                                                                                                                                                 |                       |
| <code>-h</code>        | —        | Show help and usage information                                                                                                                                                 |                       |
| <code>--help</code>    | —        | Show help and usage information                                                                                                                                                 |                       |
| <code>--version</code> | —        | Show version information                                                                                                                                                        |                       |

## Gotchas

*TODO: operational traps.*

# See also

---

[PECmd](#) , [AppCompatCacheParser](#) , [AmcacheParser](#)

# PECmd

**Kit:** FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+bde430c69ba4d97fea8b71fdddb6df7849419c10 **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

Description:

## Synopsis

```
PECmd [options]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 20 options parsed from the captured help text. The final column is filled in by review.

| Flag                 | Argument | What it does                                                                                                                                          | When you would use it |
|----------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-f</code>      | f        | File to process. Either this or -d is required                                                                                                        |                       |
| <code>-d</code>      | d        | Directory to recursively process. Either this or -f is required                                                                                       |                       |
| <code>-k</code>      | k        | Comma separated list of keywords to highlight in output. By default, 'temp' and 'tmp' are highlighted. Any additional keywords will be added to these |                       |
| <code>-o</code>      | o        | When specified, save prefetch file bytes to the given path. Useful to look at decompressed Win10 files                                                |                       |
| <code>-q</code>      | —        | Do not dump full details about each file processed. Speeds up processing when using -json or --csv                                                    |                       |
| <code>--csv</code>   | csv      | Directory to save CSV formatted results to. Be sure to include the full path in double quotes                                                         |                       |
| <code>--csvf</code>  | csvf     | File name to save CSV formatted results to. When present, overrides default name                                                                      |                       |
| <code>--json</code>  | json     | Directory to save JSON formatted results to. Be sure to include the full path in double quotes                                                        |                       |
| <code>--jsonf</code> | jsonf    | File name to save JSON formatted results to. When present, overrides default name                                                                     |                       |

| Flag                   | Argument | What it does                                                                                                                                                             | When you would use it |
|------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--html</code>    | html     | Directory to save xhtml formatted results to. Be sure to include the full path in double quotes                                                                          |                       |
| <code>--dt</code>      | dt       | The custom date/time format to use when displaying time stamps. See <a href="https://goo.gl/CNVq0k">https://goo.gl/CNVq0k</a> for options [default: yyyy-MM-dd HH:mm:ss] |                       |
| <code>--mp</code>      | —        | When true, display higher precision for timestamps                                                                                                                       |                       |
| <code>--vss</code>     | —        | Process all Volume Shadow Copies that exist on drive specified by -f or -d                                                                                               |                       |
| <code>--dedupe</code>  | —        | Deduplicate -f or -d & VSCs based on SHA-1. First file found wins                                                                                                        |                       |
| <code>--debug</code>   | —        | Show debug information during processing                                                                                                                                 |                       |
| <code>--trace</code>   | —        | Show trace information during processing                                                                                                                                 |                       |
| <code>-?</code>        | —        | Show help and usage information                                                                                                                                          |                       |
| <code>-h</code>        | —        | Show help and usage information                                                                                                                                          |                       |
| <code>--help</code>    | —        | Show help and usage information                                                                                                                                          |                       |
| <code>--version</code> | —        | Show version information                                                                                                                                                 |                       |

## Gotchas

*TODO: operational traps.*

## See also

[AppCompatCacheParser](#) , [MFTECmd](#) , [AmcacheParser](#)



# RECmd

**Kit:** FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse registry hives **Version:** 2026.5.0+bcd0ac33ed98de61ea6de551eef96052bddbbd49 **Captured:** cyberlab-aio via --help on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Description:

## Synopsis

```
RECmd [options]
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 33 options parsed from the captured help text. The final column is filled in by review.

| Flag     | Argument | What it does                                                                                   | When you would use it |
|----------|----------|------------------------------------------------------------------------------------------------|-----------------------|
| -f       | f        | Hive to search. -f or -d is required                                                           |                       |
| -d       | d        | Directory to look for hives (recursively). -f or -d is required                                |                       |
| --kn     | kn       | Display details for key name. Includes subkeys and values                                      |                       |
| --vn     | vn       | Value name. Only this value will be dumped                                                     |                       |
| --bn     | bn       | Use settings from supplied file to find keys/values. See included sample file for examples     |                       |
| --csv    | csv      | Directory to save CSV formatted results to. Be sure to include the full path in double quotes  |                       |
| --csvf   | csvf     | File name to save CSV formatted results to. When present, overrides default name               |                       |
| --saveTo | saveTo   | Saves --vn value data in binary form to file. Expects path to a FILE                           |                       |
| --json   | json     | Directory to save JSON formatted results to. Be sure to include the full path in double quotes |                       |
| --jsonf  | jsonf    | File name to save JSON formatted results to. When present, overrides default name              |                       |

| Flag                   | Argument | What it does                                                                                                                                                                                                                | When you would use it |
|------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>--details</code> | —        | Show more details when displaying results                                                                                                                                                                                   |                       |
| <code>--base64</code>  | base64   | Find Base64 encoded values with size >= Base64 (specified in bytes)                                                                                                                                                         |                       |
| <code>--minSize</code> | minSize  | Find values with data size >= MinSize (specified in bytes)                                                                                                                                                                  |                       |
| <code>--sa</code>      | sa       | Search for in keys, values, data, and slack                                                                                                                                                                                 |                       |
| <code>--sk</code>      | sk       | Search for in value record's key names                                                                                                                                                                                      |                       |
| <code>--sv</code>      | sv       | Search for in value record's value names                                                                                                                                                                                    |                       |
| <code>--sd</code>      | sd       | Search for in value record's value data                                                                                                                                                                                     |                       |
| <code>--ss</code>      | ss       | Search for in value record's value slack                                                                                                                                                                                    |                       |
| <code>--literal</code> | —        | If true, --sd and --ss search value will not be interpreted as ASCII or Unicode byte strings                                                                                                                                |                       |
| <code>--nd</code>      | —        | If true, do not show data when using --sd or --ss                                                                                                                                                                           |                       |
| <code>--regex</code>   | —        | If present, treat in --sk, --sv, --sd, and --ss as a regular expression                                                                                                                                                     |                       |
| <code>--dt</code>      | dt       | The custom date/time format to use when displaying time stamps. See <a href="https://goo.gl/CNVq0k">https://goo.gl/CNVq0k</a> for options [default: yyyy-MM-dd HH:mm:ss.ffffff]                                             |                       |
| <code>--nl</code>      | —        | When true, allow transaction log files to not exist for dirty hives                                                                                                                                                         |                       |
| <code>--recover</code> | —        | If true, recover deleted keys/values. Default is true                                                                                                                                                                       |                       |
| <code>--vss</code>     | —        | Process all Volume Shadow Copies that exist on drive specified by -f or -d                                                                                                                                                  |                       |
| <code>--dedupe</code>  | —        | Deduplicate -f or -d & VSCs based on SHA-1. First file found wins                                                                                                                                                           |                       |
| <code>--sync</code>    | —        | If true, the latest batch files from <a href="https://github.com/EricZimmerman/RECmd/tree/master/BatchExamples">https://github.com/EricZimmerman/RECmd/tree/master/BatchExamples</a> are downloaded and local files updated |                       |
| <code>--debug</code>   | —        | Show debug information during processing                                                                                                                                                                                    |                       |
| <code>--trace</code>   | —        | Show trace information during processing                                                                                                                                                                                    |                       |
| <code>-?</code>        | —        | Show help and usage information                                                                                                                                                                                             |                       |
| <code>-h</code>        | —        | Show help and usage information                                                                                                                                                                                             |                       |
| <code>--help</code>    | —        | Show help and usage information                                                                                                                                                                                             |                       |
| <code>--version</code> | —        | Show version information                                                                                                                                                                                                    |                       |

## Gotchas

*TODO: operational traps.*

## See also

---

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#)

# regfexport

**Kit:** SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: regfexport

## Synopsis

```
regfexport [ -c codepage ] [ -K key_path ] [ -l logfile ]
[ -hvV ] source
```

## Common invocations

```
# from cyberlab 04-registry-analysis
regfexport -V
# from cyberlab 04-registry-analysis
regfexport exercise/SYSTEM_sample.hive > /tmp/system_dump.txt
# from cyberlab 04-registry-analysis
regfexport exercise/SYSTEM_sample.hive | grep -i "ComputerName"
```

## Options

All 6 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                                                                                                                             | When you would use it |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -c   | —        | codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255 |                       |
| -h   | —        | shows this help                                                                                                                                                                                          |                       |
| -K   | —        | show information about a specific key path.                                                                                                                                                              |                       |
| -l   | —        | logs information about the exported items                                                                                                                                                                |                       |

| Flag            | Argument | What it does             | When you would use it |
|-----------------|----------|--------------------------|-----------------------|
| <code>-v</code> | —        | verbose output to stderr |                       |
| <code>-V</code> | —        | print version            |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`rip.pl`, `regripper`, `hivexsh`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

# regfinfo

**Kit:** SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: regfinfo

## Synopsis

```
regfinfo [ -B bodyfile ] [ -c codepage ] [ -hHvV ] source
```

## Common invocations

```
# from cyberlab 04-registry-analysis
regfinfo -V
# from cyberlab 04-registry-analysis
regfinfo exercise/SYSTEM_sample.hive
# from cyberlab 04-registry-analysis
regfinfo /tmp/SYSTEM_recovered.hive
```

## Options

All 6 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                                                                                                                             | When you would use it |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -B   | —        | output key and value hierarchy as a bodyfile                                                                                                                                                             |                       |
| -c   | —        | codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255 |                       |
| -h   | —        | shows this help                                                                                                                                                                                          |                       |
| -H   | —        | shows the key and value hierarchy                                                                                                                                                                        |                       |
| -v   | —        | verbose output to stderr                                                                                                                                                                                 |                       |

| Flag            | Argument       | What it does  | When you would use it |
|-----------------|----------------|---------------|-----------------------|
| <code>-V</code> | <code>—</code> | print version |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`rip.pl`, `regripper`, `hivexsh`, `regfexport`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

# regfmount

**Kit:** SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Invalid argument: regfmount

## Synopsis

```
regfmount [ -c codepage ] [ -X extended_options ] [ -hvV ] file
mount_point
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 5 options parsed from the captured help text. The final column is filled in by review.

| Flag | Argument | What it does                                                                                                                                                                                             | When you would use it |
|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -c   | —        | codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255 |                       |
| -h   | —        | shows this help                                                                                                                                                                                          |                       |
| -v   | —        | verbose output to stderr, while regfmount will remain running in the foreground                                                                                                                          |                       |
| -V   | —        | print version                                                                                                                                                                                            |                       |
| -X   | —        | extended options to pass to sub system                                                                                                                                                                   |                       |

## Gotchas

TODO: operational traps.



## See also

---

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#)

# regipy-diff

**Kit:** SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

TODO: one-line imperative purpose.

## Synopsis

```
regipy-diff [OPTIONS] FIRST_HIVE_PATH SECOND_HIVE_PATH
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 4 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                | When you would use it |
|-----------|----------|-----------------------------|-----------------------|
| -o        | FILE     | —                           |                       |
| -v        | —        | Verbosity                   |                       |
| --verbose | —        | Verbosity                   |                       |
| --help    | —        | Show this message and exit. |                       |

## Gotchas

TODO: operational traps.

## See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#)

# regipy-dump

**Kit:** SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

identified for some reason

## Synopsis

```
regipy-dump [OPTIONS] HIVE_PATH
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 18 options parsed from the captured help text. The final column is filled in by review.

| Flag                | Argument | What it does                                                                                                                                                      | When you would use it |
|---------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -o                  | FILE     | —                                                                                                                                                                 |                       |
| -p                  | TEXT     | A registry path to start iterating from                                                                                                                           |                       |
| --registry-path     | TEXT     | A registry path to start iterating from                                                                                                                           |                       |
| -t                  | —        | Create a CSV timeline instead                                                                                                                                     |                       |
| --timeline          | —        | Create a CSV timeline instead                                                                                                                                     |                       |
| -l                  | TEXT     | Specify a hive type, if it could not be identified for some reason                                                                                                |                       |
| --hive-type         | TEXT     | Specify a hive type, if it could not be identified for some reason                                                                                                |                       |
| -r                  | TEXT     | The path from which the partial hive actually starts, for example: -t ntuser - r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software |                       |
| --partial_hive_path | TEXT     | The path from which the partial hive actually starts, for example: -t ntuser - r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software |                       |
| -v                  | —        | Verbosity                                                                                                                                                         |                       |
| --verbose           | —        | Verbosity                                                                                                                                                         |                       |

| Flag                               | Argument | What it does                                                                                                | When you would use it |
|------------------------------------|----------|-------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-d</code>                    | —        | Not fetching the values for each subkey makes the iteration way faster. Values count will still be returned |                       |
| <code>--do-not-fetch-values</code> | —        | Not fetching the values for each subkey makes the iteration way faster. Values count will still be returned |                       |
| <code>-s</code>                    | TEXT     | If "-s" was specified, fetch only values for subkeys starting this timestamp in isoformat                   |                       |
| <code>--start-date</code>          | TEXT     | If "-s" was specified, fetch only values for subkeys starting this timestamp in isoformat                   |                       |
| <code>-e</code>                    | TEXT     | If "-e" was specified, fetch only values for subkeys until this timestamp in isoformat                      |                       |
| <code>--end-date</code>            | TEXT     | If "-e" was specified, fetch only values for subkeys until this timestamp in isoformat                      |                       |
| <code>--help</code>                | —        | Show this message and exit.                                                                                 |                       |

## Gotchas

*TODO: operational traps.*

## See also

`rip.pl`, `regripper`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-parse-header`, `regipy-plugins-run`

# regipy-parse-header

**Kit:** SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

TODO: one-line imperative purpose.

## Synopsis

```
regipy-parse-header [OPTIONS] HIVE_PATH
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 3 options parsed from the captured help text. The final column is filled in by review.

| Flag      | Argument | What it does                | When you would use it |
|-----------|----------|-----------------------------|-----------------------|
| -v        | —        | Verbosity                   |                       |
| --verbose | —        | Verbosity                   |                       |
| --help    | —        | Show this message and exit. |                       |

## Gotchas

TODO: operational traps.

## See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-plugins-run](#)

# regipy-plugins-run

**Kit:** SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

identified for some reason

## Synopsis

```
regipy-plugins-run [OPTIONS] HIVE_PATH
```

## Common invocations

TODO: up to 8 task-titled invocations.

## Options

All 11 options parsed from the captured help text. The final column is filled in by review.

| Flag                  | Argument | What it does                                                                                                                                                     | When you would use it |
|-----------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| -o                    | FILE     | Output path for plugins result [required]                                                                                                                        |                       |
| -p                    | TEXT     | A plugin or list of plugins to execute command separated                                                                                                         |                       |
| --plugins             | TEXT     | A plugin or list of plugins to execute command separated                                                                                                         |                       |
| -t                    | TEXT     | Specify a hive type, if it could not be identified for some reason                                                                                               |                       |
| --hive-type           | TEXT     | Specify a hive type, if it could not be identified for some reason                                                                                               |                       |
| -r                    | TEXT     | The path from which the partial hive actually starts, for example: -t ntuser -r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software |                       |
| --partial_hive_path   | TEXT     | The path from which the partial hive actually starts, for example: -t ntuser -r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software |                       |
| -v                    | —        | Verbosity                                                                                                                                                        |                       |
| --verbose             | —        | Verbosity                                                                                                                                                        |                       |
| --include-unvalidated | —        | Include plugins that don't have validation test cases. These plugins may return incomplete or inaccurate data. Use at your own risk.                             |                       |

| Flag                | Argument | What it does                | When you would use it |
|---------------------|----------|-----------------------------|-----------------------|
| <code>--help</code> | —        | Show this message and exit. |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`rip.pl`, `regripper`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`

# regripper

---

**Kit:** Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Rip v.3.0 - CLI RegRipper tool

## Common invocations

---

*TODO: up to 8 task-titled invocations.*

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`rip.pl`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`



# rip.pl

---

**Kit:** Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `--help` on 2026-08-01 [raw](#)

[← Capability index](#) · [Kit tool list](#)

## Purpose

---

Rip v.3.0 - CLI RegRipper tool

## Common invocations

---

```
# from cyberlab 04-registry-analysis
rip.pl -h
# from cyberlab 04-registry-analysis
rip.pl -r exercise/SYSTEM_sample.hive -p compname
# from cyberlab 49-intrusion-timeline-case
rip.pl -h | head -n 3
# from cyberlab 49-intrusion-timeline-case
rip.pl -r NTUSER.DAT -p run
# from cyberlab 49-intrusion-timeline-case
rip.pl -r SYSTEM -p usbstor
```

## Options

---

*No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.*

## Gotchas

---

*TODO: operational traps.*

## See also

---

`regripper`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

# SrumECmd

**Kit:** FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse ESE / SRUM / Amcache databases **Version:** 2026.5.0+880ad26bcb011976a8fc521eea63fc5e6e65ba02 **Captured:** cyberlab-aio via --help on 2026-08-01 raw

← [Capability index](#) · [Kit tool list](#)

## Purpose

Description:

## Synopsis

```
SrumECmd [options]
```

## Common invocations

*TODO: up to 8 task-titled invocations.*

## Options

All 11 options parsed from the captured help text. The final column is filled in by review.

| Flag                 | Argument | What it does                                                                                                                                                             | When you would use it |
|----------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>-f</code>      | f        | SRUDB.dat file to parse                                                                                                                                                  |                       |
| <code>-r</code>      | r        | SOFTWARE hive to process. This is optional, but recommended                                                                                                              |                       |
| <code>-d</code>      | d        | Directory to recursively process, looking for SRUDB.dat and SOFTWARE hive. This mode is primarily used with KAPE so both SRUDB.dat and SOFTWARE hive can be located      |                       |
| <code>--csv</code>   | csv      | Directory to save CSV formatted results to. Be sure to include the full path in double quotes                                                                            |                       |
| <code>--dt</code>    | dt       | The custom date/time format to use when displaying time stamps. See <a href="https://goo.gl/CNVq0k">https://goo.gl/CNVq0k</a> for options [default: yyyy-MM-dd HH:mm:ss] |                       |
| <code>--debug</code> | —        | Show debug information during processing                                                                                                                                 |                       |
| <code>--trace</code> | —        | Show trace information during processing                                                                                                                                 |                       |
| <code>-?</code>      | —        | Show help and usage information                                                                                                                                          |                       |
| <code>-h</code>      | —        | Show help and usage information                                                                                                                                          |                       |
| <code>--help</code>  | —        | Show help and usage information                                                                                                                                          |                       |

| Flag                   | Argument | What it does             | When you would use it |
|------------------------|----------|--------------------------|-----------------------|
| <code>--version</code> | —        | Show version information |                       |

## Gotchas

---

*TODO: operational traps.*

## See also

---

`esedbexport`, `esedbinfo`, `AmcacheParser`

# Alphabetical Tool Index

---

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